Request for Proposal for Selection of System Integrator for Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C)

Issued By:

Chhattisgarh Infotech Promotion Society

(CHiPS)

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1. Disclaimer

Chhattisgarh infotech Promotion Society (CHiPS) intend to implement BharatNet Phase-II Project. This document has been prepared on the basis of available information in CHiPS and other publicly available documents which CHiPS believes to be reliable. The sole objective of this document (the Request for Proposal or the RFP) is to solicit Technical and financial bids from interested parties for taking part in the future process leading to Selection of System Integrator for Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C).

While this document has been prepared in good faith, no representation or warranty, express or implied, is or shall be made, and no responsibility or liability shall be accepted by CHiPS or any of their employees, consultants, advisors or agents as to or in relation to the accuracy or completeness of this document and any liability thereof is hereby expressly disclaimed. Interested Parties may carry out their own study/ analysis/ investigation as required before submitting their Technical and Commercial bids.

This document does not constitute an offer or invitation, or solicitation of an offer, nor does this document or anything contained herein, shall form a basis of any agreement or commitment whatsoever.

Some of the activities listed to be carried out by CHiPS subsequent to the receipt of the responses are indicative only. CHiPS has the right to continue with these activities, modify the sequence of activities, add new activities or remove some of the activities, as dictated by the best interests of CHiPS.

2. Glossary of Terms

Abbreviation	Description
ABD	As-Build Diagram
BSNL	Bharat Sanchar Nigam Limited
BG	Bank Guarantee
Bidder	Sole Bidder
BoQ	Bill of Quantity
CHiPS	Chhattisgarh infotech Promotion Society
DD	Demand Draft
DoT	Department of Telecommunication
EMD	Earnest Money Deposit
FAT	Final Acceptance Test
FDMS	Fibre Distribution Management System
GoI	Government of India
GP	Gram Panchayat
HDPE	High-Density Polyethylene
LoI	Letter of Intent
MAF	Manufacturer Authorization Form
MoU	Memorandum of Understanding
MPLS	Multiprotocol Label Switching
SI	System Integrator
NDA	Non-Disclosure Agreement
NIT	Notice Inviting Tender
SNOC	State Network Operation Centre
O&M	Operations & Maintenance
OEM	Original Equipment Manufacturer
OFC	Optical Fibre Cable
OTDR	Optical Time-Domain Reflectometer
PBH	Primary Business Hour
PLB	Permanently Lubricated
PMA	Preferential Market Access
PoP	Point of Presence
RCC	Reinforced Concrete Cement
RF	Radio Frequency
RFP	Request for Proposal
RoW	Right of Way
SLA	Service Level Agreement
SoR	Schedule of Requirement
TAC	Type Approval Certificate
TEC GR	Telecommunication Engineering Centre Generic Requirements
TSEC	Technical Specification Evaluation Certificate
DBN	Digital Bharat Nidhi (Earlier Known as Universal Service Obligation Fund (USOF))
VPN	Virtual Private Network

Table of Contents

1. Notice Inviting Tender

Chhattisgarh infotech Promotion Society (CHiPS), having its Registered Office at SDC Building, 02nd Floor, Near Civil Lines Police Control Room, Raipur, Chhattisgarh– 492001, invites responses ("Proposals"/ "Bids") to this Request for Proposal ("RFP") from eligible Bidders to be Selected as System Integrator for Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C).

Interested bidders are advised to study this RFP carefully before submitting their proposals in response to the RFP. Submission of a proposal in response to this RFP shall be deemed to have been done after careful study and examination of this document with full understanding of its terms, conditions and implications.

Interested bidders may download the RFP from the website URL mentioned in the Fact Sheet. Any subsequent corrigenda/ clarifications shall also be made available on the website URL mentioned in the fact sheet.

Proposals must be received not later than time and date mentioned in the Fact Sheet. Proposals that are received after the deadline **WILL NOT** be considered in this procurement process.

To obtain first-hand information on the assignment, Bidders are encouraged to attend a pre-bid meeting. Attending the pre-bid meeting is optional.

2. Fact Sheet

Item	Description
Tender Issued By	Chhattisgarh Infotech Promotion Society (CHiPS)
Name of the Project Work	Selection of System Integrator for Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C).
Tender Reference No.	1630/CEO/CHiPS/BHARATNET/O&M/2025
Place of availability of Tender Documents (RFP)	https://eproc.cgstate.gov.in https://www.chips.gov.in
Tender Document	Request for Proposal
Date of Issue of Tender	**15th December 2025 9:00 PM**
Tender Type (Open/ Limited/ EOI/ Auction/ Single)	Open
Tender Category (Services/ Goods/ works)	Goods & Services
Type/ Form of Contract (Work/ Supply/ Auction/ Service/ Buy/ Empanelment/ Sell)	Supply/ Service/ Buy
Is Offline Submission Allowed (Yes/No)	No
Withdrawal Allowed (Yes/ No)	Yes (on or before the last date and time of bid submission)
Bid Validity days	180 days
Cost of Tender Document	**Rupees Five Thousand only (INR 5,000/-)** To be paid online through e-Procurement portal (https://eproc.cgstate.gov.in)
Earnest Money Deposit (EMD)	**Package-A: INR 90,00,000/- (Rupees Ninety Lakh only)** **Package-B: INR 60,00,000/- (Rupees Sixty Lakh only)** **Package-C: INR 37,00,000/- (Rupees Thirty-Seven Lakh only)** **Package-D: INR 23,00,000/- (Rupees Twenty Three Lakh only)** EMD may be submitted in the form of Bank Guarantee (BG) as per format mentioned in the RFP on stamp paper of value required under law duly signed by authorized representative of Bank: 1) Scan copy of BG should be uploaded in e-Procurement portal along with actual online bid submission.

	2) Original copy of BG should be submitted to CHiPS office before 05:00 PM on the last date of bid submission
Address to send Pre-bid Queries	The Chief Executive Officer Office of CHiPS, SDC Building, 02nd floor, Near Police Control Room, Civil Lines, Raipur, Chhattisgarh–492001 Phone: 0771-4014158 FAX No.: 0771-4066205 Website: www.chips.gov.in Email: ceochips@nic.in
Nature of Bid Process	Two stage bidding in two envelopes **Envelope/File – I** 1.1 EMD 1.2 Pre qualification/Technical Bid **Envelope/File – II** (To Be Filled online Over E-Proc Only) 2.1 Commercial Bid
Method of Selection	Package wise Lowest Cost (L1) - each bidder can quote for one or more packages. Based on the lowest cost quoted by the bidder only one package will be awarded to the bidder
Last Date for Submission of written queries by bidders	24/12/2025 by 05:00 PM (Queries received in writing or email in the format as per **Annexure-1**, by last date for submission of queries shall be discussed during the pre-bid meeting)
Date of Pre-Bid Meeting	24/12/2025 at 12:00 PM
Place for Pre-bid Meeting	CHiPS, SDC Building, 02nd floor, Near Police Control Room, Civil Lines, Raipur, Chhattisgarh–492001
Last date and time for Submission of Bids	15/01/2026 at 05:00 PM
Last date and time for Submission of Original copy of BG for EMD	16/01/2026 at 05:00 PM
Opening of Technical Bid	16/01/2026 at 06:00 PM
Opening of Commercial Bids	To be informed later through e-Procurement Portal or mail and telephone (Bidder should furnish the mobile number and e-mail of one authorized representative)
Address for Communication	The Chief Executive Officer CHiPS, SDC Building, 02nd floor, Near Police Control Room, Civil Lines, Raipur, Chhattisgarh–492001 Phone: 0771-4014158 FAX No.: 0771-4066205 Website: www.chips.gov.in Email: ceochips@nic.in

3. General

3.1 About CHiPS

CHiPS, a Registered Society promoted by the Government of Chhattisgarh, is the nodal agency and prime mover for propelling IT growth and implementation of IT plans in the State. The Hon'ble Chief Minister heads the High-Powered Governing Council of CHiPS. It includes Minister for Finance & Commercial Taxes, Minister for Commerce & Industry, Minister for Education, and Minister for Panchayat & Rural Development, Chief Secretary, representative from the Ministry of Information Technology in Government of India and eminent persons from IT industry.

CHiPS is involved as State Designated Agency (SDA) in NeGP MMP's implementation of some mega IT Projects like e-Procurement, SDC, SWAN, Bharatnet, e-District, GIS and CSCs. A professional approach is being adopted for the implementation of IT Projects using the services of e-governance experts and consultants from corporate and academia.

3.2 About Chhattisgarh

Chhattisgarh, a 21st century State, came into being on November 1, 2000. Good Governance is the highest priority in this Fast Track State. There is both policy stability as well as political stability. Chhattisgarh is truly a land of opportunities. With all major minerals including diamonds in abundance, it is the richest State in mineral resources. There are mega industries in Steel, Aluminium and Cement. Chhattisgarh contributes substantially to the Human Resources of India. Several hundred students from the State qualify for admissions in prestigious academic institutions every year.

It has large power surplus is attracting power-intensive industries, and the State is poised to become the power-hub of the nation. Its central location helps easy power transmission to any part of the country. 12% of India's forests are in Chhattisgarh, and 44% of the State's land is under forests. Identified as one of the richest bio-diversity habitats, the Green State of Chhattisgarh has the densest forests in India, rich wildlife, and above all, over 200 non-timber forest produces, with tremendous potential for value addition. One third of Chhattisgarh's population is of tribes, mostly in the thickly forested areas in the North and South. The central plains of Chhattisgarh are known as the ―Rice Bowl‖ of Central India. Female literacy has doubled in the last decade, and male literacy is higher than India's average. Gender ratio is next only to Kerala. Bastar is known the world over for its unique and distinctive tribal heritage. The Bastar Dashera is the traditional celebration of the gaiety of tribal. Many virgin, unexplored tourism destinations are there is all the parts of Chhattisgarh.

Chhattisgarh is the 29th state of India, formed on 1 November 2001. With an area of 135,194 sq. km (52,199 sq. miles), it is the 10th largest state in India by area and with a population of around 25.541 million, Chhattisgarh is the 17th most-populated state in the country. Chhattisgarh is a resource-rich state; it is a source of electricity and steel for the country, accounting for 15% of the total steel produced in India. The recorded forest area in the state is 59,722 sq.km which is around 44.21% of its geographical coverage.

3.3 About BharatNet Phase-II

The BharatNet Phase-II project was initially envisaged in 2018 to be executed under the State-Led Model in Chhattisgarh. The scope included connectivity to 5,964 Gram

Panchayats (GPs) across 83 Blocks. As of now, connectivity has been established and integrated for 5,540 GPs have been integrated with the State Network Operations Centre (S-NOC), and Optical Fibre Cable (OFC) laying is to be completed in the remaining 424 GPs.

The project adopts a ring architecture based on Internet Protocol – Multi Protocol Label Switching (IP-MPLS) technology, enabling scalable bandwidth delivery to households, institutions, and enterprises.

Under this tender, the selected System Integrator (SI) will be responsible for:

- Implementation of OFC connectivity including supply, installation, integration, commissioning and testing of underground OFC through IP-MPLS network.
- Operations & Maintenance (O&M) of the BharatNet Phase-II infrastructure already integrated into the network, starting immediately after assessment and restoration by the authorized third-party auditor.
- O&M for the newly implemented will commence after successful end-to-end integration and verification by CHiPS.
- The SI shall ensure complete end-to-end connectivity, monitoring, and service provisioning for the created network throughout the contract period. For Gram Panchayats (GPs) where implementation has not yet commenced, the SI shall complete the implementation within six (6) months from the date of signing the agreement by the Successful Bidder. The contract period shall include six (6) months for implementation and one (1) year of Operations & Maintenance (O&M), which is extendable by an additional one (1) year based on performance and requirements.
- The implementation of new and the O&M of GPs whose network has already been created shall be carried out in parallel to ensure timely completion and uninterrupted service delivery.
- The package wise details of GPs under BharatNet Phase II project is available under Annexure 2.

#	Package	Cluster	District	No of Blocks	Scope of GP's	S-NOC Scope Applicable	Existing O&M GP's	Nos GP’s to be Implemented
1	Package-A	A	10	32	2827	Yes	2684	143
2	Package-B	B	6	21	1729	No	1551	178
3	Package-C	C	7	18	718	No	678	40
4	Package-D	D	5	12	690	No	627	63
Total			**28**	**83**	**5964**		**5540**	**424**

Cluster Wise Division

- The cluster wise districts location is tabulated as below-

Cluster A	Cluster B	Cluster C	Cluster D
Balod	Bilaspur	Bastar	Jashpur
Balodabazar	Janjgir–Champa	Bijapur	Koriya
Bemetara	Korba	Dantewada	Surajpur
Dhamtari	Mungeli	Kondagaon	Surguja
Durg	Raigarh	Narayanpur	Manendragarh-Chirmiri-Bharatpur (MCB)
Gariabandh	Gaurela-Pendra-Marwahi	Sukma	
Kabirdham	Sakti	Kanker	----
Mahasamund	Sarangarh-Bhilaigarh	----	----
Raipur	---	----	----
Rajnandgaon	---	----	----
Mohla-Manpur - Ambagarh Chowki			

- The List of Gram Panchayats is provided in Annexure 2.

4. Instructions to the Bidders

4.1 General

4.1.1 While every effort has been made to provide comprehensive and accurate background Information, requirements and specifications, Bidders must form their own conclusions about the services required. Bidders and recipients of this TENDER may wish to consult their own legal advisers in relation to this TENDER.s

4.1.2 All information supplied by Bidders may be treated as contractually binding on the Bidders, on successful award of the assignment by the CHiPS on the basis of this TENDER.

4.1.3 No commitment of any kind, contractual or otherwise shall exist unless and until a formal written contract has been executed by or on behalf of the CHiPS. Any notification of preferred bidder status by the CHiPS shall not give rise to any enforceable rights by the Bidder. CHiPS may cancel this public procurement at any time prior to a formal written contract being executed by or on behalf of the CHiPS.

4.1.4 This TENDER supersedes and replaces any previous public documentation & Communications, and Bidders should place no reliance on such communications.

4.1.5 Please refer Annexure-3 for e-procurement guidelines.

4.2 Compliant Tenders/Completeness of Response

4.2.1 Bidders are advised to study all instructions, forms, terms, requirements, appendices and other information in the TENDER documents carefully. Online submission of the bid/ proposal shall be deemed to have been done after careful study and examination of the TENDER document with full understanding of its implications.

4.2.2 Prior to evaluation of Bids, CHiPS shall determine whether each Bid is responsive to the requirements of this RFP. A Bid shall be considered responsive if:

a) It is received as per the relevant format set out under this RFP.

b) It is received by the Bid Due Date including any extension thereof pursuant to the terms of this RFP.

c) It is accompanied by the EMD as specified in Clause 4.17.

d) It is accompanied by the power(s) of attorney or board resolution as relevant as per the terms of this RFP.

e) It contains all the information (complete in all respects) as requested in this RFP and/or Bidding Documents (in formats same as those specified).

f) It does not contain any condition or qualification limiting or affecting the manner or substance of performance of obligations of the Bidder under this RFP, unless specifically required as per the terms of this RFP.

g) If a Bidder imposes conditions, which is in addition to or in conflict with the conditions mentioned herein, its Bid shall be liable to be summarily rejected. In any case none of such conditions will be deemed to have been accepted unless specifically mentioned in the LOA of the Bid issued by CHiPS.

h) It does not contain any suppression of information or facts as required to be furnished by the Bidder under this RFP or as may be relevant to be furnished by the Bidder in relation to the Project.

i) It does not comprise any incomplete information, subjective / conditional / contingent / partial offers.

j) Validity period of Bid is 180 (one hundred and eighty days);

k) It is not non-responsive as per any other terms of this RFP.

4.2.3 CHiPS reserves the right to reject any Bid which is non-responsive and no request for alteration, modification, substitution or withdrawal shall be entertained by CHiPS in respect of such Bid. Provided, however, that CHiPS may, in its discretion, allow the Bidder to rectify any infirmities or omissions if the same do not constitute a material modification of the Bid, subject to its written approval and confirmation in this regard.

4.2.4 Failure to comply with the requirements of this paragraph may render the Proposal non-compliant and the Proposal may be rejected. Bidders must:

- Comply with all requirements asset out within this TENDER.
- Include all supporting documentations specified in this TENDER

4.3 Procedure for Submission of Bids

4.3.1 Bids (Technical bid & Commercial bid) shall be submitted online on website https://eproc.cgstate.gov.in

4.3.2 The participating Bidders in the tender should register themselves on e-procurement portal.

4.3.3 The Bidders can login to e-procurement portal in secure mode only by signing through the Digital certificates.

4.3.4 The Bidders should scan and upload the respective documentary evidence as mentioned in Eligibility Criteria.

4.3.5 The bidders shall sign on all the statements, documents, certificates uploaded by them, owning responsibility for their correctness/authenticity.

4.3.6 The rates should only be filled online whose commercial bid format is as per Annexure – 12

4.4 Other Conditions of Bid Submission

4.4.1 After uploading the documents online, original copy of Bank Guarantee/ EMD shall be submitted by the bidder in sealed envelope to the CHiPS office as per date & time mentioned in the fact sheet. The cover thus prepared should also clearly indicate the name, address, telephone number, E-mail ID and fax number of the Bidder to enable the Bid to be returned unopened in case it is declared "Late".

4.4.2 Failure to furnish original copy of Bank Guarantee in respect of EMD shall result in rejection of the bid. CHiPS shall not be liable for postal delay. Similarly, if any of the certificates, documents, etc., uploaded by the Bidder online are found to be false/fabricated/bogus, the Bidder shall be disqualified, blacklisted and action shall be initiated as deemed fit and the Bid Security shall be forfeited.

4.4.3 CHiPS shall not hold any risk and responsibility regarding non-clarity / non-legibility of the scanned and uploaded documents.

4.4.4 The documents that are uploaded online on the portal shall only be considered for Evaluation of bids.

4.4.5 In-case of any discrepancy found between the documents submitted online and as hardcopy to CHiPS as part of the bid response, the documents submitted online shall be considered for evaluation.

4.5 Cost to Bid

4.5.1 The Bidder shall bear all costs associated with the preparation and submission of its bid, including cost of presentation for the purposes of clarification of the bid, if so desired by the CHiPS. CHiPS shall in no case be responsible or liable for those costs, regardless of the conduct or outcome of the Tendering process.

4.6 Contents of the RFP

4.6.1 The Bidder is expected to examine all Sections, Annexures & corrigenda in the RFP and furnish all information as stipulated therein

4.7 Clarification on RFP

4.7.1 A prospective Bidder requiring any clarification on the RFP may submit queries, in writing, at the Purchaser's mailing address and email ID mentioned in Fact Sheet. Queries must be submitted in the format mentioned in Annexure-1.

4.7.2 The Purchaser shall not respond to any queries not adhering as per the format mentioned in Annexure-1.

4.7.3 All queries on the RFP should be received in Spreadsheet format (.xls) only on or before the last date as mentioned by the Purchaser in Fact Sheet.

4.8 Responses to Pre-Bid Queries and Issue of Corrigendum

4.8.1 CHiPS at its sole discretion may provide response to the queries. However, CHiPS makes no representation or warranty as to the completeness or accuracy of any response made in good faith, nor does CHiPS undertake to answer all the queries that have been posed by the bidders.

4.8.2 At any time prior to the last date for receipt of bids, CHiPS may, for any reason, whether at its own initiative or in response to a clarification requested by a prospective Bidder, modify the TENDER Document by a corrigendum.

4.8.3 The Corrigendum (if any) & clarifications to the queries from all bidders will be posted on the CHiPS website www.chips.gov.in and CG-e-procurement portal https://eproc.cgstate.gov.in.

4.8.4 Any such corrigendum shall be deemed to be incorporated into this tender.

4.8.5 In order to provide prospective bidders reasonable time for taking the corrigendum into account, CHiPS may, at its discretion, extend the last date for the receipt of bids.

4.8.6 Verbal clarifications and information given by CHiPS, or its employees or representatives shall not in any way or manner be binding on CHiPS.

4.9 Amendment of RFP

4.9.1 At any time prior to the last date for receipt of bids, the purchaser, may, for any reason, whether at its own initiative or in response to a clarification requested by a prospective Bidder, modify the RFP by an amendment. The amendment shall be notified on CHiPS website www.chips.gov.in and CG-e-procurement portal https://eproc.cgstate.gov.in and should be taken into consideration by the prospective agencies while preparing their bids.

4.9.2 In order to provide prospective Bidders reasonable time to take the amendment into account to prepare their bids, the Purchaser may, at its discretion, extend the last date for the receipt of bids.

4.10 Language of Bids

4.10.1 The Bids prepared by the Bidder, all correspondence and documents relating to the bids exchanged by the Bidder and the Purchaser, shall be written in English language, provided that any printed literature furnished by the Bidder may be written in another language so long the same is accompanied by an English translation in which case, for purposes of interpretation of the bid, the English translation shall govern

4.11 Documents Comprising the Bids

4.11.1 The bid prepared by the Bidder shall comprise of the following components:

4.11.2 **Prequalification/ Technical Bid** - The Technical bid shall comprise of the following:

- EMD in the form of Refundable & Irrevocable Bank Guarantee
- Annexure-4: Technical Bid Cover Letter (Company Letter head)
- Annexure-5: Technical Bid Compliance Checklist with all supporting documents required for eligibility
- Annexure-6: Certificate for No Conflict of Interest
- Annexure-7: Power of Attorney executed in favor of the Authorized Signatory OR Board Resolution in grating the requisite authority in favor of the person granting power of attorney to the authorised signatory
- Annexure-8: Undertaking for not being Blacklisted or debarred
- Annexure-9: Schedule of Requirements
- Annexure-10: Pre-Contract Integrity Pact

4.11.3 **Commercial Bid** - The Commercial Bid shall comprise of the following:

- Annexure-11: Commercial Bid Cover Letter (Company Letter head)
- Annexure-12: Commercial Bid Format

4.12 Procedure for Submission of Bids

4.12.1 The bid prepared by the Bidder shall comprise of the following cover (to be uploaded at e-procurement portal as individual files):

4.12.2 **Envelope / File - I (Pre Qualification/ Technical) to be filled online:**

- Envelope / File – I: shall comprise of all the documents (in PDF format):
  - Scanned copy of EMD in the form of 'Refundable & Irrevocable Bank Guarantee'
  - Documents mentioned in Clause 4.11.1
- Bidder shall upload these documents on or before the last date of bid submission as mentioned in Fact Sheet
- Bidder shall submit original copy of EMD in the form of Refundable & Irrevocable Bank Guarantee', on or before the last date of bid submission as mentioned in Fact Sheet. The address of CHiPS, name and address of the bidder and the Tender Reference Number shall be marked on the envelope. The envelope shall also be marked with a sentence "NOT TO BE OPENED BEFORE the Date and Time of Bid Opening". If the envelope is not marked

as specified above, CHiPS shall not assume any responsibility for its misplacement, pre-mature opening etc.

4.12.3 **Envelope / File - II (Commercial Bid) to be filled online: -**

- **Envelope / File - II** shall comprise of all the documents (in PDF format) mentioned in Clause 4.11.3 and uploaded on the e-proc portal
- All prices under the Commercial Bid should be in Indian National Rupee (INR) and should comply with all the rules and regulation of RBI, and the relevant Government instrumentality at their own cost. In the event of any discrepancy between the Commercial Bid quoted in words and in figures, words shall prevail.
- CHiPS may at its sole discretion and in accordance with the terms of this RFP, declare the Bidder quoting the lowest Price Bid as the lowest bidder ("L1 Bidder"). CHiPS shall not be responsible for any erroneous calculation of Taxes. Any upward/downward revision of GST shall be applicable at the time of invoicing. However, to arrive at the bid value of the respective Bid, Bidder has to quote the charges inclusive of all taxes, duties, levies etc. excluding GST. The Commercial Bid shall be inclusive of all freight, forwarding, transit insurance and installation charges.
- CHiPS shall reserve the right to negotiate with the L1 Bidder for further reduction of the price.
- The price quoted by the bidder excluding GST as per the commercial format shall be considered for deciding the L1 price
- In the event of any difference between figures and words, the amount indicated in words in the Price Bid shall be taken into account.

Note: *Prices should not be indicated/ mentioned in the Technical Bid but should only be mentioned in the Commercial Bid online only.*

4.12.4 The Bidder shall submit only one (1) bid in response to the RFP. If the Bidder submits more than one bid, it shall lead to disqualification of bidder and shall also cause the rejection of all the bids which such Bidder has submitted.

4.13 Bid Prices

4.13.1 The Bidder shall indicate in the format prescribed, the unit rates and total Bid Price of the equipment / services, it proposes to provide under the Contract. Prices shall only be filled online as per template provided on e-procurement portal (indicative BOQ (Bill of Quantity and relative calculations are in Annexure-12: Commercial Bid Format)

4.13.2 Bidder to ensure that Package wise quote is submitted for one or more package that bidder wants to bid.

4.13.3 In absence of information requested in above Clause, a bid may be considered incomplete and be summarily rejected.

4.13.4 The Bidder shall prepare the bid based on details provided in the RFP. It must be clearly understood that the Scope of Work is intended to give the Bidder an idea about the order and magnitude of the work and is not in any way exhaustive and guaranteed by the Purchaser. The Bidder shall carry out all the tasks in accordance with the requirement of the RFP and it shall be the responsibility of the Bidder to fully meet all the requirements of the RFP.

4.14 Firm Prices

4.14.1 Prices quoted in the bid must be firm and final and shall not be subject to any upward modifications, on any account whatsoever. However, the Purchaser reserves the right to negotiate the prices quoted by the SI to effect downward modification. The Bid Prices shall be indicated in Indian Rupees (INR) only.

4.14.2 The Commercial bid should clearly indicate the price to be charged and the applicable Taxes as per actuals. It is mandatory that such charges wherever applicable/ payable should be indicated separately. However, should there be a change in the upward or downward revision in GST, the same may apply.

4.15 Discount

4.15.1 The Bidders are advised not to indicate any separate discount in the Commercial Bid. Discount, if any, should be merged with the quoted prices. Discount of any type, indicated separately, shall not be taken into account for evaluation purpose. However, in the event of such an offer is found to be the lowest without taking into account the discount, the Purchaser shall avail such discount at the time of award of Contract.

4.16 Bidder Qualification

4.16.1 The "Bidder" as used in the RFP shall mean the one who has signed the Tender Form. The Bidder may be either the Principal Officer or his duly Authorized Representative, in either case he/ she shall submit a certificate of authority. All certificates and documents (including any clarifications sought and any subsequent correspondences) received hereby, shall be furnished and signed by the authorized representative and the Principal Officer.

4.16.2 It is further clarified that the individual signing the tender or other documents in connection with the tender must certify whether he/ she signs as the Constituted attorney of the firm, or a company.

4.16.3 The authorization shall be indicated by written Power-of-Attorney accompanying the bid.

4.16.4 The power or authorization and any other document consisting of adequate proof of the ability of the signatory to bind the Bidder shall be annexed to the bid.

4.16.5 Any change in the Principal Officer or his duly Authorized Representative shall be intimated to CHiPS in advance.

4.17 Earnest Money Deposit (EMD)

4.17.1 The Bidder shall furnish, as part of its bid, an Earnest Money Deposit (EMD) of the amount mentioned in the Fact Sheet

4.17.2 The EMD must be submitted, by Bank Guarantee of any Scheduled Commercial Bank/ Nationalized Bank drawn in favour of CHiPS, payable at Raipur.

4.17.3 The EMD of unsuccessful bidders shall be discharged/ returned within 60 days of award of Contract to the successful Bidder.

4.17.4 The EMD of successful bidder shall be discharged/returned upon the bidder executing the Contract, pursuant to Clause 4.31: Award of Contract and furnishing the Bank Guarantee/Security Deposit, pursuant to Clause 4.36: Performance Bank Guarantee.

4.17.5 CHiPS may, in its sole discretion and at the Successful Bidder's option, adjust the amount of EMD in the amount of PBG to be provided by him in accordance with the provisions of the Agreement.

4.17.6 CHiPS shall be entitled to forfeit and appropriate the EMD as damages inter alia in any of the events specified in Clause 4.17.7 herein. The Bidder, by submitting its Bid pursuant to this RFP, shall be deemed to have acknowledged and confirmed that CHiPS will suffer loss and Damages on account of withdrawal of its Bid or for any other default by the Bidder during the period of validity of the Bid as specified in this RFP. No relaxation of any kind on EMD shall be given to any Bidder.

4.17.7 The EMD shall be forfeited as Damages without prejudice to any other right or remedy that may be available to CHiPS under the Bidding Documents and/ or under the Agreement, or otherwise, if:

a. Bidder submits a non-responsive Bid.
b. Bidder engages in a corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice as specified in Clause 4.37 of this RFP.
c. Bidder withdraws its Bid during the period of Bid validity as specified in this RFP and as extended by mutual consent of the respective Bidder(s) and CHiPS; and
d. the Successful Bidder fails within the specified time limit:
    - to sign and return the duplicate copy of LOA; or
    - to execute the Agreement; or
    - to furnish the PBG within the period prescribed therefore in the Agreement.
e. the Successful Bidder, having executed the Agreement, commits any breach thereof prior to furnishing the PBG.

4.17.8 No interest shall be paid by the Purchaser on the EMD

4.18 Period of Validity of Bids

4.18.1 Bids shall have validity period as mentioned in Fact Sheet after the date of opening of Technical Bid. A bid valid for a shorter period may be summarily rejected by the Purchaser as non-responsive.

4.18.2 Prior to the expiry of the validity of the period of validity of the bids, the Purchaser may request the Bidder(s) for an extension of the period of validity up to 180 days or more. The request and the responses thereto shall be made in writing (or through e-mail). The period of validity of their Bids shall require extension in the period of validity of EMD submitted by the bidders or submission of new EMD to cover the extended period of validity of their Bids. A Bidder whose EMD is not extended, or that has not submitted a new EMD, shall be considered to have refused the request to extend the period of validity of its Bid.

4.19 Format and Signing of Bid

4.19.1 The original documents of the bid shall be typed or written in indelible ink. The original document shall be signed by the Bidder or a person or persons duly authorized to bind the Bidder to the Contract. All pages of the bid, except for un-amended printed literature, shall be initialled and stamped by the person(s) persons signing the bid.

4.19.2 The response to the bid should be submitted along with legible, appropriately indexed, duly filled Information sheets and sufficient documentary evidence as per Checklist. Responses with illegible, incomplete Information sheets or insufficient documentary evidence shall be rejected.

4.19.3 The bid shall contain no interlineations, erasures or overwriting except as necessary to correct errors made by the Bidder, in which case such corrections shall be initialled by the person(s) signing the bid.

4.19.4 The Bidder shall duly sign and seal its bid with the exact name of the firm/company to whom the contract is to be issued.

4.20 Revelation of Prices

4.20.1 Prices in any form or by any reason before opening the Commercial Bid should not be revealed, failing which the offer shall be liable to be rejected.

4.21 Terms and Conditions of Bidders

4.21.1 Any terms and conditions of the Bidders shall not be considered as forming part of their Bids.

4.22 Local Conditions

4.22.1 It shall be incumbent upon each Bidder to fully acquaint himself with the local conditions and other relevant factors at the proposed site which would have any effect on the performance of the contract and / or the cost.

4.22.2 The Bidder is expected to make a site visit to obtain for himself on his own responsibility, all information that may be necessary for preparing the bid and entering into contract. Obtaining such information shall be at Bidder's own cost.

4.22.3 Failure to obtain the information necessary for preparing the bids and/or failure to perform the activities that may be necessary for the providing services before entering into contract shall in no way relieve the SI from performing any work in accordance with the RFP.

4.22.4 It shall be imperative for each Bidder to fully inform themselves of all legal conditions and factors which may have any effect on the execution of the contract as described in the bidding documents.

4.22.5 It is the responsibility of the Bidder that such factors have properly been investigated and considered while submitting the bid proposals and that no claim whatsoever including those for financial adjustment to the contract awarded under the bidding documents shall be entertained by the Purchaser and that neither any change in the time schedule of the contract nor any financial adjustments arising thereof shall be permitted by the Purchaser on account of failure of the Bidder to appraise themselves of local laws and site conditions

4.22.6 It shall be deemed that by submitting a Bid, the Bidder has:

a. made a complete and careful examination of the Bidding Documents.
b. received all relevant information requested from CHiPS.
c. accepted the risk of inadequacy, error or mistake in the information provided in the Bidding Documents or furnished by or on behalf of CHiPS relating to any of the matters.

4.23 Last Date for Receipt of Bids

4.23.1 Bids shall be submitted by the bidder no later than the time and date specified in Fact Sheet

4.23.2 Original documents as per Clause 4.12 shall be received by the Purchaser at the address specified under Fact Sheet no later than the time and date specified in Fact Sheet

4.23.3 The Purchaser may, at its discretion, extend the last date for submission of bids by amending the RFP, in which case all rights and obligations of the Purchaser and Bidders previously subject to the last date shall thereafter be subject to the last date as extended.

4.24 Late Bids

4.24.1 Any bid submitted by the bidder after the last date and time for submission of bids pursuant to Fact Sheet or Original Documents received by purchaser after the last date and time for submission pursuant to Clause 4.12, shall be rejected.

4.25 Modification and Withdrawal of Bids

4.25.1 No bid may be altered/ modified subsequent to the closing time and date for receipt of bids. Unsolicited correspondences from Bidders shall not be considered.

4.25.2 No bid may be withdrawn in the interval between the last date for receipt of bids and the expiry of the bid validity period specified by the Bidder in the Bid. Withdrawal of a bid during this interval may result in the Bidder's forfeiture of its EMD and shall be declared a "defaulting bidder". In such situation the tendering process shall be continued with the remaining bidders as per their ranking.

4.25.3 If the bidder relents after being declared as selected bidder, it shall be declared as defaulting bidder and EMD of such defaulting bidder shall be forfeited and CHiPS reserves right to blacklist/ debarred such bidder for next 5 years from participating in any CHiPS tender. In such situation, L2 bidder shall be invited to match the prices of L1 bidder. However, in-case of refusal of acceptance by the L2 bidder to match the price of L1 bidder, CHiPS would carry out discussions with the subsequent bidders.

4.26 Contacting the Purchaser

4.26.1 No Bidder shall contact the Purchaser on any matter relating to its bid, from the time of the bid submission to the time the Contract is awarded.

4.26.2 Any effort by a Bidder to influence the Purchaser's bid evaluation, bid comparison or Contract award decisions may result in the rejection of the Bidder's bid.

4.27 Opening of Technical Bids by Purchaser

4.27.1 The Bids received form the Bidders shall be opened online. The opening of the Bids shall be carried out online in the physical presence of the designated representatives of CHiPS and the Bidders. However, this RFP does not mandate the physical presence of the Bidders. The absence of the physical presence of the Bidders shall in no way affect the outcome of the evaluation of the Bids.

4.27.2 CHiPS shall subsequently examine and evaluate the Bids in accordance with the provisions set out in this Clause 6.

4.27.3 To facilitate evaluation of Bids, CHiPS may, at its sole discretion, seek clarifications in writing from any Bidder regarding its Bid.

4.28 Purchaser's Right to Vary Scope of Contract

4.28.1 The Purchaser may at any time, by a written order given to the Bidder, make changes to the scope of the Contract as specified.

4.29 Purchaser's Right to Accept Any Bid & to Reject Any or All Bids

4.29.1 The Purchaser reserves the right to accept any or all bid, and to annul the Tendering process or reject all bids at any time prior to award of Contract, without thereby incurring any liability to the affected Bidder or Bidders or any obligation to inform the affected Bidder or Bidders of the grounds for the Purchaser's action

4.29.2 Further, in the event that the RFP has been terminated on account of any default of the Bidder/Successful Bidder, CHiPS reserves the right to encash the EMD and/or PBG to the extent required to make good the losses suffered by it on account of such default having been committed by the Bidder

4.29.3 This RFP does not constitute an offer by CHiPS. The Bidder’s participation in the Bidding Process may result CHiPS selecting the Bidder to engage towards execution of the Scope of Work of the Project

4.30 Notification of Award

4.30.1 Prior to the expiry of the period of bid validity, pursuant to Fact Sheet and Clause 4.18: Period of Validity of Bid, the Purchaser shall notify the successful Bidder in writing by registered letter/ courier, that its bid has been accepted.

4.30.2 The notification of award shall constitute the formation of the Contract.

4.30.3 Upon furnishing of Performance Bank Guarantee of amount equivalent to EMD of Package to be awarded to Bidder.

4.31 Award of Contract

4.31.1 There shall be only one successful bidder.

4.31.2 At the same time as the Purchaser notifies the SI that its bid has been accepted, the Purchaser shall send the SI the pro forma for Contract, incorporating all agreements between the parties.

4.31.3 Within 7 days of receipt of the Contract, the SI shall sign and date the Contract and return it to the Purchaser.

4.31.4 The SI shall ensure complete end-to-end connectivity, monitoring, and service provisioning for the created network throughout the contract period. For Gram Panchayats (GPs) where implementation has not yet commenced, the SI shall complete the implementation within six (6) months from the date of signing the agreement by the Successful Bidder. The contract period shall include six (6) months for implementation and one (1) year of Operations & Maintenance (O&M), may be extendable by an additional one (1) year based on performance and requirements.

4.31.5 The Bidder whose bid is accepted shall be required to submit Performance Bank Guarantee as mentioned in the Clause: 4.36. PBG shall be in the form of Bank Guarantee (BG) from any nationalized or scheduled bank.

4.31.6 Bidder has to agree for honouring all RFP conditions and adherence to all aspects of fair-trade practices in executing the work orders placed by CHiPS.

4.31.7 If the name of the system/ service/ process is changed for describing substantially the same in a renamed form; then all techno-fiscal benefits agreed with respect to the original product, shall be passed on to CHiPS and the obligations with CHiPS taken by the Vendor with respect to the product with the old name shall be passed on along with the product so renamed.

4.31.8 CHiPS may, at any time, terminate the contact by giving written notice to the bidder without any compensation, if the bidder becomes bankrupt or otherwise insolvent, provided that such termination shall not prejudice or affect any right of action or remedy which has accrued or shall accrue thereafter to CHiPS.

4.31.9 If at any point during the Contract, the Bidder fails to deliver as per the RFP terms and conditions or any other reason amounting to disruption in service, the Termination and Exit Management clause shall be invoked.

4.32 Annulment & Re-award

4.32.1 Failure of the Successful Bidder to comply with all the requirements shall constitute sufficient grounds for the annulment of the award of the Agreement, in which event CHiPS may make the award to the next lowest evaluated Bidder or call for new bids (as per clause 4.39.3 b).

4.33 Placing of Work Order

4.33.1 Qua**ntities mentioned Annexure-12:** Commercial Bid Format are indicative, and CHiPS reserves the right at the time of issuance of work order to increase or decrease the quantity of goods and / or services from the original requirements as specified in the RFP.

During the entire contract period that is irrelevant to the extension of the contract (Inclusion of the extended contract period) Price quoted by the bidder against good and services will be same and CHiPs may place additional order for goods and services quoted package wise.

4.33.2 For procurement of Hardware/service, Work Order shall be placed to SI.

4.33.3 Objection, if any, to the Work Order must be reported to CHiPS by the SI within five (5) working days counted from the Date of issuance of work Order for modifications, otherwise it is assumed that the SI has accepted the Work Order.

4.33.4 If the SI is not able to do the complete work as mentioned in the scope of work within stipulated timeline / adherence of the SLA then penalty clause shall be invoked.

4.33.5 The O&M phase for a GP shall begin once the GP is eligible, either:

- After takeover from CHiPS of the existing GPs will fall under O&M Phase and SLA will be applicable.

- Post Integration and Acceptance, including commissioning, NMS discovery, and issuance of End-to-End Acceptance Test certificate for New GP installation.

4.33.6 The O&M of created infrastructure (Inclusion of Active and Passive Infrastructure) will be for 1 year initially, may be extendable by another 1 year.

4.33.7 The decision of CHiPS shall be final and binding on the SI. CHiPS reserve the right to accept or reject an offer without assigning any reason whatsoever.

4.34 Tender Related Conditions

4.34.1 The SI shall confirm unconditional acceptance of full responsibility of completion of job and for executing the 'Scope of Work' of this RFP. This confirmation should be submitted as part of the Technical Bid. The bidder shall also be the sole point of contact for all purposes of the Contract.

4.34.2 The SI should not be involved in any litigation that may have an impact of affecting or compromising the delivery of services as required under this Contract. If at any stage of Tendering process or during the currency of the Contract, any suppression/ falsification of such information is brought to the knowledge of the Purchaser, the Purchaser shall have the right to reject the bid or terminate the Contract, as the case may be, without any compensation to the SI.

4.35 Rejection Criteria

4.35.1 Besides other conditions and terms highlighted in the RFP, bids may be rejected under following circumstances:

a. Technical Rejection Criteria

- Bids submitted without or improper EMD
- Technical Bid containing commercial details
- Bids received through Telex/ Telegraphic/ Fax/ E-Mail/ post etc. except, wherever required, shall not be considered for evaluation
- Bids which do not confirm unconditional validity of the bid as prescribed in the RFP
- If the information provided by the Bidder is found to be incorrect/ misleading at any stage/ time during the Tendering Process
- Any effort on the part of a Bidder to influence the Purchaser's bid evaluation, bid comparison or Contract award decisions
- Bids submitted by the Bidder after the last date and time of bid submission prescribed by the Purchaser, pursuant to Fact Sheet
- Bids without power of attorney and any other document consisting of adequate proof of the ability of the signatory to bind the Bidder
- Failure to furnish all information required by the RFP or submission of a bid not substantially responsive to the RFP in every respect
- Bidders not quoting for the complete Scope of Work as indicated in the RFP, addendum (if any) and any subsequent information given to the bidders
- Bidders not complying with the functionality, specifications and other Terms and Conditions as stated in the RFP

- The Bidder not conforming unconditional acceptance of full responsibility of providing Goods and Services in accordance with the Section 5: General Conditions of Contract and Section 7: Scope of Work
- If the bid does not conform to the timelines indicated in this RFP

b. Commercial Rejection Criteria

- Incomplete Commercial Bid
- Commercial Bids that do not conform to this RFP's Commercial Bid format
- If there is an arithmetic discrepancy in the commercial bid calculations, the Purchaser shall rectify the same. If the Bidder does not accept the correction of the errors, bid may be rejected.
- If bidder quotes NIL charges/ consideration, the bid shall be treated as unresponsive and shall not be considered

4.36 Performance Bank Guarantee

4.36.1 Within 15 days of receipt of the notification of award from the Purchaser, the successful bidder shall submit a single Performance Bank Guarantee (PBG) equivalent to amount of EMD of Package to be awarded to Bidder. This PBG shall remain valid up to 60 days after completion of the entire contract period, including implementation and O&M, as per the issued work order.

4.36.2 In addition to submitting the Performance Bank Guarantee (PBG) equivalent to the EMD amount for the awarded package, the bidder must also provide an additional PBG equal to the difference between 10% of the work order value and the PBG already submitted.

4.36.3 The PBG shall be in the form of a Bank Guarantee issued by any nationalized or scheduled bank and shall comply with the General Conditions of the Contract, in the prescribed format for the PBG shall be as per Annexure-13: Format for Performance Bank Guarantee.

4.36.4 Failure of the SI to comply with the requirement of Clause 4.36.1 shall constitute sufficient grounds for annulment of the contract and forfeiture of the EMD.

4.37 Fraud and Corrupt Practices

4.37.1 The SI and their respective officers, employees, agents and advisers shall observe the highest standards of ethics during the Bidding Process and subsequent to the issue of the Award of Work and during the subsistence of the Contract. Notwithstanding anything to the contrary contained herein, or in the Award of Work or the Contract, the Purchaser shall reject a Bid, withdraw the Award of Work, or terminate the Contract, as the case may be, without being liable in any manner whatsoever to the SI as the case may be, if it determines that the SI, as the case may be, has, directly or indirectly or through an agent, engaged in corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice in the Bidding Process. In such an event, the Purchaser shall forfeit and appropriate the Bid Security or Performance Security, as the case may be, as mutually agreed genuine pre-estimated compensation and damages payable to the Purchaser towards, inter alia, time, cost and effort of the Purchaser, without prejudice to any other right or remedy that may be available to the Authority hereunder or otherwise.

4.37.2 Without prejudice to the rights of the Purchaser under RFP hereinabove and the rights and remedies which the Purchaser may have under the Award of Work or the Contract, if a SI, as the case may be, is found by the Authority to have directly or indirectly or through an agent, engaged or indulged in any corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practice during the Bidding Process, or after the issue of the Award of Work or the execution of the Contract, such SI shall not be eligible to participate in any RFP issued by the Purchaser during a period of 5 (five) years from the date such SI is found by the Purchaser to have directly or indirectly or through an agent, engaged or indulged in any corrupt practice, fraudulent practice, coercive practice, undesirable practice or restrictive practices, as the case may be.

4.37.3 For the purposes of the Clause 4.37.1, the following terms shall have the meaning hereinafter respectively assigned to them:

a. “corrupt practice” means (i) the offering, giving, receiving, or soliciting, directly or indirectly, of anything of value to influence the actions of any person connected with the Bidding Process (for avoidance of doubt, offering of employment to or employing or engaging in any manner whatsoever, directly or indirectly, any official of the Authority who is or has been associated in any manner, directly or indirectly, with the Bidding Process or the Work Order or has dealt with matters concerning the Contract or arising therefrom, before or after the execution thereof, at any time prior to the expiry of one year from the date such official resigns or retires from or otherwise ceases to be in the service of the Authority, shall be deemed to constitute influencing the actions of a person connected with the Bidding Process); or (ii) save, engaging in any manner whatsoever, whether during the Bidding Process or after the issue of the Work Order or after the execution of the Contract, as the case may be, any person in respect of any matter relating to the Project or the Work Order or the Contract, who at any time has been or is a legal, financial or technical adviser of the Purchaser in relation to any matter concerning the Project;

b. “fraudulent practice” means a misrepresentation or omission of facts or suppression of facts or disclosure of incomplete facts, in order to influence the Bidding Process.

c. “coercive practice” means impairing or harming, or threatening to impair or harm, directly or indirectly, any person or property to influence any person’s participation or action in the Bidding Process;

d. “undesirable practice” means (i) establishing contact with any person connected with or employed or engaged by the Authority with the objective of canvassing, lobbying or in any manner influencing or attempting to influence the Bidding Process; or (ii) having a Conflict of Interest; and

e. "restrictive practice" means forming a cartel or arriving at any understanding or arrangement among SI with the objective of restricting or manipulating a full and fair competition in the Bidding Process

4.38 Disqualification of the Bidder

4.38.1 A Bidder / SI shall be liable for disqualification and forfeiture of the EMD submitted by it, if any legal, financial or technical adviser of CHiPS in relation to the Project is engaged by the Bidder, its members or any Associate thereof, as the case may be, in any manner for matters related to or incidental to such Project during the Bidding Process or subsequent to the (i) issuance of the LOA or (ii) execution of the Agreement

4.38.2 In the event any such adviser is engaged by the Successful Bidder or the Contractor, as the case may be, after issuance of the LOA or execution of the Agreement for matters related or incidental to the Project, then notwithstanding anything to the contrary contained herein or in the LOA or the Agreement and without prejudice to any other right or remedy of CHiPS, including the forfeiture and appropriation of the EMD or PBG, as the case may be, the LOA and/or the Agreement shall be liable to be terminated without CHiPS being liable in any manner whatsoever to the Successful Bidder or Contractor for the same

4.38.3 For the avoidance of doubt, this disqualification shall not apply where such adviser was engaged by the Bidder, its member or Associate in the past but its assignment expired or was terminated prior to the Bid Due Date. Nor will this disqualification apply where such adviser is engaged after a period of 3 (three) years from the date of completion of Scope of Work of the Project.

4.39 Verification and Disqualification

4.39.1 CHiPS reserves the right to verify all statements, information and documents submitted by the Bidder in response to the Bidding Documents and the Bidder shall, when so required by CHiPS, make available all such information, evidence and documents as may be necessary for such verification. Any such verification, or lack of such verification, by CHiPS shall not relieve the Bidder of its obligations or liabilities hereunder nor will it affect any rights of CHiPS hereunder

4.39.2 CHiPS reserves the right to reject any Bid and appropriate the EMD if:

a. at any time, a material misrepresentation is made or uncovered, or
b. L1 expresses its desire to or does withdraw its Bid, or
c. The Bidder does not provide, within the time specified by CHiPS, the supplemental information sought by CHiPS for evaluation of the Bid within a reasonable time as determined by CHiPS in its sole discretion

4.39.3 Such misrepresentation/ improper response/intent to withdraw shall lead to the disqualification of the Bidder. If such disqualification/rejection occurs after the Bids have been opened and the L1 Bidder gets disqualified/rejected, then CHiPS reserves the right to:

a. invite the remaining Bidders to submit fresh Bids in accordance with this RFP; or
b. approach the next lowest Bidder (L2) for matching the price of L1. Should L2 choose to match the price bid of L1, CHiPS may issue the LOA to L2. However, in-case of refusal of acceptance by the L2 bidder to match the price of L1 bidder, CHiPS would carry out discussions with the subsequent bidders
c. take any such measure as may be deemed fit in the sole discretion of CHiPS, including annulment of the Bidding Process

4.39.4 In case it is found during the evaluation or at any time before signing of the Agreement or after its execution and during the period of subsistence thereof, that one or more of the qualification conditions have not been met by the Successful Bidder/Contractor (as may be the case), or the Bidder has made material misrepresentation or has given any materially incorrect or false information, the Successful Bidder/Contractor shall be disqualified forthwith if not yet appointed as the Contractor either by issue of the LOA or entering into of the Agreement, and if the Successful Bidder has already been issued the LOA or has entered into the Agreement, as the case may be, the same shall, notwithstanding anything to the contrary contained therein or in this RFP, be liable to be terminated, by a communication in writing, by CHiPS to the Successful Bidder/Contractor, as the case may be, without CHiPS being liable in any manner whatsoever to the Successful Bidder or Contractor. However, the bidder shall be given a notice and thereafter a chance of representation in lieu of the charges.

4.39.5 Bidders are advised to study all instructions, forms, terms, requirements, appendices and other information in the RFP carefully. Online submission of the Bid shall be deemed to have been done after careful study and examination of the RFP with full understanding of its implications and failure to comply with the requirements of the RFP, in addition to submission of all the documents as required hereunder, may render the Bid non-compliant and the same may be rejected

5. General Conditions of the Contract

5.1 Definitions

5.1.1 In this RFP, unless the context otherwise requires

5.1.1.1 **“Applicable Laws”** includes all applicable statutes, enactments, acts of legislature or laws, ordinances, rules, by-laws, regulations, notifications, guidelines, policies, directions, directives, requirement or other governmental restriction and orders or judgements of any Governmental authority, tribunal, board, court or other quasi-judicial authority or other governmental restriction or any similar form of decision applicable to the relevant Party and as may be in effect on the date of execution of Agreement and during the subsistence thereof, applicable to the Project;

5.1.1.2 **“Confidential Information"** means all information as defined in Section 5 - General Conditions of the Contract

5.1.1.3 **“Contract”** or means the Agreement entered into between the SI and the Purchaser as recorded in the Contract form signed by the Purchaser and the SI including all attachments and Annexes thereto, the RFP and all Annexes thereto and the agreed terms as set out in the bid, all documents incorporated by reference therein and amendments and modifications to the above from time to time;

5.1.1.4 **“Contract Value”** means the price payable to the SI under this Contract for the full and proper performance of its contractual obligations

5.1.1.5 **“Deliverables”** means the products, infrastructure, licenses and services agreed to be delivered by the SI in pursuance of the Contract as elaborated in the RFP and includes all documents related to the user manual, technical manual, designs, process documentations, the artefacts, the training materials, process and operating manuals, service mechanisms, policies and guidelines , inter alia payment and/or process related etc. and all their respective modifications;

5.1.1.6 **“Effective Date”** means the date on which the Contract is executed by both the Parties

5.1.1.7 **“End-to-End Connectivity”** means Complete supply, Installation, Integration, Testing and Commissioning of the created network (duly approved by the agency appointed by CHiPS) which shall include OTDR link test, Power On & Self-Testing, IP-MPLS ring fail over testing for both cases i.e. in case of path/equipment failure, Radio equipment testing (If applicable), As Build Diagram (ABD reports), integration equipment at GP with State and Central BBNL NOC and Final acceptance certificate;

5.1.1.8 **“OEM”** or **“Original Equipment Manufacturer”** means the original manufacturer and owner of the Intellectual Property Rights of any Software or Equipment to be used in the Project and to which CHiPS has been granted license to use

5.1.1.9 **“Contract Performance Guarantee”** or “Performance Bank Guarantee” shall mean the guarantee provided by a Scheduled Commercial Bank/ Nationalized Bank to CHiPS by the successful bidder

5.1.1.10 **“Project”** means BharatNet Project for Supply, Installation, Integration, Testing, Commissioning and Maintenance of OFC (Underground) IP-MPLS Network, Radio Network and State Network Operations Centre (S-NOC) including Operations & Maintenance and facilitating service provisioning of the created network for the period of 2 years on turnkey basis as per the terms and conditions laid in the RFP and provision of Services in conformance to the SLA

5.1.1.11 **“Purchaser”** means Chhattisgarh infotech Promotion Society (CHiPS)

5.1.1.12 **“Purchaser’s Representative” or “Purchaser’s Technical Representative”** means the person or the persons appointed by the Purchaser from time to time to act on its behalf for overall co-ordination, supervision, and project management

5.1.1.13 **“Project Implementing Agency is also termed as System Integrator (SI)”** means the successful bidder whose bid has been accepted by the Purchaser and with whom the order for “Selection of System Integrator for BharatNet Phase – II” has been placed as per requirements and terms & conditions specified in this RFP and shall be deemed to include the Bidder’s successors, representatives (approved by the Purchaser), heirs, executors, administrators and permitted assigns, as the case may be, unless excluded by the terms of the contract

5.1.2 **“SIs’ Team”** means the Successful Bidder along with all of its partners / OEMs, who have to provide goods & services to the Purchaser under the scope of this RFP / Contract. This definition shall also include any authorized service providers/partners/agents and representatives, or other personnel employed or engaged either directly or indirectly by the SI for the purposes of this SI / Contract

5.1.2.1 **“Project Store”** means list of locations provided by the SI to store the material as required in District/Block.

5.1.2.2 **“Request for Proposal/ (RFP)”** means the documents containing the general, technical, functional, commercial and legal specifications for the implementation of the BharatNet Phase-II Project including different Annexures and includes the clarifications, explanations, minutes of the meetings, corrigendum(s) and amendment(s) issued from time to time during the bidding process and on the basis of which bidder has submitted its Proposal

5.1.2.3 **“Schedule of Requirements” or “SoR”** means the requirement regarding BharatNet Project provided by bidder in its bid, stating the prices and the quantity of the materials to be procured by the bidder (on behalf of CHiPS)

5.1.2.4 **“State Level Committee”** means a committee involving representatives of the Purchaser and senior officials of the SI shall be formed for the purpose of this Contract. This committee shall meet at intervals, as decided by the Purchaser later, to oversee the progress of the Project

5.1.2.5 **“Tender” or “Tender Document”** means RFP

5.1.2.6 **“Timelines”** means the duration of the contract as described in the RFP

5.1.2.7 **"Working Day"** means any day on which any of the office of CHiPS shall be functioning, including gazetted holidays, restricted holidays or other holidays, Saturdays and Sundays

5.2 Interpretations

5.2.1 In this RFP, unless otherwise specified:

a. Unless otherwise specified, a reference to clauses, sub-clauses, or Section is a reference to clauses, sub-clauses, or Section of this RFP including any amendments or modifications to the same from time to time

b. Words denoting the singular include the plural and vice versa and use of any gender includes the other genders

c. References to a "company" shall be construed so as to include any company, corporation or other body corporate, wherever and however incorporated or established

d. Words denoting to a "person" shall be construed to include any individual, partnerships, firms, companies, public sector units, corporations, joint ventures, trusts, associations, organizations, executors, administrators, successors, agents, substitutes and any permitted assignees or other entities (whether or not having a separate legal entity). A reference to a group of persons is a reference to all of them collectively, to any two or more of them collectively and to each of them individually

e. A reference to any statute or statutory provision shall be construed as a reference to the same as it may have been, or may from time to time be, amended, modified or re-enacted

f. Any reference to a "day" (including within the phrase "business day") shall mean a period of 24 hours running from midnight to midnight

g. References to a "business day" shall be construed as a reference to a day (other than a Sunday) on which CHiPS office is generally open for business

h. References to times are to Indian Standard Time

i. Reference to any other document referred to in this RFP is a reference to that other document as amended, varied, novated or supplemented at any time

j. All headings and titles are inserted for convenience only, they are to be ignored in the interpretation of this Contract

k. Unless otherwise expressly stated, the words "herein", "hereof", "hereunder" and similar words refer to this RFP as a whole and not to any particular Section or Annexure and the words "include" and "including" shall not be construed as terms of limitation

l. The words "in writing" and "written" mean "in documented form", whether electronic or hard copy, unless otherwise stated.

m. Unless otherwise stated, any reference to any period commencing "from" a specified day or date and "till" or "until" a specified day or date shall include either such days or date

Note: Bidders must read these conditions carefully and comply strictly while submitting their bids.

5.3 Expected To Examine All Instructions

5.3.1 The bidder is expected to examine all instructions, forms, terms, and specifications in the bidding documents. Failure to furnish all information required in the bidding documents or submitting a Bid not substantially responsive to the bidding documents in any respect may result in the rejection of the Bid.

5.4 All Costs

5.4.1 The bidder shall bear all the costs associated with the preparation and submission of its bid, and CHiPS in no case will be responsible or liable for these costs, regardless of conduct or outcome of Bidding Process

5.5 Professional Excellence and Ethics

5.5.1 CHiPS requires that all Bidders participating in this Bid adhere to the highest ethical standards, both during the selection process and throughout the execution of the contract

5.6 Failure of Successful Bidder

5.6.1 Failure of the successful bidder to comply with all the requirements shall constitute sufficient grounds for the annulment of the award, in which event CHiPS may, make the award to the next lowest evaluated Bidder or call for new bids (as per clause 4.39.3 b)

5.7 Amendment/Cancellation

5.7.1 The CHiPS reserve the right to cancel this Tender at any time without any obligation to the Bidders. The CHiPS at any time, prior to the deadline for submission of Proposals, may amend the Tender by issuing an addendum in writing or by standard electronic means. The addendum will be binding on all the Bidders. Bidders shall acknowledge receipt of all amendments. To give Bidders reasonable time to take an amendment into account in their Proposals, the CHiPS may, if the amendment is substantial, extend the deadline for the submission of Proposals.

5.8 Right to accept or reject any or all Bids

5.8.1 CHiPS reserves the right to accept any bid, and to annul the bid process and reject all bids at any time prior to award of contract, without assigning any reason & without thereby incurring any liability to the affected Bidder or Bidders or any obligation to inform the affected Bidder or Bidders of the grounds for the action

5.9 Conditional Bids

5.9.1 If a Bidder imposes conditions, which is in addition to or in conflict with the conditions mentioned herein, his bid is liable to be summarily rejected. In any case none of such conditions will be deemed to have been accepted unless specifically mentioned in the letter of acceptance of bid issued by the CHiPS

5.10 Acceptance Testing

5.10.1 All active equipment’s and passive components may be tested by CHiPS or its nominated agency

5.11 Proprietary Rights

5.11.1 The Bidder shall indemnify CHiPS against all third party claims of infringement of patent, copy right, trademark, license or industrial design rights, software piracy arising from use of goods or any part thereof within India

5.12 Delays in performance of supplier's obligation

5.12.1 Any delay by the Successful Bidder in the performance of its delivery obligations shall render the Successful Bidder liable to any or all of the following sanctions – forfeiture of its performance Bank Guarantee, imposition of Damages and / or termination of the work order for default.

5.13 Applicable Law

5.13.1 The work order issued under the Agreement shall be interpreted in accordance with the laws of India, irrespective of the place of delivery, the place of performance or place of payment under the Agreement. The Agreement shall deem to have made at the place in India from where the contract has been issued

5.14 Notices

5.14.1 Any notice given by one party to the other pursuant to this contract shall be sent in writing and confirmed in writing to CEO, CHiPS, SDC Building, 02nd floor, Near Police Control Room, Civil Lines, Raipur, Chhattisgarh–492001 Fax: - 0771-4066205, Ph. No. 0771-4014158

5.14.2 A notice shall be effective when delivered or on the notice's effective date whichever is later

5.15 Taxes & Duties

5.15.1 The Bidder shall be entirely responsible for all taxes, duties, license fee etc.

5.15.2 Any upward/downward revision of GST shall be applicable at the time of invoicing. However, to arrive at the bid value of the respective Bidder, Bidder has to quote the charges inclusive of all taxes, duties, levies etc. excluding GST

5.16 Defence of Suits

5.16.1 If any action in court is brought against the CHiPS/ Consignee for failure or neglect on the part of the Bidder to perform any acts, matters, covenants or things under the contract or for the damage or injury caused by the alleged omission of neglect on part of the SI, his agents, representatives or sub-contractors, workmen supplier or employees, the contractor in all such cases shall indemnify and keep CHiPS harmless from all costs, Damages, expenses or decrees arising out of such action

5.17 Conditions Precedent

5.17.1 This Contract is subject to the fulfilment of the following conditions precedent by the SI

5.17.1.1 Furnishing by the SI, an unconditional, irrevocable and continuing Performance Bank Guarantee, which shall be as per Clause 4.36, in a form and manner acceptable to the Purchaser

5.17.1.2 Execution of a Deed of Indemnity in terms of Clause 5.32 - Indemnity

5.17.1.3 Obtaining of all statutory and other approvals required for the performance of the Services under this Contract

5.17.1.4 Furnishing of such other documents as the Purchaser may specify

5.17.1.5 The Purchaser reserves the right to waive any or all of the conditions specified above in writing and no such waiver shall affect or impair any right, power or remedy that the Purchaser may otherwise have

5.18 Representations & Warranties

5.18.1 In order to induce the Purchaser to enter into this Contract, the Bidder hereby represents and warrants as of the date hereof, the following:

a. That the SI has the power and the authority that would be required to enter into this Contract and the requisite experience, the technical know-how and the financial wherewithal required to successfully execute the terms of this Contract and to provide Services sought by the Purchaser under this Contract

b. That the SI is not involved in any litigation or legal proceedings, pending, existing, potential or threatened, that may have an impact of affecting or compromising the performance or delivery of Services under this Contract

c. That the representations and warranties made by the SI in its Bid, RFP and Contract are and shall continue to remain true and fulfil all the requirements as are necessary for executing the obligations and responsibilities as laid down in the Contract and the RFP and unless the Purchaser specifies to the contrary, the SI shall be bound by all the terms of the Bid and the Contract through the term of the Contract

d. That the SI has the professional skills, personnel, infrastructure and resources/ authorizations that are necessary for providing all such Services as are necessary to fulfil the Scope of Work stipulated in the RFP and the Contract

e. That the SI shall ensure that all assets/ components including but not limited to equipment, documents, etc. installed, procured and created during the term of this Contract are duly maintained and suitably updated, upgraded, replaced with regard to contemporary requirements

f. That the SI shall use such assets of the Purchaser as the Purchaser may permit for the sole purpose of execution of its obligations under the terms of the Bid, RFP or this Contract. The SI shall however, have no claim to any right, title, lien or other interest in any such property, and any possession of property for any duration whatsoever shall not create any right in equity or otherwise, merely by fact of such use or possession during or after the term hereof

g. That there shall not be any privilege, claim or assertion made by a third party with respect to right or interest in, ownership, mortgage or disposal of any asset, property, movable or immovable as mentioned in any Intellectual Property Rights, licenses and permits

h. That the execution of the scope of work and the Services herein is and shall be in accordance and in compliance with all applicable laws

i. That all conditions precedent under the Contract have been satisfied

j. That neither the execution and delivery by the SI of the Contract nor the SIs' compliance with or performance of the terms and provisions of the Contract (i) shall contravene any provision of any Applicable Laws or any order, writ, injunction or decree of any court or Governmental Authority binding on the SI, (ii) shall conflict or be inconsistent with or result in any breach of any or the terms, covenants, conditions or provisions of, or constitute a default under any Contract, Contract or instrument to which the SI is a party or by which it or any of its property or assets is bound or to which it may be subject or (iii) shall violate any provision of the Memorandum and Articles of Association of the SI.

k. That the SI certifies that all registrations, recordings, filings and notarizations of the Contract and all payments of any tax or duty, including but not limited to stamp duty, registration charges or similar amounts which are required to be effected or made by the SI which is necessary to ensure the legality, validity, enforceability or admissibility in evidence of the Contract have been made

l. That the SI confirms that there has not and shall not occur any execution, amendment or modification of any agreement/ contract without the prior written consent of the Purchaser, which may directly or indirectly have a bearing on the Contract with the Purchaser.

m. That the SI owns or has good, legal or beneficial title, or other interest in, to the property, assets and revenues of the SI on which it grants or purports to grant or create any interest pursuant to the Contract, in each case free and clear of any encumbrance and further confirms that such interests created or expressed to be created are valid and enforceable

n. That the SI owns, has license to use or otherwise has the right to use, which are required or desirable for performance of its services under this contract. All Intellectual Property Rights (owned by the SI or which the SI is licensed to use) required by the SI for the performance of the contract are valid and subsisting. All actions (including registration, payment of all registration and renewal fees) required to maintain the same in full force and effect have been taken thereon and shall keep the Purchaser indemnified in relation thereto

o. That the SI shall at all times maintain sufficient manpower, resources, and facilities, to provide the Services in a workmanlike manner on a timely basis

p. That its security measures, policies and procedures are adequate to protect and maintain the confidentiality of the Confidential Information

q. That in providing the Services or deliverables or materials, neither SI or its agent, nor any of its employees, shall utilize information which may be considered confidential information of, or proprietary to, any prior employer or any other person or entity.

r. That the SI shall provide adequate and appropriate support and participation, on a continuous basis

5.19 Key Performance Measurements

5.19.1 Unless specified by the Purchaser to the contrary, the SI shall execute the services and carry out the scope of work in accordance with the terms & conditions of this Contract, and the service specifications as laid down in this RFP

5.19.2 If the Contract, Scope of Work, Service Specification includes more than one document, then unless the Purchaser specifies to the contrary, the later in time shall prevail over a document of earlier date to the extent of any inconsistency

5.19.3 The Purchaser reserves the right to amend any of the terms & conditions in relation to the Contract and may issue any such directions which are not necessarily stipulated therein if it deems necessary for the fulfilment of the scope of work

5.20 Commencement and progress

5.20.1 The SI shall be subject to the fulfilment of the conditions precedent set out in Clause 5.17 - Conditions Precedent of this Section, commence the performance of its obligations in a manner as specified in the Scope of Work

5.20.2 The SI shall proceed to carry out the activities / services with diligence and expedition in accordance with the stipulation as to the time, manner, mode, and method of execution contained in this RFP

5.20.3 The SI shall be responsible for and shall ensure that all Services are performed in accordance with the Contract, Scope of Work & Specifications. SI shall comply with such Specifications and all other standards, terms and other stipulations/conditions set out hereunder.

5.20.4 The SI shall perform the activities/ services and carry out its obligations under the Contract with due diligence, efficiency and economy, in accordance with generally accepted techniques and practices used in the industry and with professional engineering. It shall employ safe and effective equipment, machinery, material and methods. The SI shall always act, in respect of any matter relating to this RFP, as faithful advisors to the Purchaser and shall, at all times, support and safeguard the Purchaser's legitimate interests in any dealings with Third Parties.

5.20.5 The items supplied under this Contract shall conform to the standards mentioned in Annexure-14. In other cases where no applicable standard is available such standards to be approved/Validated by the CHiPS .

5.21 Standards of Performance

5.21.1 The SI shall perform the activities/ Services and carry out its obligations under the Contract with due diligence, efficiency and economy, in accordance with generally accepted techniques and practices used in the industry and with professional engineering. It shall employ safe and effective equipment, machinery, material and methods. The SI shall always act, in respect of any matter relating to this RFP, as faithful advisors to the Purchaser and shall, at all times, support and safeguard the Purchaser's legitimate interests in any dealings with Third Parties

5.22 Sub-contract

5.22.1 SI shall notify the Purchaser in writing of all subcontracts awarded under this contract if not already specified in its bid. Such notification, in its original bid or later shall not relieve the SI from any liability or obligation under the Contract

5.23 Bidder's Obligations

5.23.1 The SI's obligations shall include Supply, Installation, Integration, Testing, Commissioning and Maintenance of OFC (Underground) IP-MPLS Network, Radio Network and State Network Operations Centre (S-NOC) (for the bidder selected for package-A) including Operations & Maintenance and service provisioning of the created network for a period of 1 years on turnkey basis as specified by the Purchaser in the Scope of Work and other sections of the RFP and Contract and changes thereof to enable the Purchaser to meet their objectives and operational requirements. It shall be the SI's responsibility to ensure the proper and successful implementation, performance and continued operations of the project in accordance with and in strict adherence to the terms of the Bid, the RFP and this Contract

5.23.2 In addition to the aforementioned, the SI shall perform the Services specified by the 'Scope of Work' requirements as specified in the RFP and changes thereof

5.23.3 The SI shall ensure that the SI's Team is competent, professional and possesses the requisite qualifications and experience appropriate to the task they are required to perform under this Contract. The SI shall ensure that the Services are performed through the efforts of the SI's Team, in accordance with the terms hereof and to the satisfaction of the Purchaser. Nothing in this Contract relieves the SI from its liabilities or obligations under this Contract to provide the Services in accordance with the Contract and the Bid to the extent accepted by the Purchaser

5.23.4 The SI shall be responsible on an ongoing basis for coordination with other vendors and agencies of the Purchaser in order to resolve issues and oversee implementation of the same

5.23.5 **Obligations related to IT Infrastructure**

a. The SI shall Design, supply equipment/ components including associated accessories as under this Contract and Configuration, Implementation, Testing, Commissioning, Operations & Maintenance of those components during the entire period of Contract
b. In case of any dissatisfaction or default on part of the SI in providing the level of support desired by the Purchaser or Purchaser's Technical Representative in relation to the Infrastructure supplied by the SI, the SI shall extend the necessary support required to meet the commitments without any financial liability to the Purchaser
c. It is expected that SI and OEM shall ensure that the equipment/ components being supplied by them shall be supported for the entire contract period. If the same is de-supported by the OEM for any reason whatsoever, the SI shall replace it with an equivalent or better substitute that is acceptable to Purchaser without any additional cost to the Purchaser and without impacting the performance of the Solution in any manner whatsoever
d. In case of any problems/ issues arising due to integration of the Infrastructure supplied by the SI with any other component(s)/ product(s) under the purview of the overall solution, the SI shall replace the required component(s) with an equivalent or better substitute that is acceptable to Purchaser without any additional cost to the Purchaser and without impacting the performance of the solution in any manner whatsoever
e. The SI should perform necessary preventive maintenance and repair to ensure adherence of SLA mentioned in the RFP in accordance with the specifications of the components and the best practices followed in the industry without any additional costs to the Purchaser.
f. The SI shall extend necessary assistance, consultancy and services to the Purchaser beyond the defined Scope of Work to resolve issues related to the components

supplied by him, under critical and unforeseen situations. The SI shall provide required information to the Purchaser as and when the OEM comes out with the same

g. The SI shall ensure that it is in compliance with all Applicable Laws at all times while discharging its Scope of Work.
h. The SI shall ensure that it has procured all necessary permits and consents that may be required to discharge its Scope of Work effectively

5.23.6 **Bidder's (SI) Representative**: The SI's Representative shall have all the powers requisite for the performance of Services under this Contract. The SI's Representative shall liaise with the Purchaser's Representative for the proper coordination and timely completion of the works and on any other matters pertaining to the works. He shall extend full co-operation to Purchaser's representative in the manner required by them for supervision/ inspection/ observation of the Infrastructure, procedures, performance, reports and records pertaining to the works. He shall also have complete charge of the SI's personnel engaged in the performance of the works and to ensure internal discipline, compliance of rules, regulations and safety practice. He shall also co-ordinate and co-operate with other service providers of the Purchaser working at the site/ offsite for activities related to planning, execution of Scope of Work and providing Services under this Contract.

5.23.7 **Reporting Progress**

a. The SI shall monitor progress of all the activities related to the execution of this Contract and shall submit to the Purchaser, at no extra cost, progress reports (in MS Project) with reference to all related work, milestones and their progress during the implementation phase regularly.
b. Periodic meetings shall be held between the representatives of the Purchaser and the SI once in every 15 days during the implementation phase to discuss the progress of implementation. After the implementation phase is over, the meeting shall be held on an ongoing basis, once in every 30 days to discuss the performance of the Contract
c. The SI shall ensure that the respective teams involved in the execution of work are part of such meetings.
d. The equipment, Services and manpower to be provided/ installed/deployed by the SI under the Contract and the manner and speed of implementation & maintenance of the work and Services are to be conducted in a manner to the satisfaction of Purchaser's representative in accordance with the Contract
e. The Purchaser reserves the right to inspect and monitor/ assess the progress/ performance of the work/ services at any time during the course of the Contract. The Purchaser may demand and upon such demand being made, the SI shall provide documents, data, material or any other information which the Purchaser may require, to enable it to assess the progress/ performance of the Work/ Service
f. At any time during the course of the Contract, the Purchaser shall also have the right to conduct, either itself or through an independent audit firm appointed by the Purchaser as it may deem fit, an audit to monitor the performance by the SI of its obligations/ functions in accordance with the standards committed to or required by

the Purchaser and the SI undertakes to cooperate with and provide to the Purchaser/ any other agency appointed by the Purchaser, all documents and other details as may be required by them for this purpose

g. If the rate of progress of the works or any part of them at any time fall behind the stipulated time for completion or is found to be too slow to ensure completion of the works by the stipulated time, or is in deviation to RFP requirements/ standards, the Purchaser's representative shall so notify the SI in writing

h. The SI shall reply to the written notice giving details of the measures he proposes to take to expedite the progress so as to complete the works by the prescribed time or to ensure compliance to RFP requirements. The SI shall not be entitled to any additional payment for taking such steps. If at any time it should appear to the Purchaser or Purchaser's Representative that the actual progress of work does not conform to the approved programme the SI shall produce at the request of the Purchaser's Representative a revised programme showing the modification to the approved programme necessary to ensure completion of the works within the time for completion or steps initiated to ensure compliance to the stipulated requirements

i. The submission seeking approval by the Purchaser or Purchaser's Representative of such programme shall not relieve the SI of any of his duties or responsibilities under the Contract

j. In case during execution of works, the progress falls behind schedule or does not meet the RFP requirements, SI shall deploy extra manpower/ resources to make up the progress or to meet the RFP requirements. Programme for deployment of extra man power/ resources shall be submitted. All time and cost effect in this respect shall be borne, by the SI within the Contract value.

5.23.8 **Knowledge of Site Conditions**

a. The SI shall be deemed to have understood the requirements and have satisfied himself with the Data contained in the Bidding Documents, the quantities and nature of the works and materials necessary for the completion of the works, etc., and in-general to have obtained himself all necessary information of all risks, contingencies and circumstances affecting his obligations and responsibilities therewith under the Contract and his ability to perform it. However, if during delivery or installation, SI observes physical conditions and/or obstructions affecting the work, the SI shall take all measures to overcome them

b. SI shall be deemed to have satisfied himself as to the correctness and sufficiency of the Contract Price for the works. The consideration provided in the Contract for the SI undertaking the works shall cover all the SI's obligation and all matters and things necessary for proper execution and maintenance of the works in accordance with the Contract and for complying with any instructions which the Purchaser's Representative may issue in accordance with the connection therewith and of any proper and reasonable measures which the SI takes in the absence of specific instructions from the Purchaser's Representative

c. The SI shall have conducted its own due diligence with regard to the information contained in the RFP. The Purchaser does not make any representation or warranty as to the accuracy, reliability or completeness of the information in this RFP and it is not possible for the Purchaser to consider particular needs of each Bidder who reads or uses this RFP. Each prospective Bidder should conduct its own investigations and analyses and check the accuracy, reliability and completeness of the information provided in this RFP and obtain independent advice from appropriate sources

d. The Purchaser shall not have any liability to any prospective SI or any other person under any laws (including without limitation the law of contract or tort), the principles of equity, restitution or unjust enrichment or otherwise for any loss, expense or damage which may arise from or be incurred or suffered in connection with anything contained in this RFP, any matter deemed to form part of this RFP, the declaration of the SI, the information supplied by or on behalf of the Purchaser or its employees, any consultants, or otherwise arising in any way from the bidding process. The Purchaser shall also not be liable in any manner whether resulting from negligence or otherwise however caused arising from reliance of any Bidder upon any statements contained in this RFP

5.23.9 **Program of Work**

a. Within 10 days after the award of work under this Contract or prior to kick-off meeting whichever is earlier, the SI shall submit to the Purchaser for its approval a detailed programme showing the sequence, procedure and method in which he proposes to carry out the works as stipulated in the Contract and shall, whenever reasonably required by the Purchaser's Representative furnish in writing the arrangements and methods proposed to be made for carrying out the works. The programme so submitted by the SI shall conform to the duties and periods specified in the Contract. The Purchaser and the SI shall discuss and agree upon the work procedures to be followed for effective execution of the works, which the SI intends to deploy and shall be clearly specified. Approval by the Purchaser's Representative of a programme shall not relieve the SI of any of his duties or responsibilities under the Contract
b. If the SI's work plans necessitate a disruption/ shutdown in Purchaser's operation, the plan shall be mutually discussed and developed so as to keep such disruption/ shutdown to the barest unavoidable minimum. Any time and cost arising due to failure of the SI to develop/ adhere such a work plan shall be to his account

5.23.10 **Bidder's (SI) Organization**

a. The SI's Team shall be deployed for execution of work and provision of Services under this Contract as mentioned in Section 7 - Scope of Work.
b. Within 10 calendar days after the release of Work Order under this Contract or prior to the kick-off meeting whichever is earlier, an organization chart showing the proposed organization/ manpower to be deployed by the SI for execution of the work including the identities and Curriculum-Vitae of the key personnel to be deployed. And same shall be approved by Purchaser.
c. The SI should to the best of his efforts, avoid any change in the organization structure proposed for execution of this Contract or replacement of any manpower resource appointed for the project. If the same is however unavoidable, due to circumstances such as the resource leaving the SI's organization, the outgoing

resource shall be replaced with an equally competent resource or better on approval from the Purchaser. The SI shall promptly inform the Purchaser in writing, if any such revision or change is necessary

d. In case of replacement of any manpower resource, the SI should ensure efficient knowledge transfer from the outgoing resource to the incoming resource and adequate hand-holding period and training for the incoming resource in order to maintain the continued level of service
e. All manpower resources deployed by the SI for execution of this Contract must be available for the entire contract period.
f. The SI shall provide necessary supervision during the execution of work and as long thereafter as the Purchaser may consider necessary for the proper fulfilment of the SI's obligations under the Contract. The SI or his competent and authorized representative(s) shall be constantly present at the site whole time for supervision. The SI shall authorize his representative to receive directions and instructions from the Purchaser's Representative
g. The SI shall be responsible for the deployment, transportation, accommodation and other requirements of all its employees required for the execution of the work and provision of Services for all costs/ charges in connection thereof.
h. The SI shall provide and deploy only those manpower resources who are qualified/ skilled and experienced in their respective trades and who are competent to deliver in a proper and timely manner the work they are required to perform or to manage/ supervise the work
i. The Purchaser's Representative may at any time object to and require the SI to remove forthwith from the Project any authorized representative or employee of the SI or any person(s) of the SI's Team, if, in the opinion of the Purchaser's Representative the person in question has misconducted or his/ her deployment is otherwise considered undesirable by the Purchaser's Representative. The SI shall forthwith remove and shall not again deploy the person without the written consent of the Purchaser's Representative
j. The Purchaser's Representative may at any time object to and request the SI to remove from the Project any of SI's authorized representative including any employee of the SI or his team or any person(s) deployed by SI or his team for professional incompetence or negligence or for being deployed for work for which he is not suited. The SI shall consider the Purchaser's Representative request and may accede to or disregard it. The Purchaser's Representative, having made a request, as aforesaid in the case of any person, which the SI has disregarded, may in the case of the same person at any time but on a different occasion, and for a different instance of one of the reasons referred to above in this clause object to and require the SI to remove that person from deployment on the work, which the SI shall then forthwith do and shall not again deploy any person so objected to on the

work or on the sort of work in question (as the case may be) without the written consent of the Purchaser's Representative

k. The Purchaser's Representative shall state to the SI in writing his reasons for any request or requirement pursuant to this clause

l. The SI shall maintain backup personnel and shall promptly replace every person removed, pursuant to this section, with an equally competent substitute from the pool of backup personnel.

m. The SI and its sub-vendors shall deploy only eligible personnel for the execution of this project. Under no circumstances shall personnel previously associated with any direct vendor and Manpower of CHiPS involved in the implementation of BharatNet Phase II be engaged for this assignment.

5.23.11 **Adherence to applicable laws, safety procedures, rules regulations and restrictions**

a. SI shall comply with the provision of all laws including labour laws, rules, regulations and notifications issued there under from time to time. All safety and labour laws enforced by statutory agencies and by Purchaser shall be applicable in the performance of this Contract and SI shall abide by these laws

b. SI shall take all measures necessary or proper to protect the personnel, work and facilities and shall observe all reasonable safety rules and instructions. Purchaser's employee also shall comply with safety procedures/ policy

c. The SI shall report as soon as possible any evidence, which may indicate or is likely to lead to an abnormal or dangerous situation and shall take all necessary emergency control steps to avoid such abnormal situations

d. SI shall also adhere to all security requirement/ regulations of the Purchaser during the execution of the work

5.23.12 **Statutory Requirements**

a. During the tenure of this Contract nothing shall be done by the SI in contravention of any law, act and/ or rules/ regulations, there under or any amendment thereof governing inter-alia customs, stowaways etc. and shall keep Purchaser indemnified in this regard.

b. The SI and their personnel/ representative shall not alter/ change/ replace any hardware component proprietary to the Purchaser and/ or maintenance of Third Party without prior consent of the Purchaser.

c. The SI and their personnel/ representative shall not without consent of the Purchaser install any hardware or software not purchased/ owned by the Purchaser.

5.24 SI's Personnel

5.24.1 The SI shall employ and provide such qualified and experienced personnel as are required to perform the Services under the Contract.

5.24.2 All the personnel, also of the SI's partners shall be deployed only after adequate background verification check. The SI shall submit the background verification check report for the personnel before their deployment on the Project. Any deviations, if observed, would lead to removal of the personnel from the Project.

5.25 Contract Administration

5.25.1 No variation or modification of the terms & conditions of the contract shall be made except by written amendment signed by the parties.

5.25.2 Either party may appoint any individual / organization as their authorized representative through a written notice to the other party. Each Representative shall have the authority to:

a. Exercise all of the powers and functions of his/her Party under this Contract other than the power to amend this Contract and ensure the proper administration and performance of the terms hereof; and

b. Bind his or her Party in relation to any matter arising out of or in connection with this Contract

5.25.3 The SI along with other members / third parties / OEMs shall be bound by all undertakings and representations made by the authorized representative of the SI and any covenants stipulated hereunder, with respect to this Contract, for and on their behalf .

5.25.4 For the purpose of execution or performance of the obligations under this Contract, the Purchaser's representative would act as an interface with the nominated representative of the SI. The SI shall comply with all instructions that are given by the Purchaser's representative during the course of this Contract in relation to the performance of its obligations under the terms of this Contract and the RFP.

5.26 Purchaser's Right of Monitoring, Inspection and Periodic Audit

5.26.1 The Purchaser or Purchaser's Technical Representative reserves the right to inspect and monitor/ assess the progress/ performance/ maintenance of the project at any time during the course of the Contract, after providing due notice to the SI. The Purchaser may demand and upon such demand being made, the purchaser shall be provided with any document, data, material or any other information which it may require, to enable it to assess the progress of the Project

5.26.2 The Purchaser or Purchaser's Technical Representative shall also have the right to conduct, through an independent audit firm appointed by the Purchaser as it may deem fit, an audit to monitor the performance by the SI of its obligations/ functions in accordance with the standards committed to or required by the Purchaser and the SI undertakes to cooperate with and provide to the Purchaser, all documents and other details as may be required by them for this purpose. Any deviations or contravention identified as a result of such audit/ assessment would need to be rectified by the SI failing which the Purchaser may, without prejudice to any other rights that it may issue a notice of default

5.26.3 The SI shall at all times provide to the Purchaser or the Purchaser's Representative access to the Site.

5.27 Purchaser's Obligations

5.27.1 The Purchaser's Representative shall interface with the SI, to provide the required information, clarifications and to resolve any issues as may arise during the execution of the Contract. Purchaser shall provide adequate cooperation in providing details, assisting with coordinating and obtaining of approvals from various governmental agencies, in cases, where the intervention of the Purchaser is proper and necessary.

5.27.2 Purchaser shall ensure that timely approval is provided to the SI, where deemed necessary, which should include technical architecture diagrams and all the specifications related to IT infrastructure required to be provided as part of the Scope of Work. All such documents shall be approved within 15 days of the receipt of the documents by the Purchaser.

5.27.3 The Purchaser shall approve all such documents as per above Clause

5.28 Intellectual Property Rights

5.28.1 Purchaser shall own and have Intellectual Property Rights of all the deliverables which have been developed by the SI during the performance of Services and for the purposes of inter-alia use of such Services under this Contract. The SI undertakes to disclose all Intellectual Property Rights arising out of or in connection with the performance of the Services to the Purchaser and execute all such agreements/documents and file all relevant applications, effect transfers and obtain all permits and approvals that may be necessary in this regard to effectively conserve the Intellectual Property Rights of the Purchaser

5.28.2 If Purchaser desires, Further, the SI shall be obliged to ensure that all approvals, registrations which are inter-alia necessary for use of the infrastructure installed by the SI, the same shall be acquired in the name of the Purchaser, prior to termination of this Contract and which shall be assigned by the Purchaser to the SI for the purpose of execution of any of its obligations under the terms of the Bid, RFP or this Contract. However, subsequent to the term of this Contract, such approvals etc. shall endure to the exclusive benefit of the Purchaser

5.28.3 The SI shall ensure that while it uses any hardware, processes or material in the course of performing the Services, it does not infringe the Intellectual Property Rights of any person and SI shall keep the Purchaser indemnified against all costs, expenses and liabilities howsoever, arising out of any illegal or unauthorized use (piracy) or in connection with any claim or proceedings relating to any breach or violation of any permission/license terms or infringement of any Intellectual Property Rights by the SI during the course of performance of the Services

5.29 Record of Contract Documents

5.29.1 The SI shall at all-time make and keep sufficient copies of the specifications and Contract documents for him to fulfil his duties under the contract

5.29.2 The SI shall keep at least two copies of each and every specification and Contract document, in excess of his own requirement and those copies shall be available at all times for use by the Purchaser's Representative and by any other person authorized by the Purchaser's Representative

5.30 Ownership and Retention of Documents

5.30.1 The Purchaser shall own the Documents, prepared by or for the SI arising out of or in connection with this Contract

5.30.2 Forthwith upon expiry or earlier termination of this Contract and at any other time on demand by the Purchaser, the SI shall deliver to the Purchaser all Documents provided by or originating from the Purchaser and all Documents produced by or from or for the SI in the course of performing the Services, unless otherwise directed in writing by the Purchaser at no additional cost. The SI shall not, without the prior written consent of the Purchaser store, copy, distribute or retain any such Documents

5.31 Ownership of Equipment

5.31.1 The Purchaser shall own assets/ components including but not limited to equipment, documents and items supplied by the SI arising out of or in connection with this Contract

5.31.2 However, all the risk and liability arising out of or in connection with the usage of the equipment, assets/ components during the term of the Contract shall be borne by the SI

5.32 Indemnity

5.32.1 The SI shall indemnify and defend CHiPS and its representatives & employees and hold CHiPS, its representatives, employees harmless from:

5.32.2 Damages and losses caused by its negligent or intentional act or omission or any damages and losses caused by the negligent act of any third party or sub-contractor or agency engaged by the SI

5.32.3 If any third party files a claim against CHiPS or its nominated agency alleging that the Deliverables, Services, or Equipment provided by the SI infringe upon copyrights, trade secrets, patents, or other intellectual property rights, the SI shall:

- **Defend the claim** at its own expense; and
- **Bear responsibility for any costs or damages** that are finally awarded against CHiPS or its nominated agency.

However, the SI shall have no obligation to indemnify CHiPS if the infringement claim arises due to:

a) Misuse or modification of the Deliverables by CHiPS;

b) CHiPS' failure to implement corrections or enhancements provided by the SI; or

c) Use of the Deliverables by CHiPS in combination with any product or information not owned, developed, or supplied by the SI.

5.32.4 If any Deliverable is or likely to be held to be infringing, the SI shall at its expense and option either (i) procure the right for CHiPS to continue using it, or (ii) replace it with a non- infringing equivalent, or (iii) modify it to make it non-infringing.

5.32.5 Any environmental damages caused by it and/or its representatives or employees or employees of any third party or sub-contractor or agency engaged by the SI

5.32.6 Breach (either directly by it or through its representatives and/or employees) of any representation and guarantee declared herein by it;

5.32.7 From any and all claims, actions, suits, proceedings, taxes, duties, levies, costs, expenses, damages and liabilities, including attorneys' fees, arising out of, connected with, or resulting from or arising in connections with the services provided due to neglect, omission or intentional act

5.33 Confidentiality

5.33.1 Information relating to the examination, clarification, evaluation and recommendation for the Bidders shall not be disclosed to any person who is not officially concerned with the process or is not a retained professional advisor advising CHiPS in relation to, or matters arising out of, or concerning the Bidding Process or CHiPS will treat all information, submitted as part of the Bid, in confidence and will require all those who have access to such material to treat the same in confidence. CHiPS may not divulge any such information unless it is directed to do so by any statutory entity that has the power under Applicable Law to require its disclosure or is to enforce or assert any right or privilege of the statutory entity and/ or CHiPS or as may be required by Applicable Law or in connection with any legal process. Bidders are to treat all information as strictly confidential and shall not use it for any purpose other than for preparation and submission of their Bid. The provisions of this Clause shall also apply mutatis mutandis to Bids and all other documents submitted by the Bidders, and CHiPS shall be under no obligation whatsoever to return to the Bidders any document or any information provided along therewith

5.33.2 The SI shall not use Confidential Information, the name or the logo of the Purchaser except for the purposes of providing the Service as specified under this RFP;

5.33.3 The SI shall not, either during the term or 6 months after expiration of this Contract, disclose any proprietary or confidential information relating to the Services, Contract or the network architecture, Purchaser's business plan or operations without the prior written consent of the Purchaser.

5.33.4 The SI may only disclose Confidential Information in the following circumstances:

a. with the prior written consent of the Purchaser;

b. to a member of the SI s' Team ("Authorized Person") if:

- the Authorized Person needs the Confidential Information for the performance of obligations under this contract;
- the Authorized Person is aware of the confidentiality of the Confidential Information and is obliged to use it only for the performance of obligations under this contract
- If the information is already made available in any public domain

5.33.5 The SI shall do everything reasonably possible to preserve the confidentiality of the Confidential Information including execution of a confidential agreement with the members of the, subcontractors and other service provider's team members to the satisfaction of the Purchaser.

5.33.6 The SI shall sign a Non-Disclosure Agreement (NDA) with the Purchaser on mutually agreed terms & conditions. The SI and its antecedents shall be bound by the NDA. The SI shall be responsible for any breach of the NDA by its antecedents or delegates.

5.33.7 The SI shall notify the Purchaser promptly if it is aware of any disclosure of the Confidential Information otherwise than as permitted by this Contract or with the authority of the Purchaser.

5.33.8 The Purchaser reserves the right to adopt legal proceedings, civil or criminal, against the SI in relation to a dispute arising out of breach of obligation by the SI under this clause.

5.34 Taxes

5.34.1 The quoted price should be inclusive of all the applicable Taxes, levies, duties, etc. (excluding GST). GST shall be paid extra on actual, as applicable. The GST prevailing at the time of raising the invoice shall be paid. CHiPS shall deduct appropriate tax as applicable at source from the payment against the delivery & services and corresponding TDS certificate shall be issued at the end of respective quarter

5.34.2 In case of any variation (upward or downward) in GST wherever applicable up to the date of invoice, the benefit or the burden of the same shall be passed on to CHiPS. Necessary documentary evidence shall be produced for having paid the excise duty, GST, if applicable and/ or other applicable levies. Variation would also include the introduction of any new tax / cess

5.35 Warranty

5.35.1 A comprehensive on-site warranty and Operations & Maintenance on all goods supplied under this contract shall be provided by the respective Original Equipment Manufacturer (OEM) through SI till the end of the Contract

5.35.2 Technical Support shall be provided by the respective OEM till the end of the contract period

5.35.3 The SI shall warrant that the goods supplied under the Contract are new, non-refurbished, unused and recently manufactured; shall not be nearing End of Sale / End of Support; and shall be supported by the SI and respective OEM along with service and spares support to ensure its efficient and effective operation for the entire duration of the contract

5.35.4 The SI warrants that the goods supplied under this contract shall be of the reasonably acceptable grade and quality and consisted with the established and generally accepted standards for materials of this type. The goods shall be in full conformity with the specifications and shall operate properly and safely. All recent design improvements in goods, unless provided otherwise in the Contract, shall also be made available.

5.35.5 The SI further warrants that the Goods supplied under this Contract shall be free from all encumbrances and defects/faults arising from design, material, manufacture or workmanship (except insofar as the design or material is required by the Purchaser's Specifications).

5.35.6 The Purchaser shall promptly notify the SI in writing of any claims arising under this warranty.

5.35.7 Upon receipt of such notice, the SI shall, with all reasonable speed, repair or replace the defective Goods or parts thereof, without prejudice to any other rights which the Purchaser may have against the SI under the Contract

5.35.8 If the SI, having been notified, fails to remedy the defect(s) within a reasonable period, the Purchaser may proceed to take such remedial action as may be necessary, at the SI s' risk & expense and without prejudice to any other rights which the Purchaser may have against the SI under the Contract

5.36 Term and Extension of the Contract

5.36.1 Contract period shall include six (6) months for implementation and one (1) year of Operations & Maintenance (O&M), may be extendable by an additional one (1) year based on performance and requirements at discretion of the purchaser.

5.36.2 The Purchaser shall reserve the sole right to grant any extension to the term mentioned above and shall notify in writing to the SI, whether it shall grant the SI an extension of the term. The decision to grant or refuse the extension shall be at the Purchaser's discretion. Accordingly, the Bank Guarantee of the same amount shall be extended up to the extended period of the Contract.

5.36.3 Where the Purchaser is of the view that no further extension of the term be granted to the SI, the Purchaser shall notify the SI of its decision at least 1 month prior to the expiry of the Term. Upon receipt of such notice, the SI shall continue to perform all its obligations hereunder until such reasonable time beyond the Term of the Contract within which the Purchaser shall either appoint an alternative SI/Service provider or create its own infrastructure to operate such Services as are provided under this Contract.

5.37 Prices

5.37.1 Prices quoted must be firm and shall not be subject to any upward revision on any account whatsoever throughout the period of contract.

5.38 Change Orders/ Alteration/ Variation

5.38.1 The SI shall agree that the requirements/quantities/ specifications and Service requirements given in the RFP are minimum requirements and are in no way exhaustive and guaranteed by the Purchaser.

a. Any upward revision (and/or additions consequent to errors, omissions, ambiguities, discrepancies in the quantities, specifications, architecture etc.) of the RFP which the SI had not brought to the Purchaser's notice till the time of award of work and not accounted for in his Bid shall not constitute a change order and such upward revisions and/or addition shall be carried out by the SI without any time and cost effect to Purchaser

b. It shall be the responsibility of the SI to meet all the performance and other requirements of the Purchaser as stipulated in the RFP / Contract. Any upward revisions / additions of quantities, specifications, service requirements to those specified by the SI in his Bid documents, that may be required to be made during installation / commissioning of the network or at any time during the currency of the contract in order to meet the conceptual design, objective and performance levels or other requirements as defined in the RFP. These changes shall be carried out as per mutual consent.

5.38.2 The Purchaser may at any time, by a written change order given to the SI, make changes within the general Scope of Work. The Purchaser shall have an option to increase or decrease (decrease only if communicated to SI prior to availing of services / dispatch of goods / equipment) the Quantities and/or Specifications of the goods/equipment to be supplied and installed by the SI or service requirements, as mentioned in the Contract, at any time during the contract period.

5.38.3 The written advice to any change shall be issued by the Purchaser to the SI up to 4 (four) weeks prior to the due date of provisioning/supply of such goods/equipment or commencement of services.

5.38.4 In case of increase in Quantities / Specifications or Service requirements or in case of additional requirement, the rate as provided in the Contract shall be considered as benchmark rates for procurement of the additional requirement from the SI. However, based on the industry trends, Purchaser retains the right to review these rates. The additional requirement shall also be governed by the same terms & conditions as provided in the Contract except for the appropriate extension of time to be allowed for delivery/installation of such extra goods/equipment or for commencement of such services. In case of decrease in Quantities or Specifications of goods/equipment or Service requirements, the SI shall give a reduction in price at the rate given in the Contract corresponding to the said decrease.

5.38.5 In case applicable rates for the increase/decrease in question are not available in the Contract then the rates as may be mutually agreed shall apply. The SI shall not be entitled to any claim by way of change of price, damages, losses, etc. The SI shall be compensated at actual for any cancellation charges provided the claim is duly supported by documentary evidence of having incurred cancellation charges, which results from Purchaser's action in reducing/cancelling Scope of Work.

5.38.6 Conditions for Change Order

a. The change order shall be initiated only in case (i) the Purchaser or Purchaser's Technical Representative directs in writing the SI to incorporate changes to the goods or design requirements already covered in the Contract. (ii) the Purchaser or Purchaser's Technical Representative directs in writing to the SI to include any addition to the scope of work or services covered under this Contract or delete any part thereof, (iii) SI requests to delete any part of the work which shall not adversely affect the work and if the deletions proposed are agreed to by the Purchaser and for which the cost and time benefits shall be passed on to the Purchaser
b. Any change order comprising an alteration which involves change in the cost of the goods and/or services (which sort of alteration is hereinafter called a "Variation") shall be the Subject of an amendment to the Contract by way of an increase or decrease in the Contract Value and adjustment of the implementation schedule if any
c. If parties agree that the Contract does not contain applicable rates or that the said rates are inappropriate or the said rates are not precisely applicable to the variation in question, then the parties shall review the Contract Value which shall represent the change in cost of the goods and/or works caused by the Variations. Any change order shall be duly approved by the Purchaser in writing

5.38.7 **Procedures for change Order**

a. Upon receiving any revised requirement/advice, in writing, from the Purchaser or Purchaser's Technical Representative, the SI would verbally discuss the matter with Purchaser's Representative
b. In case such requirement arises from the side of the SI, he would also verbally discuss the matter with Purchaser's Representative giving reasons thereof
c. In either of the two cases as explained in Clause 5.38.7 (a) and clause 5.38.7 (b) above, the representatives of both the parties shall discuss on the revised requirement for better understanding and to mutually decide whether such requirement constitutes a change order or not
d. If it is mutually agreed that such requirement constitutes a "Change Order" then a joint memorandum shall be prepared and signed by the SI and Purchaser to confirm a "Change Order" and basic ideas of necessary agreed arrangement
e. SI shall study the revised requirement in accordance with the joint memorandum under Clause 5.38.7 (d) and assess subsequent schedule and cost effect, if any
f. Upon completion of the study referred to above under Clause 5.38.7 (e) the results of this study along with all relevant details including the estimated time and cost effect thereof with supporting documents would be submitted to the Purchaser to enable the Purchaser to give a final decision whether SI should proceed with the change order or not in the best interest of the works
g. The estimated cost and time impact indicated by SI shall be considered as a ceiling limit and shall be provisionally considered for taking a decision to implement change order
h. The time impact applicable to the Contract shall be mutually agreed, subsequently, on the basis of the detailed calculations supported with all relevant back up documents
i. In case SI fails to submit all necessary substantiation/calculations and back up documents, the decision of the Purchaser regarding time and cost impact shall be final and binding on the SI
j. If Purchaser accepts the implementation of the change order under Clause 5.38.7 (f) in writing, which would be considered as change order, then SI shall commence to proceed with the enforcement of the change order pending final agreement between the parties with regard to adjustment of the Contract Value and the Schedule

5.38.8 Conditions for Revised Work / Change Order

a. The provisions of the Contract shall apply to revised work / change order as if the revised work / Change order has been included in the original Scope of work. However, the Contract Value shall increase / decrease and the schedule shall be adjusted on account of the revised work / Change orders as may be mutually agreed in terms of provisions set forth in Clause 5.38. The SI's obligations with respect to such revised work / change order shall remain in accordance with the Contract

5.39 Suspension of Work

5.39.1 The SI shall, if ordered in writing by the Purchaser's Representative, temporarily suspend the works or any part thereof for such a period and such a time as ordered, then SI shall not be entitled to claim compensation for any loss or damage sustained by him by reason of temporary suspension of the Works as aforesaid but shall be eligible for the payment (of products/services delivered and accepted) during the suspension period as per contract. An extension of time for completion, corresponding with the delay caused by any such suspension of the works as aforesaid shall be granted to the SI, if request for same is made and that the suspension was not consequent to any default or failure on the part of the SI. Both SI and purchaser acknowledges the suspension of work by purchaser, if results in extension of contract, the extra cost shall be on account of Purchaser which shall be mutually agreed. In case the suspension of works, is not consequent to any default or failure on the part of the SI, and lasts for a period of more than 2 months, the SI shall have the option to request the Purchaser to terminate the Contract with mutual consent.

5.40 Time is of Essence

5.40.1 Time shall be of the essence in respect of any date or period specified in this RFP or any notice, demand or other communication served under or pursuant to any provision of this RFP and in particular in respect of the completion of the Services by the SI by the completion date

5.41 Completion of Contract

5.41.1 Unless terminated earlier, pursuant to Clauses 5.18, 5.29.3, 5.31, 5.32 and 5.34 the Contract shall terminate on the completion of term as specified in the Contract and only after the obligations mentioned in Clause 5.47 - Consequences of Termination are fulfilled to the satisfaction of the Purchaser

5.42 Special Conditions of Contract

5.42.1 Amendments of, and Supplements to, Clauses in the General Conditions of Contract

5.43 Event of Default by the Bidder

5.43.1 The failure on the part of the SI to perform any of its obligations or comply with any of the terms of this Contract which results in a material breach of the contract shall constitute an Event of Default on the part of the SI. The events of default as mentioned above may include inter-alia the following:

a. the SI has failed to adhere to any of the key performance indicators as laid down in the Key Performance Measures / Contract, or if the SI has fallen short of matching such standards/targets as the Purchaser may have designated with respect to any task necessary for the execution of the scope of work under this Contract which results in a material breach of the contract. The above mentioned failure on the part of the SI may be in terms of failure to adhere to timelines, specifications, requirements etc. or any other criteria as defined by the Purchaser;
b. the SI has failed to remedy a failure to perform its obligations in accordance with the directive, including specifications, timelines etc., issued by the Purchaser, despite being served with a default notice which laid down the specific deviance on the part of the SI to comply with any stipulations or standards as laid down by the Purchaser; or
c. the SI/SIs' team has failed to conform with any of the Service/Facility Specifications/standards as set out in the scope of work of this RFP or has failed to

adhere to any amended direction, modification or clarification as issued by the Purchaser during the term of this Contract and which the Purchaser deems proper and necessary for the execution of the scope of work under this Contract

d. the SI has failed to demonstrate or sustain any representation made by it in this Contract, with respect to any of the terms of its Bid, the RFP and this Contract
e. there is an order from a court of competent jurisdiction for bankruptcy, insolvency, winding up or there is an appointment of receiver, liquidator, assignee, or similar official against or in relation to the SI
f. the SI Abandons the project during the Term of the Contract

5.43.2 Where there has been an occurrence of such defaults inter alia as stated above, the Purchaser shall issue a notice of default to the SI, setting out specific defaults / deviances / omissions and providing a notice of Sixty (60) days to enable such defaulting party to remedy the default committed

5.43.3 Where despite the issuance of a default notice to the SI by the Purchaser the SI fails to remedy the default to the satisfaction of the SI, the Purchaser may, at its sole discretion where it deems fit, issue to the defaulting party another default notice or proceed to adopt such remedies as may be available to the Purchaser

5.44 Consequences of Event of Default

5.44.1 Where an Event of Default subsists or remains uncured the Purchaser may/ shall be entitled to:

5.44.2 The Bidder shall in addition take all available steps to minimize loss resulting from such event of default

5.44.3 The Purchaser may, by a written notice of suspension to the Bidder, suspend all payments to the Bidder under the Contract, provided that such notice of suspension:

a. shall specify the nature of the failure; and
b. shall request the Bidder to remedy such failure within a specified period from the date of receipt of such notice of suspension by the Bidder

5.44.4 In all cases of risk purchase, the difference in cost shall be borne by defaulting Bidder / SI

5.44.5 Terminate the Contract in Part or Full

a. Retain such amounts from the payment due and payable by the Purchaser to the Bidder as may be required to offset any losses caused to the Purchaser as a result of such event of default and the Bidder shall compensate the Purchaser for any such loss, damages or other costs, incurred by the Purchaser in this regard. Nothing herein shall affect the continued obligation of the SI and SI's Team to perform all their obligations and responsibilities under this Contract in an identical manner as were being performed before the occurrence of the default.
b. Invoke the Performance Bank Guarantee and other Guarantees furnished hereunder, recover such other costs/ losses and other amounts from the Bidder as may have

resulted from such default and pursue such other rights and/or remedies that may be available to the Purchaser under law

5.45 Termination

5.45.1 The Purchaser may, terminate this Contract in whole or in part by giving the SI a prior and written notice indicating its intention to terminate the Contract under the following circumstances:

a. Where the Purchaser is of the opinion that there has been such Event of Default on the part of the SI which would make it proper and necessary to terminate this Contract and may include failure on the part of the SI to respect any of its commitments with regard to any part of its obligations under its Bid, the RFP or under this Contract

b. Where it comes to the Purchaser's attention that the SI (or the SI s' Team) is in a position of actual conflict of interest with the interests of the Purchaser, in relation to any of terms of the SI s' Bid, the RFP or this Contract

c. Where the SI s' ability to survive as an independent corporate entity is threatened or is lost owing to any reason whatsoever, including inter-alia the filing of any bankruptcy proceedings against the SI, any failure by the SI to pay any of its dues to its creditors, the institution of any winding up proceedings against the SI or the happening of any such events that are adverse to the commercial viability of the SI. In the event of the happening of any events of the above nature, the Purchaser shall reserve the right to take any steps as are necessary, to ensure the effective transition of the project to a successor SI and to ensure business continuity

d. **Termination for Insolvency**: The Purchaser may at any time terminate the Contract by giving written notice to the SI, without compensation to the SI, if the SI becomes bankrupt or otherwise insolvent, provided that such termination shall not prejudice or affect any right of action or remedy which has accrued or shall accrue thereafter to the Purchaser.

e. **Termination for Convenience**: The Purchaser, may, by prior written notice sent to the SI at least 1 months in advance, terminate the Contract, in whole or in part at any time for its convenience. The notice of termination shall specify that termination is for the Purchaser's convenience, the extent to which performance of work under the Contract is terminated, and the date upon which such termination becomes effective

5.46 Consequences of Termination

5.46.1 In the event of termination of this contract due to any cause whatsoever, the contract with stand cancelled effective from the date of termination of this contract

5.46.2 In case of exigency, if the Purchaser gets the work done from elsewhere, the difference in the cost of getting the work done shall be borne by the SI

5.46.3 Where the termination of the Contract is prior to its stipulated term on account of a Default on the part of the SI or due to the fact that the survival of the SI as an independent corporate entity is threatened/has ceased, or for any other reason, whatsoever, the Purchaser through re-determination of the consideration payable to the SI as agreed mutually by the Purchaser and the SI for that part of the Services which have been authorized by the Purchaser and satisfactorily performed by the SI up to the date of termination. Without prejudice any other rights, the Purchaser may retain such amounts from the payment due and payable by the Purchaser to the SI as may be required to offset any losses caused to the Purchaser as a result of any act/omissions of the SI. In case of any loss or damage due to default on the part of the SI in performing any of its obligations with regard to the execution of the scope of work under this Contract, the SI shall compensate the Purchaser for any such loss, damages or other costs, incurred by the Purchaser. Additionally, other members of its team shall perform all its obligations and responsibilities under this Contract in an identical manner as were being performed before the collapse of the SI as described above in order to execute an effective transition and to maintain business continuity.

5.46.4 Nothing herein shall restrict the right of the Purchaser to invoke the Bank Guarantee and other Guarantees furnished hereunder, enforce the Deed of Indemnity and pursue such other rights and/or remedies that may be available to the Purchaser under law.

5.46.5 The termination hereof shall not affect any accrued right or liability of either Party nor affect the operation of the provisions of this Contract that are expressly or by implication intended to come into or continue in force on or after such termination.

5.47 Penalty

5.47.1 Ongoing performance and Service Levels shall be as per parameters stipulated by the Purchaser in this Contract, failing which the Purchaser may, at its discretion, impose penalties on the Bidder as defined in General Conditions of the Contract and Service Level Agreement of the RFP

5.48 Liquidated Damages

5.48.1 The SI shall perform the Services and comply in all respects with the critical dates and the parties hereby agree that failure on part of the SI to meet the critical dates without prejudice to any other rights that the Purchaser have, may lead to the imposition of such obligations as are laid down in the Delay and Deterrent Mechanism and/or levy of penalty as set and/or termination of the Contract at the discretion of the Purchaser.

5.48.2 Penalties shall be capped to maximum of 10% of total work order value issued under project/tender. Beyond 10% of total work order value, the Purchaser has the right to terminate the contract or a portion or part of the work thereof. The purchaser shall give 30 days' notice to the SI of its intention to terminate the Contract and shall so terminate the Contract unless the SI initiates remedial action acceptable to the Purchaser during the 30 days' notice period,

5.48.3 The Purchaser may without prejudice to its right to effect recovery by any other method, deduct the amount of liquidated damages from any money belonging to the SI in its hands (which includes the Purchaser's right to claim such amount against SI s' Bank Guarantee) or which may become due to the SI. Any such recovery or liquidated damages shall not in any way relieve the SI from any of its obligations to complete the Works or from any other obligations and liabilities under the Contract.

5.48.4 Delay not attributable to the SI shall be considered for exclusion for the purpose of computing liquidated damages

5.49 Dispute Resolution

5.49.1 The Purchaser and the SI shall make every effort to resolve amicably by direct informal negotiations, any disagreement, or disputes, arising between them under or in connection with the Contract

5.49.2 If, the parties fail to resolve the dispute, differences, arising out of works contract, amicably, through informal negotiations within Thirty (30) days from commencement of such negotiation, either party may refer the Dispute to be resolved through Madhyastham established under Chhattisgarh Madhyastham Adhikaran Adhiniyam, 1983, or any statutory modification or re-enactment thereof.

5.49.3 The exclusive jurisdiction for resolution of dispute, shall be Raipur, Chhattisgarh, India.

5.49.4 It is also a term of the Contract that neither party to the Contract shall be entitled for any interest on the amount of the award.

5.49.5 Continuance of the Contract: Notwithstanding the fact that settlement of dispute(s) (if any) may be pending before the tribunal, the parties hereto shall continue to be governed by and perform the work in accordance with the provisions under this Contract.

5.50 Insurance

5.50.1 The Goods supplied under this Contract shall be fully insured by the SI, against any loss or damage up to the time it is delivered to the SI -designated carrier for shipment to Purchaser or to Purchaser's designated location. The SI shall submit to the Purchaser, certificate of insurance issued by the insurance company, indicating that such insurance has been taken

5.50.2 The SI shall bear all the statutory levies like customs, insurance, freight, etc. applicable on the goods during their shipment from respective manufacturing/shipment site of the OEM to the port of landing.

5.50.3 All charges such as transportation, logistics, warehousing, security, etc. that may be applicable till the goods are delivered and upon completion of acceptance testing at the respective site of installation shall be borne by the SI.

5.50.4 The SI during the term of this contract undertakes to ensure that it has taken or shall take up all appropriate insurances for the delivery of goods that it is required to undertake under law as well as to adequately cover its obligations under this Contract: shall take out and maintain, at his own cost insurance with IRDA approved insurers against the risks, and for the coverage, as specified below: shall pay all premium in relation thereto and shall ensure that nothing is done to make such insurance policies void or voidable at the Purchaser's request, shall provide certificate of insurance to the Purchaser showing that such insurance has been taken out and maintained. Employer's liability and workers' compensation insurance in respect of the Personnel of the SI / SI s' Team, in accordance with the relevant provisions of the Applicable Law, as well as, with respect to such Personnel, any such life, health, accident, travel or other insurance as may be appropriate; and

5.50.5 Insurance against loss of or damage to (i) equipment or assets procured in full or in part for fulfilment of obligations under this Contract (ii) the SI s' assets and property used in the performance of the Services.

5.51 Transfer of Ownership

5.51.1 The SI must transfer all goods, clear and unencumbered titles to the assets and goods procured for the purpose of the project to the Purchaser upon end to end connectivity of GPs.

5.51.2 The asset(s) so created shall be a National Asset fully owned by the Government of India and non-discriminatory access, including leasing of dark fibre.

5.52 Limitation of the Bidder's (SI) Liability towards the Purchaser

5.52.1 Except in case of gross negligence, wilful misconduct, breach of Application Laws, breach of representations & warranties and breach of indemnity provisions on the part of the SI or on the part of any person or company acting on behalf of the SI in carrying out the Services, the SI, with respect to damage caused by the SI to Purchaser's property, shall not be liable to Purchaser

a. For any indirect or consequential loss or damage; and
b. For any direct loss or damage that exceeds the total payments payable under this contract to the SI hereunder.

5.52.2 This limitation of liability shall not affect the SI's liability, if any, for direct damage to Third Parties resulting in bodily injury, death or damage to physical property caused by the SI or any person or firm/company acting on behalf of the SI in carrying out the Services. Notwithstanding anything stated to the contrary in the RFP, Limitation of liability, including for direct damage to Third Parties, shall be to the extent of 100% of the total cost of the project calculated up to and as on the date when such section / clause is required to be invoked.

5.53 Conflict of Interest

5.53.1 If the Bidder / SI is found to have a conflict of interest that affects the Bidding Process. Any Bidder found to have a Conflict of Interest shall be disqualified. In the event of disqualification, CHiPS shall be entitled to forfeit and appropriate the EMD or PBG, for the Damages incurred by CHiPS, as the case may be, without prejudice to any other right or remedy that may be available to CHiPS under the Bidding Documents and/ or the Agreement or otherwise. Without limiting the generality of the above, a Bidder shall be deemed to have a Conflict of Interest affecting the Bidding Process, if ("Conflict of Interest"):

5.53.1.1 The Bidder / SI, its member or Associate (or any constituent thereof) and any other Bidder, its member or any Associate thereof (or any constituent thereof) have common controlling shareholders or other ownership interest; provided that this disqualification shall not apply in cases where the direct or indirect shareholding of a Bidder, its member or an Associate thereof (or any shareholder thereof having a shareholding of more than 5% (five percent) of the paid up and subscribed share capital of such Bidder, member or Associate, as the case may be) in the other Bidder, its member or Associate, is less than 5% (five percent) of the subscribed and paid up equity share capital thereof;

provided further that this disqualification shall not apply to any ownership by a bank, insurance company, pension fund or a public financial institution referred to in sub-section (72) of section 2 of the Companies Act, 1956/2013. For the purposes of this Clause, indirect shareholding held through one or more intermediate persons shall be computed as follows:

a. where any intermediary is controlled by a person through management control or otherwise, the entire shareholding held by such controlled intermediary in any other person (the “Subject Person”) shall be taken into account for computing the shareholding of such controlling person in the Subject Person; and
b. subject to Clause 5.54.1.1 (a) above, where a person does not exercise control over an intermediary, which has shareholding in the Subject Person, the computation of indirect shareholding of such person in the Subject Person shall be undertaken on a proportionate basis; provided, however, that no such shareholding shall be reckoned under this Clause, if the shareholding of such person in the intermediary is less than 26% (twenty six percent) of the subscribed and paid up equity shareholding of such intermediary; or
c. a constituent of such Bidder is also a constituent of another Bidder; or
d. such Bidder, its member or any Associate thereof receives or has received any direct or indirect subsidy, grant, or subordinated debt from any other Bidder, its member or Associate, or has provided any such subsidy, grant, or subordinated debt to any other Bidder, its member or any Associate thereof; or
e. such Bidder has the same legal representative for purposes of this Bid as any other Bidder; or
f. such Bidder, or any Associate thereof, has a relationship with another Bidder, or any Associate thereof, directly or through common third party/ parties, that puts either or both of them in a position to have access to each other’s information about, or to influence the Bid of either or each other; or
g. Such Bidder or any Associate thereof has participated as a consultant to CHiPS in the preparation of any documents, design or technical specifications of the Project.

5.53.1.2 For purposes of this RFP, Associate means, in relation to the Bidder, a person who controls, is controlled by, or is under the common control with such Bidder (the “Associate”). As used in this definition, the expression “control” means, with respect to a person which is a company or corporation, the ownership, directly or indirectly, of more than 50% (fifty per cent) of the voting shares of such person, and with respect to a person which is not a company or corporation, the power to direct the management and policies of such person by operation of Applicable Law.

5.53.1.3 The Purchaser requires that the SI provides services which at all times hold the Purchaser’s interests paramount, avoid conflicts with other assignments or its own interests, and act without any consideration for future work. The SI shall not accept or engage in any assignment that would be in conflict with its prior or current obligations to other clients, or that may place it in a position of not being able to carry out the assignment in the best interests of the Purchaser.

5.54 Severance

5.54.1 In the event any provision of this Contract is held to be invalid or unenforceable under the applicable law, the remaining provisions of this Contract shall remain in full force and effect

5.55 Governing Language

5.55.1 The Contract shall be written in English language. Subject to Clause 5.60.5 such language versions of the Contract shall govern its interpretation. All correspondence and other documents pertaining to the Contract that are exchanged by parties shall be written in English language only

5.56 "No Claim" Certificate

5.56.1 The SI shall not be entitled to make any claim, whatsoever against the Purchaser, under or by virtue of or arising out of, this contract, nor shall the Purchaser entertain or consider any such claim, if made by the SI after he shall have signed a "No claim" certificate in favour of the Purchaser in such forms as shall be required by the Purchaser after the works are finally accepted.

5.56.2 It shall be deemed that by submitting the Bid, the Bidder agrees and releases CHiPS, its employees, agents and advisers, irrevocably, unconditionally, fully and finally from any and all liability for claims, losses, damages, costs, expenses or liabilities in any way related to or arising from the exercise of any rights and/ or performance of any obligations hereunder, pursuant hereto and/ or in connection with the Bidding Process and waives, to the fullest extent permitted by Applicable Laws, any and all rights and/ or claims it may have in this respect, whether actual or contingent, whether present or in future.

5.57 Publicity

5.57.1 The SI shall not make or permit to be made a public announcement or media release about any aspect of this Contract unless the Purchaser first gives the SI its written consent.

5.58 Force Majeure

5.58.1 If, at any time, during the continuance of this contract, the performance in whole or in part by either party of any obligation under this contract is prevented or delayed by reasons of any war or hostility, acts of the public enemy, civil commotion, sabotage, fires, floods, explosions, epidemics, quarantine restrictions, strikes, lockouts or act of God (hereinafter referred to as events) provided notice of happenings of any such eventuality is given by either party to the other within 21 days from the date of occurrence thereof, neither party shall by reason of such event be entitled to terminate this contract nor shall either party have any claim for damages against other in respect of such non-performance or delay in performance, and deliveries under the contract shall be resumed as soon as practicable after such an event come to an end or cease to exist, and the decision of the Purchaser as to whether the deliveries have been so resumed or not shall be final and conclusive.

Further that if the performance in whole or part of any obligation under this contract is prevented or delayed by reasons of any such event for a period exceeding 60 days, either party may, at its option, terminate the contract.

5.58.2 Provided, also that if the contract is terminated under this clause, the Purchaser shall be at liberty to take over from the SI at a price to be fixed by the purchaser, which shall be final, all unused, undamaged and acceptable materials, bought out components and stores in course of manufacture which may be in possession of the SI at the time of such termination or such portion thereof as the purchaser may deem fit, except such materials, bought out components and stores as the SI may with the concurrence of the purchaser elect to retain.

5.59 General

5.59.1 Relationship between the Parties

a. Nothing in this Contract constitutes any fiduciary relationship between the Purchaser and SI/ SI's Team or any relationship of employer employee, principal and agent, or partnership, between the Purchaser and SI
b. No Party has any authority to bind the other Party in any manner whatsoever except as agreed under the terms of this Contract
c. The Purchaser has no obligations to the SI's Team except as agreed under the terms of this Contract

5.59.2 **No Assignment:** The SI shall not transfer any interest, right, benefit or obligation under this Contract without the prior written consent of the Purchaser

5.59.3 **Survival**

5.59.3.1 The provisions of the clauses of this Contract in relation to documents, property, Intellectual Property Rights, indemnity, publicity and confidentiality and ownership survive the expiry or termination of this Contract and in relation to confidentiality, the obligations continue to apply unless the Purchaser notifies the SI of its release from those obligations

5.59.3.2 Entire Contract

5.59.3.3 The terms and conditions laid down in the RFP and all annexure thereto as also the Bid and any attachments/ annexes thereto shall be read in consonance with and form an integral part of this Contract. This Contract supersedes any prior Contract, understanding or representation of the Parties on the subject matter

5.59.3.4 Governing Law and Jurisdiction of Courts

5.59.3.5 The Bidding Process shall be governed by, and construed in accordance with, the laws of India and the Courts in the State of Chhattisgarh, in which CHiPS has its headquarters shall have exclusive jurisdiction over all disputes arising under, pursuant to and/ or in connection with the Bidding Process

5.59.4 **Jurisdiction of Courts**

5.59.4.1 The High Court of Chhattisgarh at Bilaspur, Chhattisgarh will have jurisdiction for any proceeding in relation to this Contract

5.59.5 **Residuary Governing Rules**

5.59.5.1 Any matter which has not been covered under these provisions shall be governed as per the provisions of Chhattisgarh State Government Rules and the Applicable Law.

5.59.6 **Compliance with Laws**

5.59.6.1 The SI shall comply with the laws in force in India in the course of performing this Contract

5.59.7 **Notices**

a. A “notice” means:

- A notice; or
- A consent, approval or other communication required to be in writing under this Contract.
- All notices, requests or consents provided for or permitted to be given under this Contract shall be in writing and shall be deemed effectively given when personally delivered or mailed by pre-paid certified/ registered mail, return receipt requested, addressed as follows and shall be deemed received two days after mailing or on the date of delivery if personally delivered:

To Purchaser at:

Chhattisgarh infotech Promotion Society (CHiPS)

<<Attn: XXXX, XXXX, CHiPS >>

[Phone:]

[Fax:]

To SI at:

Attn:

[Phone:]

[Fax:]

- Any Party may change the address to which notices are to be directed to it by notice to the other parties in the manner specified above

A notice served on a Representative is taken to be notice to that Representative’s Party

5.59.8 **Waiver**

a. Any waiver of any provision of this Contract is ineffective unless it is in writing and signed by the Party waiving its rights
b. A waiver by either Party in respect of a breach of a provision of this Contract by the other Party is not a waiver in respect of any other breach of that or any other provision
c. The failure of either Party to enforce at any time any of the provisions of this Contract shall not be interpreted as a waiver of such provision

5.59.9 **Modification:** Any modification of this Contract shall be in writing and signed by an authorized representative of each Party

5.59.10 **Application:** These General Conditions shall apply to the extent that provisions in other parts of the Contract do not supersede them

5.59.11 **Overriding Effect:** Notwithstanding anything to the contrary contained in this RFP (including the Annexures), and without prejudice to the obligations of the Bidder set out in detail herein (which shall apply in addition to the obligations set out under the Agreement), in the event of a conflict, the detailed terms specified in the Agreement shall have overriding effect over the terms provided hereunder.

5.59.12 **Full Documents:** The Bidder shall ensure that all the documents as required under this RFP for evidencing inter-alia its technical capabilities, financial capabilities and eligibility requirements, are submitted by it in full and are not redacted or suppressed in any manner except as required pursuant to the Applicable Law (in which case the Bidder shall intimate CHiPS to this effect in writing before submission of the Bid).

5.59.13 **CHiPS Property:** This RFP and all other documents attached / annexed / appended / provided herewith, are provided by CHiPS and shall remain (or become upon their coming into existence) the property of CHiPS and are transmitted to the Bidders solely for preparation and the submission of a Bid in accordance herewith.

5.60 Exit Management Plan

5.60.1 This clause sets out the provisions which shall apply upon completion of the contract period or upon termination of the contract for default of the SI. An Exit Management plan shall be furnished by SI in writing to the Purchaser within 60 days on completion of the contract period or termination of the contract for default of the SI, which shall deal with at least the following aspects of exit management in relation to the contract as a whole and in relation to the Project Implementation and Service Level monitoring.

d. A detailed program of the transfer process that could be used in conjunction with a Replacement SI including details of the means to be used to ensure continuing provision of the services throughout the transfer process or until the cessation of the services and of the management structure to be used during the transfer;

e. Plans for provision of contingent support to Project and Replacement SI for a reasonable period after transfer.
f. Exit Management plan in case of normal termination of Contract period
g. Exit Management plan in case of any eventuality due to which Project is terminated before the contract period.
h. Exit Management plan in case of termination of the SI

5.60.2 Exit Management plan at the minimum adhere to the following:

a. Three (3) months of the support to Replacement SI post termination of the Contract
b. Complete handover of the Planning documents, bill of materials, technical specifications of all equipment, user manuals, guides, IPR, network architecture, change requests if any reports, documents and other relevant items to the Replacement SI / Purchaser
c. Certificate of Acceptance from authorized representative of Replacement SI issued to the SI on successful completion of handover and knowledge transfer
d. In the event of termination or expiry of the contract, Project Implementation or Service Level monitoring, both SI and Purchaser shall comply with the Exit Management Plan.
e. During the exit management period, the SI shall use its best efforts to deliver the services.

5.61 IT Act 2008

5.61.1 Besides the terms and conditions stated in this document, the Contract shall also be governed by the overall acts and guidelines as mentioned in IT Act 2008 (Amendment)

5.62 Pre-Contract Integrity Pact

5.62.1 A “Pre-Contract Integrity Pact” shall be signed between CHiPS and the Bidder. This is a binding agreement between CHiPS and Bidders. Under this Pact, the Bidders agree with the Purchaser to carry out the assignment in a specified manner. The format of Pre-Contract Integrity Pact will be as per Annexure 10.

5.62.2 The following set of sanctions shall be enforced for any violation by a Bidder of its commitments or undertakings under the Integrity Pact:

a. Denial or loss of contracts;
b. Forfeiture of the EMD and PBG;
c. Liability for damages to the Bidders; and
d. Debarment of the violator by CHiPS for an appropriate period of time.

5.62.3 The Bidders are also advised to have a company code of conduct (clearly rejecting the use of bribes and other unethical behaviour compliance program for the implementation of the code of conduct throughout the company.

6. Bid Evaluation

6.1 Evaluation of Technical Bids

6.1.1 A committee/s shall be formed for evaluation of the bids. Decision of the committee would be final and binding upon all the Bidders.

6.1.2 The purpose of this section is only to provide the Bidder(s) an idea of the evaluation process that the Purchaser may adopt. However, the Purchaser reserves the right to modify the evaluation process at any time during the Tender process, without assigning any reason, whatsoever, and without any requirement of intimating the Bidder(s) of any such change.

6.1.3 Bidder must possess the requisite experience, strength and capabilities in providing the services necessary to meet the Purchaser's requirements, as described in the RFP.

6.1.4 Bidder must possess the technical know-how and the financial wherewithal that would be required to successfully Supply, Installation, Integration, Testing, Commissioning and Maintenance of OFC (Underground) IP-MPLS Network, including Operations & Maintenance and service provisioning of the created network. The Bidders' bid must be complete in all respect and covering the entire scope of work as stipulated in the RFP.

6.1.5 The Purchaser shall examine the bids to determine their responsiveness, i.e. whether they are complete, whether the bid format conforms to the RFP requirements, whether any computational errors have been made, whether required EMD and Tender Fee (online) have been furnished, whether the documents have been properly signed, and whether the bids are generally in order.

6.1.6 A bid determined as not substantially responsive shall be rejected by the Purchaser and may not subsequently be made responsive by the Bidder by correction of the non-conformity.

6.1.7 In this part, the bid shall be reviewed for determining the Compliance of the general conditions of the Contract and Technical Bid as mentioned in the RFP. Any deviation for general conditions of the Contract and Technical Bid may lead to rejection of the bid at the sole discretion of CHiPS.

6.1.8 Bidders are expected to meet all the conditions of the RFP and the eligibility criteria as mentioned below. Bidders failing to meet these criteria or not submitting requisite supporting documents/ documentary evidence for supporting Technical Bid criteria are liable to be rejected summarily.

6.1.9 The invitation to the bids is open to all bidders who qualify the eligibility criteria as follows:

S. No.	Eligibility Criteria	Documents Required
1.	The Bidder should be registered in India under the Companies Act, 1956 or Companies Act, 2013 or as amended and should have at least 5 years of operations in India as on bid submission date. Bidder should have an average annual turnover as following package-wise for the last Five (5) audited financial years (2020-21, 2021-22, 2022-23, 2023-24 and 2024-25) Package A – at least 90 Crores Package B – at least 60 Crores Package C – at least 37 Crores Package D – at least 23 Crores	a) Copy of Certification of Incorporation / Registration Certificate b) PAN card c) GST Registration d) Audited financial statements for the last Five financial years (2020-21, 2021-22, 2022-23, 2023-24 and 2024-25) e) Certificate from CA on turnover details for the last five (5) financial years (2020-21, 2021-22, 2022-23, 2023-24 and 2024-25)
2.	The Bidder should have experience as per the scope of work mentioned in this RFP in last 5 years as on bid submission date. "Experience in Implementation / Operation and Maintenance of Optical Fibre Cable (OFC) networks" as per following - Package A - at least 4400 km under a single Work Order within India. Package B - at least 2800 km under a single Work Order within India. Package C - at least 1700 km under a single Work Order within India. Package D - at least 1200 km under a single Work Order within India.	1. Work order/ agreement clearly highlighting the scope of work & value of contract 2. Completion Certificate issued & signed by the competent authority of the client entity clearly mentioning the scope of work
3.	The Bidder should have experience as per the scope of work mentioned in this RFP in last 5 years as on bid submission date. "Experience in Implementation / Operation and Maintenance of network equipment (Routers, Switches and other associated accessories)" as per following - Package A - at least at least 700 IP-MPLS nodes under a single Work Order within India. Package B - at least at least 450 IP-MPLS nodes under a single Work Order within India. Package C - at least at least 200 IP-MPLS nodes under a single Work	1. Work order/ agreement clearly highlighting the scope of work & value of contract 2. Completion Certificate issued & signed by the competent authority of the client entity clearly highlighting the scope of work

S. No.	Eligibility Criteria	Documents Required
	Order within India. Package D - at least at least 200 IP-MPLS nodes under a single Work Order within India.	
4.	The Bidder should have valid ISO 9001:2008/ ISO 9001:2015 for Quality Management System which should be valid as on bid submission date	Copy of certificate valid as on bid submission date
5.	The Bidder should not have been blacklisted/debarred by any Govt. department or any PSU in India during last Five (5) years as on bid submission date	Self-Declaration for not being blacklisted by any of the Govt. or organisation of Govt. in India by the authorized signatory of the bidder.
6.	The Bidder should submit MAF for the passive and active material to be supplied under this tender. Bidder to provide the MAF from the OEMs confirming that the OEMs shall ensure that all equipment / components / sub-components being supplied under this tender shall be supported for the entire contract period If the same is de-supported by the OEM for any reason whatsoever, the bidder shall replace it with an equivalent or better substitute that is acceptable to Purchaser without any additional cost to the Purchaser and without impacting the performance of the solution in any manner whatsoever	Documentary evidence such as MAF(Manufacturers Authorization Form) from OEM for specified products (Active & Passive) being quoted by the Bidder OEM Compliance document of TEC-GR for all applicable products from OEM whose products are being quoted by the Bidder
7.	The Bidder should have a project office in Raipur. However, if the local presence is not there in Raipur, the selected bidder should provide an undertaking for establishment of a project office, within 30 days of award of the contract.	Self-certification duly signed by authorized signatory on company letter head.
8.	The Bidder should have positive net worth as on 31st March, 2025	Certificate from CA

Note 1: Any items proposed in tender e.g. OFC, Accessories, equipment (active and passive) etc. should have valid Type Approval Certificate (TAC) from Telecom Engineering Centre (TEC), New Delhi or Technical Specification Evaluation Certificate (TSEC) from Quality Assurance Circle, BSNL, Bengaluru against the respective

technical specifications of this RFP. Tender items such as access routers, which if not covered under TEC-GR guidelines, may be excluded from following the guidelines

OEM Compliance document of TEC-GR Obtained by the OEMs of bidders for the various tenders floated by BBNL/BSNL in the last two years shall be acceptable for the items which are part of this RFP provided that the technical specifications remains unchanged. In case of OFC, any change in the source of raw material shall be subject to fresh TSEC process of QA.

6.2 Evaluation of Commercial Bids

a. Commercial bids submitted of only those bidders, who are technically qualified shall be opened and shall be eligible for further evaluation.
b. Evaluation of bids shall be done on Least Cost/Lowest Cost (L1) criteria as detailed. The bidder have to qualify the Technical Bid for further evaluation for being eligible for opening of commercial bid. The bids quoted as per the commercial bid format shall be considered for commercial evaluation.
c. Bidders quoting unrealistic cost of items shall be rejected summarily by the committee and EMD of such bidder shall be forfeited. Any bid found to have unsatisfactory response in any of the technical criteria as mentioned may be rejected and shall not be considered for further evaluation.

6.3 Final Bid Evaluation

a. If any bidder withdraws his bid, at any stage after the last date and time of bid submission till the final evaluation or declaration of the selected bidder, it shall be declared a "defaulting bidder" and amongst other measures, EMD of such defaulting bidder shall be forfeited. In such situation the tendering process shall be continued with the remaining bidders as per their ranking.
b. If the bidder declines after being declared as selected bidder, it shall be declared as defaulting bidder and EMD of such defaulting bidder shall be forfeited and CHiPS reserves right to blacklist/debarred any such bidder for next 3 Years from participating in any tender floated by CHiPS. In such situation, the tendering process shall be continued with the L2 bidder matching the L1 price. However, in-case of refusal of acceptance by the L2 bidder to match the price of L1 bidder, CHiPS would carry out discussions with the subsequent bidders.
c. A bidder can bid on one or more Packages.
d. Any bidder can be selected as System Integrator for only one single package at max.
e. The bid for 4 packages will be opened and evaluated sequentially as below
   - Package C
   - Package D
   - Package B
   - Package A
f. Commercial bid of the Bidder already selected as the SI for any of the Package shall not be opened in the subsequent packages.

7. Scope of Work

7.1 Project Background & Objectives

7.1.1 The Government of Chhattisgarh (GoC), through its nodal agency CHiPS, envisaged the BharatNet Phase-II project in 2018 under the State-Led Model to create a digital highway by laying Underground Optical Fibre Cable (OFC) from Block Headquarters to Gram Panchayats (GPs). The objective is to enable high-speed broadband and digital services (e.g., internet, VoIP, e-services, and value-added services) for households, institutions, and enterprises across the State.

7.1.2 The project was conceived to reach previously unconnected areas and entails significant, long-life infrastructure investment. Choices on network design and technology specifications will influence immediate service quality, reliability, and the future scalability of the BharatNet backbone for years to come.

7.1.3 The original scope included 5,964 GPs across 83 Blocks. As of date, 5,540 GPs is integrated with the State Network Operations Centre (S-NOC). As per current status OFC laying, Installation, Integration, Commissioning and acceptance testing has not been initiated for 424 GPs.

For this tender Cluster/package wise scope are tabulated below:

#	Package	Cluster	District	No of Blocks	Scope of GP's	S-NOC Scope Applicable	Existing O&M GP's	Nos GP's to be Implemented
1	Package-A	A	10	32	2827	Yes	2684	143
2	Package-B	B	6	21	1729	No	1551	178
3	Package-C	C	7	18	718	No	678	40
4	Package-D	D	5	12	690	No	627	63
Total			**28**	**83**	**5964**		**5540**	**424**

- The successful bidder shall be responsible for implementation of Underground OFC connectivity for respective GP is segregated across the respective packages/Cluster.
- The bidder shall be responsible for Operations & Maintenance (O&M) of GPs already integrated into the network as mentioned in table above. O&M will commence immediately after handover of the asset from Purchaser to the bidder post the said GPs are service ready.
- For the newly implemented 424 GPs, O&M will start after successful end-to-end integration, testing and verification by CHiPS.
- The successful bidder for **Cluster A/Package A** shall be responsible for the operation and maintenance of the State Network Operations Centre (S-NOC), as well as for monitoring all network elements across the state under the BharatNet Phase II Project. In addition, the bidder for **Cluster A/Package A** shall also be responsible for the following:

7.1.4 Key considerations for the network design and service delivery include:

- Underground OFC ring architecture from Block to GPs wherever feasible and viable.

- On-demand bandwidth for Government and public institutions (Departments, PSUs, Schools, Hospitals, and other enterprises) on a non-discriminatory basis.
- For wired media (OFC), minimum 1 Gbps per GP, scalable up to 10 Gbps.
- Provisioning of minimum 6 dark fibre cores at each GP, as mandated by BBNL.
- Rolling out e-Services to increase uptake of internet connectivity and digitalization in the State.

7.1.5 Purchaser has engaged the SI for Design, Supply, Installation, Integration, Testing, Commissioning and Maintenance of OFC (Underground) IP-MPLS Network including Operations & Maintenance and service provisioning of the created network for a period of 2 years. The SI is required to provide such services, support and infrastructure as the Purchaser or Purchaser's representative may deem proper and necessary, during the term of this Contract, and includes all such processes and activities which are consistent with the proposals set forth in the Bid, the RFP and this Contract and are deemed necessary by the Purchaser, in order to meet its business requirements (hereinafter 'scope of work')

7.1.6 If any services, functions or responsibilities not specifically described in this RFP are an inherent, necessary or customary part of the Services or are required for proper performance or provision of the Services in accordance with this RFP, they shall be deemed to be included within the Scope of Work to be delivered for the Charges, as if such services, functions or responsibilities were specifically described in this RFP

7.1.7 The Purchaser or Purchaser's representative reserves the right to amend any of the terms & conditions with mutual agreement in relation to the Scope of Work and may issue any such directions which are not necessarily stipulated therein if it deems necessary for the fulfilment of the Scope of Work pursuant to Clause 5.38- Change Orders/ Alteration/ Variation

7.1.8 The selected System Integrator (SI) shall be responsible for end-to-end Supply, Installation, Integration, Testing, Commissioning and Maintenance of the passive and active OFC-based infrastructure in scope, including (but not limited to):

- Preparation of GP-wise and Block-wise detailed BOQ and validation by CHiPS (or its appointed agency) prior to implementation.
- Seeking CHiPS approval for any upward/downward BOQ variations arising during implementation.
- Supply, installation, and commissioning of 48F OFC (Underground) and associated accessories forming the IP-MPLS backbone from Block to GP (strictly OFC-based; no RF connectivity forms part of this SI scope).
- Integration with Chhattisgarh State Wide Area Network (SWAN) to establish seamless end-to-end connectivity up to S-NOC; CHiPS will facilitate necessary details and access for smooth integration.
- Continuous monitoring and service provisioning over the created network as per applicable Service Level Agreements.

7.1.9 The provisioned bandwidth and network design shall support a wide range of services including operational and social e-services, VPN networking, internet access (web, email), VoIP, high-quality live video transmission, two-way video conferencing, remote monitoring, and other digital applications.

7.1.10 An indicative BOQ is provided in this RFP. The final BOQ shall be determined at execution time based on the route survey validated by CHiPS (or its appointed agency).

7.1.11 The period for completion of implementation of all the pending GPs in respective package wise shall be 6 months from the date of contract signing. The Operations & Maintenance period shall be 1 year, extendable by 1 additional year (applicable contract Clauses).

7.1.12 Cost vs. Scalability considerations – While creating excess capacity to meet future needs is important, the infrastructure mix should balance cost efficiency with scalability, ensuring budgets are respected without compromising growth potential.

7.1.13 Laying HDPE ducts and OFC infrastructure is the most laborious and time-intensive component of fibre networks; therefore, sizing and design should be future-proof, considering expected demand and long asset life.

7.1.14 Active components can be scaled dynamically; hence initial estimates should be realistic and aligned with near-term demand to optimize total project cost without limiting future scalability.

7.1.15 Interoperability & Competition – The backbone must be open and interoperable across vendors and technologies to promote competition and avoid bias towards any particular service provider.

7.1.16 Multi-service readiness – The backbone must support diverse services (e-Governance, Education, Health, etc.) with differentiated Quality of Service (QoS) parameters; architecture decisions should reflect these multi-service requirements.

7.1.17 Monitored network & maintainability – The S-NOC shall have capabilities to identify, localize, and coordinate fault rectification, thereby minimizing downtime and repair cost.

7.1.18 Security considerations – As a state backbone, the network must ensure data security, integrity, and compliance with applicable standards and policies throughout its lifecycle.

7.2 Route Survey and ROW Clearance

7.2.1 The SI shall conduct a survey for identified OFC routes. These reports will form the basis for finalizing the route alignment and preparing the detailed Bill of Quantity (BoQ).

7.2.2 The SI shall:

- Validate and finalize OFC routes for the 424 GPs in the respective packages as per this RFP.
- Ensure all survey data is authenticated and approved by CHiPS (or its appointed agency) before execution.
- Prepare Route Index Diagrams and Joint Location Diagrams with accurate GIS coordinates (captured at intervals of 250 meters and at all critical points such as manholes, chambers, joints, and equipment sites).
- Record all landmarks along the OFC route corridor (up to 25 meters on either side of the centerline) with latitude/longitude details in decimal degrees (up to eight decimal places).

7.2.3 The SI shall comply with all statutory requirements for **Right of Way (RoW)** permissions and coordinate with local authorities, CHiPS, and other stakeholders for timely approvals.

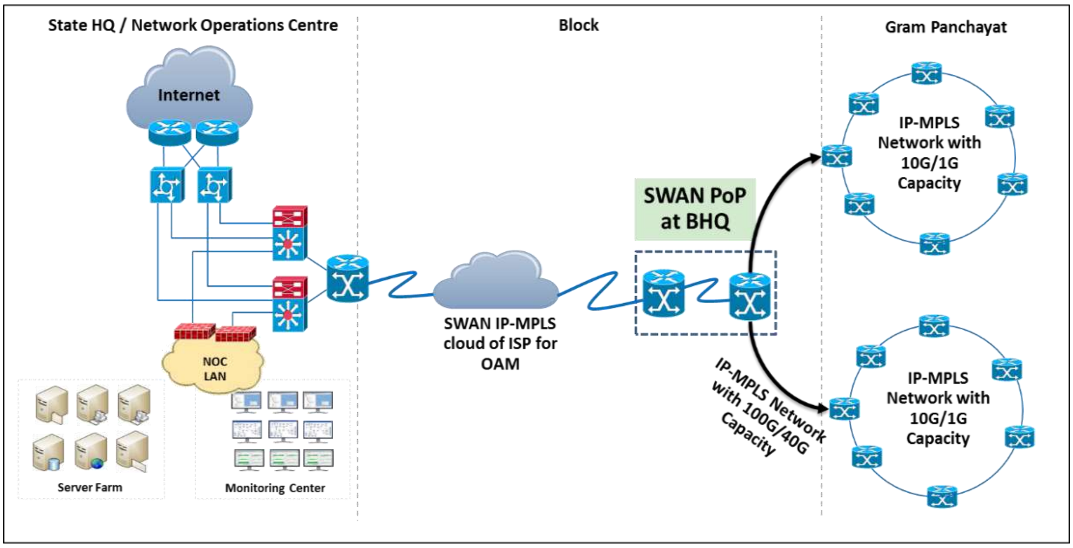
7.2.4 Safety and traffic control measures shall be implemented during trenching and laying activities, including:

- Proper barricading and warning signage with red background and reflective lights.
- Night work safety arrangements with red warning lamps and white work lights.
- Provision of pedestrian passages and footbridges wherever required.
- Compliance with local traffic regulations and coordination with police authorities for traffic control.

7.2.5 All works near railway crossings shall be carried out under the supervision of railway authorities and in compliance with their regulations.

7.3 Network Design & Architecture

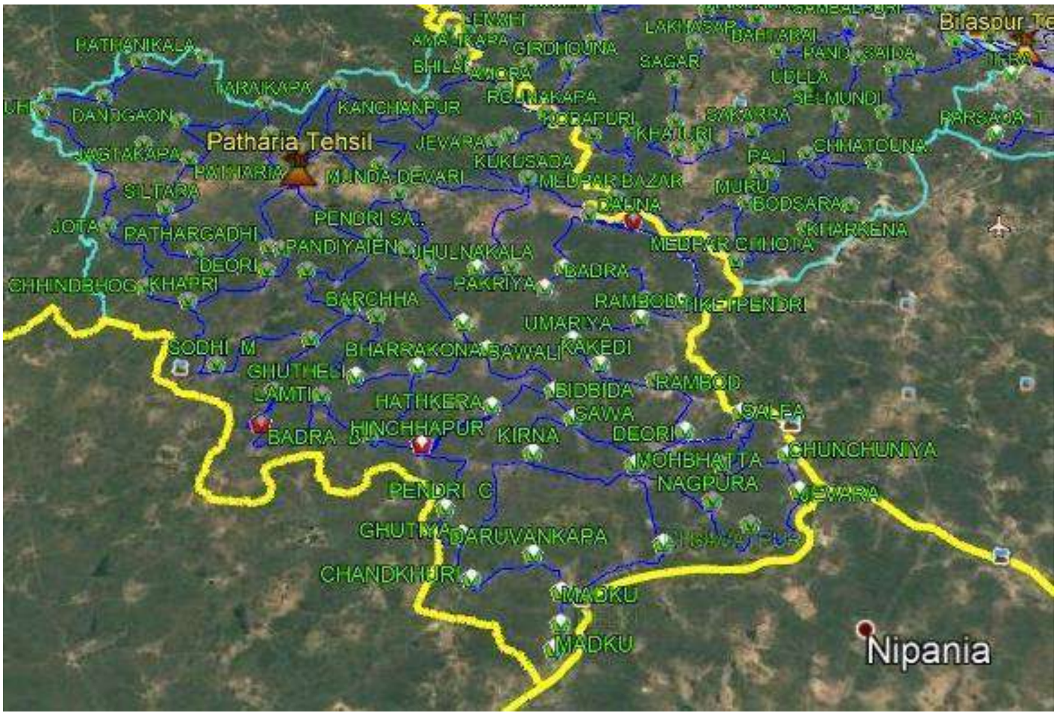
7.3.1 The network design and architecture is based on the key design considerations that in-turn would ensure accomplishment of core objectives of Chhattisgarh BharatNet Phase-II project. The network topology for Chhattisgarh BharatNet Phase-II project follows ring architecture with internet Protocol – Multi Protocol Label Switching (IP-MPLS) technology. As part of Chhattisgarh BharatNet project a high-speed IP-MPLS link shall be established between a Block to the respective Gram Panchayats that would be further integrated with the Chhattisgarh State Wide Area Network and BastarNet network to establish end-to-end monitoring link from the S-NOC up to the Gram Panchayat. At all levels a service provider for data / video / voice may be on-boarded in the future for provision of services, such as Internet, IP-telephony, e-services, etc. The logical architecture of the network is shown below for reference



**This is a representation of Indicative Logical Architecture. There may be variations such as excluding Blocks/GPs (which are covered under NOFN). There may be changes in design, topology, architecture etc. after the route survey report and design document* *finalisation. Laying of Fibre by the SI would be according to the approved physical route survey report*

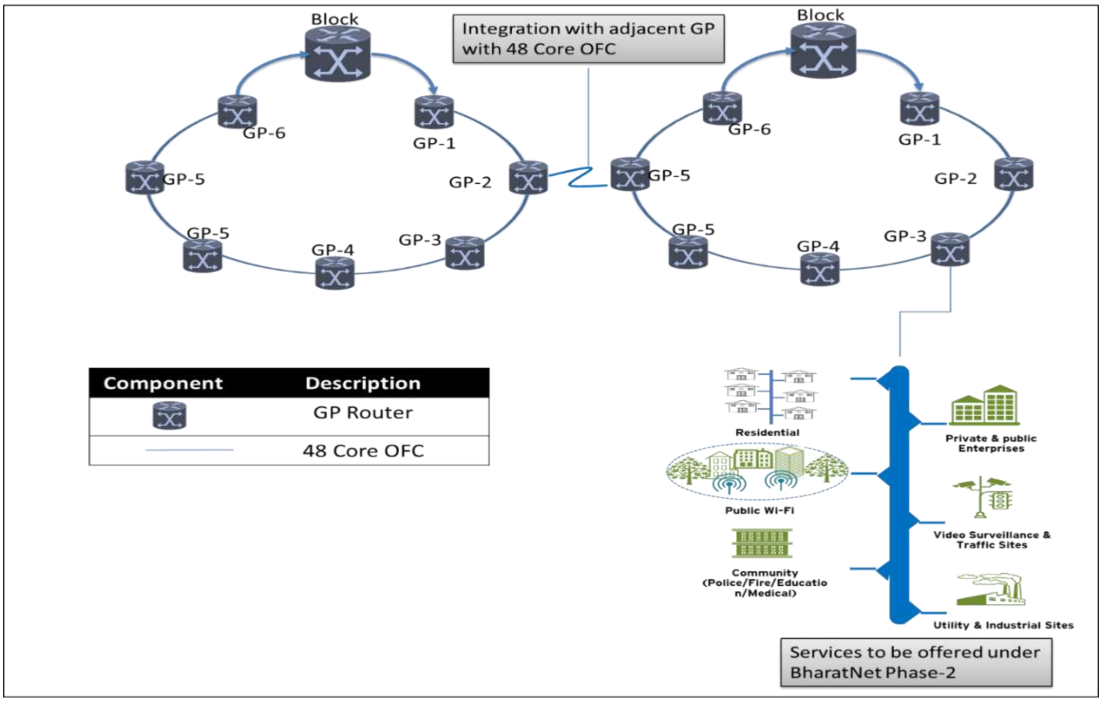
7.3.2 The assessment for the fibre requirement was carried out using GIS maps by plotting the locations of the point of presence on the GIS maps and thereby identifying the best possible route options for fibre laying as evident from the graphical view of the geography.

7.3.3 The key plan, FTP for both fresh implementation and already implemented network shall be provided by CHiPS.



GP rings at Block and GP for Patharia Block

7.3.4 The Network connectivity between Block to Gram Panchayat shall ensure a minimum of 1 Gbps per GP and the minimum availability of bandwidth for each GP ring shall be 10 Gbps. The logical representation of the gram panchayat ring along with active network components at GP PoPs and Block PoPs is as shown below:



10 G GP ring logical architecture

7.4 Medium of Connectivity

7.4.1 The preferred and only medium of connectivity between Block Headquarters and Gram Panchayats (GPs) under this tender shall be Underground Optical Fibre Cable (OFC).

7.4.1.1 Option 1 – Optical Fibre connectivity up to GP

- Inter-GP connectivity shall be established via Underground OFC, and GPs shall in turn be connected to the Block Headquarters through OFC to aggregate traffic at the Block Point of Presence (PoP).
- Each Block HQ to GP link shall use 48-core fibre.
- The Bill of Quantities (BOQ) shall be finalized upon completion of the physical survey and detailed planning. The finalized BOQ will form the basis for preparation and submission of the commercial quotation.
- Ring topology shall be formed to connect multiple GPs, ensuring redundancy and high availability. Each GP shall have minimum 1 Gbps bandwidth, scalable up to 10 Gbps.
- The 48-core fibre shall originate from the Block PoP, with 2 cores allocated for connecting GPs within each ring.
- As per design standards, 6 dedicated fibre cores shall be provisioned at each GP. To achieve this mandate, multiple 48-core OFC cables may be laid on common paths; however, no more than two cables shall be placed in a single HDPE duct.
- To ensure service continuity and redundancy at GP level, adjacent GPs of two neighboring blocks may be interconnected via OFC where feasible.
- Ideally, each ring shall include 8–10 GPs, subject to constraints such as bandwidth demand, terrain, forest cover, soil strata, and population density.

- Redundancy planning shall consider factors like terrain conditions, power availability, expected traffic volumes, distances between nodes, and GP distribution.

Scope	Bandwidth Capacity of Ring	Fibre Cores in Ring	Total Length (Approx.)
Already Implemented & Integrated (5540 GPs)	1/10 Gbps	48 cores	**~28,000 km**
New Implementation (424 GPs under this tender)	1/10 Gbps	48 cores	**~ 1600 km**

7.4.2 Prerequisites for PoPs provisioned at Block & GP location

- Point of Presence shall be strategically provisioned at various locations at GP level that are predefined and shall be made available to establish the required network infrastructure to enable Chhattisgarh BharatNet services.
- The SI of respective package/Cluster shall also be responsible for coordinating with the concerned Block Administration and Gram Panchayats, as required, for the smooth execution of project activities.

Point of Presence	Quantity	Remarks
Block PoPs	**83 (for 5540 GPs already integrated)**	Space for Block-level PoPs has been provided at existing SWAN PoPs at Block Headquarters. CHiPS shall ensure availability of air-conditioning, raw power, and backup for active network components.
GP PoPs	**424 (new implementation under this tender)**	Space for GP PoPs shall be provided by CHiPS at the respective Panchayat Bhawan. The SI shall provision necessary infrastructure including fibre management system, racks, UPS, networking equipment, and proper earthing. In cases where space is not available at Panchayat Bhawan, the SI shall coordinate with CHiPS for an alternate location.

7.4.2.1 In case of PoP where the equipment has to be installed, does not have appropriate plug points/ wiring, the SI shall draw the electrical wiring from the nearest junction box via network rack with proper fittings and avoid loosely hung cables that would impact services (with requisite plug points etc.) at no additional cost

7.5 Chhattisgarh infotech Promotion Society (CHiPS)

7.5.1 CHiPS shall be the owner and nodal agency for the project and shall be responsible for formulating project implementation strategies in line with overall objectives, appointing and enabling relevant agencies for project execution, monitoring the project and executive decision making. CHiPS will manage the project and will be Responsible for:

- Executive decision making
- Appointment of SI for execution of work
- Creation of project manual comprising:
  - Project governance structure
  - Monitoring and evaluation mechanism
  - Project level procedures including.
    - Implementation guidelines, standards and checklists
    - Testing and acceptance process
    - Issues reporting and management process
    - Other project management and coordination procedures
- Project management and coordination
- Final escalation point for issues resolution
- Centralized planning and information sharing between relevant stakeholders
- Defining and monitoring of implementation and maintenance SLAs
- Facilitating handshake with the agency responsible for obtaining Rights of Way (RoW) clearances from the State Government bodies and other necessary agencies such as Railways, NHAI, forest etc. where fibre laying work is to be done Provision of power, security & other site requirements at Block / Gram Panchayat
- Final Signoffs and acceptance of completed sites, processing vendor invoices and releasing payments

7.6 System Integrator (SI)

7.6.1 SI of respective package/Cluster would be responsible for procurement and carrying out laying of the OFC network, including digging, trenching, OFC laying, splicing etc., procurement & installation of Access equipment (Equipment / Hardware / Software) and other passive infrastructure components

7.6.2 The SI of respective package/Cluster will also be responsible for maintaining the OFC laid and Block/GP network equipment installed as part of the BharatNet Phase II project for a period of 1 year (extendable by 1 year). The activities are to be performed as per the laid down standards, guidelines and procedures in this RFP document.

7.6.3 **Overview of scope of the SI as shown below:**

7.6.3.1 The overall scope of work is divided into three phases:

- Planning & Survey Phase
- Project Implementation Phase
- Operations & Maintenance Phase

Implementation of new GP's and O&M of existing handed over GPs shall run in parallel, with planning aligned accordingly.

7.6.3.2 Obtain and validate survey reports and As-Built Diagrams (ABDs) for identified OFC routes generated by the designated survey agency.

7.6.3.3 Shall be responsible for Validate and finalize OFC routes for the 424 GPs across the package/Cluster under this project, prepare and submit the final Bill of Quantity (BoQ) for CHiPS approval.

7.6.3.4 Execute underground OFC laying works as per approved routes and engineering instruction as per Annexture -17.

7.6.3.5 Perform activities including but not limited to:

- Trenching and duct laying
- OFC cable blowing in laid ducts
- Splicing and termination
- Installation of joint markers and route markers
- Construction of manholes/joint chambers
- Providing protection to ducts using RCC/DWC pipes and GI pipes wherever required.
- Installation and Termination of Indoor/Outdoor FDMS

7.6.3.6 Procure and install active and passive equipment for end-to-end OFC network connectivity for the 424 GPs.

7.6.3.7 Integrate newly implemented GPs with existing BharatNet backbone and State Network Operations Centre (S-NOC) for monitoring and service provisioning.

7.6.3.8 Conduct link testing (OTDR, power meter, etc.) and commissioning of the implemented OFC network.

7.6.3.9 Update all drawings and documentation provided to SI to reflect actual implementation details.

7.6.3.10 Coordinate with CHiPS, local administration, and other stakeholders for site access, RoW permissions, and end-to-end implementation.

7.6.3.11 Ensure seamless integration of the established network with S-NOC and interoperability with Chhattisgarh SWAN.

7.6.3.12 Enter project data into the Project Monitoring Tool, GIS platform, and mobile application as required.

7.6.3.13 Perform end-to-end testing of OFC network (from core/aggregation router to access router).

7.6.3.14 Maintain and monitor the State Network Operations Centre (S-NOC), including fault reporting and rectification for active and passive components (Applicable for Package/Cluster A).

7.6.3.15 Manage warehouse operations, including inventory control and material handling.

7.6.3.16 Provide field data, reports, and certificates in predefined formats as per CHiPS requirements.

7.6.3.17 Undertake any additional field implementation work related to BharatNet Phase-II as directed by CHiPS.

7.6.3.18 Submit monthly and quarterly reports to CHiPS on network uptime and performance as per Section 10 – Service Level Agreements.

7.6.3.19 Submit ABDs validated and approved by CHiPS post-completion of each milestone as per project timelines.

7.6.3.20 SI may be required to procure additional material and provide services for GPs beyond listed all the 424 GPs for implementation and 5540 GPs for O&M, as per CHiPS instructions and approved rates under this tender.

7.6.3.21 Submit a project plan (in MS Project) for implementation of new GPs as per timeline mentioned in the RFP.

7.6.3.22 Identify OFC routes for the 424 GPs in Package/Cluster as per RFP.

7.6.3.23 Submit the final Bill of Quantity (BoQ) for CHiPS approval.

7.6.3.24 Plan for stores and warehouses for material procured by SI and share details with CHiPS.

7.7 Project Implementation Phase

7.7.1 General

7.7.1.1 The SI shall procure and supply all materials required under the Package/Cluster for successful implementation of OFC connectivity for 424 GPs, including:

- Optical Fibre Cable (OFC)
- HDPE ducts (PLB)
- Fibre Distribution Management System (FDMS)
- Associated accessories, active and passive components

Any additional material required for successful commissioning of GP connectivity shall be provided by the SI at its own cost.

7.7.1.2 Transport and logistics:

- SI shall ensure onward transportation of all procured material to its warehouse and work sites.
- SI shall bear all costs, taxes, duties, and levies related to material movement.

7.7.1.3 Site-level coordination:

- Coordinate with CHiPS and SWAN teams for integration with existing PoPs.
- Liaise with local authorities for RoW permissions and OFC laying.
- The selected SI shall be responsible for issuing formal communication to PWD, NHAI, PMGSY, and Nagar Nigam regarding their upcoming projects. The

communication must clearly state that certain Gram Panchayat (GP) implementation works are planned to be initiated along the same routes. The intent of this coordination is to ensure that these authorities consider the proposed works during their planning and execution phases to prevent duplication of efforts and to maintain effective inter-agency coordination.

7.7.1.4 Implementation:

- Carry out OFC laying as per engineering instructions (Annexure-17) and approved design.
- Connect laid OFC to designated endpoints at GP and Block PoPs.

7.7.1.5 Install active & Passive equipment and associated hardware/software at GP level.

7.7.1.6 Testing tools: Arrange necessary tools for testing, including OTDR machines, power meters, and other required instruments.

7.7.1.7 Documentation:

- Submit As-Built Diagrams (ABDs) on completion of each block.

7.7.2 Laying of HDPE PLB Duct and OFC Blowing

- Based on approved route survey, BoQ, and network design, SI shall initiate trenching, duct laying, OFC blowing, splicing, termination, and commissioning for approximately 1600 km of OFC covering 424 GPs.
- Ensure compliance with safety and engineering instruction.
- Conduct end-to-end testing from new GP access router to Block aggregation router to ensure the network is configured according to the architecture of the existing Bharatnet phase II network and discover the new GP at S-NOC.
- Test each fibre core and share OTDR/LSPM reports with CHiPS.

7.7.3 Active Equipment Installation & Commissioning

- SI shall supply, install, configure, and operationalize networking equipment at GPs under their respective packages.

7.8 Operations & Maintenance (O&M) Phase

7.8.1. General Scope and Start Dates

- Scope:
  The Managed Service Integrator (SI) is responsible for providing O&M for the entire network, covering end-to-end active and passive infrastructure, including both IT and Non-IT components. This ensures uninterrupted connectivity and service quality across all Gram Panchayats (GPs).
- Start Dates:
  - New Implementation (424 GPs in all packages/Cluster):
    O&M begins after successful end-to-end integration and CHiPS verification.

  - Already Integrated (5540 GPs across 83 Blocks):
    O&M starts immediately after physical assessment and restoration of the existing GP network and handover by CHiPS to SI.

- Service Period:
  O&M services for all GPs will be delivered on a pro-rata basis from their respective start dates until the overall O&M period defined in the contract is completed.

7.8.2. Maintenance and Responsibility

- SI Responsibilities:
  - Routine Maintenance Cost: SI bears the cost of pre-notified planned shifting and provides all required materials to maintain SLA compliance.
  - Component Failure Cost: SI is responsible for rectification or replacement of issues related to OFC, PLBE, accessories, active and passive network elements. All costs for such events are borne by SI.
  - Manpower & Testing: SI must provide skilled manpower, testing instruments, and equipment necessary for maintenance and SLA compliance.
- CHiPS Responsibilities:
  - Network/OFC Route Shifting: If shifting is required by CHiPS, the cost is borne by CHiPS.
  - Theft/Physical Damage (Repair Not Possible): Replacement costs are borne by CHiPS after due recommendation and FIR lodging.

7.8.3. Categorization of Maintenance Responsibilities

Scenario	Description	Applicability
1	Maintenance of existing GPs (5540) assessed and restored by a third-party and handed over to SI.	Existing integrated GPs
2	Restoration and maintenance of defunct/damaged network sections due to neglect or civil works.	Pre-existing damaged sections
3	Maintenance of new network sections deployed by SI (424 GPs) after CHiPS acceptance.	Newly deployed sections
4	Public liaising and coordination with government departments to safeguard OFC routes during future works.	All OFC routes

7.8.4. Types of Maintenance

7.8.4.1. Corrective Maintenance

- Fault Reporting: Issues reported by GPs/Blocks are directed to the State S-NOC help desk.

- Root Cause Analysis (RCA): SI must conduct RCA for every fault and submit a detailed report with corrective actions to CHiPS.
- **Key Activities:**
  - Localize faults using OTDR
  - Restore OFC continuity via fusion splicing
  - Rectify power issues (UPS replacement)
  - Replace defective cards/modules
  - Update network diagrams and inventory

7.8.4.2. Preventive Maintenance

- **Procedure:** Conducted with prior approval during planned shutdowns.
- **Key Activities:**
  - Regular OFC route surveillance
  - Weekly dusting of network equipment at each Gram Panchayat (GP)
  - UPS battery maintenance and monthly draining
  - Check earthing voltage at intervals of every two weeks or monthly.
  - Structured cabling upkeep and asset tagging

7.8.5. Operation and Maintenance procedure

7.8.5.1. O&M of Network equipment

Operation and Maintenance Phase (O&M) includes routine maintenance/First line maintenance, diagnosis & rectification of faults in EMS, Network Equipment etc. The skilled manpower, testing instruments and equipment & material required for proper maintenance and meeting the SLA obligation shall be the sole responsibility of the SI. The SI should have proper documentation to guarantee quality, timely supply, support, performance and O&M during the full life cycle of the contract.

The Operation & Maintenance shall include, but not limited to the following:

- The SI shall quote for charges for one year (extendable by 1 year) comprehensive O&M (including labour, spares, maintenance materials, SFP’s, Cards, power cable, pig tail, patch cord, connectors, consumables, UPS, visit of the engineers as and when required to meet the conditions of O&M).
- The SI shall maintain an up-to-date inventory of all FRU’s (Field Replaceable unit) required for the integrated solution. The same shall be liable for inspection by CHiPS or its designated agency. The SI must submit complete address details of nodal/repair centres.
- SI shall perform the first Line Maintenance as well as Second line maintenance of all equipment (active and passive) supplied under this tender
- As a part of First Line Maintenance (routine maintenance), the SI’s personnel will make at least one visit to each PoP location in 2 weeks for carrying out routine

maintenance and upkeep activities by providing a report of the visit. However, detailed set of maintenance procedures, periodic test/maintenance schedules, Fault Report generation & analysis and remedial measures to be taken in each occasion shall be finalised by SI and approved by CHiPS within a month of award of work. The detailed Terms and Conditions of the Routine Maintenance Agreement shall accordingly be mutually decided and signed with SI.

- Normal place of duty of Person deployed shall be at BHQ location or as decided by CHiPS, from where he will first observe the proper functioning of all the equipment under his jurisdiction, through fault reporting mechanism of EMS/NMS.
- The SI shall equip maintenance personnel with minimum necessary testing equipment and spares required for the maintenance. The same shall be liable for inspection by CHiPS.
- In case any equipment is down, either reported by CHiPS or its designated agency or monitored through EMS/NMS, he will attend the fault and rectify the same and/or report for Second Line maintenance team.
- Failure of SI to perform maintenance activities as stipulated in the SLA, shall levy penalty as mentioned in this tender.

7.8.5.2. O&M of Optical Fibre Infrastructure

The optical network will comprise of 48 Core OFC as per execution plan in this RFP, the SI should have capabilities for long term maintenance & sufficient maintenance team with tools, equipment, accessories, spares, OF cable material etc. SI should have sufficient route inspection parties for preventive maintenance.

The SI will maintain the Optical Fibre Cable laid from Block to GP with high level of availability of services & transmission losses within limit.

CHiPS shall appoint an agency, who shall be responsible for ensuring that all field works covered under the tender are carried out in accordance with approved designs, drawings & specifications and conditions of tender as agreed to.

The Operation & Maintenance shall include, but not limited to the following:

- Routine inspection, viz, patrolling on the routes, to identify areas where OFC / Duct is exposed due to natural wear and tear etc.
- Faulty rectification of OFC cuts along routes.
- During OFC cut restoration, the SI shall first splice the live fiber cores to immediately restore services, followed by splicing all remaining cores to ensure full continuity for future use.
- Replacement of OFC routes due to non-viability of transmission link.
- Ensure availability of OFC route marker along the route at regular intervals.
- Maintain proper condition of joint closure.
- Prevent third party damages viz. theft or damage/s caused by other underground utility service provider
- Maintain condition of OFC with casing or with special arrangements near critical areas viz, major bridges, railway crossing, pipeline crossing, Forest, etc.
- Visual inspection of joint closure to check ingress of water, foreign particles etc.

- Periodic measurement of the link attenuation loss to ensure that the link is free from any splice loss or point loss defects etc.
- Maintenance and update of as-built drawing, information along the OFC route.
- Maintaining history of events, analysis and reporting.
- Public liaising with concerned authorities.

7.8.6. Fault Restoration Services

- The SI shall deploy Maintenance Teams at the designated locations as decided by the CHiPS. The Maintenance teams shall comprise of manpower, logistics, required tools/tackles/machinery & equipment.
- The SI shall provide OFC maintenance service on round the clock basis for attending & rectifying the OFC fault in minimum downtime (including travel time) from the time of lodging the complaint to the representative of SI. The SI shall provide all assistance including providing manpower, transportation of men and materials etc. in the event of failure.
- The SI shall provide conveyance facilities for maintenance, for transporting the manpower, tools/tackles, Test/Measuring equipment and consumables like: OFC cable, Joint Closures, Jointing Pit, duct, couplers, etc. Suitable vehicle shall be available round the clock with each of the maintenance team. Vehicle should be in good working condition.
- The SI shall provide communication facilities viz. mobile phones to members of the maintenance teams.
- The SI shall be required to carry out maintenance activities which include identification of OFC fault/cut on ground, obtaining permission from local authorities if required, excavation of earth to expose cable, laying of required length of OFC with protection wherever required, splicing of OFC, installation of Jointing pit & back filling of pit with

sand, supply and installation of cable Route Markers and Joint Markers as per specification, testing of OFC and updating of OFC as-built drawings etc.

- The SI shall arrange for logistics to provide facilities such as AC/DC power source, lighting arrangement, dewatering facility, DG sets etc., which may be required during the execution of maintenance job at site.
- Optimum functionality of maintenance teams is a prime necessity to carry out day to day maintenance of OFC links. OFC and accessories spares to cater for repair of at least 10 fibre cuts shall be always maintained with each of these teams.
- SI shall take insurance for all the workmen engaged under this contract as per labour laws applicable from time to time.

7.8.6.1. Methodology for Fault Restoration/Fibre Cuts

Under OFC link cut condition, the following minimum activities shall have to be taken up by the SI for its restoration for the end-to-end restoration of the network traffic/service:

- On receipt of information of fault in OFC the team stationed shall move immediately for locating and rectifying fault as per the response time given below. The working fibres shall be restored first. Sufficient labour shall be engaged for speedy restoration. Adequate care shall be taken not to damage any other cable if laid in the same trench.
- For the identification of exact fault location on immediate basis, the OTDR measurement of spare fibre shall be made from the nearest PoP. For better clarity, the OTDR measurement on spare fibre shall be taken at PoP / nearest OFC joints situated at both ends of cut and using dummy fibre spool of 1km, in case required.
- After the OTDR measurement, the as-built drawing shall be referred and the physical site of fault on ground shall be located. It may be possible that data in as- built document may not be correct for the accuracy purpose. As-built drawing shall be taken as reference only. No claim of SI will be entertained on account of this. Accordingly, locating the OFC fault, the job of excavation in all types of soil, identification of OFC, blowing of cable, construction of jointing pits, splicing of OFC, back filling of trench & jointing-pit shall be taken up as per the standard procedure. This should be incorporated in the cable route plan also. The splicing of fibres is to be carried out in line with the installed fibre and measurements are to be taken on spare fibres. In case the active fibres are to be used, precautions are to be taken with regard to the power launched on to the fibre. Restoration of site shall be done to the entire satisfaction of CHiPS.
- In case of OFC cut where it is not possible to pull the cable from either end, the SI has to make two pits/ splicing joints between the required lengths of new OFC to be laid between the two joints.
- The spacing of joints/ pits shall be depending up on situation at site and shall be as decided by Site Engineer. Remaining OFC has to be coiled in both the pits.
- Wherever new joint is provided, or existing joint is attended for rectification during the maintenance period, joint shall be buried to the depth of 1.2 Mtr. from the ground level in joint chamber.
- After the completion of site activities, the SI shall ensure the restoration of the traffic from the associated Network Operations Center (S-NOC) and thereafter fresh OTDR measurement & traces shall be taken for all fibres & submitted to CHiPS representative.

- After the completion of site activities & hop test, the As-built drawing shall be updated by incorporating the new details like OFC loop used, Joint-pit location, etc. The length of loop in joint pit after fault restoration shall be incorporated in as built drawings.
- After attending the fault & permanent restoration, a Fault-Rectification report jointly signed by CHiPS & SI, shall be generated for the closure of the complaint.
- Any other job required for the restoration of the OFC fault/cut in totality is to be taken up by the SI.
- In case the site condition is not favourable for the immediate restoration of the fault, the temporary restoration of the service fibres shall be taken up immediately with the approval of CHiPS. Permanent restoration work will not be considered in downtime unless there is link break again during restoration job. Permanent restoration of joint pits is to be carried out by SI within reasonable time of fault / OFC cut. In case the site is not conducive for permanent restoration, some arrangement of manpower has to be done by SI for safeguarding exposed OFC till permanent restoration. No extra payment shall be given to SI on account of deployment of additional manpower. In case of further cuts at exposed OFC location, SI will be accountable for this additional downtime of OFC link.
- It is mandatory for the SIs to install the jointing chambers after permanent restoration is done.
- In case of any breakdown in the OFC network, SI shall be responsible for obtaining approval at his own cost from statutory authorities like Municipal Corporation, Development Authorities, Electricity Department PWD, NHAI and any other concerned authority as required for carrying out the repair. CHiPS can assist in getting permission for repair in few cases where there is urgency.
- Drains, pipes, cables, overhead wires and similar services encountered in the course of the works shall be guarded by the SI at his own cost, so that they may continue in full and uninterrupted use to the satisfaction of the Owners thereof.
- Should any damage be done by the SI to any AC Power mains, utility pipelines cables or lines (whether above or below ground etc.) whether or not shown on the drawings, the SI must make good or bear the cost of making good the same without delay to the satisfaction of the CHiPS.
- SI shall observe all national and local laws, ordinances, rules and regulations and requirements pertaining to the work and shall be responsible for extra costs arising from violations of the same.
- SI shall have at all times during the performance of the work, a competent supervisor at block level. Any instruction given to such Supervisor shall be considered as having been given to the SI.
- The SI shall employ as many personnel as deemed necessary to comply with the local rules and administrative orders governing the Working Hours of Employment. The SI shall be responsible for compliance with all statutory requirements including personnel related matters.
- In the event of urgency, the team has to move to the adjacent section.
- The minimum down time shall include time taken in restoration of fault/cut caused by any means like miscreant activity at day or night, due to work done by any other

organization, development of high losses / break at existing joints, fault caused due to rodent, ant etc.

- In case of partial damage of the cable or development of high loss in the working and spare fibre or cable cut at any time (day/night) by miscreants or by any agency, the responsibility of repairing the defective fibre lies with the SI.
- In case SI fails to completely restore the fault (as per original condition) or submit OTDR & test (power level in live equipment) records to establish completion of work, a penalty shall be levied for the work involved at site.
- When finished work is taken down for the purpose of inspection for any reason, the SI shall bear the entire expenses incidental thereto in the event that the said work is found to be defective. This situation may be applicable to both planned works as also to emergency restoration.
- During the maintenance or fault rectification work, should any damage occur to the other cables, SI is liable to pay compensation as demanded by the respective authorities

7.8.7. Audit and Security services

- The SI shall be required to provide comprehensive support to CHiPS during the Audit Agency and Security Audit etc. The SI shall be responsible in getting the required readiness built in the network during audit for security solutions.
- CHiPS reserves the right to inspect, monitor and assess the progress and performance of the project either itself or through its designated Audit Agency or any other agency appointed by DBN(DoT) as it may deem fit, throughout the course of the Contract. CHiPS may demand and upon such a demand being made, CHiPS shall be provided with any document, data material or any other information which it may require, to enable it to assess the progress of the project.
- CHiPS shall also have the right to conduct, either itself or through its designated Audit Agency as it may deem fit, an audit to monitor the performance of the SI of its obligations/ functions in accordance with the standards committed to or required by CHiPS and the SI undertakes to cooperate with and provide to CHiPS or its designated Audit Agency or any other agency appointed by DBN (DoT), all documents and other details as may be required by them for this purpose.

7.8.8. MIS Reports

- The SI shall submit the periodical reports on a regular basis as per the prescribed formats provided by the CHiPS.
- Indicative report may inter-alia includes the following reports:
  - Summary of issues/ complaints
  - Summary of resolved/ unresolved and escalated issues/ complaints
  - Summary of resolved/ unresolved and escalated issues/ complaints to vendors
  - Summary fault/complain reported and pending at CHiPS end
  - Component wise IT infrastructure availability and resource utilization
  - Consolidated SLA/ (non) conformance report

7.9 Operations & Maintenance (O&M) of SNOC

Operation & Maintenance (O&M) of the S-NOC shall be done by the System Integrator of Package A only.

1) Continuous Operations (24×7×365)

- **Network supervision & monitoring:** Maintain stable operations by continuously monitoring WAN/LAN devices, servers, storage, UPS (SNMP), and links; use alarms/thresholds and video wall dashboards for real-time visibility.
- **Incident management:** Log all incidents in the service-desk tool, categorize/severity-tag, triage, restore service, and escalate per defined matrix with automated SLA tracking and hierarchical escalation. Maintain knowledge base and RCA reporting.
- **Service & change requests:** Handle service provisioning requests and daytoday changes; synchronize changes with maintenance windows; enforce configuration, change control, and compliance policies.
- Integration of discovered network with S-NOC and BBNL Central NOC.

2) Configuration, Asset & Inventory Management

- Discover and maintain inventory of routers, L2/L3 switches, servers/VMs, ports, and IP devices (IPv4/IPv6); identify dormant/unused ports; capture and version device configs; compare against baselines; monitor compliance.
- Maintain **network inventory**, **asset inventory**, and **spare parts** logistics to minimize MTTR and control CAPEX on spares/consumables; manage repair/return cycles.

3) Performance & Capacity Management

- Real-time performance monitoring (latency, errors, availability, bandwidth utilization); baseline and threshold management; capacity planning and long-term traffic analysis; identify over/under-utilized links; generate customizable reports from a central console.
- Allocate adequate resources for the monitoring of all network elements under the BharatNet Phase II Project.

4) EMS/NMS Administration

- **Design/implement/operate EMS** covering configuration, fault, incident/problem/change, asset, remote control, SLA, performance, backup, event, server/storage/network management; single-pane dashboards and northbound interfaces as needed.
- **NMS features:** Auto-discover network elements; GUI maps for end-to-end view; fault correlation; diagnostics & schedulers; event logging; software management (load/activate/backup/restore); high availability & geo-redundancy; AAA integration (RADIUS/TACACS+); firewall/NAT support; usage accounting.

5) Security & Quality Management

- Define and enforce **security policies**; role-based access, domain partitioning, user action logging; detect Local Craft Terminal sessions; ensure continuity access paths in case of management system failure.
- Establish **quality management** practices and periodic audits to sustain competitive QoS.

6) Helpdesk Operations (Centralized L1/L2)

- Operate a 24×7 centralized helpdesk with multi-channel intake (toll-free, landline, tool, email, walk-in); log/assign incidents and service requests; provide L1 resolution and L2 centralized support; dispatch field engineers for emergencies; analyze monthly incident metrics.

- Maintain FAQs/solutions; ensure on-premise coverage for **three shifts** with on-call field support.

7) IT & DB / Backup Administration

- Day-to-day **IT/DB management** for OSS/EMS/IT systems and databases; backups and restorations; monitor backup jobs and retention policies; ensure recovery objectives.

8) Workforce & Field Operations

- Workforce planning and dispatch; adherence to **preventive maintenance plans**; ticket-driven field interventions; track parts/time/expenses; parent/child ticket correlation and collaborative resolution across teams.

9) Reporting & SLA Governance

- End-to-end **SLA/QoS management**: response & resolution SLAs (customer/internal), violation handling, escalations, periodic reporting (daily/weekly/monthly/annual), and dashboards; measure metrics like average call response, mean time to resolve/provide services.

10) Video Wall Operations

- Operate/maintain the **6×3 laser DLP video wall** and controller; project operator screens; display EMS/NMS alarms, status, and reports; ensure input/output capacity and uptime.

11) Site & Infrastructure Upkeep

- Maintain NOC workspace (~3,000 sq ft) and supporting infra: furniture, electrical, raw/UPS power, AC, fire detection/suppression, lighting, CCTV/access control/biometrics, rodent control, phones, structured cabling (per-workstation data/voice/power provisioning).

Team to be Deployed (Roles, Counts & Shift Coverage)

The qualification of **resources deployed at** SNOC shall be as below.

S.No	Profile	Total Exp	Qualification & Certification
1.	Project Manager	10+	B.E/ B.Tech/ MCA/M-Tech with ITILv3 experience, PMP/PRINCE2 certified
2.	NOC Expert	5+	B.E/ B.Tech/MCA, M.Sc (Computer science) experience must in setting up and maintenance on NOC
3.	System Administrator	3+	B.E/ B.Tech/MCA/M.Sc (Computer science), Microsoft certification (MCSE)
4.	Backup administrator	5+	B.E/B.Tech/MCA, sound knowledge of backup software like NetVault/Network/Netbackup
5.	Network Administrator	5+	B.E/B.Tech/MCA, Certified network administrator with minimum 2 years' experience as a network administrator in large data centre

S.No	Profile	Total Exp	Qualification & Certification
6	Helpdesk Technical Support	3+	B.Tech/MCA in computer science, minimum 2 years' experience in similar profile in any data centre

Notes:

- The **Helpdesk** and **NOC** are staffed to ensure three on-premise shifts (Shift-1, Shift-2, Shift-3/Night) with on-call support, as required.
- If SLAs or ticket volumes demand more capacity, the SI must **augment at no additional cost** per the document. Supervisors are **over and above** minimum operators.

8. Payment Schedule

8.1 Implementation Payment (Applicable for all packages)

Payments shall be released only on satisfactory acceptance of the deliverables for each activity/task as per the following schedule:

Payment Milestones	Proposed Payment Schedule
Supply of 100% of material in a block i.e. Duct, OFC, electronics and related accessories on project store	30% of total workorder value
All Field Acceptance Testing (Except End to End Testing of Optical Fibre Cable Route from GP PoP to Block PoP and integration with S-NOC) of a Block	40% of total workorder value
End to End Testing of all GPs under a block and integration of these GPs with S-NOC	20% of Delivered material value
Completion of all documentation with regards to implementation work	10 % of the total CAPEX

- **Note 1**: All payments shall be released after certification of Delivery and Implementation Milestones by CHiPS or its appointed agency as per actual consumption against the workorder.
- **Note 2**: All Payments shall be made in Indian Rupees Only
- **Note 3**: Payment shall be released by the purchaser against the invoices raised by SI within 30 calendar days on providing all the relevant documents, timely and complete in all aspects.
- **Note 4:**
    - All payments shall be made through RTGS only.
    - Payments should be subject to deductions of any amount for which the SI is liable under the RFP conditions. Further, all payments shall be made subject to deduction of TDS (Tax deduction at Source) as per the current Income-Tax Act.

8.2 O&M Payment (Applicable for all packages)

Payments will be made quarterly, subject to satisfactory SLA compliance.

Quarterly Guaranteed Remuneration (QGR) for O&M services will be released by CHiPS upon SLA compliance, validated through performance reports and third-party audits.

Activity	Payment Frequency	Basis of Payment	Release Conditions
Operations & Maintenance (O&M)	Quarterly (at the end of each 3-month period)	QGR amount as per approved financial bid, adjusted for SLA compliance	Subject to verification, certification, and approval by CHiPS

8.3 SNOC Payment (Applicable for Package A Only)

Payments for the O&M of SNOC will be made quarterly, subject to satisfactory SLA compliance.

Quarterly Guaranteed Remuneration (QGR) for O&M services will be released by CHiPS upon SLA compliance, validated through performance reports and third-party audits.

Payment for implementation of any new tools shall be made as a one time payment as per requirement.

Activity	Payment Frequency	Basis of Payment	Release Conditions
Operations & Maintenance (S-NOC)	Quarterly (at the end of each 3-month period)	QGR amount as per approved financial bid, adjusted for SLA compliance	Subject to verification, certification, and approval by CHiPS
Implementation of specific tool as per requirement.	One time	Integration of specific tool with SNOC	Subject to verification and approval by CHiPS

8.4 O&M payment process

QGR Calculation and Payment Schedule

1. Any **SLA-linked deductions** shall be applied as per the penalty matrix specified in Section 9 of this RFP before release of payment.

Prerequisites for Quarterly Payment Release

1. Submission of the following documents:
   - Quarterly invoices
   - Quarterly network performance report (System Generated)
   - Preventive maintenance completion certificate for all sites
   - Asset verification and inventory reconciliation record
   - Copy of fault management and resolution logs
   - SLA compliance reports
   - Fault rectification reports
   - Third-party certification (if applicable).
2. Confirmation of no outstanding penalties or unresolved faults exceeding the permissible limit for the period.

SLA-Linked Payment Adjustment

1. The **Quarterly Payment (QGR)** shall be subject to downward adjustment in case of non-compliance of SLA parameters as detailed in Section 9.
2. If the **average uptime** for the quarter falls below the defined threshold (e.g., 98%), proportional deductions shall be applied as per the penalty formula.
3. Deductions shall also be applicable for:
   - Delay in fault rectification beyond defined timelines,
   - Non-conduct of scheduled preventive maintenance, or
   - Incomplete or inaccurate reporting.
4. If uptime is below the minimum acceptable level (e.g., 90%), the quarter shall be treated as **non-performing**, and no QGR payment shall be released for that period.

Adjustment Clause

1. Any pending penalties or adjustments from previous quarters may be offset against subsequent QGR payments.

4.1.1 **Note 1**: All payments shall be released after certification of O&M Milestones by CHiPS or its appointed agency

5.1.1 **Note 2**: All Payments shall be made in Indian Rupees Only

6.1.1 **Note 3**: Payment shall be released by the purchaser against the invoices raised by SI within 30 calendar days on providing all the relevant documents, timely and complete in all aspects.

7.1.1 Note 4:

8.1.1 All payments shall be made through RTGS only.

9.1.1 Payments should be subject to deductions of any amount for which the SI is liable under the RFP conditions. Further, all payments shall be made subject to deduction of TDS (Tax deduction at Source) as per the current Income-Tax Act.

9. Implementation Plan, Deliverables and Penalties

9.1 The key milestone dates ("critical dates")* as anticipated by the Purchaser are

S. No.	Deliverable	Timelines for completion	Penalties*	Remarks
1.	Issuance of Award of Contract and work order	T0	N.A.	On selection of the SI, Award of Contract and Work Order is issued
2	Resource deployment plan for Implementation & O&M	T+30 Days	1% of the Workorder value per week of delay	Approval of deployment Plan by CHiPS
3.	Supply of 100% of material in a block i.e. Duct, OFC, electronics and related accessories on project store	T+60 Days	If the SI fails to deliver material for respective Block within stipulated time a penalty of 1% of work order value per week	Copy of duly signed and stamped Delivery Challan with serial number of material (for material without serial no, count shall be verified by field team)
4.	Completion of Field Acceptance Testing (Except End to End Testing of Optical Fibre Cable Route from GP PoP to Block PoP and integration with S-NOC) of a Block	T+120 Days	If the SI fails to complete Field Acceptance Testing of all GPs under the Block, then a penalty of 1% of work order value per week (in case of dependency on other agencies, relaxation may be given)	Copy of duly signed and stamped Installation, Commissioning and testing report approved by State, OTDR link test reports, power on, post, and certificate issued along with configuration reports by State, As-Build Diagram (ABD report), Monthly Progress Report
5.	Completion of End to End Testing of all GPs under a block and integration of these GPs with S-NOC	T+150 Days	If the SI fails to complete End to End Testing of all GPs under a block and integration of these GPs with S-NOC, then a penalty of 1% of work order value per week (in case of dependency on other agencies, relaxation may be given)	Copy of duly signed and stamped SNOC report along with Logical Ring failover testing report issued along with configuration reports by State, Monthly Progress Report
6.	End to End Connectivity	T+180 Days	If the SI fails to complete the End to	Copy of duly signed reports By CHiPS

S. No.	Deliverable	Timelines for completion	Penalties*	Remarks
	along with Documentation (Material Reconciliation Report, ITP/ MB Book, and other documents as per RFP)		End Connectivity along with its documentation for all GPs in the Blocks then a penalty of 1% of work order value per week	

The Payment & Penalties shall be against the workorder for respective Blocks

Note: A maximum penalty should be capped 10% of the work order value.

10. Service Level Agreement

10.1 Purpose

10.1.1 The purpose is to define the levels of service provided by SI to the CHiPS for the duration of the contract. The benefits of this are:

a) Start a process that applies to CHiPS and SI attention to some aspect of performance, only when that aspect drops below the threshold defined by the CHiPS

b) Help the CHiPS control the levels and performance of SI's services

10.1.2 The Service Levels are between the CHiPS and SI

10.2 Service Level Agreements & Targets

10.2.1 This section is agreed to by CHiPS and SI as the key performance indicator for the project. This may be reviewed and revised according to the procedures detailed in Clause 10.8 SLA Change Control.

10.2.2 The following section reflects the measurements to be used to track and report system's performance on a regular basis. The targets shown in the following tables are for the period of contact.

10.2.3 The procedures in Clause 10.7 shall be used if there is a dispute between CHiPS and SI on what the permanent targets should be.

10.3 General principles of Service Level Agreements

10.3.1 The Service Level agreements have been logically segregated in the following categories:

10.3.2 **Service Level Agreement**

10.3.2.1 SLA would be applicable in operations and maintenance phase of the project. The penalties shall be applicable on Operations & Maintenance cost of the project calculated quarterly. SLA would be applicable on:

a) Network availability at GP.

b) MTTR (Mean time to restore Fiber).

10.4 Service Levels Monitoring

10.4.1 The Service Level parameters defined in Clause 10.2 shall be monitored on a periodic basis, as per the individual parameter requirements. SI shall be responsible for providing appropriate web based online SLA measurement and monitoring tools for the same. SI shall be expected to take immediate corrective action for any breach in SLA. In case issues are not rectified to the complete satisfaction of CHiPS, within a reasonable period of time defined in this RFP, then the CHiPS shall have the right to take appropriate penalizing actions, or termination of the contract.

10.5 Measurements & Targets

10.5.1 Network related performance levels

S. No	Measurement	Definition	Quarterly Target	Penalty as %age of QGR	Remark
1	Availability of SHQ	All the Network Equipment installed and commissioned under Bharatnet Phase-II project at SNOC/Data Centre	≥99.99 %	No Penalty	Applicable for Package A/Cluster A only
			≥99.5 % to <99.99 %	2.0 %	
			≥99.0 % to <99.5 %	5.0 %	
			<99.0 %	1% penalty on account of each 0.5% reduction in uptime	
2	Network Availability in Block (Aggregation)	The definition for equipment availability is common for all network equipment i.e. availability in Block (Aggregation) and Gram Panchayat (Access) level Network equipment availability for a quarter is defined as total time (in minutes) in a quarter - total down time (in minutes) in a quarter excluding planned network downtime. The network is considered available when all the services in full capacity are available. Network Availability (%) = (Total minutes during the quarter – Planned downtime - Downtime minutes during the quarter) *100 / Total minutes during the quarter Total Time shall be measured on 24*7 basis. Planned Network Component Downtime refers to unavailability of network services due to infrastructure	≥99.5 %	No Penalty	Applicable for all package/cluster
			≥99.5 % to <97.5 %	1.0 %	
			≥97.5 % to < 95.0 %	2.0 %	
			≥95.0 % to <90.0 %	3.5 %	
			<90.0 %	5.0 %	
3	Network Availability at Gram Panchayat (Access) level		≥98.0 %	No Penalty	Applicable for all package/cluster
			≥98.0 % to <95.0 %	1.0 %	
			≥95.0 % to < 90.0%	2.0 %	
			≥90.0 % to < 85.0%	3.5 %	

S. No	Measurement	Definition	Quarterly Target	Penalty as %age of QGR	Remark
		maintenance activities like configuration changes, up gradation or changes to any supporting infrastructure. Details related to such planned outage shall be agreed with CHiPS. The downtime of the Block and GPs commissioned during the implementation phase shall be calculated on prorate basis. Measurement Tool: Reports from EMS duly approved by CHiPS or its appointed agency. SI shall submit quarterly reports on the performance and adherence to the SLA			
			≥85.0 % to < 80.0%	5.0 %	
4	Mean Time To Repair (MTTR) for Fibre for Medium priority Links i.e. 25% of total route length	Mean Time To Repair (MTTR) shall be monitored on the time taken between logging of complaint against the network and its closure Measurement Tool: Reports from EMS duly approved by CHiPS or its appointed agency. SI shall submit quarterly reports on the performance and adherence to the SLA while the penalties shall be charged on quarterly basis	up to 12 hrs	No Penalty	Applicable for all package/cluster
			≥ 12 hrs to 18 hrs	0.5 %	
			≥ 18 hrs to 24 hrs	1.0 %	
			≥ 24 hrs	2.0 %	
5	RCA (Root Cause analysis)	Details root cause analysis of any Incident/Fault in the network	up to 12 hrs	Rs.500 for every hour's delay on an incremental basis.	Applicable for Package A/Cluster A only

During post-implementation period, in case any field equipment is damaged by a vehicular accident (or due to any other reason outside the control of SI) and needs repair/replacement, then the corresponding equipment to be replaced by SI as per the SLAs defined in this section. In such cases, damages are to be borne by SI through proper comprehensive insurance for all the equipment (in the field or at S-NOC) during contract period.

10.5.2 Helpdesk SLA

Service	Measurement	Definition	Target	Penalty as %age of QGR
Incident Reporting	Any failure/incident on any part of the network infrastructure or its facilities shall be communicated immediately to Helpdesk as an exceptional report giving details of downtime, if any.	Reports generated from Ticket logging system	100% incidents to be reported within 1 hour with the cause, action and remedy for the incident.	No Penalty
			Delay beyond an hour	Rs.500 for every hour's delay on an incremental basis.
			100% incident log to be submitted that comprises exceptional & normal reportable activities by 5th of every Quarter for the previous quarter.	No Penalty
			Delay beyond the date of submission of log	Rs. 1000 for every days delay on an incremental basis above the hourly penalty prescribed

10.5.2.1 Medium Priority Links covering approximately 25% of route length

No. of Fibre Cuts per 1000 Km/Month	Penalty
Up to 15	Nil
16 to 25	INR 1500 per cut
26 to 40	INR 2000 per cut
41 and above	INR 2500 per cut

10.6 Reporting Procedures

10.6.1 SI representative shall prepare and distribute Service level performance reports in a mutually agreed format by the 5th working day of subsequent month. The reports shall include "actual versus target" Service Level Performance, a variance analysis and discussion of appropriate issues or significant events. Performance reports shall be distributed to CHiPS management personnel as directed by CHiPS.

10.6.2 Also, SI may be required to get the Service Level performance report audited by the agency appointed by CHiPS

10.7 Issue Management Procedures

10.7.1 General

10.7.1.1 This process provides an appropriate management structure for the orderly consideration and resolution of business and operational issues in the event that quick consensus is not reached between CHiPS and SI.

10.7.1.2 Implementing such a process at the beginning of the outsourcing engagement significantly improves the probability of successful issue resolution. It is expected that this pre-defined process shall only be used on an exception basis if issues are not resolved at lower management levels.

10.7.2 Issue Management Process

10.7.2.1 Either CHiPS or SI may raise an issue by documenting the business or technical problem, which presents a reasonably objective summary of both points of view and identifies specific points of disagreement with possible solutions.

10.7.2.2 Any unresolved issues/disputes concerning the Project/Contract between the Parties shall first be referred in writing to the Project Manager for his consideration and resolution. If the Project Manager is unable to resolve any issue/dispute within 5 days of reference to them, the Project Manager shall refer the matter to the Project Management Unit. If the Project Management Unit is unable to resolve the issues/disputes referred to them within 15 days the unresolved issue/dispute shall be escalated as per the agreed escalation matrix for resolution. CHiPS within 30 days of reference to them shall try to resolve the issue/dispute.

10.7.2.3 If CHiPS fails to resolve a dispute as per the above clause, the same shall be referred to Clause 5.49 of this RFP.

10.8 Service Level Change Control

10.8.1 General

10.8.1.1 It is acknowledged that this Service levels may change as CHiPS's business needs evolve over the course of the contract period. As such, this document also defines the following change management procedures:

- A process for negotiating changes to the Service Levels
- An issue management process for documenting and resolving particularly difficult issues.

- CHiPS and SI management escalation process to be used in the event that an issue is not being resolved in a timely manner by the lowest possible level of management.

10.8.1.2 Any changes to the levels of service provided during the term of this Contract shall be requested, documented and negotiated in good faith by both parties. Either party can request a change.

10.8.2 **Service Level Change Process:** The parties may amend Service Level by mutual agreement in accordance. Changes can be proposed by either party .Unresolved issues shall also be addressed. SI's representative shall maintain and distribute current copies of the Service Level document as directed by CHiPS. Additional copies of the current Service Levels shall be available at all times to authorized parties.

10.8.3 **Version Control / Release Management**: All negotiated changes shall require changing the version control number. As appropriate, minor changes may be accumulated for periodic release or for release when a critical threshold of change has occurred.

10.8.4 **Change Control Note**

Change Control Note		**CCN Number:**
Part A: Initiation		
Title:		
Originator:		
Sponsor:		
Date of Initiation:		
Details of Proposed Change		
(To include reason for change and appropriate details/specifications. Identify any attachments as A1, A2, and A3 etc.)		
Authorized by CHiPS	Date:	
Name:		
Signature:		
Received by the SI	Date:	
Name:		
Signature:		
Change Control Note		CCN Number:
Part B : Evaluation		
(Identify any attachments as B1, B2, and B3 etc.) Changes to Services, payment terms, payment profile, documentation, training, service levels and component working arrangements and any other contractual issue.		
Brief Description of Solution:		
Impact:		
Deliverables:		

Timetable:	
Charges for Implementation: (including a schedule of payments)	
Other Relevant Information: (including value-added and acceptance criteria)	
Authorized by the SI	Date:
Name:	
Signature:	
Change Control Note	**CCN Number :**
Part C : Authority to Proceed	
Implementation of this CCN as submitted in Part A, in accordance with Part B is: (tick as appropriate)	
Approved Rejected Requires Further Information (as follows, or as Attachment 1 etc.)	
For SI and its nominated agencies	For SI
Signature	Signature
Name	Name
Title	Title
Date	Date

10.9 Exclusions

10.9.1 For the purpose of calculating SLA, the following faults or outage hours shall be excluded:

- Periods where any link is switched off at CHiPS or Govt. Office due its own reasons. The onus lies on CHiPS or respective Govt. establishment to ensure that the on-site equipment are powered ON
- Periods where the failure of any components or equipment belonging to CHiPS/Govt. office.
- The time lost in attending to a complaint due to delay in entering CHiPS/Govt. institutions premises shall not be considered as a down time
- Any or all other reasons not mentioned above for which SI is not accountable

11. Annexures

11.1 Annexure-1: Request for Clarifications/ Pre-bid queries

11.1.1 SI requiring specific points of clarification may communicate with CHiPS during the specified period using the following format to be submitted in (.xls):

<table>
<tr><th colspan="4">Bidder's Request for clarification/ pre-bid queries</th></tr>
<tr><td colspan="2">Name of Organization submitting request</td><td>Name & position of person submitting request</td><td>Full address of the Organization including phone, fax and email points of contact</td></tr>
<tr><td colspan="2" rowspan="3"></td><td rowspan="3"></td><td>Tel:</td></tr>
<tr><td>Fax:</td></tr>
<tr><td>Email:</td></tr>
<tr><td>S. No.</td><td>Bidding Document Reference(s) (Section number/ page)</td><td>Content of RFP requiring Clarification</td><td>Points of clarification Required</td></tr>
<tr><td>1.</td><td></td><td></td><td></td></tr>
<tr><td>2.</td><td></td><td></td><td></td></tr>
</table>

***Note*: The name of the organization and the date shall appear in each page of such as document/ email in the header or footer portion.**

Yours sincerely,

Authorized Signature [In full and initials]: ___________________

Name and Title of Signatory: _____________________

Name of Firm: __________________

Address: __________________________

Location: ____ Date: ___________________

11.2 Annexure-2: List of GPs under BharatNet Phase II project in Chhattisgarh

(List of GPs attached as a separate file in eProcurement website)

11.3 Annexure-3: Guidelines for E-Procurement

(Guidelines for bidders on using integrated e-Procurement System Govt. of Chhattisgarh. https://eproc.cgstate.gov.in)

Note: These conditions will over-rule the conditions stated in the tender document(s), wherever relevant and applicable.

1. Vendor / Bidder Registration on the e-Procurement System:

All the Users / Bidders (Manufacturers / Contractors / Suppliers / Vendors / Distributors etc.) registered with and intending to participate in the Tenders of various Govt. Departments / Agencies / Corporations / Boards / Undertakings under Govt. of Chhattisgarh processed using the Integrated e-Procurement System are required to get registered on the centralized portal https://eproc.cgstate.gov.in and get approval on specific class (e.g. A, B, C, D, UGE, UDE, Others/Open) from Public Works Department (in case to participate in tenders restricted to vendors / bidders in a particular class).

The non – registered users / bidders who are also eligible to participate in the tenders floated using the e-Procurement system are also required to be registered online on the e-Procurement system.

Vendors are advised to complete their online enrolment / registration process on the portal well in advance to avoid last minute hassle, it is suggested to complete enrolment at least four days before the last date of bid submission date, failing which may result in non-submission of bids on time for which vendor/end user shall be solely responsible.

For more details, please get in touch with e-Procurement system integrator, M/s. MJunction Services Limited, Raipur – 492 001 on Toll free 1800 258 2502 or email helpdesk.eproc@cgswan.gov.in.

2. Digital Certificates:

The bids submitted online must be signed digitally with a valid Class II / Class – III Digital Signature Certificate to establish the identity of the bidders submitting the bids online. The bidders may obtain pair of Encryption & Signing Class – II / Class – III Digital Certificate issued by an approved Certifying Authority (CA) authorized by the Controller of Certifying Authorities (CCA), Government of India.

Note: It may take up to 7 to 10 working days for issuance of Class-II / Class-III Digital Certificate, Therefore the bidders are advised to obtain it at the earliest. It is compulsory to possess a valid Class-II / Class-III Digital Certificate while registering online on the above mentioned e-Procurement portal. A Digital Certificate once mapped to an account / registration cannot be remapped with any other account / registration however it may be inactivated / deactivated.

Important Note: bid under preparation / creation for a particular tender may only be submitted using the same digital certificate that is used for encryption to encrypt the bid data during the bid preparation / creation / responding stage. However bidder may prepare / create and submit a fresh bid using his/her another / reissued / renewed Digital Certificate only within the stipulated date and time as specified in the tender.

In case, during the process of a particular bid preparation / responding for a tender, the bidder loses his/her Digital Certificate because of any reason they may not be able to submit the same bid under preparation online, Hence the bidders are advised to keep their Digital Certificates secure to be used whenever required and comply with IT Act 2000 & its amendments and CVC guidelines.

The digital certificate issued to the authorized user of an individual / partnership firm / private limited company / public limited company / joint venture and used for online bidding will be considered as equivalent to a no-objection certificate / power of attorney to the user.

Unless the certificate is revoked, it will be assumed to represent adequate authority of the specific individual to bid on behalf of the organization / firm for online tenders as per Information Technology Act 2000. This authorized user will be required to obtain a valid Class-II / Class-III Digital Certificate. The Digital Signature executed through the use of Digital Certificate of this authorized user will be binding on the organization / firm. It shall be the responsibility of management / partners of the concerned organization / firm to inform the Certifying Authority, if the authorized user changes, and apply for a fresh digital certificate for the new authorized user.

3. Online Payment

As the bid is to be submitted only online, bidders are required to make online payment(s) of the Registration fee / Transaction or Service fees using the online payments gateway services integrated into the e-Procurement system using various payment modes like Credit Card / Debit Card / Internet Banking / Cash Card / NEFT / RTGS etc.

For the list of available online modes of electronic payments that are presently accepted on the online payments gateway services, please refer the link ‘Payments accepted online’ on the e-Procurement portal https://eproc.cgstate.gov.in.

4. Setup of User’s Computer System

In order to operate on the e-Procurement system for a bidder / user, the computer system / desktop / laptop of the bidder is required to have Java ver. 765 , Internet explorer 9 / 11, latest Mozila, firefox with IE Tab V2 (Enhanced IE Tab) or any other latest browser. A detailed step by step document on the same is available on the home page. Also internet connectivity should be minimum one MBPS.

5. Publishing of N.I.T.:

For the tenders processed using the e-Procurement system, only a brief advertisement notice related to the tender shall be published in the newspapers and the detailed notice shall be published only on the e-Procurement system. Bidders can view the detailed notice, tender document and the activity time schedule for all the tenders processed using the e-Procurement system on the portal https://eproc.cgstate.gov.in.

6. Tender Time Schedule:

The bidders are strictly advised to follow the tender time for their side for tasks / activities and responsibilities to participate in the tender, as all the activities / tasks of each tender are locked before the start time & date and after the end time & date for the relevant activity of the tender as set by the concerned department official.

7. Download Tender Document(s):

The tender document and supporting document(s) if any can be downloaded only online. The tender document(s) will be available for download to concerned bidders after online publishing of the tender and up to the stipulated date & time as set in the tender.

8. Submit Online Bids:

Bidders have to submit their bid online after successful filling of forms within the specified date and time as set in the tender.

The encrypted bid data of only those bidders who have submitted their bids within the stipulated date & time will be accepted by the e-Procurement system. It is expected that the bidder complete his bid ad submit within timeline, a bidder who has not submitted his bid within the stipulated date & time will not be available during opening.

Bid documents uploading during bid preparation should be less than five MB (for individual document) and over all bid documents should be less than fifty MB.

9. Submission of Earnest Money Deposit:

The bidders shall submit their Earnest Money Deposit Either as usual in a physically sealed Earnest Money Deposit envelope and the same should reach the concerned office OR Online using payment gateway as stated in the Notice Inviting Tender. Bidders also have to upload scanned copy of Earnest Money Deposit instrument.

10. Opening of Tenders:

The concerned department official receiving the tenders or his duly authorized officer shall first open the online Earnest Money Deposit envelope of all the bidders and verify the same uploaded by the bidders. He / She shall check for the validity of Earnest Money Deposit as required. He / She shall also verify the scanned documents uploaded by the bidders, if any, as required. In case, the requirements are incomplete, the next i.e. technical and commercial envelopes of the concerned bidders received online shall not be opened.

The concerned official shall then open the other subsequent envelopes submitted online by the bidders in the presence of the bidders or their authorized representatives who choose to be present in the bid opening process or may view opened details online.

11. Briefcase:

Bidders are privileged to have an online briefcase to keep their documents online and the same can be attached to multiple tenders while responding, this will facilitate bidders to upload their documents once in the briefcase and attach the same document to multiple bids submitting.

For any further queries / assistance, bidders may contact:

The Service Integrator of e-Procurement system, M/s. Mjunction Service Ltd. on Help Desk Toll free No. 1800 258 2502 or email helpdesk.eproc@cgswan.gov.in, or Tel. No. 0771 - 4014158 or email: pro-chips@nic.in.

11.4 Annexure-4: Technical Bid Cover Letter (Company Letter head)

To,

Chief Executive Officer
Chhattisgarh Infotech Promotion Society (CHiPS)
SDC Building, 02nd floor, Near Police Control Room,
Civil Lines, Raipur, Chhattisgarh–492001

Sub: Submission of the response to the RFP No 1630/CEO/CHiPS/BHARATNET/O&M/2025 dated 15/12/2025 for Selection of System Integrator Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C)

Dear Sir,

We, the undersigned, offer to provide Implementation & Maintenance of BharatNet Phase-II for CHIPS in response to the Request for Proposal dated 15.12.2025 and Tender Reference No 1630/CEO/CHiPS/BharatNet/O&M/2025 for "Selection of Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C)". We are hereby submitting our Proposal online, which includes the Technical Bid and Commercial Bid.

We hereby declare that all the information and statements made in this Technical Bid are true and accept that any misinterpretation contained in it may lead to our disqualification.

We undertake, if our Proposal is accepted, to initiate the Implementation services related to the assignment not later than the date indicated in this tender.

We agree to abide by all the terms and conditions of the RFP and related corrigendum(s)/ addendum(s). We would hold the terms of our bid valid for 180 days as stipulated in the RFP.

We hereby declare that as per RFP requirement, we have not been black listed/ debarred by any Central/ State Government and we are not the subject of legal proceedings for any of the foregoing.

We understand you are not bound to accept any Proposal you receive.

Yours sincerely,

Authorized Signature [In full and initials]: ____________________

Name and Title of Signatory: ______________________

Name of Firm: ___________________

Address: ___________________________

Location: ________________________________ Date: _______________

11.5 Annexure-5: Technical Bid Compliance Checklist

Following are the compliance and reference documents for "Selection of System Integrator Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C)" against Tender Reference No.. 1630/CEO/CHiPS/BHARATNET/O&M/2025 Dated 15/12/2025

S. No.	Eligibility Criteria	Documents Required	Compliance
1.	The Bidder should be registered in India under the Companies Act, 1956 or Companies Act, 2013 or as amended and should have at least 5 years of operations in India as on bid submission date. Bidder should have an average annual turnover as following package-wise for the last Five (5) audited financial years (2020-21, 2021-22, 2022-23, 2023-24 and 2024-25) Package A – at least 90 Crores Package B – at least 60 Crores Package C – at least 38 Crores Package D – at least 23 Crores	a.Copy of Certification of Incorporation / Registration Certificate b.PAN card c.GST Registration d.Audited financial statements for the last Five financial years (2020-21, 2021-22, 2022-23, 2023-24 and 2024-25) e.Certificate from CA on turnover details for the last five (5) financial years (2020-21, 2021-22, 2022-23, 2023-24 and 2024-25)	
2.	The Bidder should have experience as per the scope of work mentioned in this RFP in last 5 years as on bid submission date. "Experience in Implementation / Operation and Maintenance of Optical Fibre Cable (OFC) networks" as per following - Package A - at least 4400 km under a single Work Order within India.	1. Work order/ agreement clearly highlighting the scope of work & value of contract 2. Completion Certificate issued & signed by the competent authority of the client entity clearly mentioning the scope of work	

S. No.	Eligibility Criteria	Documents Required	Compliance
	Package B - at least 2800 km under a single Work Order within India. Package C - at least 1700 km under a single Work Order within India. Package D - at least 1200 km under a single Work Order within India.		
3.	The Bidder should have experience as per the scope of work mentioned in this RFP in last 5 years as on bid submission date. “Experience in Implementation / Operation and Maintenance of network equipment (Routers, Switches and other associated accessories)” as per following - Package A - at least at least 700 IP-MPLS nodes under a single Work Order within India. Package B - at least at least 450 IP-MPLS nodes under a single Work Order within India. Package C - at least at least 200 IP-MPLS nodes under a single Work Order within India. Package D - at least at least 200 IP-MPLS nodes under a single Work Order within India.	1. Work order/ agreement clearly highlighting the scope of work & value of contract 2. Completion Certificate issued & signed by the competent authority of the client entity clearly highlighting the scope of work	
4.	The Bidder should have valid ISO 9001:2008/ ISO 9001:2015 for Quality Management System which should be valid as on bid submission date	Copy of certificate valid as on bid submission date	
5.	The Bidder should not have been blacklisted/debarred by any Govt. department or any PSU in India during last five (5) years as on bid submission date	Self-Declaration for not being Black Listed by any of the Govt. or organisation of Govt. in India by the authorized signatory of the bidder and consortium partner	

S. No.	Eligibility Criteria	Documents Required	Compliance
6.	The Bidder should submit MAF for the passive and active material to be supplied under this tender. Bidder to provide the MAF from the OEMs confirming that the OEMs shall ensure that all equipment / components / sub-components being supplied under this tender shall be supported for the entire contract period If the same is de-supported by the OEM for any reason whatsoever, the bidder shall replace it with an equivalent or better substitute that is acceptable to Purchaser without any additional cost to the Purchaser and without impacting the performance of the solution in any manner whatsoever	Documentary evidence such as MAF(Manufacturers Authorization Form) from OEM for specified products (Active & Passive) being quoted by the Bidder OEM Compliance document of TEC-GR for all applicable products from OEM whose products are being quoted by the Bidder	
7.	The Bidder should have a project office in Raipur. However, if the local presence is not there in Raipur, the selected bidder should provide an undertaking for establishment of a project office, within 30 days of award of the contract.	Self-certification duly signed by authorized signatory on company letter head.	
8.	The Bidder should have positive net worth as on 31st March, 2025	Certificate from CA	

Note: Bidder to also submit the signed copy of the RFP and all its published corrigendum as part of the submission

11.6 Annexure-6: Certificate for No Conflict of Interest Certificate (Company Letter head)

To,
Chief Executive Officer
Chhattisgarh Infotech Promotion Society (CHiPS)
SDC Building, 02nd floor, Near Police Control Room,
Civil Lines, Raipur, Chhattisgarh–492001

Sub: Undertaking on No Conflict of Interest Certificate regarding Selection Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C)

Dear Sir,

I/We do hereby undertake that there is absence of, actual or potential conflict of interest on the part of the SI or any prospective subcontractor due to prior, current, or proposed contracts, engagements, or affiliations with CHiPS.

I/We also confirm that there are no potential elements (time-frame for service delivery, resource, financial or other) that would adversely impact our ability to complete the requirements as given in the RFP.

We undertake and agree to indemnify and hold CHiPS harmless against all claims, losses, damages, costs, expenses, proceeding fees of legal advisors (on a reimbursement basis) and fees of other professionals incurred (in the case of legal fees and fees of professionals, reasonably) by CHiPS and/ or its representatives, if any such conflict arises later.

Yours sincerely,

Authorized Signature [In full and initials]: __________________

Name and Title of Signatory: ____________________

Name of Firm: _________________

Address: ________________________

Location: _______________________________________ Date: __________________

11.7 Annexure-7: Format for Power of Attorney

[To be executed on stamp paper of appropriate value]

Know all men by these presents, We, *[Insert full legal name of the bidding entity]*, having registered office at *[Insert registered office address]* (hereinafter referred to as the "Principal") do hereby constitute, nominate, appoint and authorize *[Insert full name of authorized signatory]* son of *[Insert father's name]* presently residing at *[Insert address of authorized signatory]* who is presently employed with us and holding the position of *[Insert position / designation of the authorized signatory]* as our true and lawful attorney (hereinafter referred to as the "Authorized Attorney") to do in our name and on our behalf, all such acts, deeds and things as are necessary or required in connection with or incidental to the submission of our proposal in response to the RFP bearing number ______ for '<RFP Name>' dated ______, including but not limited to signing and submission of all applications, proposals and other documents and writings, participating in pre-Bid and other conferences and providing information/ responses to the Chhattisgarh Infotech Promotion Society (hereinafter referred to as the "CHiPS"), representing us in all matters before the CHiPS, signing and execution of all contracts and undertakings/ declarations consequent to acceptance of our Proposal and generally dealing with the CHiPS in all matters in connection with or relating to or arising out of our Proposal for the said assignment and/ or upon award thereof to us till the execution of appropriate Agreement/s with the CHiPS.

AND, we do hereby agree to ratify and confirm all acts, deeds and things lawfully done or caused to be done by our said Authorized Attorney pursuant to and in exercise of the powers conferred by this deed of Power of Attorney and that all acts, deeds and things done by our said Authorized Attorney in exercise of the powers hereby conferred shall always be deemed to have been done by us.

IN WITNESS THEREOF WE, ___________________________________ THE ABOVE NAMED PRINCIPAL HAVE EXECUTED THIS POWER OF ATTORNEY ON THIS DAY OF ...., 2025

For_______________

(Signature, name, designation and address)

[Please put company seal if required]

[Notarize the signatures]

Witness 1: Witness 2:

Name:	Name:
Designation:	Designation:
Address:	Address:
Signature:	Signature:

11.8 Annexure-8: Declaration Pro-forma (Company Letter head)

{Place}

{Date}

To,

Chief Executive Officer

Chhattisgarh infotech Promotion Society (CHiPS)

SDC Building, 02nd Floor, Near Police Control Room, Raipur,

Chhattisgarh– 492001

Ref: Tender Reference No. : 1630/CEO/CHiPS/ BHARATNET/O&M/2025 dated 15/12/2025

Subject: Declaration for not being under an ineligibility for corrupt or fraudulent practices or blacklisted/debarred with any of the Government or Public Sector Units

Dear Sir/Madam,

We, the undersigned, hereby declare that we are not under a declaration of in eligibility for corrupt or fraudulent practices or blacklisted/debarred with any of the Government or Public Sector Units as on bid submission date.

Yours sincerely,

Authorized Signature [In full and initials]: ___________________

Name and Title of Signatory: ______________________

Name of Firm: __________________

Address: __________________________

Location: ____ Date: ___________________

11.9 Annexure-9: Schedule of Requirements

Supply (As per technical specifications given in Annexure- 14 & 15)				
Sr. No.	**Item Description**	**Make**	**Model**	**Complied (Yes/No)**
A	**Passive infrastructure**			
1	48 core Optical Fibre Cable			
2	Supply of PLB HDPE Duct			
3	Supply of FDMS outdoor (Joint Closure)			
4	Supply of Route / joint indicator			
5	Manhole			
6	FDMS at GP			
7	Fibre Patch cord (1M)			
8	Fibre Patch cord (3M)			
9	Rack - 6U wall mount with all accessories for GPs			
10	1G SFP compatible with current infrastructure (Compatible with Cisco ASR-920-12CZ-A)			
11	1G SFP Compatible with Proposed GP Access Routers			
12	10G SFP compatible with current infrastructure (Compatible with Cisco ASR-920-12CZ-A)			
13	10G SFP Compatible with Proposed GP Access Routers			
B	**Active infrastructure**			
1	Access Routers at GPs			
C	**Power infrastructure**			
1	UPS of minimum 600 VA at GP level to be SNMP enabled or equivalent for SLA monitoring.			

11.10 Annexure-10: Pre-Contract Integrity Pact

INTEGRITY PACT

1. GENERAL

1.1 This pre-bid contract Agreement (hereinafter called the Integrity Pact) is made on ....day of the month........... 20...... between, the Government of Chhattisgarh acting through Shri.. (Designation of the officer, Department) Government of Chhattisgarh (hereinafter called the "**Tendering Authority**", which expression shall mean and include, unless the context otherwise requires, his successors in the office and assigns) and the First Party, proposes to procure (name of the Stores/Equipment/Work/Service) and M/s...represented by Shri.. (hereinafter called the "Bidder/Seller", which expression shall mean and include, unless the context otherwise requires, his successors and permitted assigns) as the Second Party, is willing to offer/ has offered.

1.2. WHEREAS the BIDDER is a Private Company/Public Company/ Government Undertaking/ Partnership firm, constituted in accordance with the Applicable Laws in the matter and the TENDERING AUTHORITY is a Ministry/Department of the Government, performing its function on behalf of the Government of Chhattisgarh.

2. OBJECTIVES

2.1. Enabling the TENDERING AUTHORITY to obtain the desired Stores/Equipment/Work/Service at a competitive price in conformity with the defined specifications by avoiding the high cost and the distortionary impact of corruption on public procurement, and

2.2. Enabling BIDDERs to abstain from bribing or indulging in any corrupt practices in order to secure the contract by providing assurance to them that their competitors will also abstain from bribing any corrupt practices and the TENDERING AUTHORITY will commit to prevent corruption, in any form, by its official by following transparent procedures.

3. COMMITMENTS OF THE TENDERING AUTHORITY

3.1. The TENDERING AUTHORITY undertakes that no official of the TENDERING AUTHORITY, connected directly or indirectly with the contract, will demand, take promise for or accept, directly or through intermediaries, any bribe, consideration, gift, reward, favour or any material or immaterial benefit or any other advantage from the BIDDER, either for themselves or for any person, organization or third party related to the contract in exchange for an advantage in the Bidding Process, bid evaluation, contracting or implementation process related to the contract.

3.2. The TENDERING AUTHORITY will, during the pre-contract stage, treat BIDDERs alike, and will provide to all BIDDERs the same information and will not provide any such information to any particular BIDDER which could afford an advantage to that particular BIDDER in comparison to the other BIDDERs.

3.3. All the officials of the TENDERING AUTHORITY will report the appropriate Government office any attempted or completed breaches of the above commitments as well as any substantial suspicion of such a breach. In case any such preceding misconduct on the part of such official(s) is reported by the BIDDER to the TENDERING AUTHORITY with the full and verifiable facts and the same prima fade found to be correct by the TENDERING AUTHORITY, necessary disciplinary proceedings, or any other action as deemed, fit, including criminal proceedings may be initiated by the TENDERING AUTHORITY and such a person shall be debarred from further dealings related to the contract process. In such a case while an enquiry is being conducted by the TENDERING AUTHORITY the proceedings under the contract would not be stalled.

4. COMMITMENTS OF BIDDERS

4.1. The BIDDER will not offer, directly or through intermediaries, any bribe, gift, consideration, reward, favour, any material or immaterial benefit or other advantage, commission, fees, brokerage or inducement to any official of the TENDERING AUTHORITY, connected directly or indirectly with the biding process, or to any person, organization or third party related to the contract in exchange for any advantage in the bidding, evaluation, contracting and implementation of the contract.

4.2. The BIDDER further undertakes that it has not given, offered or promised to give, directly or indirectly any bribe, gift, consideration, reward, favour, any material or immaterial benefit or other advantage, commission, fees, brokerage, or inducement to any official of the TENDERING AUTHORITY or otherwise in procuring the Agreement or forbearing to do or having done any act in relation to the obtaining or execution of the contract or any other contract with the Government for showing or forbearing to show favour or dis-favour to any person in relation to the contract or any other contract with the Government.

4.3. The BIDDER further confirms and declares to the TENDERING AUTHORITY that the BIDDER in the original Manufacture/Integrator/Authorized government sponsored export entity of the stores and has not engaged any individual or firm or company whether Indian or foreign to intercede, facilitate or in any way to recommend to the TENDERING AUTHORITY or any of its functionaries, whether officially or unofficially to the award of the contract to the BIDDER, nor has any amount been paid, promised or intended to be paid to any such individual, firm or company in respect of any such intercession, facilitation or recommendation.

4.4. The BIDDER, either while presenting the bid or during pre-contract negotiations or before signing the Agreement, shall disclose any payment he has made, is committed to or intends to make to officials of the TENDERING AUTHORITY or their family members, agents, brokers or any other intermediaries in connection with the contract and the details of services agreed upon for such payments.

4.5. The BIDDER will not collude with other parties interested in the contract to impair the transparency, fairness and progress of the Bidding Process, bid evaluation, contracting and implementation of the contract.

4.6. The BIDDER will not accept any advantage in exchange for any corrupt practice, unfair means and illegal activities.

4.7. The BIDDER shall not use improperly, for purpose of competition or personal gain, or pass on to others, any information provided by the TENDERING AUTHORITY as part of the business relationship, regarding plans, technical proposal and business details, including information contained in any electronic data carrier. The BIDDER also undertakes to exercise due and adequate care lest any such information is divulged.

4.8. The BIDDER commits to refrain from giving any complaint directly or through any other manner without supporting it with full and verifiable facts.

4.9. The BIDDER shall not instigate or cause to instigate any third person to commit any of the acts mentioned above.

5. PREVIOUS TRANSGRESSION

5.1. The BIDDER declares that no previous transgression occurred in the last three years immediately before signing of this Integrity Pact with any other company in any country in respect of any corrupt practices envisaged hereunder or with any Public Sector Enterprise in India or any Government Department in India that could justify BIDDER's exclusion from the tender process.

5.2. If the BIDDER makes incorrect statement on this subject, BIDDER can be disqualified from the tender process or the contract, if already awarded, can be terminated for such reason.

6. EARNEST MONEY (SECURITY DEPOSIT)

Every BIDDER while submitting commercial bid, shall deposit an amount as specified in RFP as Earnest Money/Security Deposit online through e-procurement portal

6.1. No interest shall be payable by the TENDERING AUTHORITY to the BIDDER on Earnest Money/Security Deposit for the period of its currency.

7. SANCTIONS FOR VIOLATIONS

7.1. Any breach of the aforesaid provisions by the BIDDER or any one employed by it or acting on its behalf (whether with or without the knowledge of the BIDDER) shall entitle the TENDERING AUTHORITY to take all or any one of the following actions, wherever required:-

(i) To immediately call off the pre contract negotiations without assigning any reason or giving any compensation to the BIDDER. However, the proceedings with the other BIDDER(s) would continue.

(ii) To forfeit fully or partially the Earnest Money Deposit (in pre-contract stage) and/or Security Deposit/Performance Bond (after the contract is signed), as decided by the TENDERING AUTHORITY and the TENDERING AUTHORITY shall not be required to assign any reason therefore.

(iii)To immediately cancel the contract, if already signed, without giving any compensation to the BIDDER.

(iv) To recover all sums already paid by the TENDERING AUTHORITY, and in case of the Indian BIDDER with interest thereon at 2% higher than the prevailing Prime Lending Rate while in case of a BIDDER from a country other than India with interest thereon at 2% higher than the LIBOR. If any outstanding payment is due to the BIDDER from the TENDERING AUTHORITY in connection with any other contract such outstanding payment could also be utilized to recover the aforesaid sum and interest.

(v) To encash the advance bank guarantee and performance bond/warranty bond, if furnished by the BIDDER, in order to recover the payments, already made by the TENDERING AUTHORITY, along with interest.

(vi) To cancel all or any other contracts with the BIDDER and the BIDDER shall be liable to pay compensation for any loss or damage to the TENDERING AUTHORITY resulting from such cancellation/rescission and the TENDERING AUTHORITY shall be entitled to deduct the amount so payable from the money(s) due to the BIDDER.

(vii) To debar the BIDDER from participating in future Bidding Processes of the Government of Chhattisgarh for a minimum period of five years, which may be further extended at the discretion of the TENDERING AUTHORITY.

(viii) To recover all sums paid in violation of this Pact by BIDDER(s) to any middlemen or agent or broken with a view to securing the contract.

(ix) In cases where irrevocable Letters of Credit have been received in respect of any contract signed by the TENDERING AUTHORITY with the BIDDER, the same shall not be opened.

(x) If the BIDDER or any employee of the BIDDER or any person acting on behalf of the BIDDER, either directly or indirectly, is closely related to any of the officers of the TENDERING AUTHORITY, or alternatively, if any close relative of an officer of the TENDERING AUTHORITY has financial interest/stake in the BIDDER's firm, the same shall be disclosed by the BIDDER at the time of filling of tender. Any failure to disclose the interest involved shall entitle the TENDERING AUTHORITY to rescind the contract without payment of any compensation to the BIDDER

The term 'close relative' for this purpose would mean spouse whether residing with the Government servant or not, but not include a spouse separated from the Government servant by a decree or order of a competent court; son or daughter or step son or step daughter and wholly dependent upon Government servant, but does not include a child or step child who is no longer in any way dependent upon the Government servant or of whose custody the Government servant has been deprived of by or under any Applicable Law; any other person related, whether by blood or marriage, to the Government servant or to the Government servant's wife or husband and wholly dependent upon Government servant.

(xi) The BIDDER shall not lend to or borrow any money from or enter into any monetary dealings or transactions, directly or indirectly, with any employee of the TENDERING AUTHORITY, and if he does so, the TENDERING AUTHORITY shall be entitled forthwith to rescind the contract and all other contracts with the BIDDER. The BIDDER shall be liable to pay compensation for any loss or damage to the TENDERING AUTHORITY resulting from such rescission and the TENDERING AUTHORITY shall be entitled to deduct the amount so payable from the money(s) due to the BIDDER.

7.2. The decision of the TENDERING AUTHORITY to the effect that a breach of the provisions of this pact has been committed by the BIDDER shall be final and conclusive on the BIDDER. However, the BIDDER can approach the Monitor(s) appointed for the purposes of this Pact.

8. FALL CLAUSE

8.1. The BIDDER undertakes that he has not supplied/is not supplying similar product/systems or subsystems at a price lower than that offered in the present bid in respect of any other Department of the Government of Chhattisgarh or PSU and if it is found at any stage that similar product/systems or sub systems was supplied by the BIDDER to any other Department of the Government of Chhattisgarh or a PSU at a lower price, then that very price, with due allowance for elapsed time, will be applicable to the present case and the difference in the cost would be refunded by the BIDDER to the TENDERING AUTHORITY, if the contract has already been concluded.

9. INDEPENDENT MONITORS

9.1. The TENDERING AUTHORITY will appoint Independent Monitors (hereinafter referred to as Monitors) for this Pact.

9.2. The task of the Monitors shall be to review independently and objectively, whether and to what extent the parties comply with the obligations under this Pact.

9.3. The Monitors shall not be subject to instructions by the representatives of the parties and perform their functions neutrally and independently.

9.4. Both the parties accept that the Monitors have the right to access all the documents relating to the project/procurement, including minutes of meetings. The Monitor shall be under contractual obligation to treat the information and documents of the BIDDER/Sub-Selected Bidder(s) with confidentiality.

9.5. As soon as the Monitor notices, or has reason to believe, a violation of this Pact, he will so inform the Authority designated by the TENDERING AUTHORITY.

9.6. The Monitor will submit a written report to the designated Authority of TENDERING AUTHORITY/Secretary in the Department/within 8 to 10 weeks from the date of reference or intimation to him by the TENDERING AUTHORITY/BIDDER and, should the occasion arise, submit proposals for correcting problematic situations

10. FACILITATION OF INVESTIGATION

In case of any allegation of violation of any provisions of this Pact or payment of commission, the TENDERING AUTHORITY or its agencies shall be entitled to examine all the documents including the Books of Accounts of the BIDDER and the BIDDER shall provide necessary information of the relevant documents and shall extend all possible help for the purpose of such examination.

11. LAW AND PLACE OF JURISDICTION

This Pact is subject to Indian Law, the place of performance and jurisdiction shall be the seat of the TENDERING AUTHORITY.

12. OTHER LEGAL ACTIONS

The actions stipulated in this Integrity Pact are without prejudice to any other legal action that may follow in accordance with the provisions of the any other Applicable Law in force relating to any civil or criminal proceedings.

13. VALIDITY

13.1. The validity of this Integrity Pact shall be from the date of its signing and extend up to 2 years or the complete execution of the contract to the satisfaction of both the TENDERING AUTHORITY and the BIDDER/Seller whichever is later. In case BIDDER is unsuccessful, this Integrity Pact shall expire after six months from the date of the signing of the contract.

13.2. If one or several provisions of this Pact turn out to be invalid; the remainder of this Pact shall remain valid. In such case, the parties will strive to come to an agreement to their original intentions.

14. The parties hereby sign this

Integrity Pact

at...on.........................

CHiPS	**BIDDER**
___________ [*Signature*]	___________ [*Signature*]
___________ [*Name*]	___________ [*Name*]
___________ [*Designation*]	___________ [*Designation*]
Witness	**Witness**
1.	1.
2.	2.

11.11 Annexure-11: Commercial Bid Cover Letter

To,

Chief Executive Officer
Chhattisgarh infotech Promotion Society (CHiPS)

SDC Building, 02nd floor, Near Police Control Room,
Civil Lines, Raipur, Chhattisgarh–492001

Sub: Selection of System Integrator Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C)

Ref: **Tender Reference No: < > dated < >**

Dear Sir,

We, the undersigned SI, having read and examined in detail all the Tender documents in respect of **Selection of System Integrator Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C)** do hereby propose to provide services as specified in the Tender Reference No. <1630/CEO/CHiPS/ BHARATNET/O&M/2025 > dated <15/12/2025 >

I. PRICE AND VALIDITY

11.1.1 All the prices mentioned in our Bid are in accordance with the terms & conditions as specified in the RFP. Prices of all hardware items under this RFP are valid for a period of 1 year from the date of opening of the commercial bid. Prices of Operations & Maintenance are valid for a period of 2 years. The validity of bid is 180 days from the date of bid submission.

a. We are an Indian Firm and do hereby confirm that our prices are inclusive of all duties, levies etc. excluding GST.

b. We have studied the clause relating to Indian Income Tax and hereby declare that if any income tax, surcharge on Income Tax, Professional and any other Corporate Tax in altercated under the law, we shall pay the same.

II. UNIT RATES

We have indicated in the relevant schedules enclosed, the unit rates for the purpose of on account of payment as well as for price adjustment in case of any increase to/ decrease from the Scope of Work under the Contract.

III. DEVIATIONS

We declare that all the services shall be performed strictly in accordance with the RFP irrespective of whatever has been stated to the contrary anywhere else in our bid.

Further we agree that additional conditions, if any, found in our bid documents, shall not be given effect to.

IV. EARNEST MONEY DEPOSIT (EMD)

We have enclosed an EMD in the form of Non-refundable & Irrevocable Bank Guarantee for a sum of INR **<value as per Factsheet>/- (<value in words as per Factsheet >)**. This EMD is liable to be forfeited as per clause 4.17

V. TENDER PRICING

We further confirm that the prices stated in our bid are in accordance with your Instruction to Bidders included in Tender documents.

VI. QUALIFYING DATA

We confirm having submitted the information as required by you in your Instruction to Bidders. In case you require any other further information/ documentary proof in this regard before evaluation of our Tender, we agree to furnish the same in time to your satisfaction.

VII. BID PRICE

We declare that our Bid Price is for the entire scope of the work as specified in the RFP. These prices are indicated in Annexure-12: Commercial Bid Format attached with our Tender as part of the Tender.

VIII. PERFORMANCE BANK GUARANTEE

We hereby declare that in case the Contract is awarded to us, we shall submit the Performance Bank Guarantee in the form prescribed in Annexure-13: Format for Performance Bank Guarantee

We hereby declare that our Tender is made in good faith, without collusion or fraud and the information contained in the Tender is true and correct to the best of our knowledge and belief.

We understand that our Tender is binding on us and that you are not bound to accept a Tender you receive.

We confirm that no Technical deviations are attached here with this commercial offer.

Yours sincerely,

Authorized Signature [In full and initials]: __________________

Name and Title of Signatory: ____________________

Name of Firm: _________________

Address: _________________________

Location: ____ Date: __________________

11.12 Annexure-12: Commercial Bid Format

Table A: Summary of Cost Component – Overall * The financial bid has to be filled online

Sr. No.	Item Description	Total Cost (Excluding GST)
		INR
1	Total CAPEX (New GPs)	X1
2	Total OPEX (New & Exiting GPs)	X2
3	OPEX For SNOC (For Package A only)	X3
Total INR in Figures (Excluding GST)		Z = (X1+X2+X3)
Total INR in words (Excluding GST)		Z = (X1+X2+X3)

<u>Total Cost (Z= X1+X2+X3 i.e. CAPEX + OPEX) would be considered for commercial evaluation of the bids.</u>

Instructions to fill the Commercial Bid:

i. Bidder should provide all prices as per the prescribed format under this Annexure. Bidder should not leave any field blank.
ii. All the prices are to be entered in Indian Rupees ONLY (% age values are not allowed)
iii. Purchaser reserves the right to ask the Bidder to submit proof of payment against any of the taxes, duties, levies indicated.
iv. Purchaser shall take into account that all Taxes, Duties & Levies shall be paid as per actual.
v. For the purpose of evaluation of Financial Bids the Purchaser shall make appropriate assumptions to arrive at a common bid price for all the Bidders. This however shall have no co-relation with the Contract value or actual payment to be made to the Bidder.
vi. Bidder should fill the Commercial Bid for at least a single Package. The commercial bid filled for all components under a package shall be considered a valid bid.
vii. The Bidders need to submit the commercial bid on the eproc portal as well as, as an attachment under the corresponding commercial package.

Per year OPEX value quoted by bidder should be minimum 5% of total CAPEX value. If quoted less than 5%, the financial bid of such bidder shall not be considered

Table B: Summary of Cost Component – CAPEX# * The financial bid has to be filled online

Sr. No.	Item Description	Unit	Indicative Quantity	Unit Cost (INR) (Excluding GST)	Total Cost (INR) (Excluding GST)	GST (%)	Total GST (INR)	Total Cost (INR) (including GST)	HSN Code	Part Code
			A	B	C=A*B	D	E=C*D	F=C+E		
A	***Passive infrastructure***									
1	48 core Optical Fibre Cable	Km			-		-	-		
2	Supply of PLB HDPE Duct	Km			-		-	-		
3	Supply of FDMS outdoor (Joint Closure)	Nos.			-		-	-		
4	Supply of Route / joint indicator	Nos.			-		-	-		
5	Manhole	Nos.			-		-	-		
6	FDMS at GP	Nos.			-		-	-		
7	Fibre Patch cord (1M)	Nos.			-		-	-		
8	Fibre Patch cord (3M)	Nos.			-		-	-		
9	Rack - 6U wall mount with all accessories for GPs	Nos.			-		-	-		
10	1G SFP compatible with current infrastructure (Compatible with Cisco ASR-920-12CZ-A)	Nos.			-		-	-		
11	1G SFP Compatible with Proposed GP Access Routers	Nos.								
12	10G SFP compatible with current infrastructure (Compatible with Cisco ASR-920-12CZ-A)	Nos.								
13	10G SFP Compatible with Proposed GP Access Routers	Nos.								
B	***Active infrastructure***									
1	Access Routers at GPs	Nos.								
C	***Power infrastructure***									

Sr. No.	Item Description	Unit	Indicative Quantity	Unit Cost (INR) (Excluding GST)	Total Cost (INR) (Excluding GST)	GST (%)	Total GST (INR)	Total Cost (INR) (including GST)	HSN Code	Part Code
			A	B	C=A*B	D	E=C*D	F=C+E		
1	Offline UPS (600 VA) **at Gram Panchayat level**	Nos.			-		-	-		
D	***Services (As per technical specifications given in Annexure-XX TEC GR No.) and Engineering Instruction***									
E	***Passive infrastructure***									
1	Excavation of trench for PLB pipe laying, PLB pipe laying, Backfilling, Reinstatement and Compaction after laying of PLB pipe, Pulling/laying / blowing of optical Fibre Cable inside laid PLB pipe, splicing and jointing of Optical Fibre Cable including supply of As Built Diagram (ABD) of constructed OFC Route and Acceptance Testing, commissioning and makeover of the routes. The work also includes road / bridge crossing, laying of PLB inside DWC pipe, wherever required. Commissioning of 48 core optical fibre connectivity from the Block PoP to GP and termination at GP. The cost shall include fixing of joint chamber, manhole, splice closure, Fixing, painting and sign writing of Route /Joint Indicators, termination at FTB and all the relevant accessories etc. and end to end testing of dark and lit fibre per GP (For Underground	Km			-		-	-		

Sr. No.	Item Description	Unit	Indicative Quantity	Unit Cost (INR) (Excluding GST)	Total Cost (INR) (Excluding GST)	GST (%)	Total GST (INR)	Total Cost (INR) (including GST)	HSN Code	Part Code
			A	B	C=A*B	D	E=C*D	F=C+E		
	laying) (As per the Engineering Instruction issued by BBNL)									
F	***Active infrastructure***									
1	Installation, integration, testing and commissioning of Electronics at GP Level	Nos.			-		-	-		
G	***Site Preparation Cost***									
1	One Time Site Preparation cost for all Civil & Electrical Works including proper earthing at GP	Nos.			-		-	-		
	Total INR in Figures	**INR**			**X1**					
	Total INR in words									

Note 1: The quantities listed in the BoQ sheet online are tentative and may change at the time of implementation.

Table C: Summary of Cost Component – OPEX# * The financial bid has to be filled online

Sr. No.	Item Description	Unit	Indicative Quantity	Unit Cost (INR) (Excluding GST)	Total Cost (INR) (Excluding GST)	GST (%)	Total GST (INR)	Total Cost (INR) (including GST)	HSN Code	Part Code
			A	**B**	**C=A*B**	**D**	**E=C*D**	**F=C+E**		
1	OPEX Cost of one (1) GP on quarterly basis (Including warranty period) *The unit cost has to be mentioned for one (1) GP Only	Quarter	No Unit x 4 Quarter							
Total INR in Figures (Excluding GST)					***X2***					
Total INR in words (Excluding GST)					***X2***					

Note:

- ***The total cost would be considered for commercial evaluation of the bids, CHiPS reserves the right at the time of award of Contract to increase or decrease goods and/ or services from what was originally specified while floating the RFP without any change in unit price or any other terms and conditions.***
- ***Prices in Commercial Bid should be quoted in the provided format. All prices should be quoted in Indian Rupees and indicated both in figures and words. Price in words shall prevail, in the event of any mismatch.***
- ***O&M / AMC Rates discovered are for 1 year only, and the same rates shall be applicable for the extension period as well.***

Table D: Summary of Cost Component – OPEX for SNOC# * The financial bid has to be filled online

Sr. No.	Item Description	Unit	Indicative Quantity	Unit Cost (INR) (Excluding GST)	Total Cost (INR) (Excluding GST)	GST (%)	Total GST (INR)	Total Cost (INR) (including GST)	SAC Code	Part Code
			A	B	C=A*B	D	E=C*D	F=C+E		
1	Cost of AMC of Core Routers	Quarter	2 Unit *4 Quarter							
2	Cost of AMC of L2 & L3 Switches (Edge/ Spine / Leaf)	Quarter	6+2+6 Unit *4 Quarter							
3	Cost of AMC of Firewall	Quarter	2 Unit *4 Quarter							
4	Cost of AMC of IPS	Quarter	2 Unit *4 Quarter							
5	Cost of AMC of Hyperflex Server	Quarter	1 Unit *4 Quarter							
6	Cost of AMC of Video Wall (Projector + Controller)	Quarter	1 Unit *4 Quarter							
7	Cost of AMC of UPS 60 KVA	Quarter	2 Unit *4 Quarter							
8	Cost of AMC of UPS 6 KVA	Quarter	1 Unit *4 Quarter							
9	Cost of AMC of Rodent Repellent system	Quarter	1 Unit *4 Quarter							
10	Cost of AMC of Biometric access control system	Quarter	1 Unit *4 Quarter							
11	Cost of AMC of Fire Alarm System	Quarter	1 Unit *4 Quarter							

12	Cost of AMC of EMS Solution (MotaData)	Quarter	1 Unit *4 Quarter							
13	Cost of AMC of EMS Solution (C-DOT)	Quarter	1 Unit *4 Quarter							
14	Cost of AMC of GIS Solution	Quarter	1 Unit *4 Quarter							
15	Cost of AMC of Log Retention Solution	Quarter	1 Unit *4 Quarter							
16	Cost of AMC of Helpdesk Solution	Quarter	1 Unit *4 Quarter							
17	Cost of AMC of all other electrical components at SNOC	Quarter	1 Unit *4 Quarter							
18	Cost of AMC of Thin/Desktop Clients at SNOC	Quarter	1 Unit *4 Quarter							
19	Operation Cost of overall SNOC as per the defined scope of work excluding AMC cost of above mentioned components	Quarter	1 Unit *4 Quarter							
20	Cost of Implementation of EMS Solution (MotaData)	Onetime	1							
21	Cost of Implementation of GIS Solution	Onetime	1							
22	Cost of Implementation of Log Retention Solution	Onetime	1							
23	Cost of Implementation of Helpdesk Solution	Onetime	1							
24	Cost of Implementation of Ticketing Tool	Onetime	1							
Total INR in Figures (Excluding GST)					***X3***					
Total INR in words (Excluding GST)					***X3***					

Note 1: Mentioned components are for BharatNet Phase II project for the operation of SNOC

Instructions to fill the Commercial Bid:

i. Bidder should provide all prices as per the prescribed format under this Annexure. Bidder should not leave any field blank.

ii. All the prices are to be entered in Indian Rupees ONLY (%age values are not allowed)

iii. Purchaser reserves the right to ask the Bidder to submit proof of payment against any of the taxes, duties, levies indicated.

iv. Purchaser shall take into account that all Taxes, Duties & Levies shall be paid as per actual.

v. For the purpose of evaluation of Financial Bids the Purchaser shall make appropriate assumptions to arrive at a common bid price for all the Bidders. This however shall have no co-relation with the Contract value or actual payment to be made to the Bidder.

vi. O&M / AMC Rates discovered are for 1 year only, and the same rates shall be applicable for the extension period as well.

11.13 Annexure-13: Format for Performance Bank Guarantee

(To be issued by a Bank)

This Deed of Guarantee executed at ----------- by ———————— (Name of the Bank) having its Head/ Registered office at —————————————— (hereinafter referred to as "the Guarantor") which expression shall unless it be repugnant to the subject or context thereof include its heirs, executors, administrators, successors and assigns;

In favour of Chhattisgarh infotech Promotion Society (hereinafter called "CHiPS" which expression shall unless it be repugnant to the subject or context thereof include its heirs, executors, administrators, successors and assigns);

<Organization name > a company registered in India under the Companies Act, 1956 or Companies Act 2013 or as amended and having its Registered Office at ----- , India (herein referred to as the "O&M Agency" for Implementation and Operation & Maintenance of BharatNet Phase-II Project in the State of Chhattisgarh (Package-C), for the work order number ---- dated ---- issued by CHiPS and selected < Organization name > (hereinafter referred to as the SI) for the Contract by CHiPS as more specifically defined in the aforementioned Document including statement of work and the Contract executed between the CHiPS and SI. The Contract requires the SI to furnish an unconditional and irrevocable Bank Guarantee for an amount of INR ----/- (Rupees -----------------) by way of security for guaranteeing the due and faithful compliance of its obligations under the Contract.

Whereas, the SI approached the Guarantor and the Guarantor has agreed to provide a Guarantee being these presents:

Now this Deed witnessed that in consideration of the premises, we, _______Bank hereby guarantee as follows:

1. The SI shall implement the Project, in accordance with the terms and subject to the conditions of the Contract, and fulfil its obligations there under
2. We, the Guarantor, shall, without demur, pay to CHiPS, an amount not exceeding of INR ----/- (Rupees ----) within 21 (Twenty One) days of receipt of a written demand therefore from CHiPS stating that the SI has failed to fulfil its obligations as stated in Clause 1 above
3. The above payment shall be made by us without any reference to the SI or any other person and irrespective of whether the claim of the CHiPS is disputed by the SI or not
4. The Guarantee shall come into effect from ________ (Start Date) and shall continue to be in full force and effect till the earlier of its expiry at 1700 hours Indian Standard Time on (Expiry Date) (both dates inclusive) or till the receipt of a claim, from Chhattisgarh infotech Promotion Society under this Guarantee, whichever is earlier. Any demand received by the Guarantor from CHiPS prior to the Expiry Date shall survive the expiry of this Guarantee till such time that all the moneys payable under this Guarantee by the Guarantor to CHiPS.
5. In order to give effect to this Guarantee, CHiPS shall be entitled to treat the Guarantor as the principal debtor and the obligations of the Guarantor shall not be affected by any variations in the terms and conditions of the Contract or other documents by CHiPS or by the extension of time of performance granted to the SI or any postponement for any time of the power exercisable by CHiPS against the SI or forebear or enforce any of the terms

and conditions of the Contract and we shall not be relieved from our obligations under this Guarantee on account of any such variation, extension, forbearance or omission on the part of SI or any indulgence by CHiPS to the SI to give such matter or thing whatsoever which under the law relating to sureties would but for this provision have effect of so relieving us.

6. This Guarantee shall be irrevocable and shall remain in full force and effect until all our obligations under this guarantee are duly discharged
7. The Guarantor has power to issue this guarantee and the undersigned is duly authorized to execute this Guarantee pursuant to the power granted under ______________

In witness, whereof the Guarantor has set its hands hereunto on the day, month and year first here-in-above written.

Signed and Delivered by ____________ Bank by the hand of Shri_______________ its _____and authorised office.

Authorised Signatory ____________Bank

11.14 Annexure-14: TEC-GR Nos for Technical Specifications

Sr. No.	Item Description	TEC GR Ref. No.	Complied (Yes/No)
A	**Passive infrastructure**		
1	48 core Optical Fibre Cable	GR No. TEC/GR/TX/ORM-01/04 SEP.09	
2	Supply of PLB HDPE Duct	TEC GR No. TEC/GR/TX/CDS-008/03/MAR-11	
3	Supply of FDMS outdoor (Joint Closure)	TEC/GR/TX/FDM-003/01 MAR 2012	
4	Supply of Route / joint indicator	NA	
5	Manhole	NA	
6	FDMS at GP	GR/FDM-01/02 APR 2007	
7	Fibre Patch cord (1M)	NA	
8	Fibre Patch cord (3M)	NA	
9	Rack - 6U wall mount with all accessories for GPs	NA	
10	1G SFP compatible with current infrastructure (Compatible with Cisco ASR-920-12CZ-A)	NA	
11	1G SFP Compatible with Proposed GP Access Routers	NA	
12	10G SFP compatible with current infrastructure (Compatible with Cisco ASR-920-12CZ-A)	NA	
13	10G SFP Compatible with Proposed GP Access Routers	NA	
B	**Active infrastructure**		
1	Access Routers at GPs	TEC GR No. TEC/GR/IT/TCP-006/01/AUG-16	
E	**Power infrastructure**		
1	Offline UPS (600 VA) at GP Level	TEC/GR/TX/UPS-01/04 DEC 2013	

11.15Annexure-15: Technical Specifications – Minimum Requirement

SI shall provide compliance to technical specifications on letter head of the OEM

11.15.1GP Access Router

SI shall provide compliance to technical specifications on letter head of the OEM

Technical Requirements Specifications- GP Access Router Product Name- GP Access Router				
Sr. No.	**Item**	**Minimum Requirement Description**	**Compliance (Yes / No)**	**Reference (Document /Page no)**
GPACR.REQ.001	Architecture & Performance	The router should be Standalone / modular		
GPACR.REQ.002	Architecture & Performance	The OEM is allowed to offer TEC-GR: TEC/GR/IT/TCP-006/01/AUG-16 complied or CE 2.0 complied products subject to OEM submitting CE 2.0 certificate before commencement of shipment.SLF		
GPACR.REQ.003	Architecture & Performance	Router should have 2 x 1 GE SFP Uplink (10 Kms/40 kms) and/or scalable to 2 x 10G Ports (10 Kms/40 kms) for future scalability. It should also have minimum 4 x 1G SFP (fiber) Ports and 4 x 1G SFP (copper) Ports.		
GPACR.REQ.004	Architecture & Performance	Minimum back-plane capacity of 40Gbps		
GPACR.REQ.005	Architecture & Performance	The router should support 10/100/1000 Ethernet, 10-Gig Interfaces.		
GPACR.REQ.006	High Availability Features	It should support ring capabilities with 1-Gig today and 10-Gig in future with only replacement of Optics		
GPACR.REQ.007	High Availability Features	It should have flexibility features to support both Metro-Ethernet and IP/MPLS features		
GPACR.REQ.008	High Availability Features	It should support SyncE (1588)		
GPACR.REQ.009	Protocols – IPv4, IPv6, MPLS	Border Gateway Protocol (BGP) V4		

Technical Requirements Specifications- GP Access Router Product Name- GP Access Router				
Sr. No.	**Item**	**Minimum Requirement Description**	**Compliance (Yes / No)**	**Reference (Document /Page no)**
GPACR.REQ.010	Protocols – IPv4, IPv6, MPLS	IP Multicast, PIM IGMP, IGMP Snooping		
GPACR.REQ.011	Protocols – IPv4, IPv6, MPLS	MPLS for QoS, reliability and resiliency.		
GPACR.REQ.012	Protocols – IPv4, IPv6, MPLS	Label Distribution Protocol (LDP)		
GPACR.REQ.013	Protocols – IPv4, IPv6, MPLS	Multi-protocol Label Switching (MPLS)		
GPACR.REQ.014	Protocols – IPv4, IPv6, MPLS	Should support the following L2 VPN, L3 VPN & VPLS/HVPLS		
GPACR.REQ.015	Protocols – IPv4, IPv6, MPLS	Should support the following Open Shortest Path First (OSPF), Resource Reservation Protocol (RSVP), Fast reroute Link Node and path protection		
GPACR.REQ.016	Security features	DHCP Relay on L3 interfaces, DHCP snooping		
GPACR.REQ.017	Security features	ARP support		
GPACR.REQ.018	Security features	Protection against broadcast, multicast & unicast storms		
GPACR.REQ.019	Ethernet Features	The supported features should include: 802.3ad, 802.1Q, VLAN Stacking (QinQ), Outer/Inner VLAN tagging , 802.1p, LACP, VLAN-ID manipulation-POP/PUSH, STP, RSTP, MSTP (802.1d, 802.1s, 802.1w), 802.1ad, L2 Ethernet Switching, Bridge Learning, L2 ACLs, 802.3ah , 802.1ag, ERPS -G.8032v1/v2, Carrier Ethernet features – E-LAN, E-Line and E-Tree		
GPACR.REQ.020	QoS Classifications	The supported feature shall include: 802.1Q, Source and Destination IP address/subnet mask, DSCP value, IP precedence classification, Protocol type (TCP, UDP, etc.), Default markings per port (ingress), 802.1p/DSCP (ingress), Per port and per queue shaping		

Technical Requirements Specifications- GP Access Router Product Name- GP Access Router				
Sr. No.	**Item**	**Minimum Requirement Description**	**Compliance (Yes / No)**	**Reference (Document /Page no)**
GPACR.REQ.021	QoS Classifications	Router should support Eight No of hardware queues per port for flow treatment of traffic, Policy Based QOS, WRED, Ethernet OAM and Y.1731 performance management		
GPACR.REQ.022	Performance Numbers	Shall support minimum 15k MAC addresses, 20k IPv4 routing entries per system, 5k IPv6 routing entries per system, Minimum 30 VRF’s, 1000 VLAN’s		
GPACR.REQ.023	OAM & Monitoring Features	The router should support Zero touch provisioning for ease of management		
GPACR.REQ.024	Network Management	The router shall be manageable from a Network management System (NMS) platform. The Router shall support the following management features:		
GPACR.REQ.025	Network Management	Console or Out-of-band Management		
GPACR.REQ.026	Network Management	The Router shall support Network Time Protocol (NTP) as per RFC 1305		
GPACR.REQ.027	Network Management	Role based privileges for the system access		
GPACR.REQ.028	Network Management	Radius authentication for the System admin.		
GPACR.REQ.029	Network Management	Device management: NETCONF, CLI, SNMP v1/v2/v3		
GPACR.REQ.030	Port requirement from Day 1	GP Access router should be loaded with 2 x 1 GE SFP Uplink loaded with range 10 Kms/40 kms and 8 x 1G (SFP / Copper) Ports		
GPACR.REQ.031	Physical specifications	19” standard Rack mountable,		
GPACR.REQ.032	Physical specifications	Operating Temperature 0 to 55 deg (0 to +55)		
GPACR.REQ.033	Physical specifications	Relative Humidity : 10% to 90% non-condensing		

11.15.2 Rack 6U

SI shall provide compliance to technical specifications on letter head of the OEM

Technical Requirements Specifications- Rack (6U - Wall Mount) Product Name- Rack (6U)				
Sr. No.	**Item**	**Minimum Requirement Description**	**Compliance (Yes / No)**	**Reference (Document /Page no)**
RK6.REQ.001	General Requirements	6U 19" wall mount rack 600W/500mmD, Front glass door with lock & key Bolted construction, Top and Bottom Cable entry provision, Maximum load carrying capacity of 40Kg.		
RK6.REQ.002	Accessories	Cooling Fan 230v,90cfm, Power strip (6 x 5A), Cable manger 1u 19”Cantilever Shelf 1U 19” mtg, CKD type completely knock down condition		

11.15.3 **Technical Specification for OFC & Accessories**

SECTION-A

TECHNICAL SPECIFICATIONS: TECHNICAL SPECIFICATIONS: GENERIC REQUIREMENT FOR 48F METAL FREE RIBBON OPTICAL FIBRE CABLE WITH DOUBLE HDPE SHEATH (G.652.D Fibre)

1. This section consists of detailed Technical Specifications of 48F Metal free ribbon Optical Fibre Cable items and Standards of sub-items. A details about the specifications of OF Cable.

2.	GR No. TEC/GR/TX/ORM-01/04 SEP.09	Specification for Raw Material used in manufacturing of OF Cables.
3.	GR No. GR/OFT-01/03. APR 2006	Tools for installation & Operating the OFC & for assembly of the OF Splice Closures.
4.	GR No. G/CBD-01/02. NOV 94	Drum specifications for Cable ends.

5.**OTHER STANDARDS (EIA/IEC/Bell Core/CISPR/ISO etc.)**

5.1	ITU-T G.652D	ITU-T Recommendations
5.2	GR-20-CORE Issue4, 2013	Generic Requirement for Optical Fibre Cable (Bell Core)
5.3	IEC 811-5-1, IEC 794-1-2-E1 IEC 794-1-2-E2, IEC 794-1-2-E3 IEC 794-1-2-E4, IEC 794-1-2-E7, IEC 794-1-2-E10, IEC 794-1-E11 IEC 794-1-2-F1, IEC 794-1-2-F3, IEC 794-1-2-F5, IEC 60793-1-30, IEC 60793-1-31, IEC 60793-1-32, IEC 60793-1-33, IEC 60793-1-34 IEC 60793-1-47, IEC 60793-1-51, IEC 60793-1-52, IEC-60793-1-53 IEC -60793-2-50	Test Methods for Optical Fibres
5.4	EIA 598-C	Colour Standard
5.5	EIA 455-104, EIA/TIA-455-181, EIA/TIA-455-73	Test Method for Optical Fibre
5.6	ISO 175, ISO 9001-2000	Test Methods for Optical Fibres International Quality Management System
5.7	FOTP-89, FOTP-181	Test Methods
5.8	ASTM D-566, ASTM D-790 ASTM D-1248, ASTM D-4565	Test Methods

SECTION-B

DETAILED TECHNICAL SPECIFICATIONS / REQUIREMENTS FOR 48F METAL FREE RIBBON OPTICAL FIBRE CABLE WITH DOUBLE HDPE SHEATH (G.652D Fibre)

1. Introduction:

This document describes the generic requirements of Metal free Ribbon Optical fibre cable (Non-Nylon Ribbon type) for underground installation in ducts. The cable shall have double HDPE jacketing anti-termite & anti-rodent (Optional) with glass yarn in between as reinforcement. The optical fibre cable shall be suitably protected for the ingress of moisture by WS yarn and WS tape. The raw material used in the cable shall meet the requirements of the GR for raw materials (GR No. TEC/GR/TX/ORM-01/04 SEP-09).

2. Functional Requirement:

2.1 The design and construction of Optical fibre cable shall be inherently robust and rigid under all conditions of installation, operation, adjustment, replacement, storage and transport.

2.2 The Ribbon Optical fibre cable shall be able to work in a saline atmosphere in coastal areas and should be protected against corrosion.

2.3 Life of cable shall be at least 25 years. Necessary statistical calculations shall be submitted by the manufacturer, based upon life of the fibre and other component parts of the cable. The cable shall meet the cable aging test requirement.

2.4 It shall be possible to operate and handle the Ribbon Optical fibre cable with tools as per GR No. GR/OFT-01/03 APR 2006 and subsequent amendment, if any. If any special tool is required for operating and handling the optical fibre cable the same shall be provided along with the cable.

2.5 The Optical fibre cable supplied shall be suitable and compatible to match with the dimensions, fixing, terminating & splicing arrangement of the splice closure. The cable supplied shall also meet other requirement of splice closure (GR No. TEC/GR/TX/OJC-002/03/APR-2010) and subsequent amendments, if any.

2.6 The manufacturer shall submit an undertaking that the optical and mechanical fibre characteristics shall not change during the lifetime of the cable against the manufacturing defects.

2.7 It is mandatory that the Optical fibre cable supplied in a particular route is manufactured from a single source of optical fibres.

2.8 The Optical fibre cable shall be manufactured so as to protect the cable from rodent and termite.

3. Technical Requirements of Optical Fibres:

Single Mode Optical Fibre used in manufacturing optical fibre cables shall be as per ITU-T Rec.G.652.D. The specifications of optical fibres are mentioned below:

3.1	Type of fibre: (Wavelength band optimized nominal 1310 nm)	Single mode (Section -I of the GR No. TEC/GR/TX/ORM-01/04/SEP-09 and subsequent amendments, if any)
3.2	Geometrical Characteristics:	
3.2.1	MFD	8.8-9.8 µm
3.2.2	Cladding Diameter	125 µm ± 1.0 µm
3.2.3	Cladding Non-circularity	< 1%
3.2.4	Core Clad concentricity error	≤ 0.6 µm
3.2.5	Diameter over primary coated with double UV cured acrylate. (Shall be measured on uncoloured fibre)	245 µm ± 10 µm

Note: The thickness of colour coating may be over and above the values specified above, if the manufacturer adopts separate UV cured colouring process (to colour the un coloured fibres) other than the on line integrated colouring process (of secondary layer of primary coating) of the fibres, during fibre manufacturing.

3.2.6	Coating / Cladding Concentricity	≤12µm
3.3	**Transmission Characteristics:**	
3.3.1	**Attenuation:**	
a)	Fibre attenuation before Cabling	
i)	At 1310 nm	≤ 0.34 dB/Km
ii)	Between 1285 to 1380 nm	≤ 0.37 dB/Km
iii)	Between 1390 to 1525 nm	≤ Value at 1310nm
iv)	At 1550 nm	≤ 0.21 dB/Km
V)	Between 1525 to 1625 nm	≤ 0.24 dB/Km
b)	Water Peak Attenuation before cabling Between 1380-1390nm	≤ Value at 1310nm

Note:

1. Attenuation in the band 1380-1390nm shall be checked at every 2nm after Hydrogen ageing as per IEC 60793-2-50. Hydrogen ageing test is to be carried out by CACT, Bangalore or any other recognized laboratory for type test.

2. Sudden irregularity in attenuation shall be less than 0.1 dB
3. The spectral attenuation shall be measured on un-cabled fibre.
4. The Spectral attenuation in the 1250 nm–1625 nm band shall be measured at an interval of **10nm** and the test results shall be submitted.

c)	Fibre attenuation after cabling

i)	At 1310 nm	≤ 0.36 dB/km
ii)	At 1550 nm	≤ 0.23 dB/Km
iii)	At 1625 nm	≤ 0.26 dB/Km

3.3.2	Dispersion:	
a)	Total Dispersion	
i)	In 1285-1330 nm band	≤ 3.5 ps/nm.km
ii)	In 1270-1340 nm band	≤ 5.3 ps/nm. Km
iii)	At 1550 nm	≤ 18.0 ps/nm. Km
iv)	At 1625 nm	≤ 22.0 ps/nm. Km

Note: The dispersion in the 1250 nm–1625 nm band shall be measured on un-cabled fibre at an interval of10nm and the test results shall be submitted.

b)	Polarization mode dispersion at 1310 & 1550 nm	
i)	Fibre	≤0.2 ps/√km
ii)	Cabled Fibre	≤0.3 ps/√'km

Note: Measurement on un-cabled fibre may be used to generate cabled fibre statistics and correlation established.

c)	Zero Dispersion Slope	≤ 0.092 ps/(nm^2 Km)
ii)	Zero dispersion wave length range	1300 -1324 nm

3.3.3	Cut off wavelength for fibres used in cables	1320 nm Max.

Note: The above cut off wavelength is w.r.t. 2M sample length of fibre.

3.3.4	Cable Cut off wavelength	1260nm Max.
3.4	**Mechanical Characteristics**:	
3.4.1	Proof test for minimum strain level (Test method IEC–60793-1– 30)	1%
3.4.2	Peak Stripability force to remove primary N coating of the fibre. (Test method IEC–60793-1-32)	1.3 ≤ F ≤ 8.9

Note: The force required to remove 30 mm ± 3 mm of the fibre coating shall not exceed 8.9 N and shall not be less than 1.3 N.

3.4.3	Dynamic Tensile Strength (Test method IEC–60793-1-31)	
a)	Un-aged	≥ 550 KPSI (3.80 GPa)

b)	Aged	≥ 440 KPSI (3.00 GPa)
3.4.4	Dynamic Fatigue (Test method IEC- 60793 - 1-33)	≥ 20
3.4.5	Static Fatigue (Test method IEC- 60793 - 1-33)	≥ 20
3.4.6	Fibre Macro bend (Test method FOTP– 62/ IEC- 60793-1-47)	
a)	Change in attenuation when fibre is coiled with	≤ 0.05 dB at 1550nm
	100 turns on 30 ± 1.0 mm radius mandrel	≤ 0.5 dB at 1625nm
b)	Change in attenuation when fibre is coiled with	≤ 0.5 dB at 1550nm
	1 turn around 32 ± 0.5 mm diameter mandrel	≤ 1.0 dB at 1625nm
3.4.7	Fibre Curl (Test method as per IEC 60793-1-34)	≥ 4 meters radius of curvature

3.5 Material Properties:

3.5.1 **Fibre Materials:**

a)	The substances of which the fibres are made	To be indicated by the manufacturer
b)	Protective material requirement:	
i)	The physical and chemical properties of the material used for the fibre primary coating	It shall meet the requirement of fibre
ii)	Coating and for single jacket fibre	Stripping force as per clause No. 3.4.2
	The best way of removing protective coating material.	To be indicated by the manufacturer
c)	Group refractive Index of fibre	To be indicated by the manufacturer

.6	Environmental Characteristic of Fibre (Type test):	
3.6.1	Operating Temperature (Test Method IEC – 60793 – 1-52)	
	Temperature Dependence of Attenuation	- 60°C to +85°C
	Induced Attenuation at 1550 nm at -60°C to +85° C	≤ 0.05 dB/km
3.6.2	Temperature – Humidity Cycling (Test method /EIA/TIA-455-73)	
	Induced Attenuation at 1550 nm at -10°C to +85°C	≤ 0.05 dB/km and 95% relative humidity
3.6.3	Water Immersion 23°C (Test method IEC- 60793 – 1 -53)	
	Induced Attenuation at 1550 nm due to Water Immersion at 23+- 2°C	≤ 0.05 dB/km
3.6.4	Accelerated Aging (Temperature) 85°C (Test method IEC- 60793 – 1- 51)	

	Induced Attenuation at 1550 nm due to Temperature Aging at 85 +- 2°C	≤ 0.05 dB/km
3.6.5	Retention of Coating Color *(Test method IEC- 60793 – 1 - 51)*	
	Coated Fibre shall show no discernible change in color, when aged for relative humidity	30 days at 85°C with 95° Humidity and then 20 days in 85°C dry heat

3.7 Colour Qualification and Primary coating Test:

3.7.1 Colour Qualification Test:

a) MEK Rub Test (Methyle Ethyl Ketone Test):

To be tested by using soaked (Solvent) tissue paper for ten strokes unidirectional on 10 cm length of fibre. No colour traces shall be observed on the tissue paper after testing.

b) Water immersion Test (Type Test):

To be tested for coloured fibre for 30 days. After the test Colour qualification, Attenuation measurement & Strippability test are to be taken.

3.7.2 Primary coating Test:

a) Fourier Transform Infrared Spectroscopy (FTIR) Test:

To be tested to check the curing level of coating on the surface of natural fibre. The curing level shall be better than 90%.

b) Adhesion Test:

To be tested by using soaked (Solvent) tissue paper for ten strokes unidirectional on 10 cm length of fibre. No coating shall be observed on the tissue paper after testing.

3.8 Ribbon Structure:

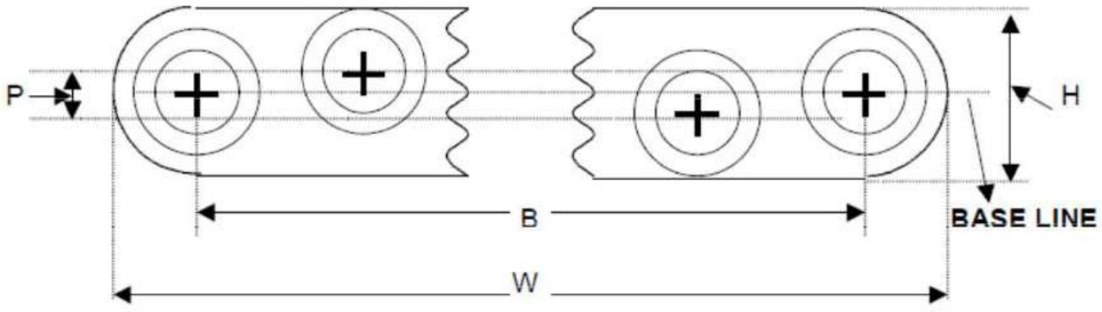
3.8.1 A group of primary coated fibres shall be arranged in ribbon structure. The fibres in the structure shall be parallel and shall not cross over each other along the entire length of the ribbon. The dimensions of 6 fibres ribbon shall be as per the Bell Core document no. GR-20-Core issue 4, 2013 with latest version and as given below:

3.8.2 **Ribbon Dimensions:**

The maximum dimensions of fibre ribbon shall be as follows and the cross section geometry of the fibre ribbon shall be as shown in the following figure

Table - Maximum Ribbon Dimensions

Number of Fibers per Ribbon	Ribbon Width w (µm)	Ribbon Height h (µm)	Fiber Alignment	
			Extreme Fibers b (µm)	Planarity p (µm)
6	1648	360	1310	50



Cross section of Fibre Ribbon

3.8.3 **Ribbon Material**:

The ribbon shall be manufactured using single mode optical fibres coloured with UV cured resin and the ribbon shall be encapsulated with a further layer of UV cured acrylate. The fibres and the ribbons shall confirm to the colour requirement as per clause no. 4.4 of this GR.

3.8.4 **Ribbon mechanical properties**:-

3.8.4.1	**Ribbon Macro-bend**	
	Change in attenuation when wrapped on a 60 mm diameter mandrel for100 turns at 1310 & 1550 nm	≤ 0.05 dB
3.8.4.2	**Ribbon Compression Resistance**	
	Change in attenuation when subjected to a compressive load of 500 N at 1310nm & at 1550nm.	≤ 0.05 dB
3.8.4.3	**Ribbon Torsion Resistance**	
	Change in attenuation (At 1310 nm & 1550 nm)	≤ 0.05 dB

3.9 Ribbon Optical Fibre Cable Construction Specifications for Dry core which will be procured through this tender:

General: The Metal Free optical fibre cable shall be designed to the parameters mentioned in Section-C. The manufacturer shall submit designed calculations and the same shall be studied and checked.

TYPICAL STRUCTURAL DRAWING FOR 48 FIBRE OF DRY CORE CABLE

3.9.1 Secondary Protection: The primary coated Ribbon fibres may be protected by loose packaging within tube, which shall be filled with thixotropic jelly. The dimensions of tube shall be as per Section-C.

3.9.2	Number of fibres in cable:	48
a.	Number of fibres per ribbon	Six (6) Fibres

a. The number of ribbons per loose tube in ribbon optical fibre cable shall be as follows:

Sl. No.	No. of Fibres	Multi Loose type
a.	48 Fibres	4 tubes and 1 Filler 2 ribbon per tube

(Type approval for a cable shall be issued depending on the no. of fibres in the cable)

3.9.3 Strength Member:

3.9.3.1 Solid FRP non - metallic strength member shall be used in the center of the cable core. The strength member in the cable shall be for strength and flexibility of the cable and shall have anti buckling properties. The FRP shall keep the fibre strain within permissible values. The size of FRP shall be as per Section-C.

3.9.3.2 Impregnated Glass Fibre Reinforcement are used to achieve the required tensile strength of the optic fibre cables over the cable inner sheath to provide peripheral reinforcement along with Solid Rigid FRP Rod in the centre at cable core. These flexible strength members shall be **Non-water blocking type.** The use of Solid Rigid FRP Rod(s) is mandatory in Optical Fibre cable design. Impregnated Glass Fibre Reinforcement used shall be equally distributed over the periphery of the cable inner sheath. It shall be applied **helically** and shall provide full coverage to inner sheath to provide rodent protection. The quantity of the Impregnated Glass fibre Reinforcement used per km length of the cable along with its dimensions shall be as per Section-C. The specification of the glass roving shall be as per Section XII of GR No. TEC/GR/TX/ORM-01/04 SEP.09 and as per other details given in the Section-C.

3.9.4 Cable Core Assembly: Primary coated fibres in loose tubes stranded together around a central strength member using helical or reverse lay techniques shall form the cable core. The dimensions of FRP and stranding pitch shall be as per Section-C.

3.9.5 Core Wrapping: The main cable core containing Ribbon fibres shall be wrapped by a layer / layers of Water Swellable tape. The nylon/polyester binder tape/thread/PP tape shall be used to hold the tape, if required. The core wrapping shall not adhere to the secondary fibre coating and shall not leave any kink marks over the loose tube.

3.9.6 Moisture barrier (protection):. The main cable core (containing loose tubes stranded around central strength member) shall be protected by water swellable yarns over FRP (central strength member) and by a layer of water swellable tape.

3.9.7 Filling compound: The filling compound used in the loose tube shall be compatible to fibre, secondary protection of fibre, core wrapping and other component part of the cable. The drip point shall not be lower than +70°C. The fibre movement shall not be constrained by stickiness & shall be removable easily for splicing. Reference test method to measure drop point shall be as per ASTM D 566**.** The thixotropic filling compound (jelly) shall be as per the GR No. TEC/GR/TX/ORM-01/04/SEP-09 and subsequent amendment issued, if any.

3.9.8 Inner Sheath: A non-metallic moisture barrier sheath may be applied over and above the cable core. The core shall be covered with tough weather resistant High Density Polyethylene (HDPE) sheath, black in colour (UV Stabilized) and colour shall confirm to Munsell colour standards. Thickness of the sheath shall be uniform & shall not be less than **0.9 mm.** The sheath shall be circular, smooth, free from pin holes, joints, mended pieces and other defects. Reference test method to measure thickness shall be as per IEC 189 para 2.2.1 and para 2.2.2.

Note: HDPE material, black in colour, from the finished cable shall be subjected to following tests (on sample basis) and shall confirm to the requirement of the material as per GR No. TEC/GR/TX/ORM-01/04 SEP.09.

i) Density
ii) Melt flow index
iii) Oxidative Induction time
iv) Carbon black content
v) Carbon black dispersion
vi) ESCR
vii) Moisture content
viii) Tensile strength and elongation at break

3.9.9 Glass Reinforcement: Impregnated Glass Fibre Reinforcement are used to achieve the required tensile strength of the optic fibre cables over the cable inner sheath to provide peripheral reinforcement along with Solid Rigid FRP Rod in the centre at cable core. These flexible strength members shall be **Non-water blocking type.** The use of Solid Rigid FRP Rod(s) is mandatory in Optical Fibre cable design. Impregnated Colour Coated Glass Fibre Reinforcement used shall be equally distributed over the periphery of the cable inner sheath. It shall be applied **helically** and shall provide full coverage to inner sheath to provide rodent protection. The quantity of the Impregnated Glass fibre Reinforcement used per km length of the cable along with its dimensions shall be as per Section-C. The specification of the glass roving shall be as per Section XII of GR No. TEC/GR/TX/ORM-01/04 SEP.09 and as per other details given in the Section-C.

3.9.10 Outer Sheath: A non-metallic moisture barrier sheath (black in colour) shall be applied over the inner sheath, which shall consist of tough weather resistant made High Density Polyethylene compound (HDPE) which is Anti-termite. The outer sheath shall be UV stabilized and the colour shall confirm to Munsell colour standards. The thickness of the outer sheath shall not be less than **1.5 mm** .The outer sheath shall be uniform, circular, smooth, free from pin holes, joints mended pieces and other defects The reference test method to measure thickness shall be as per IEC 811-5-1.

Note: HDPE material from finished product shall be subjected to following tests (on sample basis) and shall confirm to the requirement of the material as per the GR no. TEC/GR/TX/ORM-01/04/SEP-09 (Section-D):

i) Density
ii) Melt flow index
iii) Oxidative Induction time
iv) Carbon black content
v) Carbon black dispersion
vi) ESCR
vii) Moisture content
viii) Tensile strength and elongation at break

Note: The outer jacket of HDPE shall be able to protect the cable from attack by termites. Manufacturer shall provide the details of doping material used and same shall be verified during bulk testing. The outer sheath shall be termite protected. The surface of the sheath shall be smooth and free of defects such as cracks, blisters, etc. The cable shall be rodent protected. As specified in various clauses of the Technical specifications of the OF cable, it is to be clarified that the HDPE Outer Jacket shall be anti-termite with/without dopants. Addition of dopants for anti-rodent property is optional. The tests as per clause 4.28 shall be carried out as applicable.

3.9.11 Cable diameter: The finished cable diameter shall be as per Section-C

3.9.12 Cable Weight: The nominal cable weight shall be as per Section-C.

3.9.13 RIP Cord:

a) Four suitable ripcords shall be provided in the cable, which shall be used to open the both HDPE sheath of the cable. Two ripcords shall be placed diametrically opposite to the each other at below the outer Jacket & two ripcords shall be placed at below inner sheath. It shall be capable of consistently slitting the sheath without breaking for a length of 1meter at the installation temperature. The ripcords (3ply & twisted for outer sheath and suitable ripcord for inner sheath) shall be properly waxed to avoid wicking action and shall not work as water carrier.
b) The ripcords used in the cable shall be readily distinguishable from any other components utilized in the cable construction.

4. Mechanical Characteristics and Tests on Optical Fibre Cable:

(Note: All observations are to be taken at 1550nm wavelengths. Change in attenuation value at 1550nm to be taken after conducting all mechanical tests.)

4.1 Tensile strength Test:

Objective: This measuring method applies to optical fibre cables which are tested at a particular tensile strength in order to examine the behaviour of the attenuation as a function of the load on a cable which may occur during installation.

Method: IEC 60794-1-2-E1.

Test Specs : The cable shall have sufficient strength to withstand a load of value T(N) = 3000 Newton . The load shall be sustained for 10 minutes and the strain of the fibre monitored.

Requirement: The load shall not produce a strain exceeding 0.25% in the fibre and shall not cause any permanent physical and optical damage to any component of the cable. The attenuation shall be noted before strain and after the release of strain. The change in attenuation of each fibre after the test shall be ≤ 0.05 dB both for 1310 nm and 1550 nm wavelength.

4.2 Abrasion Test:

Objective: To test the abrasion resistance of the sheath and the marking printed on the surface of the cable.

Method: IEC-60794-1-2-E2 or by any other international test method

Test Specs: The cable surface shall be abraded with needle (wt. 150 gm) having diameter of 1 mm with 500 grams weight (Total weight more than equal 650 gms.)

No. of cycles	100
Duration	One minute (Nominal)

Requirement: There shall be no perforation & loss of eligibility of the marking on the sheath.

4.3 Crush Test (Compressive Test):

Objective: The purpose of this test is to determine the ability of an optical fibre cable to withstand crushing.

Method: IEC 60794-1-2-E3.

Test Specs.: The fibres and component parts of the cable shall not suffer permanent damage when subjected to a compressive load of 2000 Newton applied between the plates of dimension 100 x 100 mm. The load shall be applied for 60 Secs. The attenuation shall be noted before and after the completion of the test.

Requirement: The change in attenuation of the fibre after the test shall be ≤ 0.05 dB both for 1310 nm and 1550 nm wavelength.

4.4 Impact Test:

Objective: The purpose of this test is to determine the ability of an optical fibre cable to withstand impact.

Method: IEC 60794-1-2-E4.

Test Specs: The cable shall have sufficient strength to withstand an impact caused by a mass weight of 50 Newton, when falls freely from a height of 0.5 meters. The radius R of the surface causing impact shall be 300 mm. 10 such impacts shall be applied at the same place. The attenuation shall be noted before and after the completion of the test.

Requirement: The change in attenuation of the fibre after the test shall be ≤0.05dB both for 1310 nm and 1550 nm wavelength.

4.5 Repeated Bending:

Objective: The purpose of this test is to determine the ability of an optical fibre cable to withstand repeated bending.

Method: EIA-455-104.

Test Specs: The cable sample shall be of sufficient length (5 m minimum) to permit radiant power measurements as required by this test. Longer lengths may be used if required.

Parameters:	
Weight	5 Kg
Minimum distance from Pulley centre to holding device	216mm
Minimum distance from Wt. to Pulley centre	457mm
Pulley Diameter	20 D (D - cable diameter)
Angle of Turning	90o
No. of cycles	30
Time Required for 30 cycles	2 min

Requirement: During the test no fibre shall break and the attenuation shall be noted before and after the completion of the test. The change in attenuation of the fibre after the test shall be ≤ 0.05 dB both for 1310 nm and 1550 nm wavelength.

4.6 Torsion Test:

Objective: The purpose of this test is to determine the ability of an optical fibre cable to withstand torsion.

Method: IEC 60794-1-2-E7.

Test Specs.: The length of the specimen under rest shall be 2 meters and the load shall be 100 N. The sample shall be mounted in the test apparatus with cable clamped in the fixed clamp sufficiently tight to prevent the movement of cable sheath during the test. One end of the cable shall be fixed to the rotating clamp which shall be rotated in a clock wise direction for one turn. The sample shall then be returned to the starting position and then rotated in an anti-clock wise direction for one turn and returned to the starting position. This complete movement constitutes one cycle. The cable shall withstand 10 such complete cycles. The attenuation shall be noted before and after the completion of the test.

Requirement: The cable shall be examined physically for any cracks tearing on the outer sheath and for the damage to other component ports of the cable. The twist mark shall not be taken as damage. The change in attenuation of the fibre after the test shall be ≤ 0.05 dB both for 1310 nm and 1550 nm wavelength.

4.7 Kink Test:

Objective: The purpose of this test is to verify whether kinking of an optical fibre cable results in breakage of any fibre, when a loop is formed of dimension small enough to induce a kink on the sheath.

Method: IEC 60794-1-2-E10.

Test Specs.: The sample length shall be 10 times the minimum bending radius of the cable The sample is held in both hands, a loop is made of a bigger diameter and by stretching both the ends of the cable in opposite direction, the loop is made to the minimum bend radius so that no kink shall form. After the cable comes in normal condition, the attenuation reading is taken.

Requirement: The kink should disappear after the cable comes in normal condition. The change in attenuation of the fibre after test shall be ≤0.05 dB both for 1310 nm & 1550 nm wavelength.

4.8 Cable Bend Test:

Objective: The purpose of this test is to determine the ability of an optical fibre cable to withstand repeated flexing. The procedure is designed to measure optical transmittance changes and requires an assessment of any damage occurring to other cable components.

Method: IEC 60794-1-2-E11 (Procedure-I).

Test Specs.: The fibre and the component parts of the cable shall not suffer permanent damage when the cable is repeatedly wrapped and unwrapped 4 complete turns of 10 complete cycles around a mandrel of 20 D, where D is the diameter of the cable. The attenuation shall be noted before and after the completion of the test.

Requirement: The change in attenuation of the fibre after the test shall be <=0.05 dB both for 1310 nm and 1550 nm wavelength. Sheath shall not show any cracks visible to the naked eye when examined whilst still wrapped on the mandrel.

4.9 Temperature Cycling (Type Test):

Objective: To determine the stability behaviour of the attenuation of a cable subjected to temperature changes which may occur during storage, transportation and usage.

Method: IEC 60794-1-2-F1 (To be tested on Standard cable length & drum i.e. 2Km. ± 10%)

Test Specs.: The permissible temperature range for storage and operation will be from -20°C to +70°C. The rate of change of temperature during the test shall be 1°C per minute approx. The cable shall be subjected to temperature cycling for 12 Hrs. at each temperature as given below:

TA2 temp.	-20°C
TAt temp.	- 10°C.
TB1 temp.	+ 60°C.
TB2 temp.	+ 70°C.

The test shall be conducted for 2 cycles at the above temperatures.

Requirement: The change in attenuation of the fibre under test shall be ≤0.05 dB for 1310 nm and 1550 nm wave length respectively for the entire range of temperature.

4.10 Cable aging Test (Type Test):

Objective: To check the cable material change dimensionally as the cable ages.

Method: At the completion of temperature cycle test, the test cable shall be exposed to 85 ± 2 degree C for 168 hours. The attenuation measurement at 1310 & 1550 nm wave length to be made after stabilization of the test cable at ambient temperature for 24 hours.

Requirement: The increase in attenuation allowed: ≤ 0.05 dB at 1310 and 1550 nm.

Note: The attenuation changes are to be calculated with respect to the base line attenuation values measured at room temperature before temperature cycling.

4.11 Water Penetration Test (Type Test):

Objective: The aim of this test is to determine the ability of a cable to block water migration along a specified length.

Method: IEC 60794-1-2-F5

Test Specs. A circumferential portion of the inner HDPE cable end shall face the water head. The water tight sleeve shall be applied over the cable. The cable shall be supported horizontally and one meter head of water, containing a sufficient quantity of water soluble fluorescent dye for the detection of seepage, shall be applied on the inner HDPE sheath for a period of 7 days at ambient temperature. No other coloured dye is permitted.

Note: For bulk testing, 24 hours as duration of test shall be considered.

Requirement: No dye shall be detected when the end of the 3m length of the cable is examined with ultraviolet light detector. The cable sample under test shall be ripped open after the test and then it shall be examined for seepage of water into the cable and the distance to be noted.

4.12 Test of Figure of 8 (Eight) on the cable (Type Test):

Objective: Check of easiness in formation of figure of 8 of the cable during installation in the field.

Test Method: 1000 meter of the cable shall be uncoiled from the cable reel and shall be arranged in figure of 8 (eight) shape. The diameter of each loop of the figure of 8 shall be maximum 2 meters.

Requirement: It shall be possible to make figure of 8 of minimum 1000 meters of the cable uncoiled from the cable reel without any difficulty. No visible damage shall occur.

4.13 Flexural Rigidity Test on the optical fibre cable (Type Test):

Objective: To check the Flexural Rigidity of the metal free optical fibre cable.

Method: To be tested as per ASTM D –790

Test Specs: The fibre and the component parts of the cable shall not suffer permanent damage in the cable subjected to Flexural Rigidity Test as per the above method. The attenuation shall be noted after and before the completion of the test.

Requirement: The change in attenuation of the fibre after the test shall be < 0.05 dB at 1310nm and 1550nm wavelengths. The sheath shall not show any cracks visible to the naked eye.

4.14 Static Bend test (Type Test):

Objective: To check the Cable under Static Bend.

Method: As per the clause no 4.8 of the GR alternatively as per ASTM D790.

Test Specs: The cable shall be subjected to static bend test. The optical fibre cable shall be bend on a mandrel having a Diameter of 10 D (D is diameter of the cable).

Requirement: The change in attenuation of the fibre after the test shall be ≤0.05 dB for 1310 nm and 1550 nm wavelengths. Sheath shall not show any cracks visible to the naked eye when examined whilst still wrapped on the mandrel.

4.15 Cable Jacket Yield Strength and Ultimate Elongation:

Objective: To determine the yield strength and elongation of the polyethylene (HDPE) cable sheath (jacket).

Test Method: FOTP-89 or ASTM 1248 Type III class

Test Condition: 1. Sample shall be taken from a completed cable. The aged sample shall be conditioned at 100 ± 2°C for 120 hours before testing.

2. The cross-head speed shall be 50 mm per minute.

Requirement:

Jacket Material	Minimum Yield Strength		Minimum Elongation (%)
	(MPa)	(psi)	
HDPE un-aged	16.5	2400	400

HDPE aged	12.4	1800	375

4.16 To Check of the quality of the loose tube (containing ribbon optical fibre) (Type Test):

Embrittlement Test

This test method is based on bending by compression and reflects embrittlement much better than the other tensile tests. This test is independent of wall thickness of the loose tube.

Sample: The minimum length of the test sample depends on the outside diameter of the loose tube and should be 85 mm for tubes up to 2.5 mm outside dia. The length of the bigger tubes should be calculated by using the following equation:

$$Lo > 100 \times \left[\frac{(D^2 + d^2)}{4}\right]^{1/2} \text{ where}$$

Lo = Length of tube under test.

D = Outside dia of loose tube.

d = Inside dia of loose tube.

Procedure: Both the ends of a buffer tube test sample may be mounted in a tool, which is clamped in jaws of a tensile machine which exerts a constant rate of movement. The movable jaw may move at a rate of 50 mm per minute toward the fixed jaw. Under load, the tube will bend so that it is subjected to tensile and compressive stresses. The fixture for holding the tube should be designed in a manner that the tube might bend in all directions without further loading.

Requirement: The tube should not get embrittled. No kink should appear on the tube up to the safe bend diameter of tube (15 D), where D is the outside diameter of the loose tube. There should also not be any physical damage or mark on the tube surface.

b. Kink Resistance Test on the Loose Tube

Objective: To safeguard the delicate optical fibres, the quality of the loose tube material should be such that no kink or damage to the tube occur while it is being handled during installation and in splicing operations.

Procedure: To check the kink resistance of the loose tube, a longer length of the loose tube is taken (with fibre and gel), a loop is made and loop is reduced to the minimum bend radius of loose tube i.e. 15 D (where D is the outside diameter of the loose tube). This test is to be repeated 4 times on the same sample length of the loose tube.

Requirement: No damage or kink should appear on the surface of the tube.

4.17 Ribbon Dimension Measurements test (Type Test):

Objective: To check the fibres in ribbon structure, fibres cross over and fibre identity to ensure the transmission performance and mechanical service life of the fibre in the ribbon structure.

Test method: FOTP-123 (Video Gray Scale Analysis (VGSA) or Microscopic method).

Requirement: It shall meet the dimensional requirements given in clause no.3.8.2 of this GR. The fibres in the entire length of the ribbon shall not cross over at any point.

4.18 Ribbon Resistance to Twist (Robustness) test (Type Test):

Objective: To check the robustness of the fibre ribbons to withstand the twist during installation conditions and to check the structural integrity of the ribbon over the deployed length for mid-span entry, maintenance purposes, consideration in rearrangements and housekeeping.

Test method: FOTP-141.

Requirement: The un-aged and aged (at 85 ± 2°C with uncontrolled humidity for a period of 30 days) completed ribbon shall not show any separation of individual fibres from the ribbon structure after completion of the twist test when observed under 5X magnification.

4.19 Ribbon Residual Twist (Flatness) Test (Type Test):

Objective: To check the dimensional integrity of the ribbon without twisting to allow rearrangements and to limit the potential attenuation increases due to a macro-bending caused by twisting of the fibre ribbon.

Test Method: FOTP-131.

Requirement: The aged (at 85 ± 2º C with uncontrolled humidity for a period of 30 days) ribbon residual twist (if any) shall have a pitch : > 400 mm.

4.20 Ribbon Separation Test (Type Test):

Objective:

a. To check the separation of individual fibres, separation of sub-unit of fibres and mid span separation from a fibre ribbon.
b. To check the retention of sufficient colorant for identification for any 2.5 cm length of fibre after separation for individual and sub-unit of fibres.

Test to be conducted for:

a) Separation of any single fibre or a multi-fibre subgroup by a tool or by hand from a ribbon for a length of 1 meter. Midspan separation from a 2 meter sample, separated close to middle for at least 0.5 meter (both single fibre and the three fibre sub – units) for un-aged ribbon.

Requirement: The un-aged ribbon of minimum length of a 0.3 meter (1.0foot) of an individual fibre and a sub group of three fibres shall be separated from the ribbon without breaking the fibres or damaging the fibre coating. The force required to perform separation shall not exceed 4.4 N. The area at the separation shall not show any damage to the fibre coating when examined under 5X magnification.

b) Retention of the Colour and Fibre Identification after separation.

Requirement: Individual fibre colour identification shall be maintained after the separation test. It shall retain sufficient colorant that any 2.5 cm length is readily identifiable.

c) Removal of Ribbon matrix material to access individual fibres.

Requirement: No damage shall occur either to fibre coating or the fibres. The coating shall not sustain any swelling self-stripping, cracking or splitting when examined under 5X magnification.

Note: The manufacturer shall recommend the procedure for the removal of ribbon matrix.

4.21 Ribbon Stripability Test (Type Test):

Objective: Check of removal of the matrix material and the fibres protective coating mechanically with commercial stripping tools from un-aged and aged ribbons.

Test Method: GR-20-CORE issue 4, 2013

Pre Conditioning:

a. **Aged samples:** The humidity of aged ribbons shall be soaked at 85 ± 2 °C and a non-condensing humidity of 85 ± 5% for a period of 30days.

b. **Water aged samples:** The water aged ribbons shall be soaked in deionizador distilled water at a temperature of 23 ± 5 °C for a period of 14 days. The fibre ribbon strip-ability testing shall be conducted at standard atmospheric conditions. The un-aged, humidity – aged, and water aged ribbons shall be tested within eight hours after aging.

Requirement: There shall be no fibre breakage and any coating residue shall be removable with a single isopropyl alcohol wipe when at least 25 mm of the matrix material and the fibre Protective coating is mechanically removed with commercial stripling tools from un-aged and aged ribbons.

4.22 Ribbon Macro-bend Performance

Objective: To check the macro-bend performance of a ribbon.

Method: One hundred turns of ribbon are wound around a 60 mm diameter ribbon and the loss increase at 1310 nm & 1550 nm shall be measured.

Requirement: The change in attenuation of the fibre shall be ≤ 0.05 dB, for 1310 nm and 1550 nm wavelengths.

4.23 Torsion Resistance of the ribbon Test (Type Test):

Objective: To check the torsion resistance of the ribbon.

Method: One meter length of ribbon is twisted to through five revolutions of 360° and measurement is taken.

Requirement: The change in attenuation of the fibre shall be ≤ 0.05 dB, for1310 nm and 1550 nm wavelengths.

4.24 Crush Resistance of Ribbon (Type Test):

Objective: To check the crush resistance of the ribbon.

Method: A 50 mm^2 sample is subjected to a load of 500 N and the attenuation measurement taken for both 1310 nm & 1550 nm wavelengths.

Requirement: The change in attenuation of the fibre shall be ≤ 0.05 dB, for 1310 nm and 1550 nm wavelengths.

4.25 Drainage Test for Loose Tube and Drip test on the cable (Type Test):

a. Drainage Test for loose Tube

Sample Size: 30 cm tube length.

Test procedure:

a) Cut the tube length to 40 cm.
b) Fill the tube with the tube filling gel ensuring that there are no air bubbles and the tube is completely full.
c) Place the filled tube in a horizontal position on a clean worktop and cut 5 cm from either end so that the finished length of the sample is 30 cm.
d) Leave the filled tube in a horizontal position at an ambient temperature for 24 hrs. (This is necessary because the gel has been sheared and the viscosity has been reduced during the filling process).

e) The sample tube is then suspended vertically in an environment heat oven over a weighed beaker. It is left in the oven at a temperature of 70° C for a period of 24 Hrs.

f) At the end of the 24 hrs period the beaker is checked and weighed to see if there is any gel in the beaker.

Results:

1. If there is no gel or oil in the beaker the tube has PASSED the drainage test.
2. If there is gel or oil in the beaker the tube has FAILED the drainage test.

b. Drip test on the cable

Objective: The purpose of this test is to determine the ability of jelly in the O.F. cable to withstand a temperature of 70 degree C.

Method: Take a sample of 30 cm. length of the cable with one end sealed by end cap. Remove outer black sheath, binder tapes for 5 cm from open end of the sample. Clean the jelly. Then the sample is kept vertically with open end downwards in the oven for 24 hours at 70^{o} C with a paper under the sample.

Test Specs: Examine the paper placed below the cable inside the oven for dripping of the jelly after 24 hours. There should be no jelly drip or oily impression on the paper.

4.26 Check of easy removal of sheath:

Objective: Check of the easy removal of sheath of the fibre optic cable by using normal sheath removal tool.

Procedure: To check easy removal, the sheath shall be cut in circular way and the about 300 mm length of the sheath should be removed in one operation. It should be observed during sheath removal process that no undue extra force is applied and no component part of the cable is damaged. One should be able to remove the sheath easily.

4.27 Check of the effect of aggressive media on the cable (Acidic and Alkaline Behaviour) (Type Test):

Procedure: To check the effect of aggressive media, solution of PH4 and PH10 shall be made. The two test samples of the finished cable, each of 600 mm in length, are taken and the ends of the samples are sealed. These test samples are put in the PH4 and PH10 solutions separately. After 30 days these samples are taken out from the solutions and examined for any corrosion etc on the sheath and other markings of the cables. (Test method no. ISO175).

Requirement: The sample should not show any effect of these solutions on the sheath and other marking of the cable.

4.28 Termite & Rodent Test (Type Test):

The Anti-rodent (Optional) test shall be checking the uniformity of application of glass rowing yarns around the periphery of inner sheath. The test shall be carried out during type testing or can be carried out at any recognized lab on finished cable and in addition, manufacturer shall also give an undertaking in this respect.

The reports shall be submitted by the manufacturers. Termite resistance shall be provided with an additive/without additive in outer sheath and rodent protection shall be provided with Glass roving yarns around the periphery of inner sheath and these yarns should be spread uniformly around the periphery of inner sheath.

The following minimum parametric tests on Anti termite / Anti rodent dopants shall be carried out during the TSEC testing

1. Non-toxicity
2. Thermal Stability
3. Long life Span / half-life
4. Compatibility
5. Efficacy

The thermal stability of the dopant should not deteriorate during cable execution process. The life of the dopant should be equal or better than the life of the cable specified in the technical specification herein. Appropriate certificate in this regard from any neutral lab accredited with NABL / Government Laboratory / Institute should be produced.

Similarly other parameters such as non-toxicity, efficacy and compatibility shall be certified in any neutral lab accredited with NABL / Government Laboratory / Institute and test report is to be submitted during TSEC testing.

PART II – GENERAL REQUIREMENTS

5 Engineering Requirements:

5.1 Cable Marking:

5.1.1 A long lasting suitable marking shall be applied in order to identify this cable from other cables. The cable marking shall be imprinted (indented). The marking on the cable shall be indelible of durable quality and at regular intervals of one meter length. The accuracy of the sequential marking must be within -0.25% to +0.5% of the actual measured length. The sequential length markings must not rub off during normal installation and in life time of optical fibre cable. The total length of the cable supplied shall not be in negative tolerance.

5.1.2 The marking shall be in contrast colour over the black HDPE Sheath (jacket) and shall be one by hot foil indentation method. The colour used must withstand the environmental influences experienced in the field.

5.1.3 The type of legend marking on O.F. cable shall be as follows:

a) Company Legend
b) Legend containing telephone mark & international acceptable Laser symbol
c) Type of Fibre– G.652.D
d) Number of Fibres
e) Type of cable
f) Year of manufacture
g) Sequential length marking
h) User's Identification
i) Cable ID

5.2 Cable Ends:

5.2.1 Both cable ends (the beginning end and end of the cable reel) shall be sealed and readily accessible. Minimum 5 meter of the cable of the beginning end of the reel shall accessible for testing. Both ends of the cable shall be kept inside the drums and shall be located so as to be easily accessible for the test. The drum (confirming to GR No. G/CBD-O1/02 Nov. 94 and subsequent amendment) should be marked to identify the direction of rotation of the drum. Both ends of cable shall be provided with cable pulling (grip) stocking and the anti-twist device (free head hook). The wooden drums shall be properly treated against termites and other insects during transportation and storage. The manufacturer shall submit the methodology used for the same

5.2.2 An anti-twist device (Free head hook) shall be provided, attached to the both the ends of the cable pulling arrangement. The arrangement of the pulling eye and its coupling system along with the anti-twist system shall withstand the prescribed tensile load applicable to the cable.

5.3 The nominal drum length:

5.3.1 Length of OF Cable in each drum shall be 2 Km 10% / 4Km 10% / 8Km

±10% / 10Km ± 10% and shall be supplied as per the order. The variation in length of optical fibre cable, as specified above (in each drum), shall be acceptable.

5.3.2 The fibres in cable length shall not have any joint.

5.3.3 The drum shall be marked with arrows to indicate the direction of rotation.

5.3.4 Packing list supplied with each drum shall have at least the following information:

a) Drum No.
b) Type of cables
c) Physical Cable length
d) No. of fibres
e) Length of each fibre as measured by OTDR
f) The Cable factor - ratio of fibre/cable length
g) Attenuation per Km. of each fibre at 1310 & 1550 nm
h) Users / Consignee's Name
i) Manufacturers Name, Month, Year and Batch No.
j) Group refractive index of fibre.
j) Purchase Order No.
k) Cable ID

5.4 Colour coding in O.F. Cables:

5.4.1 The colorant applied to individual fibres shall be readily identifiable throughout the lifetime of the cable and shall match and conform to the Munsell Colour Standards (EIA- 598-C).

5.4.2 **Colour Coding Scheme:** When the loose tubes are placed in circular format, the marking to indicate the loose tube no. "1" shall be in blue colour followed by loose tube no.2 of orange and so on for other tubes as per the colour scheme given below and complete the circular format by placing the dummy /fillers at the end.

Depending upon the number of fibres in a Ribbon (which depends on the cable capacity), the colour of fibres are serially chosen from the column no. II of the following table-1.

Table -1 :Colour Coding scheme of the Optical Fibre & Loose tube

No.of Fibres/Buffer Tube I	Fibre identification in Ribbon II	Loose tube identification III
1	Blue	Blue
2	Orange	Orange
3	Green	Green
4	Brown	Brown
5	Slate	Slate
6	White	White

5.4.3 Identification of Ribbon:

No. of fibre in a Cable	No. of Tubes	No. of Ribbon per Tube	Fibre Per Ribbon	Marking on Ribbon
48 Fibres	4 Tubes 1 Filler	2	6	Individual Ribbon shall be printed with respective number as 1, 2.

Note: The individual number marking shall be at regular interval of every 300 mm or lesser on natural color ribbon and shall be legible. The printing on the ribbon shall also be of durable quality and shall be compatible with coating of the ribbon and Thixotropic Jelly (filled in the loose tube of the cable).

6 Quality Requirements:

6.1 The cable shall be manufactured in accordance with the international quality standards ISO 9001-2008 (latest issue) for which the manufacturer should be duly accredited. The Quality Manual shall be submitted by the manufacturer.

6.2 Raw Material:

6.2.1 The cable shall use the raw materials approved against the GR No. TEC/GR/TX/ORM-01/04 SEP.09 and the subsequent amendment issued, if any.

6.2.2 Any other material used shall be clearly indicated by the manufacturer. The detailed technical specifications of such raw materials used shall be furnished by the manufacturer at the time of evaluation/testing.

6.2.3 The raw materials used from multiple sources is permitted and the source / sources of raw materials (Type and grade) from where these have been procured shall be submitted by the manufacturer.

6.2.4 The manufacturer can change the raw material from one approved source to other approved source with the approval of QA wing of purchaser. In case of change of source/grade of SM Optical Fibre, the call for fresh evaluation/testing shall be decided by QA wing of purchaser.

6.2.5 The raw material used (HDPE black in colour) for outer sheath shall protect the cable from attack by termite & rodent. The manufacturer shall specify anti-termite and anti-rodent (optional) additives and submit the detail characteristics of the material and additives used to make it termite & rodent (optional) proof. The additives shall also be non-toxic. The cable shall be tested for the presence of Anti termites & Anti rodent (optional) additives by recognized laboratory or institute. The cable shall also be tested for its termite & rodent (optional) proven-ness by NABL accredited Lab / Govt laboratory or institute.

6.2.6 The HDPE black in colour used for sheath shall be UV stabilised.

Note: A test certificate from a recognised laboratory or institute may be acceptable for the UV stability of the HDPE sheath material

6.2.7 The material used in optical fibre cable must not evolve hydrogen that will affect the characteristics of optical fibres.

Note: A test certificate from a recognised laboratory or institute may be acceptable.

6.3 Cable Material Compatibility:

Optical fibre, buffers/core tubes, and other core components shall meet the requirements of the compatibility with buffer/core tube filling material(s) and/or water-blocking materials that are in direct contact with identified components within the cable structure (This shall be tested as per clause no. 6.3.3 of Telecordia document GR-20-CORE issue 4, 2013.

Note: The tests may be conducted in house (if facility exist) or may be conducted at CACT or any other recognized laboratory. The test certificate may be accepted and the tests may not be repeated subsequently, in next type approvals, if the raw material used is of same make and grade.

7 Documentation:

7.1 Complete technical literature in English with detailed cable construction diagram of various sub-components with dimensions, weight & test data and other details of the cable shall be provided.

7.2 All aspects of installation, operation, maintenance and fibre splicing shall also be covered in the handbook. The pictorial diagrams of the accessories (with model no. and manufacturer name) supplied along with the cable as package shall be also be submitted. A hard as well as soft copy of the manuals shall be provided.

8 Safety:

8.1 The material used in the manufacturing of the Optical fibre cables shall be non-toxic and dermatologically safe in its life time and shall not be hazardous to health. The manufacturer shall submit MSDS (Material safety Data Sheet) for all the material used in manufacturing of OF Cable to substantiate the statement.

SECTION-C

The following parameters of the component parts of the cable are to be taken in to account while designing and manufacturing the optical fibre cables of the required fibre count. These parameters shall be checked during evaluation of the OF cables.

SN	Parameter	Unit	48F Ribbon Fibre cable
1	FRP Rod EAA Coated	mm	2.5 +0.1/-0.0
2	FRP Up-coating thickness	mm	0
3	Tube ID (min)	mm	2.6
4	Tube OD	mm	3.4 ± 0.2
5	No of fibre / ribbon	No	6
6	No of Ribbon in a tube	No	2
7	Color of fibre per Ribbon		BL,OR,GR, BR,SL,WH
8	No of loose tubes	No	4 with 2 Ribbons
9	Colour of loose tubes		BL,OR,GR,BR
10	No of dummy cord	No	1
11	Tube stranding lay over length	mm	> 200
12	Inner Sheath Thickness (min)	mm	0.9
13	Qty. of Impregnated Glass roving (min.)	Kg	27
14	Outer Sheath Thickness(min)	mm	1.5
15	Cable diameter	mm	16.0 +1.0
16	Cable weight	Kg/km	185 ± 8%
17	Cable to be designed to Fibre strain value of	%	0.1
18	Cable to be tested at defined load for fibre strain value of	%	0.25

Note: The manufacturer shall submit the design calculations which shall be cross checked.

SECTION-D

TECHNICAL SPECIFICATIONS: TECHNICAL SPECIFICATIONS: GENERIC REQUIREMENT FOR DUCT

1. Specifications of Materials to be used

1.1 PLB HDPE Duct

Optical Fibre Cables should be pulled though Permanently Lubricated HDPE Duct of 40mm/33 mm size conforming to the specifications as per TEC GR No. TEC/GR/TX/CDS-008/03/MAR-11 with latest Amendments. The Ducts shall be blue in colour and have the identification markings as per TEC GR wherein CHiPS logo shall be marked as purchaser's name.

1.2 PLB HDPE Duct Accessories

a) Push fit Coupler

Push Fit couplers shall be used for coupling PLB HDPE ducts/coils. The specifications of the couplers shall be as per TECGR no TEC/GR/TX/CDS-008/03/Mar11 with latest amendments.

b) End Cap

End Cap shall be used for sealing the ends of the empty ducts, prior to installation of the OF Cable and shall be fitted immediately after laying the duct to prevent the entry of any dirt, water, moisture, insects/rodents etc. It should confirm to TEC GR No. TEC/GR/TX/CDS-008/03/MAR-11with latest amendments. The ends of the PLB HDPE ducts/coilslaid in the manholes should be closed with End Caps. The End Caps used should be suitable for closing40mm/33mm PLB HDPE ducts/coils. A suitable arrangement should be provided in the End Cap to tie PP Rope. (See figure-1 for details)

c) Cable sealing Plug

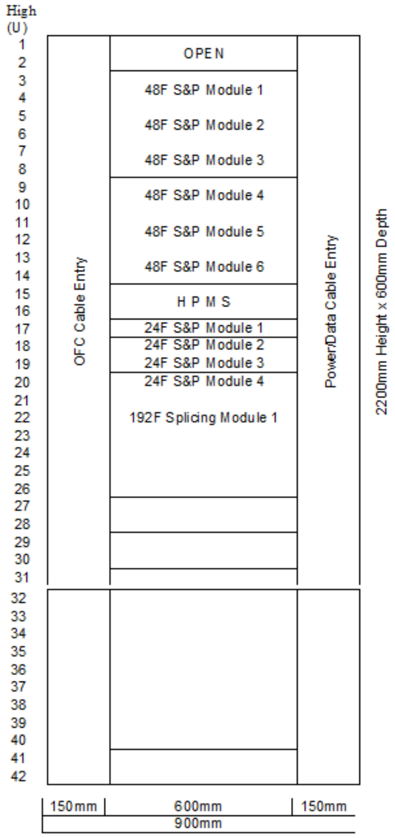
This shall be used to seal the end of the ducts perfectly, after the OF cable is pulled in the duct. For pulling the cable through the ducts, it is necessary to provide manholes at that location and also at bends and corners wherever required. The ends of the PLB HDPE ducts/coilsare closed with cable sealing Plugs. The End Plugs used should be suitable for closing40mm/33mm PLB HDPE ducts/coils. The Cable seal-ing plug shall confirm to TEC GR No. TEC/GR/TX/CDS-008/03/ MAR-11 with latest amendments.(Wherever blowing technique is used for laying OF Cable, at the discretion of the CPSUs concerned, the hand holes/manholes required for accessing the cable during cable laying can be at longer distances depending upon requirement.)

SECTION-E

(A) Fibre Distribution Management System (Exchange) Type – I

1. **TEC GR Reference - GR/FDM-01/02 APR 2007.** Design and additional specification of FDMS Type-I (at Exchange). Wherever Specifications under this section shall contradict with TEC GR, then specifications under this section shall prevail.
2. FDMS Manufacturers who have got TSEC/TAC for FDMS ((Section VI)) as per BSNL Tender no. CA/CNP/NFS OFC/T-441/2013 are not required to submit QF-103 against this Tender. However they have to obtain TSEC/TAC with Incremental Tests, as per Tender requirement, within the lead time period of 2 months from the date of issue of APO to bidder.

   If the bidder has obtained TSEC against CA/NOFN/OFC/T-609/2017 Dated: 15.12.2017 the same would be accepted for supplies to this tender as well.
3. Schematic Diagram and Tray Termination Capacity High



1. **Specifications**

5.1	FDMS Outer Dimensions	Height 2200mm, Width 900mm, Depth 600mm, +/- 5%
5.2	48F S & P Module (6 Nos.)	19 inch Rack Mountable, 2U high, with 4 nos. of 2x6FRibbon Splicing & Patching Trays, each preloaded with 2Nos. 6F SC-PC type Ribbon Fan-out & 12 Nos. of SCPC Adaptors.
5.3	24F S & P Module (4 Nos.)	19 inch Rack Mountable, 1U high, with 2 nos. of 12F Splicing & Patching Trays, each preloaded with 12 Nos.single fibre pigtails of 0.9mm SCPC connectrised and 1.5 meters in length & 12 Nos. of SCPC Adaptors. The pigtails shall be color coded for easy identification.
5.4	FDMS Exchange Type - I capacity	Total 384F (6 Nos. Modules of 48F and 4 Nos. Modules of 24F)
5.5	Splicing Module (1 No.)	19 inch Rack Mountable, 3U high(splice tray shall be 16 nos with each tray having a capacity of 2 x 6F Ribbon) = 192 F.
5.6	Constructed from	Powder-Coated CRCA Steel Sheet (min.1.6mm thick). Shelves can be constructed from CRCA 1.2mm thick
5.7	Mounting Method	Floor Mounting using Leveling Screws (4 Nos.)
5.8	Ventilation Method	'Perforated Rear and Side Walls for efficient heat management
5.9	Type of Ribbon	6F as per Bellcore document GR-20 Issue-4 2013
5.10	Type of Adaptor, (CACT Approved)	SC-PC as per TEC GR TX/OFJ-01/05 NOV 2009
5.11	Ribbon Connector	Fan-out
5.12	Type Fan-out	of
5.13	Patch	Cords Supplied
5.14	OFC Entry Ports (Top)	4 Nos. for Ribbon OFC Diameter up to 24mm, 6 Nos. for 24F OFC Diameter up to 14mm
5.15	Patch (Top)	Cord
5.16	Provisions for (on Top)	8 Nos. for Entry of Power Cable/ Data Cable Diameter up to 14mm to MCBs
5.17.	MCB Mount	Din rail for mounting MCB to be supplied along with the rack
5.18	Test Procedure	As per TEC GR referred.
5.19	Splice Protection Sleeves	24F FDMS S&P Shelf: As per GR TEC/GR/TX/PTS-02/03. JAN-2011 - 4 nos. + 25% extra 48F FDMS S&P Shelf: As per GR TEC/GR/TX/PTS-02/03. JAN-2011 - 8 nos. + 25% extra 192F FDMS Splicing Shelf: As per GR TEC/GR/TX/PTS-02/03. JAN-2011 - 32 nos. + 25% extra

SECTION-F

Fibre Distribution Management System Outdoor

1. **TEC GR Reference - TEC/GR/TX/FDM-003/01 MAR 2012.** Additional specification of FDMS Outdoor. Wherever Specifications under this section shall contradict with TEC GR, then specifications under this section shall prevail.

2. FDMS Manufacturers who have got TSEC/TAC for FDMS ((Section VI)) as per BSNL Tender no. CA/CNP/NFS OFC/T-441/2013 are not required to submit QF-103 against this Tender. However they have to obtain TSEC/TAC with Incremental Tests, as per Tender requirement, within the lead time period of 2 months from the date of issue of APO to bidder.

3. If the bidder has obtained TSEC against CA/NOFN/OFC/T-609/2017 Dated: 15.12.2017 the same would be accepted for supplies to this tender as well.

4. **Fibre Distribution Management System (Outdoor) Schematic Diagram and specifications (GR No. TEC/GR/TX/FDM-003/01 MAR 2012)**

5. **Specification details:**

OFC Entry Ports	4 Nos. Round Port, suitable for Ribbon O F Cable of Diameter 14-22mm; 1 No. Oval Port (100mm x 40mm) suitable for Ribbon O F Cable of Diameter 14-24mm
Splicing Tray Capacity	14 Nos. Splice Trays of 4 x 6F Ribbon each = 336 Fibres
Type of Ribbon	6F as per Bellcore document GR-20 Issue 4, 2013
Cable Entry Port Sealing	Supplied with 4 Sets suitable for Ribbon O F Cable of Diameter 14-22mm and 1 Set suitable for Ribbon O F Cable of Diameter 14-24mm
Splice Protection Sleeves	As per GR TEC/GR/TX/PTS-02/03. JAN-2011 - 56 nos. + 25 extra
Test Procedure	As per TEC GR referred.

SECTION-G

Fibre Distribution Management System (Exchange) Type – I at GP

1. **TEC GR Reference - GR/FDM-01/02 APR 2007.** Design and additional specification of FDMS Type-I (at Exchange) at GP. Wherever Specifications under this section shall contradict with TEC GR, then specifications under this section shall prevail.
2. FDMS Manufacturers who have got TSEC/TAC for FDMS ((Section VI)) as per BSNL Tender no. CA/CNP/NFS OFC/T-441/2013 are not required to submit QF-103 against this Tender. However they have to obtain TSEC/TAC with Incremental Tests, as per Tender requirement, within the lead time period of 2 months from the date of issue of APO to bidder.
3. If the bidder has obtained TSEC against CA/NOFN/OFC/T-609/2017 Dated: 15.12.2017 the same would be accepted for supplies to this tender as well.
4. This will be mounted in the rack at GP sites

5.1	48F S & P Module (2 Nos.)	19 inch Rack Mountable, 2U high, with 4 nos. of 2x6FRibbon Splicing & Patching Trays, each preloaded with 2Nos. 6F SC-PC type Ribbon Fan-out & 12 Nos. of SCPC Adaptors.
5.2	Type of Ribbon	6F as per Bellcore document GR-20 Issue-4 2013
5.3	Type of Adaptor, (CACT Approved)	SC-PC as per TEC GR TX/OFJ-01/05 NOV 2009
5.4	Ribbon Fan-out Connector	SC-PC as per TEC GR TX/OFJ-01/05 NOV 2009 (CACT Approved source)
5.5	Type Fan-out of Ribbon	6F Ribbon, G-652D, Ribbon Length 1.5m, Fan out Length - 0.5m with 6 Nos. of SC-PC Connectors.
5.6	Splice Protection Sleeves	As per GR TEC/GR/TX/PTS-02/03. JAN-2011 - 8 nos. + 25% extra

11.16 Annexure-16: Acceptance Test Procedure

CHiPS or its appointed agency, **"hereinafter known as Audit Agency"**, would be responsible to oversee the work being done by SI to ensure quality of work as well as quantity verification. SI need to coordinate with AUDIT AGENCY for getting the acceptance certification done and the schedules for the same need to be planned by the SI in consultation with the CHiPS and AUDIT AGENCY. All records and testing output conducted by SI has to be verified by AUDIT AGENCY. SI shall be responsible to conduct all the acceptance tests but not limited to, on the basis of acceptance test procedures defined in this document. All the acceptance test reports must be duly verified by AUDIT AGENCY before its submission to CHiPS.

AUDIT AGENCY shall be broadly responsible for below responsibilities but not limited to:

- Site survey agency shall be responsible for identification of route length on the basis of priority i.e. High, Medium and low, however it is the responsibility of AUDIT AGENCY to verify the same.
- SI shall make sure that design document, architecture and BoQ shall be made on the basis of verified survey reports and same shall be approved by AUDIT AGENCY
- Audit and inspection of material at warehouse and certifying the invoices as well
- Quality Review of the work done by SI on day to day basis for all 100% sites
- Complete passive infrastructure shall be validated as per the engineering instructions provided in this RFP
- Complete acceptance testing of the active and passive equipment shall be done as per the technical specifications mentioned in Annexure-15
- Acceptance Testing of the State Network Operation Center (S-NOC)
- Acceptance Testing of GPs over RF
- GIS co-ordinates of the entire OFC route shall be captured by SI, will be updated in the GIS tool at an interval of minimum 250 meters (including Equipment Site, Manhole, chambers, Jointing / Splicing, etc.). The GIS tool along with its licenses need to be bought in the name of CHiPS and will belong to purchaser
- Acceptance testing shall be performed on IP-MPLS ring fail over along with Network convergence for both cases i.e. in case of path and equipment failure
- AUDIT AGENCY shall do verification of SI's final deliverables including ABDs prepared by SI

TPA shall conduct all pre-defined tests for ensuring quality of work done by SIs including:

d. Field Acceptance Test – of OFC laid
e. End-to-End testing which means Complete Installation, Integration, Commissioning and Testing of the created network which shall include OTDR link test, Power On & Self-Testing, IP-MPLS ring fail over testing for both cases i.e. in case of path/equipment failure, Radio equipment testing (If applicable), As Build Diagram (ABD reports with optical power loss budget), integration equipment at GP with State and BBNL NOC and Final acceptance certificate

For all such testing's, the requisite tools shall be provided by the SI to the TPA.

The purchaser shall have the right to perform an audit and technical examination of the work and the final bills of the SI including all supporting vouchers, abstract etc. If as a result of such audit and Technical examination, any sum is found to have been overpaid in respect of any work done by the SI under the contract or any work claimed by him to have been done by him under the contract and found not to have been executed, the SI shall be liable for refund of the amount of over payment and it shall be lawful for the purchaser to recover the same from SI.

I. The purchaser shall be entitled to recover any sum overpaid.
II. Any sum of money due and payable to the SI (including security deposit returnable to him) under this contract may be appropriated by the purchaser for the payment of a sum of money arising out or under any other contract made by the SI with the purchaser.

All payments shall be released after certification of Delivery and Implementation Milestones by CHiPS or its appointed agency.

1. **Measurement & Inspection**

f. **Measurement:** The measurement books are to be prepared by SI block-wise and are to be certified by AUDIT AGENCY. One hard bound copy (duly signed on each page by SI and AUDIT AGENCY) and soft copies (scanned) in three CDs will be handed over by AUDIT AGENCY to the CHiPS every month

g. The measurements of various items of work shall be taken and recorded in the measurements Books. The measurements shall be taken and recorded by SI which will be countersigned by the AUDIT AGENCY. AUDIT AGENCY shall be directly responsible for supervision of work, shall be responsible for accuracy of 100% of measurements. All the support in terms of tools, availability of manpower at sites and all other assistance etc. shall be provided by the SI. The purchaser, without any prejudice, reserves the right to carry out any kind of inspection of the works being carried out by AUDIT AGENCY and SI at any time to ascertain its quantity and quality

h. Site Images (Photographs) for PoP locations and others shall be taken by SI through Digital Cameras during Acceptance Testing and verification of measurement book. The images should be minimum resolution of 1024 x 768 pixels. The images must display the date and time of capture of the image on it bottom right corner. The digital camera should also have the capability to record the GPS co-ordinates of the location and embed the co-ordinates (Latitude, Longitude and Altitude) as EXIF data in the image. No alteration, fabrication or makeover of any kind should be made to the Site images being submitted. Site images shall be captured at every 500 meters and at PoP site. The site images are to be captured in such a manner so that the object being captured is clearly visible and the surrounding areas are also identifiable or distinguishable. The photographs shall be printed in sizes 5” x 7” (matte) and attached along with the Field Acceptance Test Report. The site images, in soft copies, are to be maintained in separate folders / directories and nomenclature as per the PoP. The site images should be so arranged such that they are easily locatable and identifiable in the folder where they are stored. The site images are to be monthly recorded on a non-erasable, good quality Compact Disk and sent to CHiPS along with the Monthly Status Reports. The site images are to be uploaded and GIS co-ordinates are to be entered by SI in the project management tool and GIS tool. Printing of site images to be arranged by the SI

i. **Method of measurement:** The measurement of the work shall be done activity-wise as and when the item of work is ready for measurement. The method of measurement of various items are enumerated as under:

a. Measurement of length of cable

The length of cables laid underground shall be measured by use of PON OTDR. The length should be cross-verified with the marking of lengths on the cables. The lengths shall be recorded in sheet provided in the measurement book.

b. Measurement of other items

The measurement/ numerical details of other items shall be recorded in the sheets provided for respective items viz.

- Termination of Cable in equipment room
- The number of joints
- Record splice loss details for each joint

c. The AUDIT AGENCY & SI shall sign all the measurement recorded in the measurement sheet/book. This will be considered as an acceptance by the AUDIT AGENCY of measurements recorded in the MB by SI.

d. Measurement of the work of cable for calculation of services portion will be taken equal to the route length on which the cable has been laid (as measured in the Roadometer) and not the total length of the cable laid.

e. Measurement Book (MB)

The SI shall also maintain a Measurement Book for each Block. This will be maintained as compilation of copies of the measurement sheets verified by AUDIT AGENCY. This book is one of the primary records to be maintained by the SI carrying out the work during the course of execution of works. The SI shall remove all the defects pointed out by AUDIT AGENCY in the Measurement Sheet. The AUDIT AGENCY / SI or their authorized representatives shall also be at liberty to note their difficulties etc. in these Sheets. The AUDIT AGENCY approved hard-bounded measurement sheets shall be submitted to CHiPS at the time of making milestone payments to the SI.

2. Procedure for preparation, processing and payment of bill for works

I. For claiming the payment on successful completion of the milestones defined in payments Section, the SI shall prepare the bill along with Block wise testing and acceptance document of all the works and submit the same to AUDIT AGENCY. The final bill shall be prepared as per measurements of all items involved in execution of complete route

II. The AUDIT AGENCY scrutinize the final bill against the works entrusted and accord necessary certificates stating that the work has been executed satisfactorily in accordance with Specification and terms and conditions of the contract. The AUDIT AGENCY shall verify the quantities of items of work done by SI with reference to measurement recorded in the measurement sheet and Acceptance Testing procedure as defined in this Clause shall be followed

a. The AUDIT AGENCY shall verify and submit the bills (for work carried out by SI in Block which is accepted by AUDIT AGENCY) provided by SI along with all self-certifications, test reports and measurement records to the CHiPS.

b. The CHiPS shall release the payment to SI accordingly based on certificates received from AUDIT AGENCY

c. AUDIT AGENCY shall submit its report to the CHiPS, however the payment shall be made as per payment schedule. The CHiPS shall exercise the prescribed checks on the bills provided by AUDIT AGENCY and make payments

3. Procedure for Payment for Sub-Standard Works

The SI is required to execute all works satisfactorily and in accordance with the Specification. If certain items of work are executed with unsound, imperfect or unskilled workmanship or with materials of any inferior description or that any materials or articles provided by him for execution of work or unsound or of a quality inferior to that contracted for or otherwise not in accordance with the contract (referred to as substandard work hereinafter), the CHiPS shall make a demand in writing specifying the work, materials or articles about which there is a complaint

4. Timely Action by AUDIT AGENCY

I. Timely reporting and action, to a great extent, can prevent occurrence of sub-standard work, which will be difficult or impossible to rectify later on. It is incumbent on the part of AUDIT AGENCY for supervision of work to point out the defects in work in time during progress of the work. The AUDIT AGENCY responsible for supervision of work shall without any loss of time submit a report of occurrence of any sub-standard work to the CHiPS besides making an entry in the site order book. A notice in respect of defective work shall be given to the SI in writing during the progress of work asking the SI to rectify/replace/remove the sub-standard item of work and also definite time period within which such rectification/removal/replacement has to be done. After expiry of the notice period, if the SI fails to rectify/ replace/ remove the sub-standard items, the defects shall be rectified/replaced/removed by the CHiPS, at its sole discretion, through some other agency at the risk and cost of the SI

II. Non-reporting of the sub-standard work in time on the part of AUDIT AGENCY shall not in any way entitle the SI to claim that the defects were not pointed out during execution and as such the SI cannot be absolved of the responsibility for sub-standard work and associated liabilities

5. Quality Control of Works

I. The importance of quality of Fibre optic cable splicing works cannot be over-emphasized. The quality and availability of long distance media and efficiency of the reliable media connectivity between terminal equipment depends upon quality of optic fibre cable plant. The quality of fibre optic cable plant depends upon the quality of individual items of work involved viz. laying, Protection, Jointing of cables and Terminations in equipment room and also on documentation of cable network. The work shall be carried out strictly in accordance with Specifications laid down to achieve the requisite quality aim

II. The CHiPS shall be the final judge of the quality of the work and the satisfaction of the CHiPS in respect thereof set forth in the contract document. Laxity or failure to enforce compliance with the contract documents by the CHiPS and/ or its representative shall not manifest a change or intent of waiver, the intention being that, notwithstanding the same, the SI shall be and remain responsible for complete and proper compliance with the contract documents and the specifications there in. The representative of the CHiPS has the right to prohibit the use of men and any tools, materials and equipment which, in his opinion, do not produce the required work or performance meet the requirement of the contract documents.

III. It is imperative that the SI is fully conversant with the construction practices and shall be fully equipped to carry out the work in accordance with the specifications. The SI is expected and bound to ensure quality in construction works in accordance with specifications laid down. The SI shall engage adequate and experienced supervisors to ensure that work is carried out as per specifications, with due diligence and in a professional manner. A two stage testing process will be incorporated as follows:

   a. The first level of testing shall be carried out by the SI. Once the SI is confirmed about the quality assurance of their work and material then they will hand it over to AUDIT AGENCY for review and testing.

   b. The AUDIT AGENCY team shall carry out the second level of testing.

IV. In addition to Acceptance Testing being carried out by AUDIT AGENCY, all works at all times shall be open to inspection of the purchaser. The SI shall be bound, if called upon to do so, to offer the works for inspection without any extra payment. The presence of monitoring teams nominated by the CHiPS during construction shall not preclude separate acceptance testing teams to recheck adherence to all aspects as mentioned in the contract

6. Quality Control of material supplied by SI

AUDIT AGENCY has to ascertain that all the material being supplied by SI for works being carried out are in compliance with the required standard and quality and as per Quality Assurance Plan. Any instance of violation by SI shall be immediately reported to the CHiPS by AUDIT AGENCY.

7. Inspection and Testing

I. All materials furnished and all work performed under this Contract shall be inspected and tested. The SI shall furnish all manpower and materials for tests, including testing facilities, power and instrumentation, and replacement of damaged parts. The costs shall be borne by the SI and shall be deemed to be included in the contract price.

II. The entire cost of testing for factory & site acceptance, routine tests, production tests and other test during manufacture & site activities specified herein shall be treated as included in the quoted unit price of materials, except for the expenses of the CHiPS representative.

III. Prior notice of at least 15 days should be given to AUDIT AGENCY by SI for making the representative of AUDIT AGENCY available for observing the factory tests. Any cost of pertaining to making available the AUDIT AGENCY representative at the necessary site shall be solely borne by AUDIT AGENCY and non-chargeable to the CHiPS.

IV. All tests conducted by SI must be verified by AUDIT AGENCY.

V. Should any inspections or tests indicate that specific item does not meet Specification requirements; the appropriate items shall be replaced, upgraded, or added by the SI as necessary and as applicable to correct the noted deficiencies at no cost to the CHiPS. After correction of a deficiency, all necessary retests shall be performed to verify the effectiveness of the corrective action.

VI. Deliveries shall not be shipped until all required inspections and tests have been completed and all deficiencies have been corrected to comply with this specification and approved for shipment by the CHiPS.

VII. Acceptance or waiver of tests will not relieve the SI from the responsibilities to furnish material and works in accordance with the specifications and to the CHiPS's satisfaction.

VIII. Unless otherwise specified in this Contract, selection of test samples, number of specimens and acceptance of results shall be in accordance with the terms of the relevant Standards and Codes. Where no terms exist, CHiPS or its appointed agency shall instruct details in advance of the inspection and tests in response to the request of the SI.

IX. SI shall comply with various instructions / guidelines issued by CHiPS relating to testing and acceptance of the deliverables of the SI.

8. Optical Fibre Identification

I. Individual optical fibres within the fibre units shall be identifiable in accordance with EIA/TIA 598 or IEC 60304 or Bellcore GR-20 colour coding scheme. The actual colouring scheme shall be mentioned by the SI in Data Requirement Sheet and the same shall be finalized during the detailed engineering in consultation with the CHiPS.

II. Colouring utilized for colour coding optical fibre shall be integrated into the fibre coating and shall be homogenous. The colour shall not bleed from one fibre to another and shall not fade during fibre preparation for termination of splicing. Each fibre cable shall have traceability of each fibre back to the original fibre manufacturer's fibre number and parameters of the fibre.

9. Testing Methodology

I. Acceptances Tests

a. The works shall be deemed to have been completed only after the same has been accepted by AUDIT AGENCY as per the process mentioned in this tender and after it has been informed by AUDIT AGENCY to the CHiPS confirming the completion of work. The various testing will be undertaken by SI in the presence of AUDIT AGENCY. SI may conduct its own test prior for self-assessment before asking for tests to be conducted in the presence of AUDIT AGENCY. Certificate will be issued by AUDIT AGENCY representative after successful completion of testing (for each milestone).

12.1.1 The SI, after having satisfied himself of completion of work, from equipment at Block PoP to equipment at GP PoP, shall offer the work to AUDIT AGENCY for conducting Testing. The work shall be offered for Inspection as soon as link to Block is complete.

b. If the measurements (of length of OFC laid) taken by AUDIT AGENCY are found to be lesser than the measurements recorded by the SI responsible for recording the measurements, the measurement taken by AUDIT AGENCY shall prevail without prejudice to any punitive action against the SI as per provisions of the contract and the testing officer of SI recording the measurements.

13.1.1 The SI shall be obligated to remove defects/deficiencies pointed out by the AUDIT AGENCY without any additional cost. The CHiPS does not take any responsibility of return of defective used items / items previously accepted by SI.

14.1.1 **Factory Acceptance Test:** Factory Acceptance Tests shall be conducted as per relevant Standards and Codes on randomly selected final assemblies of selected equipment to be supplied. These tests shall be carried out in the presence of the CHiPS's authorized representatives unless waiver for witnessing by the CHiPS is intimated to the SI. Factory acceptance testing shall be carried out on OFC and related accessories, all active and passive equipment, Test Equipment, installation accessories and all other items to be supplied unless factory testing and inspection has been waived off by the CHiPS.

Equipment shall not be shipped to the CHiPS until required factory tests are completed satisfactorily, all variances are resolved, and the CHiPS has issued Dispatch Clearance, which may be issued after completion of FAT by the CHiPS or its authorized representatives deputed for carrying out the FAT. Successful completion of the factory tests and the CHiPS approval to ship shall in no way constitute final acceptance of the system or any portion thereof.

The Factory Acceptance Test (FAT) shall demonstrate the technical characteristics of the Fibre Optic cable & associated accessories in relation to this specifications and approved drawings and documents. The list of factory acceptance tests shall be supplemented by the SI's standard FAT testing program. In general, the FAT for other items shall include at least: Physical verification, demonstration of Technical characteristics, various operational modes, functional interfaces, alarms and diagnostics etc. For Test equipment, FAT shall include supply of proper calibration certificates, demonstration of satisfactory.

There shall be no factory splice allowed within a continuous length of cable. Only one continuous cable length shall be provided on each drum. The lengths of the cable to be supplied on each drum shall be standard or as determined by a “cable drum schedule” prepared by the SI.

c. **Scope of Acceptance Testing:** It is essential to verify the integrity and the capability of the Optical Fibre Cable and to assess its readiness for intended services. This scope defines the methodology for cable and accessories. The purpose of acceptance and testing is to verify integrity of measurement and quality of work done.

II. Fibre Optic cable link testing:

- **a.** Fibre continuity and link attenuation (Bi-directional) between FODP connectors at two ends for each fibre at 1310 and 1550 nm by OTDR
- **b.** Fibre continuity and link attenuation (Bi-directional) between FODP connectors at two ends for each fibre at 1310 and 1550 nm by Power meter & Laser source
- **c.** Average fibre attenuation and average splice loss in the link including FODP
- **d.** Proper termination and labelling of fibre and fibre optic cables at FODP
- **e.** Data loss test and ensuring that the same is within the acceptable limits

III. Termination arrangement at PoP location: The fibres of the cable shall be spliced to the pigtails for connection to the optical line systems. Pigtails shall be duly terminated at the FDMS (fibre distribution management system).

IV. Field Acceptance Test

- **a.** The field installation test shall be performed for all equipment at each location. If any equipment has been damaged or for any reason does not comply with this Specification, the SI shall provide and install replacement parts at its own cost and expense
- **b.** As per Technical requirements, the Acceptance Test is required to be carried out for all 48 Fibres in each cable section, and the CHiPS Acceptance Test schedule is to be followed for proper testing of the OF cable network

c. The OF cable sections shall be identified on ABDs attached with the Acceptance Test Report and in the GIS tool.

d. Testing shall be done in each OF cable section in one direction only and for two wavelengths viz. 1310 nm and 1550 nm using power meter and source. OTDR traces would be obtained for each OF Cable sections to measure and record the splice loss wherever applicable.

e. A minimum length of 2.0 km shall be maintained for all the OF cable between splices except as directed by the CHiPS for any intermediate T-offs

V. End to End Testing of Optical Fibre Cable Route from GP PoP to Block PoP

This document defines the procedure to be adopted for end to end testing of the OF cable route from GP PoP to Block PoP.

a. The End to End testing from GP PoP to Block PoP shall be carried out using Power meter/source and with OTDR after splicing OF cable

b. The average attenuation (dB/Km) for cable shall be recorded

c. End to end Testing shall be done in one direction only for the two wavelengths i.e. 1310 nm & 1550 nm from PoP to PoP side using Power meter and source. The Fibre connected to each port of the OTN shall also be tested using the OTDR and the traces obtained shall be recorded for future reference

d. The SI shall be responsible for co-ordination for conducting this test

e. After carrying out this test, the respective PoP shall be detected in the NMS of NOC and shown as active at the NOC implemented by the SI as part of this RFP. SI shall share the list of PoPs in each district which have gone active.

f. The format for End-to-End Testing of Cable Route is provided in this RFP.

10. Fibre Link Length

The estimated optical fibre lengths for various feeder lengths from one terminal point (FODP) to the other are given in the respective price schedules. However, the SI shall supply and install the optical fibre cable as required based on actual work requirements finalized after detailed site survey.The payment will be based on actual quantities of work carried out by the SI, as per the measurement criterion set forth in these specifications.

11. Commissioning Certificate

15.1.1 SI shall be eligible to apply for Commissioning Certificate of a Block after successful completion of End-to-End testing of a Block.

I. The End-to-End Testing Report has to be submitted for obtaining the Commissioning Certificate.

16.1.1 The CHiPS, without any prejudice, reserves the right to carry out any kind of inspection of the works being carried out by AUDIT AGENCY and SI at any time to ascertain its quantity and quality.

II. Testing will also be done for NOC, NMS and all other components supplied by the SI as per the guidelines issued by the CHiPS in this regard from time to time.

12. Final Completion Certificate

I. Only upon completion of all Works as required to be done by SI for the State, SI shall be given the Final Completion Certificate or Go-Live Certificate by the CHiPS.

A. OFC Testing & Certification Formats

Test Format 1: Physical Inspection of Underground OFC & Accessories

S. No.	Name of Items	Quantities (or Length) Installed / Used (Verified as per Physical Inspection)	Observations	Remarks
1	48 Core Optical Fibre Cable		Accessories: Length:	
2	PLB HDPE Duct		Accessories: Length:	
3	DWC Duct		Accessories: Length:	
4	GI Pipe		Accessories: Length:	
5	FDMS at Block Level		Routing & Tagging of fibres	
6	FDMS at GP Level		Routing & Tagging of Fibres	
7	FDMS Outdoor			
8	Any other items			

Section Identity/No. :__________

Section Length: __________ Kms

Splice No. :__________

Tube Colour	Fibre Colour	Fibre Number		Splice Loss (dB)	
			1310 nm		1550 nm
		1			
		2			
		3			
		4			
		To			
		45			
		46			
		47			
		48			

Name, Signature & Seal of the SI Official
(Accepted / Rejected)

Date:

Name, Signature & Seal of AUDIT AGENCY Official

Date:

Remarks, if any:

Note: Splice Loss Measurement using OTDR

1. The fibre under test is connected to the OTDR which directly displays the splice loss after suitably adjusting the markers. The observations shall be recorded for both the windows i.e. 1310 nm and 1550 nm.
2. For the splice(s) within the Fibre Cable section (in cases where the Fibre Cable section is more than 2 Kms in length) the splice loss shall be measured for all the 48 fibres. The splice no. shall be counted from POP side towards the nearest POPs. The test results shall be recorded in the format given in Form 3 of the Formats for the Test Report.
3. Specification: Max Splice Loss 0.05 dB for one fibre per splice for straight/branch joints.

Test Format 3: Attenuation Test for Fibre Cable Section in the POP using Power Meter (for each fibre)

Section Identity/No: ___________

Section Length: ___________ Kms

Transmit Power (PTx):___________ dB

Fibre No.		**Testing at 1310 nm**			**Testing at 1550 nm**	
	Level at Rec. End	**Loss (in dB)**	**Attenuation per KM (dB/Km)**	**Level at Rec. End**	**Loss (in dB)**	**Attenuation per KM (dB/Km)**
	(PRx)	**(A=PTx-PRx)**	**(A/section length)**	**(PRx)**	**(A=PTx-PRx)**	**(A/section length)**
1						
2						
TO						
47						
48						

Name, Signature & Seal of the SI Official
(Accepted / Rejected)

Date:

Name, Signature & Seal of AUDIT AGENCY Official

Date:

Remarks, if any:

Note:

1. Carry out total section attenuation loss as mentioned in table above.
2. All the cables should meet the standard for both the wavelength i.e.1310 nm and 1550 nm as per specification given above i.e. less than 0.43 dB/Km for 1310 nm and 0.30 dB/km.for1550 nm.
3. Connect standard optical source with 1310 nm and 1550 nm at particular level (say P1 dBm.) at one end of the fibre. Measure with power meter the power at the other End of the fibre (say P2 dBm.)
   Thus, attenuation of the fibre = (P1 - P2) dB.
4. Specifications

   I. At l310nm

   Total Link Loss ≤ 0.36 dB/km x Section Length + (0.05 dB/Splice) x (No. of Splices) + 0.5 dB x No. of Connectors + splitter loss

   II. At l550nm

   Total Link Loss ≤ 0.21dB/km x Section Length + (0.05 dB/Splice) x (No. of Splices) + 0.5 dB x No. of Connectors
5. All the 48 Fibres of the Fibre Cable shall be tested with the pigtail spliced to each fibre one by one for taking the test readings as per the table below.
6. Attenuation test shall also be taken with OTDR at 1550nm and 1310nm and printout for each fibre for each window shall be obtained.

17.1.1 Preparation of OTDR Traces Report

1. This method uses an optical time-domain reflectometer (OTDR). Unlike a Power Meter, the OTDR can identify and locate the position of each component in the network. The OTDR will reveal splice loss, connector loss and reflectance, and the total end to end loss.
2. For End-to-End measurements including joint enclosures must be carried out to document the characterization of the joint loss and the total link loss. The OTDR measurement must be conducted upstream (i.e., from the GP to Block).
3. Carry out OTDR measurements and take traces taken on all wavelengths (1310 nm / 1550 nm). Soft copy of this report needs to made available for updating in Test reports

B. Depth Acceptance Test of OF Cable Route

Status report of OFC Depth Acceptance Test

The details of route is as under

Name of the block: ___________________________

Estimate No.: ___________________________

Total Length: _________________________

Length of PLP Pipe laid: _________________________

Length of 48 OF cable: ________________________

Length of GI-Pipe: _______________________

Length of DWC pipe: _______________________

The above said

Acceptance Test conducted the testing samples of HDD depth was carried out on different man holes or exit pits and for trench depth at different pits was taken. Details are as follows:

The details of testing/ Samples points are found satisfactory with in the limit/beyond acceptance limit.

The GP wise test results are as per the following format for OF cable.

Name, Signature & Seal of the SI Official
(Accepted / Rejected)

Date:

Name, Signature & Seal of AUDIT AGENCY Official

Date:

Remarks, if any:

Format for Depth Acceptance Test of OFC

All the measurement & observation have to be as per Engineering Instructions (as given in Annexure-17) (With latest amendment/addendum)

S. No.	Name of the GP	Location of pit		Type of Laying		Depth on Top of pipe (m)	Observation OK/ Not Ok	Remarks (If any)
		As per MB (Measurement Book)	As per Landmark	Open Trench	HDD			

Name, Signature & Seal of the SI Official

Date:

AGENCY Official

(Accepted / Rejected)

Name, Signature & Seal of AUDIT

Date:

Remarks, if any:

C. Electronics & Network Equipment testing & certification

The AUDIT AGENCY will work in-line with the agreed execution schedule during the installation of Electronics & Network Equipment at the field level. The AUDIT AGENCY needs to test and certify the each location where electronics and network equipment is installed for its operations and acceptance.

Note: Necessary testing equipment required to meet standards compliance will be provided by the SI. After completion of the testing for each site, AUDIT AGENCY need to submit the report in prescribed format to the CHiPS.. AUDIT AGENCY will use the equipment to validate the reports submitted by the SI.

Electronics & Network Equipment Testing & Certification Format:

Test Format 4: Field Acceptance Test for PoP

Name of District:

Name of Block:

PoP Name:

Latitude & Longitude:

Test Requirements:	1310 nm	1550 nm
Cable Attenuation per km (dB)		
Total Physical Cable Length (km)		
Splice Loss Mean Value (dB)		
Total Number of Splices		
Maximum Connectors Loss (dB)		
Total Number of Connectors		
Total Section Loss (dB)		

Test Period: From ________________ to ____________________

Total Route Length of Fibre Cable laid from POP to POP: _________________ Kms

Total Fibre Cable length tested: _________________ Kms

No. of Straight Joints: _______________

No. of Branch Joints: _______________

No. of FDBs: ___________________

Name, Signature & Seal of the SI Official (Accepted / Rejected)

Date: Name, Signature & Seal of AUDIT AGENCY Official

Date:

Remarks, if any:

D. NOC & Point of Presence (PoP) Testing & Certification

AUDIT AGENCY will work in-line with the agreed execution schedule during the set-up, installation and commissioning of Network Operations Centre and Points of Presence. The AUDIT AGENCY needs to test and certify commissioning the each PoP and the centralized NOC including other components as mentioned in this RFP.

NOC & PoP Testing & Certification Formats

Test Format 5: End to End Testing of Route From PoP to PoP

1. The End to End testing from POP to POP shall be carried out using Power meter/source and with OTDR after splicing Optical Fibre Cable.
2. The average attenuation (dB/Km) for cable shall be recorded in the given format:
   a. Tested Length : ________ Km
   b. Total loss on the length : ________ dB
   c. Average Attenuation per Km. : ________ dB/Km
   d. Fibre length used : ________ Km
   e. Net loss for Fibre at FDMS : ________ dB
3. End to end Testing shall be done in one direction only for the two wavelengths i.e. 1310 nm & 1550 nm from POP to POP side using Power meter and source. The Fibre connected to each port of the OTN shall also be tested using the OTDR and the traces obtained shall be recorded for future reference.
4. The O&M Agency entrusted with the task of for laying the Optical Fibre Cable shall be responsible for co-ordination for conducting this test.
5. After carrying out this test, the respective PoP shall be detected in the NMS of NOC and shown as active at the NOC implemented by the SI as part of this RFP. SI shall share the list of PoPs in each district which have gone active.

Test Format 6: End to End Testing of Route from Block to GP

1. The End to End testing of the Fibre Cable route from the Block to GP shall be done by measuring the receive power at each GP location using the power meter and source. Thus total attenuation would be recorded.
2. The attenuation shall be tested from Block to each FDMS at GP in one direction only for two wavelengths 1310 nm and 1550 nm.
3. The tests shall be carried on the fibre connected to each of utilized ports.
4. The test results shall be recorded in the format given in the following table. Separate sheet shall be prepared for each port.
5. Specifications: The total maximum attenuation from Block to GP at each location should be about 25 dB.
6. The fibre connected to each port shall also be tested using the OTDR and the traces obtained shall be recorded for future reference.

End-to-End Network Testing & Certification

The AUDIT AGENCY will work in-line with the agreed execution schedule and needs to test and certify commissioning End-to-End Network Operations for the CHiPS. This will require testing of all installed infrastructure, hardware and software of the project, Standard Operating Procedures for NOC, Processes for Network Management, Service Provisioning & Monitoring. The End-to-End Network testing will be required to be done along with commissioning of the blocks.

Commissioning Certificate For OF Cable Network

No. : ____________________________
Dated : __________________________

The OF cable network laid under the BharatNet project for the block as listed below is hereby declared as commissioned as on <**date**>. The names of the GPs and the details of the OF cable network being commissioned are given in the tables below

State : ________________________
District : ______________________
Block : _______________________
No. of GPs completed : ___________
No. of GPs left out / incomplete __________
Type & Make of OF Cable: 48F, **<manufacturer name>**

A. Names of the Gram Panchayats completed

Sl. No.	Gram Panchayat Name	Sl. No.	Gram Panchayat Name

B. OF Cable Network details

S .No	Parameter	Qty	Unit	Remarks
1	Gram Panchayats Total		Nos	
2	Gram Panchayats Completed		Nos	
3	Gram Panchayats Left out / incomplete		Nos	
4	Total OF cable length		Meters	
5	Total HDPE PLB length		Meters	
6	FDMS 48 F at Block		Nos	

7	FDMS 48 F at GP		Nos	
8	Patch Cords (nos. per GP, 1m/3m/5m/10m)		Nos	

Name, Signature & Seal of CHiPS Official

Date:

Remarks, if any:

Commissioning Certificate for Equipment

No. : ____________________________

Dated : __________________________

The Equipment installed under the BharatNet project for the block as listed below is hereby declared as commissioned as on <**date**>. The names of the GP locations and the details of the equipment installed and being commissioned are given in the tables below

State : ________________________
District : ______________________
Block : _______________________
Block Location : ___________________
No. of GPs (GP locations) connected: ___________
No. of GPs left out / incomplete __________

A. Equipment Details

Type and Make : ______Port, ________ Make

B. Names of the Gram Panchayats connected

Sl. No.	Gram Panchayat Name	Sl. No.	Gram Panchayat Name

Name, Signature & Seal of CHiPS Official

Date:

Remarks, if any:

11.17 Annexure-17: Engineering Instructions for OFC Construction Practices

Colour, content, logo are indicative and same shall be decided by the CHiPS

1. Scope

1.1. The Engineering Instructions spelt out in this document deals with the methods to be adopted for underground Optical Fibre Cable laying in PLB HDPE ducts and termination of Optical Fibre Cables at Gram Panchayats(GPs) for BharatNet Phase- II

2. Optical Fibre Cable Laying Approach

2.1. Generally, Optical Fibre Cable may preferably be laid straight as far as possible along the road near the boundaries, away from the burrow pits. When the Optical Fibre Cable is laid along the National Highways, Cable should run along the road land boundary or at a minimum distance of 15 meters from the centre line of the road where the road land is wider as the OFC carries high capacity traffic and is planned for about 25 to 30 years of life. It is essential that the cable is laid after obtaining due permission from all the concerned authorities to avoid any damage (which may result in disruption of services / revenue loss) and shifting in near future due to their planned road widening works. For obtaining ROW clearances will be provided by purchaser.

2.2. In special cases where it may be necessary to avoid burrow pits or low lying areas, the Cable may be laid underneath the shoulders at a distance of 0.6 meter from the outer edge of the road embankment provided the same is located at least 4.5 meters away from centre line of road.

3. General

3.1. Soil Classification

Soil shall be classified under two broad categories Rocky and Non Rocky, The soil is categorized as rocky if the cable trench cannot be dug without blasting and / or chiselling. All other types of soils shall be categorized as Non Rocky including Murrum & soil mixed with stone or soft rock.

3.1.1. Rocky soil

The terrain which consists of hard rocks or boulders where blasting/ chiseling is required for trenching such as quartzite, granite, basalt in hilly areas and RCC (reinforcement to be cut through but not separated) and the like.

3.1.2. Non Rocky soils

This will include all types of soil- soft soil/hard soil/Murrum i.e. any strata, such as sand, gravel, loam, clay, mud, black cotton murrum, shingle, river or nullah bed boulders, soling of roads, paths etc. (All such soils shall be sub-classified as kachcha soil) and hard core, macadam surface of any description (water bound, grouted tarmac etc.), CC roads and pavements, bituminous roads, bridges, culverts (All such soils shall be classified as Pucca soils)

4. Laying of The Optical Fibre Cable

The Optical Fibre Cable shall be laid through PLB HDPE Ducts buried at a nominal depth of 165cm. The steps involved in Optical Fibre Cable laying are as under:

4.1. Excavation of trench up to a nominal depth of 165 cm in non-Rocky soil, according to construction specifications along National/State Highways/other roads and in built up /rural areas. Under exceptional conditions/ genuine circumstances due to site constraints/ soil conditions, relaxation can be granted by the competent authority for excavation of trench to a depth lesser than 165cm. Such relaxation shall be given as per the laid down norms/ procedures being set by ChiPS and with the approval of the competent authority. The payment in such cases shall be made on pro-rata basis as per the laid down norms adopted by the purchaser.

4.2. Laying of PLB HDPE Ducts/coils coupled by sockets in excavated trenches, on bridges and culverts, as per construction specification and sealing of PLB HDPE Ducts pipe ends at every manhole by end-plugs of appropriate size. Optical Fibre Cables should be blown through Permanently Lubricated HDPE Duct of 40 mm-OD and 33 mm ID Pipe of 500/1000 meter coil which meets the specification as given in G/R No-G/CDS-08/02 Nov 2004 with latest amendments shall only be used for laying the OFC. Wherever DWC pipe or GI pipes or R.C.C. pipes are used for protection, the two ends of the pipes should be properly sealed to protect HDPE pipe from sharp edge of GI pipe and to bar the entry of rodents. For providing additional protection Split RCC/GI pipes should be used from top instead of full RCC/GI pipes.

4.3. Providing additional protection by R.C.C. Pipes/ GI pipes and/or concreting/chambering, wherever required according to construction specification.

4.4. Fixing of GI pipes/troughs with clamps on culverts/bridges and/or chambering or concreting of G.I. Pipes/troughs, wherever necessary. Normally, RCC/DWC pipes shall be used and use of GI pipes shall be avoided. However, in case it is felt that GI pipe is unavoidable in certain circumstances this should be done with the prior approval of competent authority within the concerned purchaser. This shall be recorded appropriately.

4.5. Laying Protection Pipes on Bridges and Culverts. In case trenching and pipe laying is not possible on the culverts, the pipes shall be laid on the surface of the culverts/bridges after due permission from the competent authority within the concerned purchaser as per construction specification.

4.6. Back filling and Dressing of the Trench according to construction specifications.

4.7. Making manhole of size (1.0 m inner diameter x 1.1 m outer diameter x 0.5 Height) and thickness 5 cm at every Cable pulling location for housing the Optical Fibre Cable loop & Pulling Optical Fibre Cable using proper tools and accessories. Sealing of both ends of the PLB HDPE pipe in manhole by hard rubber bush of suitable size to avoid entry of rodents into the PLB HDPE Ducts, putting split PLB HDPE Ducts and split RCC pipes with proper fixtures over cable in the manhole to protect the bare cable.

4.8. Digging of pit of size 2 meter x 2 meter x 1.8 meter (depth) for fixing of Jointing chambered-cast RCC cover or stone of suitable size on jointing chamber to protect the Joint and backfilling of jointing chamber with excavated soil.Round base plate of 120cm of diameter and 4cm thickness.

4.9. Digging of pits 500 cm to 1000 cm towards jungle side at every manhole and jointing chamber along the route to a depth of 75 cm fixing of route Indicator/joint indicator, concreting and backfilling of pits. Painting of route indicators with Blue colour and joint Indicator by Grey colour and sign writing denoting route/joint indicator number, kilometre, and marked as decided by CHiPS, as per construction specification.

4.10. Specifications of Materials to be used

PLB HDPE Duct

Optical Fibre Cables should be pulled though Permanently Lubricated HDPE Duct of 40mm/33 mm size conforming to the specifications as per TEC GR No. TEC/GR/TX/CDS-008/03/MAR-11 with latest Amendments. The Ducts shall be in Grey colour and have the identification markings as per TEC GR wherein Chhattisgarh/CHiPS logo shall be marked as purchaser's name.

PLB HDPE Duct Accessories

g) Push fit Coupler

Push Fit couplers shall be used for coupling PLB HDPE ducts/coils. The specifications of the couplers shall be as per TECGR no TEC/GR/TX/CDS-008/03/Mar11 with latest amendments.

h) PP Rope

The PP Rope should conform to TEC GR No.TEC/GR/TX/CDS-008/03/MAR-11 with latest Amendments. However, this is optional and SI may use the same on need basis. The PP rope can be ordered along with the PLB duct as required. In this case PP ropes drawn through the HDPE/PLB pipes/coils and safely tied to the end caps at either ends with hooks to facilitate pulling of the Optical Fibre Cables at a later stage. The rope used is 3 strands Polypropylene Para Pro rope having yellow colour and size of 6 mm diameter. It should have a minimum breaking strength of 550 kg. The length of each coil of rope should be 5 meter more than the standard length of duct(or as ordered) and it should conform to (i) BS 4928 Part-II of 1974 (ii) IS 5175 of 1982. It should be of special grade and should have ISI certificate mark. It should be manufactured out of industrial quality Polypropylene.

i) End Cap

End Cap shall be used for sealing the ends of the empty ducts, prior to installation of the Optical Fibre Cable and shall be fitted immediately after laying the duct to prevent the entry of any dirt, water, moisture, insects/rodents etc. It should conform to TEC GR No. TEC/GR/TX/CDS-008/03/MAR-11with latest amendments. The ends of the PLB HDPE ducts/coils laid in the manholes should be closed with End Caps. The End Caps used should be suitable for closing40mm/33mm PLB HDPE ducts/coils. A suitable arrangement should be provided in the End Cap to tie PP Rope. (See figure-1 for details).

j) Cable sealing Plug

This shall be used to seal the end of the ducts perfectly, after the Optical Fibre cable is pulled in the duct. For pulling the cable through the ducts, it is necessary to provide manholes at that location and also at bends and corners wherever required. The ends of the PLB HDPE ducts/coils are closed with Cable sealing Plugs. The End Plugs used should be suitable for closing40mm/33mm PLB HDPE ducts/coils. The Cable sealing plug shall conform to TEC GR No. TEC/GR/TX/CDS-008/03/MAR-11 with latest amendments.(Wherever blowing technique is used for laying Optical Fibre Cable, at the discretion of the concerned SI, the hand holes/manholes required for accessing the cable during cable laying can be at longer distances depending upon requirement).

5. Duct integration Test (DIT) for HDPE ducts

5.1. SI shall ensure that the trench does not have any sharp bend and the couplers are tightened to the maximum.

5.2. SI shall ensure the backfilling and the compaction of the trench are satisfactory prior to start of DIT.

5.3. SI shall ensure to pass the compressed air at 8Kg/Sq.cm and clean the duct from deposits like mud and small stones.

5.4. SI shall insert a medium density sponge into the duct and push it with compressed air of 8Kg/Sq.cm. The sponge should eject with full force.

5.5. The mandrill made of hard rubber or polished wood in the shape of cylinder of diameter 0.75 x D in diameter of HDPE duct and 75 mm long shall be used.

5.6. On completion of test seal the ends with end plugs.

5.7. SI shall ensure that there is no pressure leakage during DIT

6. Method of Blowing of Fibre Optic Cable through HDPE duct

6.1. SI shall ensure the duct Integrity test has been completed.

6.2. Drum should be kept approximately at the center of two adjacent chambers. (I.e. if drum length is 4 Km, at 2 Km.) So that on either side 2 Km blowing can be done.

6.3. Cable drum should be mounted on jack, which shall be kept on a plain surface.

6.4. Cable blowing to be done with the help of compressor, hydraulic power pack and blowing Machine. Anti-twist tool can be used to avoid twisting of cable while blowing.

7. Stripping/Cutting of the Cable

7.1. The cable is stripped of their outer and inner sheath with each sheath staggered approximately 10mm from the one above it.

7.2. Proper care must be taken when removing the inner sheath to ensure the fibres are not scratched or cut with the stripping knife or tool to prevent this, it is best to only score the inner sheath twice on opposite sides of the cable, rather than cut completely through it. The two scores marking on either side of the cable are then stripped of the inner sheath by hand quite easily.

7.3. The fibres are then removed from cable one by one and each fibre/ribbon is cleaned individually.

8. Material for Providing Additional Protection

8.1. RCC Full Round Pipes: Reinforced cement concrete pipes (spun type) coupled with RCC collars sealed with cement mortar used to provide additional protection to PLB HDPE Ducts/coils at lessor depths should be of full round, NP-2 class and size 100 mm (internal diameter), conforming to IS standard 458-1988 with latest amendments. The pipes should have a nominal length of 2 meters.

8.2. The RCC collars should be properly sealed using cement mortar 1:3 (1:53 grade cement of reputed brand, 3: fine sand without Impurities). If case of long spans, every third joint will be embedded in a concrete block of size 60 cm (L) x 40cm (W) x 25 cm (H) of 1:2:4 cement concrete mix (1: cement, 2: coarse sand, 4: stone aggregate of 20 mm nominal size) so that the alignment of RCC pipes remain firm and intact. Also, both ends of RCC pipes spans will be sealed by providing concrete block of size 40 cm (L) x 40 cm (W) x 25 cm (H) of 1:2:4 cement concrete mix to avoid entry of rodents.

8.3. RCC Split Pipes: The split Reinforced cement concrete pipes (spun type) with in-built collars are used to provide additional protection to PLB HDPE Ducts/coils should be of 100mm internal dia.(Spotted), Class--NP-3, Thickness: 25mm, Length: 2 Meters with inbuilt collaret one end, Conforming to ISI Specification IS: 458, 1988 with latest amendment

8.4. G.I. Pipes: G.I. pipes should be of medium duty class having inner diameter of 50 mm and should conform to specifications as per IS 554/1985 (revised up to date) IS 1989 (Part-I), 1900 Sockets (revised up to date) & IS 1239 (Part-II) 1992 (revised up to date).

8.5. DWC Pipes: Use of normal duty DWC (Double walled corrugated) HDPE pipe – conforming to TEC GR no.GR/DWC-34/01 Sep.2007 with latest amendments shall be preferably utilized as first choice for protection of Optical Fibre Cable instead of GI pipes. The DWC pipes used shall be of size 75/61mm as per table 2 of the said TEC GR.

8.6. M.S. Weld Mesh: The PLB HDPE Ducts can also be protected by embedding it in concrete of size of 25 cm x25 cm reinforced with MS weld mesh. The MS weld mesh used should be of 50 mm x 100 mm size, 12 SWG, 120 cm in width in rolls of 50 m each. One meter of MS weld mesh caters to approx. 3 meters of concreting. (See figure

'2' for details). The strength of RCC/CC is dependent on proper curing, therefore, it is imperative that water content of CC/RCC mix does not drain out into the surrounding soil. In order to ensure this, the RCC/CC work should be carried out by covering all the sides by yellow PVC sheets of weight not less than 1 kg per 8 sq. metre to avoid seepage of water into the soil.

8.7. Joint Chamber: The Joint chamber shall be provided at every joint location to keep the Optical Fibre Cable joint well protected and also to house extra length of cable which may be required in the event of faults at a later date. The Joint chamber shall be of pre-cast RCC type as per construction specification. Brick chamber can also be made with prior permission of Purchaser.

8.8. Rubber Bush: To prevent entry of rodents into PLB HDPE DUCTS, the ends of PLB HDPE DUCTS are sealed at every manhole and joint using rodent resistant hard rubber bush (cap) after Optical Fibre cable is pulled. The rubber bush should be manufactured from hard rubber with grooves and holes to fit into 40 mm PLB HDPE DUCTS pipe, so that it should be able to prevent the entry of insects, rodents, mud, and rainwater into the PLB HDPE DUCTS pipe. It should conform to TEC GR with latest amendments. (See Figure-3).

8.9. Route/Joint Indicator: The Route/Joint indicators are co-located with each manhole/joint chamber. In addition Route indicators are also to be placed where route changes direction like road crossings etc. Either RCC/Pre-cast or Stone based route indicators can be used. The detailed specification and design of the same shall be as per construction specification. Generally, Stone Route indicators shall be used for the BharatNet Phase- II project.

9. Excavation of Trenches

9.1. Trenching

9.1.1. Location and Alignment of the Trench: In built up areas, the trench will normally follow the foot-path of the road except where it may have to come to the edge of the carriage way cutting across road with specific permissions from the concerned authorities maintaining the road (such permissions shall be obtained by the department as per MOU signed with respective State Govt.). Outside the built up limits the trench will normally follow the boundary of the roadside land. However, where the road side land is full of burrow pits or afforestation or when the cable has to cross culverts/ bridges or streams, the trench may come closer to the road edge or in some cases, over the embankment or shoulder of the Road (permissions for such deviations for cutting the embankment as well as shoulder of the road shall be obtained). The alignment of the trench will be decided by a responsible official of the purchaser. Once the alignment is marked, no deviation from the alignment is permissible except with the approval of purchaser. While marking the alignment only the centre line will be marked and the SI shall set out all other work to ensure that, the excavated trench is as straight as possible. The SI shall provide all necessary assistance and labour, at his own cost for marking the alignment. SI shall remove all bushes, undergrowth, stumps, rocks and other obstacles to facilitate marking the centre line without any extra charges. It is to be ensured that minimum amount of bushes and shrubs shall be removed to clear the way and the SI shall give all, consideration to the preservation of the trees.

9.1.2. The line-up of the trench must be such that PLB pipe(s) shall be laid in a straight line, both laterally as well as vertically except at locations where it has to necessarily take a bend because of change in the alignment or gradient of the trench, subject to the restrictions mentioned elsewhere.

9.1.3. Line-Up: The line-up of the trench must be such that PLB HDPE Ducts shall be laid in a straight line except at locations where it has to necessarily take a bend because of change in the alignment or gradient of the trench, subject to the restrictions mentioned elsewhere.

9.2. Method of Excavation

9.2.1. In built up areas, the SI shall resort to use of manual labour / HDD only to ensure no damage is caused to any underground or surface installations belonging to other public utility services and/or private parties.

9.2.2. However, along the Highways and cross country there shall be no objection to the SI resorting to mechanical means of excavation, provided that no underground installations existing the path of excavation, if any, are damaged.

9.2.3. There shall be no objection to resort to horizontal boring to bore a hole of required size and to push through G.I. Pipe (50 mm ID) through horizontal bore at road crossing or rail crossing or small hillocks etc.

9.2.4. All excavation operations shall include excavation and 'getting out'. 'Getting out' shall include throwing the excavated materials at a distance of at least one meter or half the depth of excavation, whichever is more, clear off the edge of excavation. In all other cases 'getting out' shall include depositing the excavated materials as specified.

9.2.5. In Rocky strata excavation shall be carried out by use of electro mechanical means like breakers/ jack hammers or by blasting wherever permissible with express permission from the competent authority. If blasting operations are prohibited or not practicable, excavation in hard rock shall be done by chiselling/ jack hammers.

9.2.6. Trenching shall as far as possible be kept ahead of the laying of pipes. SI shall exercise due care that the soil from trenching intended to be loose for back filling is not mixed with loose debris. While trenching, the SI should not cause damage to any underground installations belonging to other agencies and any damage caused should be made good at his own cost and expense.

9.2.7. Necessary barricades, night lamps, warning board and required watchman shall be provided by the SI to prevent any accident to pedestrians or vehicles. While carrying out the blasting operations, the SI shall ensure adequate safety by cautioning the vehicular and other traffic. The SI shall employ sufficient man-power for this with caution boards, flags, sign writings etc.

9.2.8. The SI should provide sufficient width at the trench at all such places, where it is likely to cave in due to soil conditions without any extra payment. A minimum free clearance of 15 cm should be maintained above or below any existing underground installation. No extra payment will be made towards this. In order to prevent damage to PLB HDPE DUCTS over a period of time, due to the growth of trees, roots, bushes, etc., the SI shall cut them when encountered in the path of alignment of trench without any additional charges.

9.2.9. In large burrow pits, excavation may be required to be carried out for more than 165 cm in-depth to keep gradient of bed less than 15 degrees with horizontal. If it is not possible as stated above, alignment of trench shall be changed to avoid burrow pit completely.

9.3. Depth and Size of the Trench

The depth of the trench from top of the surface shall not be less than 165 cm unless otherwise relaxation is granted by purchaser under genuine circumstances.

In rocky terrain, less depth shall be allowed only in exceptional circumstances with additional protection where it is not possible to achieve the normal depth due to harsh terrain/ adverse site conditions encountered. This shall be done only with the approval of the purchaser. This shall be properly documented. In all cases, the slope of the trench shall not be less than 15 degrees with the horizontal surface. The width of the trench shall normally be 45 cm at the top & 30 cm at the bottom.

In case, additional pipes (HDPE/GI/RCC Pipes) are to be laid in some stretches, the same shall be accommodated in this normal size trench.

When trenches are excavated in slopes, uneven ground and inclined portion, the lower edge shall be treated as top surface of land and depth of trench will be measured accordingly. In certain locations, such as uneven ground, hilly areas and all other Places, due to any reason whatsoever it can be ordered to excavate beyond standard depth of 165 cm to keep the bed of the trench as smooth as possible. Near the culverts, both ends of the culverts shall be excavated more than 165 cm to keep the gradient less than 15 degree with horizontal. For additional depth in excess of 165 cm, no additional payment shall be applicable.

If excavation is not possible to the minimum depth of 165 cm, as detailed above, full facts shall be brought to the notice of the purchaser in writing giving details of location and reason for not being able to excavate that particular portion to the minimum depth.

Approval shall be granted by the purchaser in writing under genuine circumstances. The decision of the purchaser shall be final and binding on the SI. All the relaxations granted as specified above shall be dealt with as per the laid down norms and procedure of purchaser.

9.3.1. Dewatering: The SI shall be responsible for all necessary arrangements to remove or pump out water from trench. The SI should survey the soil conditions encountered in the section and make his own assessment about dewatering arrangement that may be necessary. No extra payment shall be admissible for this.

9.3.2. Wetting: Wherever the soil is hard due to dry weather conditions, if watering is to be done for wetting the soil to make it loose, the same shall be done by the SI. No extra payment shall be admissible for this.

9.3.3. Blasting: For excavation in hard rock, where blasting operations are considered necessary, the SI shall obtain approval of the purchaser in writing for resorting to blasting operation. The SI shall obtain license from CHiPS for undertaking blasting work as well as for obtaining and storing the explosive as per the Explosive Act, 1884 as amended up to date and the explosive Rules, 1983. The SI shall purchase the explosives fuses, detonators, etc. only from a licensed dealer. Transportation and storage of explosive at site shall conform to the aforesaid Explosive Act and Explosive Rules. The SI shall be responsible for the safe custody and proper accounting of the explosive materials. Fuses and detonators shall be stored separately and away from the explosives. The purchaser or his authorized representative shall have the right to check the SI's store and account of explosives. The SI shall provide necessary facilities for this. The SI shall be responsible for any damage arising out of accident to workmen, public or property due to storage, transportation and use of explosive during blasting operation. Blasting operations shall be carried out under the supervision of a responsible authorized agent of the SI (referred subsequently as agent only), during specified hours as approved in writing by the purchaser. The agent shall be conversant with the rules of blasting. All procedures and safety precautions for the use of explosives drilling and loading of explosives before and after shot firing and disposal of explosives shall be taken by the SI as detailed in IS: 4081 safety code for blasting and related drilling operation.

9.3.4. Trenching Near Culverts/ Bridges: The PLB HDPE Ducts shall be laid in the bed of culvert at the depth not less than 165 cm protected by RCC pipes as decided by purchaser. Both ends of culverts shall be excavated more than 165 cm in depth to keep the gradient of not less than 15 degree with horizontal. The bed of trench should be as smooth as possible.

9.3.5. While carrying out the work on bridges and culverts, adequate arrangement for cautioning the traffic by way of caution boards during day time and danger lights at night shall be provided. In case of small bridges and culverts, where there is a likelihood of their subsequent expansion and remodelling, the cable should be laid with some curve on both sides of the culvert or the bridge to make some extra length available for readjustment of the cable at the time of reconstruction of culvert or the bridge.

10. Laying Of PLB HDPE Ducts

After the trench is excavated to the specified depth, the bottom of the trench has to be cleared of all stones or pieces of rock and levelled up properly. A layer of soft soil/or sand (in case the excavated material contains sharp pieces of rock/stones) of not less than 5 cm is required for levelling the trench to ensure that the cable when laid will follow a straight alignment. Adequate care shall be exercised while laying so that the Optical Fibre Cables are not put to undue tension/pressure after being laid as this may adversely affect the optical characteristics of cables with passage of time.

The SI shall ensure that trenching and pipe laying activities are continuous, without leaving patches or portions incomplete in between. In case intermediate patches are left, measurement of the completed portions will be taken only after work in such left over patches are also completed in all respects.

Preparatory to aligning the pipe for jointing, each length of the PLB HDPE Ducts shall be thoroughly cleaned to remove all sand, dust or any other debris that may clog, disturb or damage the Optical Fibre Cable when it is pulled at a later stage. The ends of each pipe and inside of each Socket shall be thoroughly cleaned of any dirt or other foreign materials.

After the trench is cleaned the PLB HDPE Ducts/Coil shall be laid in the cleaned trench, jointed with Sockets. Drawing up of PP rope is optional as per TEC GR. In case of use of PP Rope, at every manhole approximately at every 200 m or at bends or turns the PP rope will be tied to the HDPE end caps used for sealing the PLB HDPE Ducts, to avoid entry of rodents/mud etc.

At the end of each day work, the open ends of the pipes sections shall be tightly closed with endcaps to prevent the entry of dirt/mud, water or any foreign matter into PLB HDPE Ducts until the work is resumed. In built up area falling within Municipal/Corporation limits, the PLB HDPE Ducts shall be laid with protection using RCC Pipes/ Concreting reinforced with weld mesh (only in exceptional cases).

For lesser depths requiring additional protection in built up areas, towns and cities falling within t the municipal limits, suitable protection shall be provided to PLB HDPE pipes/coils using RCC/DWC full round/split pipes or GI pipes or cement concreting reinforced with MS weld mesh or a combination of any of these as per the site requirement. This shall be done only with the prior instructions/approval of the purchaser. The specifications for providing each of these protections are given later in this document.

Moreover, in cross country routes, if depth is less than 1.2 meters, protection by using RCC/DWC Pipe shall be provided. Purchaser shall decide about such stretches and type of protection to be provided in view of the site requirements. Normally 100 mm RCC /DWC Pipes shall be used for protecting PLB HDPE Ducts but if more than one PLB pipe is to be laid and protected, RCC/DWC Pipe of suitable size to accommodate the required number of PLB Pipes shall be used.

The PLB HDPE Ducts shall be laid in RCC Full Round spun Pipes/GI Pipes as required at Road crossings. The RCC pipes/GI pipes shall extend at least 3 meters on either side of the road at Road crossings. At Road crossings, extra GI/PLB HDPE Ducts may be laid as per the direction of the purchaser. On Rail bridges and crossings, the PLB HDPE Ducts shall be encased in suitable cast iron as prescribed by the Railway Authorities.

Wherever RCC pipes are used for protection, the gaps between the RCC collars and the RCC pipes shall be sealed using cement mortar 1:3 (1:53 grade cement of reputed brand, 3: fine sand without impurities) to bar entry of rodents. Every third collar of RCC pipes (normally of 2 meters length) and also both ends of RCC Pipes will be embedded in a concrete block of size 40 cm (L)x 40 cm(W) x 25 cm (H) of 1:2:4 cement concrete mix (1:53 grade cement of reputed brand, 2: coarse sand, 3: stone aggregate of nominal size of 20 mm) so that the alignment of RCC pipes remain firm and intact and to avoid entry of rodents.

Wherever GI pipes are used, special care should be taken to ensure that G.I. Pipes are coupled properly with the sockets so as to avoid damage to PLB pipe and eventually the Optical Fibre Cable in the event of pressure coming on the joint and G.I. Pipe joint giving its way. Rubber bushes shall be used at either ends of the GI pipes to protect PLB pipe. Both the ends of G.I. Pipe will be embedded in a concrete block of size 40 cm (L) x 40 cm ((W) x 25 cm (H) of 1:2:4 cement concrete mix (1:53 grade cement of reputed brand, 2: coarse sand, 3: stone aggregate of nominal size of 20 mm) so that the alignment of G.I. Pipes remain firm and intact and to avoid entry of rodents.

In case of protection by concreting at site, the nominal dimension of concreting shall be 250 mm x250 mm section. Cement Concrete Mixture used shall be of 1:2:4 composition i.e. 1:53 grade Cement of a reputed company, 2: Coarse Sand, 4: Graded Coarse Stone aggregate of 20 mm nominal size, reinforced with MS weld mesh. As the RCC is cast at site, it is imperative to ensure that special care is taken to see that proper curing arrangements are made with adequate supply of water. The SI shall invariably use mechanical mixer at site for providing RCC protection, to ensure consistency of the mix.

For carrying out concreting work in trenches, yellow PVC sheets of width not less than 1.0 M and of weight not less than 1 kg. Per 8 sq. meters shall be spread and nailed on sides of the trench to form trapezoidal section for concreting in the cleaned trench, to avoid seepage of water into the soil.

A bed of cement concrete mixture of appropriate width and 75 mm thickness shall be laid on the PVC sheet, before laying PLB HDPE ducts. The PLB HDPE Ducts shall then be laid above this bed of concrete. After laying the PLB HDPE Ducts, MS weld mesh is wrapped around and tied and concrete mix is poured to form the cross sectional dimensions as instructed by the purchaser.

The strength of RCC is dependent on proper curing therefore, it is imperative that water content of RCC mix does not drain out into the surrounding soil. Portions where cement concreting has been carried out shall be cured with sufficient amount of water for reasonable time to harden the surface. After curing, refilling of the balance depth of the trench has to be carried out with excavated soil.

The PLB HDPE Ducts/RCC/GI Pipes shall be laid only in trenches accepted by purchaser or his representative. The SI shall exercise due care to ensure that the PLB HDPE Ducts are not subjected to any damage or strain.

Water present in the trench at the time of laying the PLB HDPE Ducts shall be pumped out by the SI before laying the pipes in the trench to ensure that no mud or water gets into the pipes, thus choking it.

In case of nallahs, which are dry for nine months in a year, the PLB HDPE Ducts shall be laid inside the RCC Pipes laid at a minimum depth of 165 cm, as instructed by the purchaser. The mechanical protection shall extend at least 5 meters beyond the bed of nallah on either side.

Notwithstanding anything contained in clauses referred above, the purchaser may order, based on special site requirements, that the PLB HDPE Ducts may be encased in reinforced cement concrete, as detailed, in bid. While laying the pipes, a gap of 2 M is kept at convenient locations approx. 200 m apart and at the bends and turns, which will be used as manholes during Optical Fibre cable pulling. Ends of the PLB HDPE Ducts at the manholes shall be sealed using end caps after tying the PP rope to the end caps to avoid choking of the pipes. In a similar manner, manholes shall be kept while approaching bridges, road crossings etc., as instructed by the purchaser. The location of the manholes will be decided by the purchaser.

10.1. Laying Protection Pipes on Bridges and Culverts:

In case trenching and pipe laying is not possible on the culverts, the pipes shall be laid on the surface of the culverts/bridges after due permission from the purchaser. Of late the bridge construction authorities are providing channel ducts on the footpaths on the bridges for various services. The RCC/DWC/ G.I. Pipes can be laid in these ducts for pulling cables. However, for laying cables on existing bridges, where duct arrangement does not exist, one of the following methods may be adopted.

In case of the Bridges/Culverts, where there are no ducts and where the cushion on the top of the Arch is 50 cm to 100 cm or more, G.I. Pipe (Carrying PLB HDPE pipe and cable) may be buried on the top of the Arch adjoining the parapet wall, by digging close to the wheel guards. Every precaution shall be taken to see that no damage occurs to the arch of the culvert. After burying the GI pipe, the excavated surface on the arch shall be restored.

Where the thickness of the Arch is less than 50 cm, the pipe must be buried under the wheel guard masonry and the wheel guard rebuilt.

If neither of the two methods is possible, the G.I. Pipes/GI Troughs must be clamped on the parapet wall with the clamps. If necessary, the pipes may be taken through the parapet wall at the ends where the wall diverges away from the road.

Methods cited in above clauses should be carried out under close supervision of Road authorities. The surface to be concreted should be thoroughly cleaned and levelled before concreting. At both ends of the Bridges/Culverts, where the GI Pipes /GI Troughs slope down and get buried, the concreting should be extended sufficiently to ensure that no portion of the GI Pipes/GI Troughs is exposed as approved by the purchaser to protect the pipe/trough from any possible externally caused damage.

Where white wash/colour wash is existing on the Bridges/ Culverts, the same should also be carried out on the concreted portion to ensure uniformity.

11. Back Filling and Dressing of the Trench

Provided that the PLB HDPE pipes have been properly laid in the trench at the specified depth, the back filling operation shall follow as early as practicable. The earth used for filling shall be free from all roots, Grass, shrubs, vegetation, trees, saplings and any other kind of garbage or pebbles. The back filling operation shall be performed in such a manner so as to provide firm support under and above the pipes and to avoid bend or deformation of the PLB HDPE pipes when the pipes get loaded with the back filled earth.

At locations where the back filled materials contains stones/sharp objects which may cause injury to the PLB HDPE pipes and where the excavated or rock fragments are intended to refill the trench in whole or in part, the trench should be initially filled, with a layer of ordinary soil or loose earth (free from any stones/pebbles) not less than 10 cm thick over the pipes.

Back filling on public, roads, railway crossings, footpaths in city areas shall be performed immediately after laying the HDPE pipes. Back filling at such locations shall be thoroughly rammed, so as to ensure original condition so that it is safe for the road traffic. All excess soil/ material left on road/ footpath/railway crossing shall be removed by SI. However, along the highways and in country side, the excess dug up material left over after refilling should be kept in a heap above over the trench.

In city limits, at any given time not more than 50 Meters length of trench should be kept open and in all places where excavation has been done, no part of the trench should be kept open over night to avoid occurrence of any mishap or accident in darkness.

12. Restoration of Road Surface

Road restoration work to be made with bituminous macadam for semi grouting 50 mm thick and premix carpet surfacing 25 mm thick over the grouted surface (total up to 75 mm thick) including supply of asphalt etc. to evenly match the road, including consolidation and rolling as per standard specification of DSR 1997.

Road restoration work with cement concrete 1:4:8 mix for thickness varying from 150 mm to 225 mm, including supply of concrete to be made to evenly match the road.

13. Cable Pulling and Joining/Splicing

13.1. Cable Pulling

Manholes marked during PLB HDPE Ducts pipe laying of approx. size of 2.0 m length x 1.0 m width x 1.65 m depth shall be excavated for pulling the cables. There may be situations where addition manholes are required to be excavated, for some reasons, to facilitate smooth pulling of cable. Excavation of addition manholes will be carried out, without any extra cost. De-watering of the manhole, if required, will be carried out without any extra costs. Dewatering/ De-gasification of the Ducts, if required, will be carried out without any extra costs.

The Optical Fibre Cables are available in drums in lengths of approx. 2 km. The cables shall be blown / manually pulled (in exceptional cases) through already laid PLB HDPE DUCTS. This work is to be carried out under the strict supervision of site in-charge. It shall be ensured that during the blowing / pulling of Cable the tension is minimum and there is no damage to the Cable/Optical Fibres.

After pulling of the drum is completed, both ends of the PLB HDPE DUCTS pipe in each Manhole should be sealed by hard rodent resistant rubber bush, to avoid entry of rodents/mud into PLBHDPE Ducts.

The Manholes are prepared by providing 40 mm split PLB HDPE DUCTS pipe of 2.5 to 3m length and closing the split PLB HDPE Ducts by providing necessary clamps/ adhesive tape as per the directions of purchaser. Afterwards, the split/cut PLB HDPE DUCTS pipe are covered with 100 mm split RCC pipe of 2m length and sealing the ends of RCC pipe with lean cement solution for protecting bare cable in the manhole . After fixing of RCC Split Pipes necessary back filling/reinstatement and dressing of manholes should be carried out as referred under trenching. The location of the pulling manhole should be recorded for preparation of documentation.

13.2. Jointing/ Splicing

Optical Fibre Cable Joints will be at varying distances depending upon the incremental fibre to be laid for connecting Panchayats. The 48 fibres are to be spliced at every Joint & at both ends (Terminations) in the equipment room as directed by CHiPS. The Infrastructure required for cable splicing i.e. Splicing machine, OTDR, Optical talk set, Tool kit, etc. will be arranged by the SI and also any additional accessories. e. g. Engine etc. required at site for splicing will also be arranged by the SI.

The Optical Fibre Cable thus jointed end-to-end will be tested by the purchaser/ Acceptance Testing unit of the concerned purchaser for splice losses and transmission parameters as specified by CHiPS and prevalent at that time. The through Optical Fibre should meet all the technical parameters, specified and no relaxation will be granted.

13.3. Splicing of Fibres

13.3.1. Wash hands thoroughly prior to connecting this procedure.

13.3.2. Place the bare fibre inside ‘V’ groove of the splicing machine by opening clamp handle such that the end of fibre is app.1 mm. over the end of the ‘V’ groove towards the electrodes.

13.3.3. Repeat the same procedure for other fibre; however first insert heat shrink splice protector.

13.3.4. Press the start button on the splice controller.

13.3.5. The machine will pre fuse, set align both in ‘X’ and ‘Y’ direction and then finally fuse the fibre.

13.3.6. Inspect the splice on monitor if provided on the fusion splicing machine and assure no nicking, bulging is there and cores appear to be adequately aligned if the splice does not visually look good repeat the above procedure.

13.3.7. Slide the heat shrink protector over the splice and place in tube heater. Heat is complete when soft inner layer is seen to be ‘oozing’ out of the ends of the outer layer of the protector.

13.3.8. Repeat the same procedure for all the other fibres.

13.4. Organizing Fibre and Finishing Joint

13.4.1. After each fibre is spliced, the heat shrink protection sleeve must be slipped over the bare fibre before any handling of fibre takes place, as uncoated fibres are very brittle and cannot withstand small radius bends without breaking.

13.4.2. The fibre is then organized into its tray by coiling the fibres on each side of the protection sleeve using the full tray side to ensure the maximum radius possible for fibre coils.

13.4.3. The tray is placed in the position.

13.4.4. OTDR reading taken for all splices in this organized state and recorded on the test sheet to confirm that all fibres attenuation are within 0.1 db per splice. This OTDR test will confirm fibres were not subjected to excessive stress during the organizing process.

13.4.5. After this the joint can be closed with necessary sealing etc. and ready for placement in the pit

13.5. Placing of Completed Joint in Pit

13.5.1. Joint is taken out from the vehicle and placed on the tarpaulin provided near the pit.

13.5.2. The cable is laid on the ground, loop the cable such that pen mark previously placed on the cable line up. Tape these loops together at the top of the coil.

13.5.3. The joint can now be permanently closed and sealed mechanically. However, before closing, silica gels to be kept inside for moisture protection.

13.5.4. Now the joint closure is fixed to the bracket on the pit wall and pit is closed.

14. Construction of Jointing Chamber

The joint chambers are provided at every joint to keep the Optical Fibre Cable joint well protected and also to keep extra length of cable, which may be, required to attend the faults at a later date. Jointing chambers are to be prepared normally at distance of every 2 km. Actual location of jointing chamber depends on length of cable drum and appropriateness of location for carrying out jointing work. The location is finalized by purchaser.

The jointing chambers are constructed by way of fixing pre-cast RCC chambers/Brick Chambers and covers as per the instructions from purchaser.

14.1. Pre Cast RCC Chamber

For fixing pre cast RCC chamber, first a pit of size 2 m x 2 m x 1.8 m depth shall be required to be dug. Pre cast RCC chamber shall consist of three parts (i) round base plate of 120 cm diameter and 4 cm thickness in two halves (ii) full round RCC joint chamber with diameter of 110 cm and height of 50 cm and thickness of 5 cm (iii) round top cover will be in two halves with diameter of 120 cm and thickness of 4 cm having one handle for each half in centre and word 'CG OFC' engraved on it. (See figure '4'). After, fixing the pre cast RCC joint chamber, the joint chamber is filled with clean sand before closing. Back filling of joint chamber pit with excavated soil shall be carried out in the end.

Brick Chamber

For constructing brick chamber, first a pit of size 2m x2 mx1.8 m depth is shall be required to be dug, then, base of the chamber shall be made using concrete mix of 1:5:10 (1 cement, 5 coarse sand, 10 graded stone aggregate of 40mm nominal size) of size of 1.7m x 1.7 m and 0.15 m thickness. Wall of brick chamber should be constructed on this base having wall thickness of 9" using cement mortar mix of 1:5 (1: cement, 5: fine sand). The chamber should have internal dimensions of 1.2 m x 1.2 m and 1 m height. The bricks to be used for this purpose should be of size 9" x 4.5" x 3", best quality available and should have smooth rectangular shape with sharp corners and shall be uniform in colour and emit clear ringing sound when struck.

The joint chamber should be so constructed that PLB pipe ends remain protruding minimum 5 cm inside the chamber on completion of plastering. The PLB pipes should be embedded in wall in such a way that, the bottom brick should support the pipe and upper brick should be provided in a manner that PLB HDPE pipe remains free from the weight of the construction. The joint chamber should be plastered on all internal surfaces and top edges with cement mortar of 1:3 (1: cement, 3: coarse sand), 12 mm thick finished with a floating coat of complete cement as per standard. Pre-cast RCC slab with two handles to facilitate easy lifting, of size 0.7 m x 1.4 m and of thickness of 5 cm having one handle for each half in centre and word ''OFC' engraved on it are to be used to cover the joint chamber. Two numbers of such slabs are required for one joint chamber. This pre-cast slab should be made of cement concrete mix of 1:2:4 (1: cement, 2: coarse sand, 4: stone aggregate 6 mm nominal size) reinforced with steel wire fabric 75 x 25 mm mesh of weight not less than 7.75 Kg per sq. Meter. The joint chamber is filled with clean sand before closing. Back filling of joint chamber pit with excavated soil shall be carried out in the end

15. Fixing of Route Indicators / Joint Indicators

Pits shall be dug around every 250 m towards jungle side at every Manhole and Jointing chamber for fixing of Route/Joint Indicator. In addition, Route Indicators are also required to be placed where Optical Fibre Cable changes directions like road crossing etc. The pits for fixing the indicator shall be dug for a size of 60 cm x 60 cm and 75 cm (depth).

The indicator shall be secured in upright position by ramming with stone and murrum up to a depth of 60 cm and concreting in the ratio of 1:2:4 (1: cement, 2: coarse sand, 4 stone aggregate 20 mm nominal size) for the remaining portion of 15 cm. Necessary curing shall be carried out for the concreted structure with sufficient amount of water for reasonable time to harden the structure.

15.1. RCC/Pre cast Route Indicators

The route /joint indicator made of pre-cast RCC should have the following dimensions:

Base - 250 mm x 150 mm Top - 200 mm x 75 mm Height - 1250 mm (See Figure '5')

15.2. Stone based Route Indicators

The route /joint indicators made of Sand/lime Stone Should have the following dimension. The word 'CG OFC' should be engraved on the Route/Joint indicators.

i. Stone to be used (Sand/lime Stone)
ii. Indicator Top surface to be rounded
iii. Base 155 mm × 100 mm
iv. Upper 500 mm length to be Tapered width wise as shown in the drawing and homogeneously finished.
v. Height 650mm (Straight) + 400 mm (Tapered)
vi. The route indicators should be engraved with word ‘OFC’ of size 80mm length & 50mm, width.
vii. Length 3.5 Ft., top 4”x4” dressed 1Ft. from top & tapered.

(See figure ‘6’ for details of Stone Route Indicators)

The Route indicators shall painted Blue and placed at 500 to 1000 cm away from the centre of the trench towards jungle side. The Joint indicators are placed at OFC joints and placed 500 to 1000 cm away from wall of the joint chamber facing jungle side and are painted Grey. The engraved word “CG OFC” should be painted in white, on route as well as joint indicators. Numbering of route indicators/joint indicators should also be done in white paint. The numbering scheme for route indicators will be Joint No. /Route Indicator No. for that joint. For example, 2/6 marking on a route indicator means 6th route indicator after 2ndjoint. Additional joints on account of faults at a later date should be given number of preceding joint with suffix A, B, C, and D. For example sign writing 2A on a joint indicator means, additional joint between joint No. 2 and 3. The numbering of existing route/joint indicator should not be disturbed on account of additional joints. Enamel paints of reputed brand should be used for painting and sign writing of route as well joint indicators.

The route and joint indicator shall be painted with primer before painting with oil paint. The material used should bear ISI mark. The size of each written letter should be at least 3.5 cm. The colours of painting and sign writing is as under:

i. For Joint Indicator: Grey color
ii. For Route Indicator: Blue color
iii. For CHiPS OFC & Nos: White color

16. Documentation

The documentation, consisting of the following shall be prepared for each Block and Gram Panchayat connected to the Block. 4 sets of documentation shall be provided both in electronic format on CD as well as hard bound copy.

16.1. **As Build Diagrams.** The ABD shall be an essential part of the documentation process and shall ensure easy maintenance of the OFC route. The ABD shall be separately prepared for each route. The ABD will contain the following details: -

16.1.1. Cable Details

- Make and size of the cable.
- Identification of cable drums deployed.
- Test reports.

16.1.2. Joint Details

- Location of Joint Chamber (lat/long details in decimal degree format up to eight decimal places).
- Depth of Joint Chamber covers from ground level.
- Details of cable slack left at each joint chamber. Splice loss details.

16.1.3. FDMS Details

- Details of FDMS deployment.
- OTDR measurements for each fibre terminated at FDMS.

- Fibre Connectivity Details.
- Route marker Details
  - Location of Cement / Electronic Route Marker (lat/long details in decimal degree format up to eight decimal places)
  - Route Marker Identification details.

16.1.4. Optical Test results

- OTDR readings.
- Light Source Power Meter readings.
- Chromatic Dispersion measurement readings.
- Optical test results for each fibre shall be furnished.
- OTDR settings to include wavelength, pulse width and average time shall also be recorded.
- Bi-directional OTDR readings shall be furnished for each Joint Closure.
- OFC Alignment Details
- Offset of cable from centre of the road at every 10 meters.
- Details of crossings (road/rail/nala etc) should be provided.
- Depth profile of Cable at every 10 Meter interval.
- Details of protection with type of protection.
- Location of culvert and bridges with their lengths and scheme of laying of HDPE/PLB pipe thereon.
- Important landmarks to facilitate locating the cable in future to include culverts, houses, petrol pumps, schools etc.
- Location of Joints and manholes.
- Location of Cement and Electronic Route Markers.
- The OFC alignment details shall be provided at multiple Levels of Detail (LOD) to facilitate section wise identification of route. While one sheet shall contain the entire link the LOD would gradually increase to upto 400m on a single sheet.
- Diagram shall consist of Cable Route Details on Geographical Map drawn to scale with prominent land marks and alignment of cable with reference to road.
- Indexing and referencing of the diagrams shall be made to provide easy identification of route details.

16.1.5. Landmarks along route

- All the important landmarks for verification along the OFC route corridor would be recorded as part of the ABD. The corridor would include all verifiable landmarks upto 25m on each side along the center of the road.
- All landmarks recorded as part of the ABD would be provided along with accurate lat long details in decimal degree upto eight decimal places.
- The broad category of landmarks to be recorded would be as given below:-
- Milestones.
- Culverts.
- Bridges.

- Important building footprint.
- Road Features.

16.1.6. ABD shall provide the basis for carrying out GIS mapping of the entire OFC route alignment. The GIS details required to be recorded as part of the ABD shall be measured using DGPS system (providing sub meter accuracy) and Total Stations. Readings recorded through these systems would use accurate control stations and shall be digitally corrected by using appropriate software. ABD shall be generated using CAD tool(s).

16.1.7. Accuracy of details provided as part of the ABD will be very critical for successful implementation of the GIS based Optical Fibre NMS being procured through a separate tender. A penalty of Rs 50,000.00 will be levied on the Bidder for each inaccurate recording of lat/long and distance details in the ABD and discovered during the ATP.

16.1.8. All the diagrams shall bear the signatures of the contractor and the project manager as a proof of accuracy of the details. The diagrams shall be bound in A-4 size book with cover. The cover sheets shall be laminated and should have the following details.

- Name of the Project Organization.
- Name of the OFC Link with ID.
- Name of the Contractor.
- Name of CHiPS Rep as part of Acceptance Test.
- Name of PICG Rep as part of Acceptance Test.
- Date of commencement of work.
- Date of completion of work.
- For each route/section 6 sets of above mentioned document shall be submitted.

16.2. **Route Index Diagrams –** General: This diagram shall consist of Cable Route Details on Geographical Map drawn to scale with prominent land marks and alignment of cable with reference to road. This shall be prepared on A-3 sheets of 80 GSM.

16.3. Route Index Diagrams –Profile

These diagrams will contain

i. Make and size of the cable.
ii. Offset of cable from centre of the road at every 10 meters
iii. Depth profile of Cable at every 10 meter;
iv. Details of protection with type of protection depicted on it;
v. Location of culvert and bridges with their lengths and scheme of laying of PLB HDPE Ducts pipe thereon.
vi. Important landmarks to facilitated locating the cable in future; Location of Joints and pulling manholes.

These diagrams shall be prepared on A-4 sheets of 80 GSM. On one sheet profile of maximum 400 meters shall be given to ensure clarity.

16.4. Joint Location Diagram

This diagram will show

i. Geographical location of all the joints.
ii. Depth of Joint Chamber covers from ground level
iii. Type of chamber (Brick/Pre-cast)
iv. Length of Optical Fibre Cable kept inside the joint chamber from either

direction. This shall be prepared on A-4 sheets of 80 GSM.

All the diagrams (1), (2) & (3) shall bear the signatures of the SI, and the purchaser as a proof of accuracy of the details. The diagrams shall be bound in A-4 size book with cover.

The cover sheets shall be of 110 GSM and laminated. The front cover shall have the following details.

i. Name of the State/District/Block
ii. Name of the Panchayats connected
iii. Name of the CG/CHiPS with logo
58.1.1 Name of the SI
iv. Date of commencement of work
v. Date of completion of work

For each Block 1 sets of above mentioned document shall be submitted to purchaser.

17. SAFETY PRECAUTIONS

17.1. Safety Precautions when excavating or working in excavations close to electric cables

The purchaser of the work will assist SI to get full information from electricity undertaking regarding any electric cables, which are known or suspected to exist near the proposed excavation and unless this is done, excavation should not be carried out in the section concerned. The electricity undertaking should be asked to send a representative and work should be preceded with close consultation with them.

Only wooden handled hand tools should be used until the electric cables have been completely exposed. Power Cables, not laid in conduits, are usually protected from above by a cover slab of concrete, brick or stone. They may or may not be protected on the sides. It is safer, therefore, always to drive the point of the pick axe downwards then uncovering a cable, so that there is less chance of missing such warning slabs. No workman should be permitted to work alone where there are electric cables involved. At least one more man should be working nearby so that help can be given quickly in case of an accident. If disconnection of power could be arranged in that section it will be better. No electric cables shall be moved or altered without the consent of the Electric Authority and they should be contacted to do the needful. If an electric cable is damaged even slightly, it should be reported to the Electric Authority and any warning bricks disturbed during excavation should be replaced while back filling the trench. Before driving a spike into the ground, the presence of other underground properties should be checked. Information on plans regarding the location of power cables need not to be assumed as wholly accurate. Full precautions should be taken in the vicinity until the power cable is uncovered. All electric cables should be regarded as being live and consequently dangerous. Any power is generally dangerous, even low voltage proving fatal in several cases.

17.2. Electric shock-Action and treatment

Free the victim from the contact as quickly as possible. He should be jerked away from the live conductors by dry timber, dry rope or dry clothing. Care should be taken not to touch with bare hands as his body may be energized while in contact. Artificial respiration should begin immediately to restore breathing even if life appears to be extinct. Every moment of delay is serious, so, in the meanwhile, a doctor should be called for.

17.3. Safety Precautions while working in public street and along railway lines

Where a road or footpath is to be opened up in the course of work, special care should be taken to see that proper protection is provided to prevent any accidents from occurring. Excavation work should be done in such a manner that it will not unduly cause inconvenience to pedestrians or occupants of buildings or obstruct road traffic. Suitable bridges over open trenches should be so planned that these are required for the minimum possible time. Where bridges are constructed to accommodate vehicular traffic and is done near or on railway property, it should be with the full consent and knowledge of the competent railway authorities.

17.4. Danger from falling material

Care should be taken to see that apparatus, tools or other excavating implements or excavated materials are not left in a dangerous or insecure position so as to fall or be knocked into the trench thereby injuring any workman who may be working inside the trench.

17.5. Care when working in Excavations

Jumping into a trench is dangerous. If it is deep, workmen should be encouraged to lower themselves. Workers should work at safe distance so as to avoid striking each other accidentally with tools. If the walls of the trench contain glass bits, corroded wire or sharp objects they should be removed carefully. If an obstruction is encountered, it should be carefully uncovered and protected if necessary. If an obstruction is encountered, it should be carefully uncovered and protected if necessary. Care must be taken to see that excavated material is not left in such a position that it is likely to cause any accident or obstruction to a roadway or waterway. If possible the excavated material should be put between the workmen and the traffic without encroaching too much on the road.

17.6. Danger of cave in

When working in deep trenches in loose soil, timbering up/shoring the sides will prevent soil subsidence. The excavated material should be kept at sufficient distance from the edge of the trencher pit. Vehicles or heavy equipment must not be permitted to approach too close to the excavation.

When making tunnelled opening, it should be ensured that the soil is compact enough to prevent cave in even under adverse conditions of traffic. Extra care should be taken while excavating near the foundations of buildings or retaining walls. In such cases, excavation should be done gradually and as far as possible in the presence of the owners of the property.

17.7. Protection of Excavations

Excavations in populated areas, which are not likely to be filled up on the same day should be protected by barriers or other effective means of preventing accidents and the location of all such openings must in any event be indicated by red flags or other suitable warning signs. During the hours from dusk to dawn, adequate number of red warning lamps should be displayed. Supervisory officers should ensure that all excavations are adequately protected in this manner as serious risk and responsibility is involved. Notwithstanding adoption of the above mentioned precautions, works involving excavations should be so arranged as to keep the extent of opened ground and the time to open it to a minimum.

17.8. Precautions while working on roads

The period between half an hour after sun-set and half an hour before sunrise, and any period of fog or abnormal darkness may also be considered as night for the purpose of these instructions, for the purpose of providing the warning signs.

Excavation liable to cause danger to vehicles or the public must at all times be protected with fencing of rope tied to strong uprights or bamboo poles at suitable height or by some other effective means. Any such temporary erection which is likely to cause obstructions and which is not readily visible should be marked by posts carrying red flags or boards with a red background by day and by continuously lighted lamps at night.

The flags and the lamps should be placed in conspicuous positions so as to indicate the pedestrians and drivers of vehicles the full expanse i.e. both width and length of the obstruction. The distance between lamps or between floors should not generally exceed 1.25 m along the width and 6m along length of the obstruction in non-congested areas, but 4 meters along the length in congested areas.

If the excavation is extensive, sufficient notice to give adequate warning of the danger, should be displayed consciously not less than 1.25 m above the ground and close to the excavation. Where any excavation is not clearly visible for a distance of 25m to traffic approaching from any direction or any part of the carriage way of the road in which the excavation exists, a warning notice should be placed on the kerb or edge of all such roads from which the excavation or as near the distance as is practicable but not less than 10 m from the junction of an entering or intersecting road in which the excavation exists. All warnings, in these should have a red background and should be clearly visible and legible. All warning lamps should exhibit a red light, but white lights may be used in addition to facilitate working at night. Wherever required a passage for pedestrians with footbridge should be provided. At excavations, cable drums, tools and all materials likely to offer obstructions should be properly folded round and protected. This applies to jointer’s tents as well. Leads, hoses etc. stretched and across the carriageway should be guarded adequately for their own protection and also that of the public.

17.9. Traffic Control

The police authorities are normally responsible for the control of traffic and may require the setting up of traffic controls to reduce the inconvenience occasioned by establishment of a single line of traffic due to restriction in road width or any other form of obstruction caused by the work. As far as possible, such arrangements should be settled in advance. If there are any specific regulations imposed by the local authorities, these should be followed.

17.10. Work along Railway Lines

Normally all works at Railway crossing is to be done under supervision of the railway authorities concerned, but it is to be borne in mind that use of white, red or green flags by the Departmental staff is positively forbidden to be used when working along a railway line as this practice may cause an accident through engine drivers mistaking them for railway signals. When working along double line of railway, the men should be warned to keep a sharp look on both the “UP” and’ DOWN” lines to avoid the possibility of any accident when trains pass or happen to cross one another near the work spot.

17.11. Procedure and Safety Precautions for use of explosives during blasting for trenching

In areas where the cable trench cannot be done manually on account of boulders and rocks, it is necessary to blast the rocks by using suitable explosives. The quality of explosive to be used depends on the nature of the rocks and the kind of boulders. A few types of explosive fuses and detonators normally used for making trenches for cable works are detailed below:

i. Gun powder
ii. Nitrate Mixture
iii. Gelatine
iv. Safety fuse
v. Electric Detonator
vi. Ordinary Detonator

17.11.1. Procedure

A detailed survey of the route is to be done to assess the length of the section where trenching is to be done with the help of blasting. A route diagram of the rocky section may be prepared indicating the length of the route where the explosives are to be used. For the purpose of obtaining license, a longer length of route should be given in the application as in many cases, after digging, rocks appear which was not initially anticipated.

Next a license will have to be obtained for use and storing of explosive in that section. If the area falls under a police commissioner, the authority for granting such license is the police commissioner of the concerned area. When the route does not fall in the jurisdiction of a police commissioner, the authority for issuing license is the District Magistrate.

The concerned authority from purchaser should be applied in prescribed form with a route map. The concerned authority will make an enquiry and issue license for using/storing explosives for cables trenching work. Such license will be valid for 15 days only. The license should be got renewed if the blasting operation needs to be extended. Once the license is granted, it is the responsibility of the holders of the license for the proper use of explosives, its transportation and storing.

17.11.2. Method of using

The safest explosive is the Gelatin and electric detonator. Gelatin is in the form of a stick. Electric detonator is a type of fuse used for firing the explosive electrically. Holes are made at suitable intervals on rocky terrain or boulders either by air compressor or by manual chipping. The depth of the holes should be 2 to 3 ft. Fill up the holes with small quantity of sand for about 6”. First the electric detonator is to be inserted into the Gelatin and the Gelatin is to be inserted into the holes keeping the + ve and - ve wirings of electric detonators outside the holes. Again refill the holes with sand. These +ve and-ve insulated wires of detonator are than extended and finally connected to an EXPLODER kept at a distance of not less than 100 m.

Now the explosive is ready for blasting. But, before connecting wires to exploder for blasting, all necessary precautions for stopping the traffic, use of red flags, exchange of caution signals, etc. should be completed and only then Exploder should be connected and operated.

17.11.3. Operation of exploder (IDL schaffler type 350 type exploder)

The type 350 blasting machine consists of a bearing block with blasting machine system and the explosion proof light- alloy injection moulded housing. The exploder is held with the left hand. The twist handle is applied to the drive pin, clapped with the right hand turned in the clock wise direction in continuous measurements at the highest speed from the initial position until it reached to a stop. At this stage an indication lamp will glow. When the indication lamp glows, “press button switch” should be pressed. This will extend the electric current to detonator and Gelatin will be detonated. The rock will be blasted out of the trench. Number of holes can be blasted in a single stroke by connecting all such detonators in series connection and finally to the exploder. After blasting, again mazdoors are engaged on the work to clear the debris. It the result of the first blasting is not satisfactory; it should be repeated again on the same place.

17.11.4. Warning

There may be two reasons for unsatisfactory results of the blasting

I. Misfire of Gelatin due to leakage of current from detonator.

II. Over loading because of overburdens.

Never pull the broken wire pieces form the holes in such cases. Attempt should not be made to reblast the misfired Gelatin. The safest way is to make a fresh hole by its side and put fresh Gelatin in that hole and blast it.

17.12. Precautions

The abstract of Explosives Rules 1983 which are relevant to our work is given below:

17.12.1. Restriction of delivery and dispatch of explosives

No person shall deliver or dispatch any explosives to anyone other than a person who is the holder of a license to possess the explosives or the agent of a holder of such a license duly authorized by him in writing on his behalf?

OR

Is entitled under these rules to possess the explosives without a license.

The explosives so delivered or dispatched shall in no case exceed the quantity, which the person to whom they are delivered or dispatched is authorized to possess with or without a license under these rules.

No person shall receive explosives form any person other than the holder of a license granted under these rules. No person shall receive from or transfer explosives to any person for a temporary storage or safe custody in a licensed premise unless prior approval is obtained from the Chief Controller.

A person holding license for possession of explosives granted under these rules shall store the explosives only in premised specified in the license.

17.12.2. Protection from Lightening During Storing

Every magazine shall have attached there to one or more efficient lightening conductors designed and erected in accordance with the specification laid down in Indian Standard Specifications No.2309 as amended from time to time. The connections to various parts of earth resistance of the lightening conductor terminal on the building to the earth shall be tested at least once in every year by a qualified electrical engineer or any other competent person holding a certificate of competency in this behalf from the State Electricity Department. A certificate showing the results of such tests and the date of the last test shall be hung up in conspicuous place in the building.

17.12.3. Precautions during thunder-storm

When a thunder- storm appears to be imminent in the vicinity of a magazine or store house every person engaged in or around such magazine and store house shall be withdrawn to a safe distance from such magazine or store house and the magazine and store house shall be kept closed and locked until the thunder storm has ceased or the threat of it has passed.

17.12.4. Maintenance of records

Every person holding a license granted under these rules for possession, sale or use of explosives shall maintain records in the prescribed form and shall produce such record on demand to an Inspection Officer.

17.12.5. Explosives not to be kept in damaged boxes

The licensee of every magazine or store house shall ensure that, the explosives are always kept in their original outer package. In case, the outer package gets damaged so that the explosive contained therein cannot be stored or transported, such explosives shall be repacked only after the same are examined by controller of explosives.

17.12.6. Storage of explosives in excess of the licensed quantity

The quantity of any kind of explosives kept in any licensed magazine or store house shall not exceed the quantity entered in the license against such kind of explosives. No explosives in excess of the licensed quantity shall be stored in the magazine or store house unless a permit in this behalf is obtained from the licensing authority by a letter or telegram.

17.12.7. Precautions to be observed at Site

The electric power at the blasting site shall be discontinued as far as practicable before charging the explosives. No work other than that associated with the charging operations shall be carried out within 10 meters of the holes unless otherwise specified to the contrary by the licensing authority.

When charging is completed, any surplus explosive detonators and fuses shall be removed from the vicinity of the hole and stored at a distance which should prevent accidental detonation in the event of a charge detonating prematurely in any hole. The holes which have been charged with explosive shall not be left unattended till the blasting is completed. Care shall be taken to ensure that fuse or wires connected to the detonation are not damaged during the placing of stemming materials and tamping.

17.12.8. Suitable warning procedure to be maintained

The licensee or a person appointed by the licensee to be in charge of the use of explosives at the site shall lay down a clear warning procedure consisting of warning signs and suitable signals and all persons employed in the area shall be made fully conversant with such signs and signals.

17.12.9. Precautions to be observed while firing

The end of the safety fuse (if used in place of a detonator should be freshly cut before being lighted. The exploders shall be regularly tested and maintained in a fit condition for use in firing. An exploder shall not be used for firing a circuit above its rated capacity. The electric circuits shall be tested for continuity before firing. All persons other than the shot-firer and his assistant, if any, shall be withdrawn from the site before testing the continuity.

For the purpose of jointing, the ends of all wires and cables should have the insulation removed fora maximum length of 5 cm. and should, then be made clear and bright for a minimum length of 2.5 cm. and the ends to be joined should be twisted together so as to have a positive metal contact.

Then these should be taped with insulation to avoid leakage when in contact with earth. In case of blasting with dynamite or any other high explosive, the position of all the bore holes to be drilled shall be marked in circles with white paint. These shall be inspected by the SI’s agent. Bore holes shall be of a size that the cartridge can easily pass down. After the drilling operation, the agent shall inspect the holes to ensure that drilling has been done only at the marked locations and no extra hole has been drilled. The agent shall then prepare the necessary charge separately for each bore hole. The bore holes shall be thoroughly cleaned before a cartridge is inserted. Only cylindrical wooden tamping rods shall be used for tamping. Metal rods or rods having pointed end shall never be used for tamping. One cartridge shall be placed in the bore hole and gently pressed but not rammed down. Other cartridges shall then be added as may be required to make up the necessary charge for the bore hole. The top most cartridge shall be connected to the detonator which shall in turn be connected to the safety fuses of required length. All fuses shall be cut to the length required before being inserted into the holes. Joints in fuses shall be avoided.

Where joints are unavoidable, a semi-circular niche shall be cut in one piece inserted into the niche. The two pieces shall then be wrapped together with string. All joints exposed to dampness shall be wrapped with rubber tape.

The maximum of eight bore holes shall be loaded and fired at one occasion. The charges shall be fired successively and not simultaneously. Immediately before firing, warning shall be given and the agent shall see that all persons have retired to a place of safety. The fuses of the charged holes shall be ignited in the presence of the agent, who shall see that all the fuses are properly ignited.

Careful count shall be kept by the agent and other of each blast as it explodes. In case all the charged bore holes have exploded, the agent shall inspect the site soon after the blast but in case of misfire the agent shall inspect the site after half an hour and mark red crosses (X) over the holes which have not exploded. During this interval of half an hour, nobody shall approach the misfired holes. No driller shall work near such bore until either of the following operations has been done by the agent for the misfired boreholes.

The SI's agent shall very carefully (when the tamping is a damp clay) extract the tamping with a wooden scraper and withdraw the primer and detonator.

The holes shall be cleaned for 30 cm of tamping and its direction ascertained by placing a stick in the hole. Another hole shall then be drilled 15 cm away and parallel to it. This hole shall be charged and fired. The misfired holes shall also explode along with the new one.

Before leaving the site of work, the agent of one shift shall inform the agent relieving him for the next shift, of any case of misfire and each such location shall be jointly inspected and the action to be taken in the matter shall be explained to the relieving agent. CHiPS shall also be informed by the agent of all cases of misfire, their cause and steps taken in that connection.

17.12.10. General Precautions

For the safety of persons red flags shall be prominently displayed around the area where blasting operations are to be carried out. All the workers at site, except those who actually ignite the fuse, shall withdraw to a safe distance of at least 200 metres from the blasting site. Audio warning by blowing whistle shall be given before igniting the fuse.

Blasting work shall be done under careful supervision and trained personnel shall be employed. Blasting shall not be done within 200 meters of an existing structure, unless specifically permitted by the purchaser in writing.

17.12.11. Precautions against misfire

The safety fuse shall be cut in an oblique direction with a knife. All saw dust shall be cleared form inside of the detonator. This can be done by blowing down the detonator and tapping the open end. No tools shall be inserted into the detonator for this purpose. If there is water present or if the borehole is damp, the junction of the fuse and detonator shall be made water tight by means of tough grease or any other suitable material. The detonator shall be inserted into the cartridge so that about one-third of the copper tube is left exposed outside the explosive. The safety fuse just above the detonator shall be securely tied in position in the cartridge. Waster proof fuse only shall be used in the damp borehole or when water is present in the borehole. If a misfire has been found to be due to defective fuse, detonator or dynamite, the entire consignment from which the fuse, detonator or dynamite was taken shall be inspected by the purchaser or his authorized representative before resuming the blasting or returning the consignment.

17.12.12. Pre caution against stray currents

Where electrically operated equipment is used in locations having conductive ground or continuous metal objects, tests shall be made for stray current to ensure that electrical firing can proceed safely.

18. ALLIED ACTIVITIES

18.1. Storing/Warehousing of Materials: SI will be responsible for storing and warehousing of all the material and accessories, but not limited to, supplied by him at his own cost. No storing/warehouse shall be provided by purchaser.

18.2. Transportation of Materials: The SI shall be responsible for transporting the materials, to be supplied to execute the work under the contract, to site at his/ their own cost. The costs of transportation are subsumed in the standard quoted Rates and therefore no separate charges are payable on this account.

18.3. Disposal of Empty Cable Drums: The SI shall be responsible to dispose of the empty cable drums after laying of the cables. The cost of various sizes of empty cable drums recoverable from the SI will be fixed taking into account the prevailing market rates.

18.3.1. It shall be obligatory on part of the SI to dispose of the empty cable drums at his/their level and the amount fixed for various empty cable drums shall be recovered from the bill for the work for which the drum (s) was/were issued or from any other amount due to the SI or the Security Deposit.

18.3.2. The SI shall not be allowed to dump the empty cable drums in Govt. /Public place which may cause inconvenience to public. If the SI does not dispose of the empty cable drums within 3 days of becoming it empty, the purchaser shall be at liberty to dispose-off the drums in any manner deemed fit and also recover the amount fixed in this contract from the bill/security deposit/ any other amount due to the SI.

18.4. Supply of Materials: There are some materials (Accessories) other than as mention in BOQ required to be supplied by the SI for execution of work under this contract like Bricks, Cement, Wire Mesh and Steel for protection, etc., besides using other consumables which do/don’t become the part of the asset. The SI shall ensure that the materials supplied are of best quality and workmanship and shall be strictly in an accordance with the specifications.

18.5. Social auditing: While carrying out the execution work of cable/Eqpt. ,videography may be carried out on sample basis for duration of 15 to 30 minutes per Gram Panchayat which may also involve the local people of the Gram Panchayats and villages including the Gram Panchayat Pradhan (If possible) and same may be submitted in a form of CD along with the documentation sets for information.

Note: All the materials as above have to be TSEC/Type approved by BSNL QA/TEC against mentioned TEC GR or as per the approval procedure of purchaser for which TEC GR not there.

<u>Penalty for Deviation from Standard Engineering Instructions</u>

Underground Laying

Normally depth of the trench should 1.65 m in normal & mix soil and 1.2m in hard soil. Deviations due to field conditions will be required to have necessary protections in case of less depth. The cases and solutions are as following.

- Minimum depth of burial in general shall be 1.65m
- In rocky area (including Murrum & soil mixed with stone or soft rock) depth of burial shall be 1.2m at the minimum.
- In case of utility where depth is 90 to 120 cm then DWC protection is to be used in normal/mix soil case.
- In some areas where the depth is 60cm, in those cases reinforced concrete casing of 4"(Four inch) round should be provided.
- For hard strata/rock soil layer for 60 to 90 cm cases DWC with wire mess and PCC is to be used. However, for depth relaxation photograph (with GPS) proof and justification is required.

Above ground installation of ducts shall be limited to culvert and bridge crossings only. At such locations, ducts shall be installed inside GI pipe or HDPE DWC pipes with metal sheet protection (GI sheet wrapping) of appropriate size (4” to 6”) suitable for number of ducts to be installed

The relaxation by the competent authority prescribed below shall be obtained giving reasons for not achieving standard depth;

Size/Type of Cable	Standard Depth cms.	Minimum acceptable Depth without Relaxation	Powers delegated for Relaxation, For depth upto	
			Designated CHiPS appointed Officer -1	Designated CHiPS appointed Officer -2
OFC	165	90%	80%	30% As per latest EI and latest instructions with protection

In case, SI does not adhere to the mentioned Engineering Instructions (Annexure 17) and does not provide requisite protection, then the SI is liable to penalty as per below;

Depth between	Reduction in rates	Rate Payable
<165cm. to ≥ 150cm.	5% of approved rates	95% of the approved rates for the achieved depth
<150cm. to ≥ 130cm.	12.5% of approved rates	87.5% of the approved rates for the achieved depth
<130cm. to ≥ 100cm.	25% of approved rates	75% of the approved rates for the achieved depth
<(Bellow)100 cm.	40% of approved rates	60% of the approved rates for the achieved depth

Note: In case of depth below 1.2m, instructions as per latest EI and instructions for protection etc. will be followed.

Assuming that the standard depth required is 165 cm and the rate approved is Rs.100/- for the standard depth, then as per the above formula, for a depth of 100 cm the rate worked out is

Rate Applicable = 100X0.75 = Rs.75/- per running meter

Actual amount to be paid = (100/165) X75 = Rs.45.455/-

= Rs.45.5/- per running meter

Figure 1

HDPE END CAPS

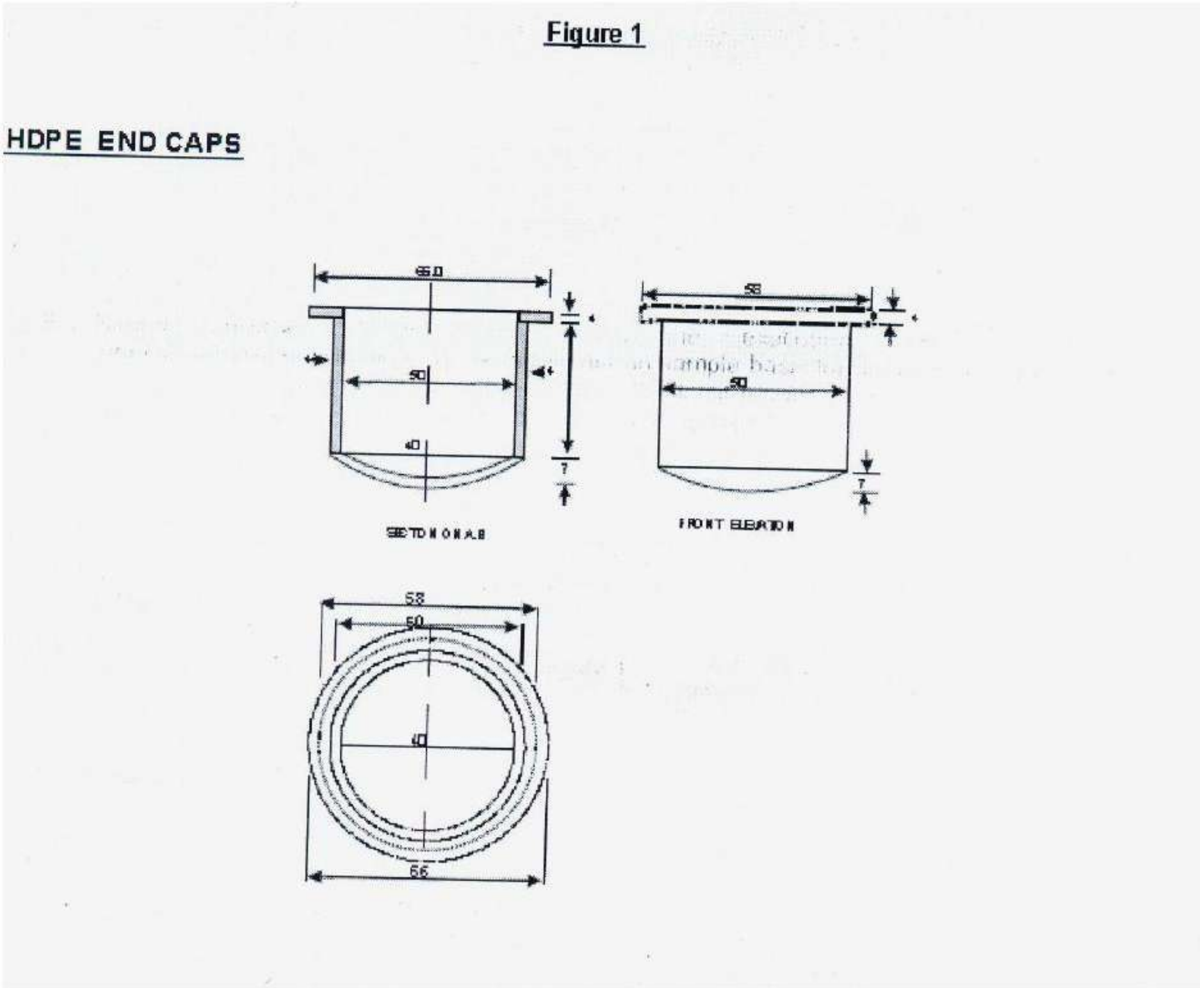
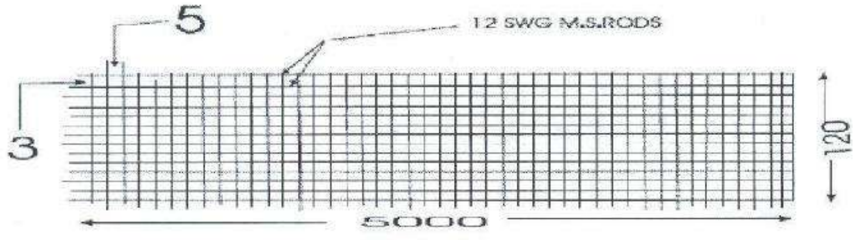


Figure2

M.S. WELDMESH

DETAILS OF 100 MM X 50 MM, 12 SWG MILD STEEL WELD MESH HAVING WIDTH OF 120 CM.



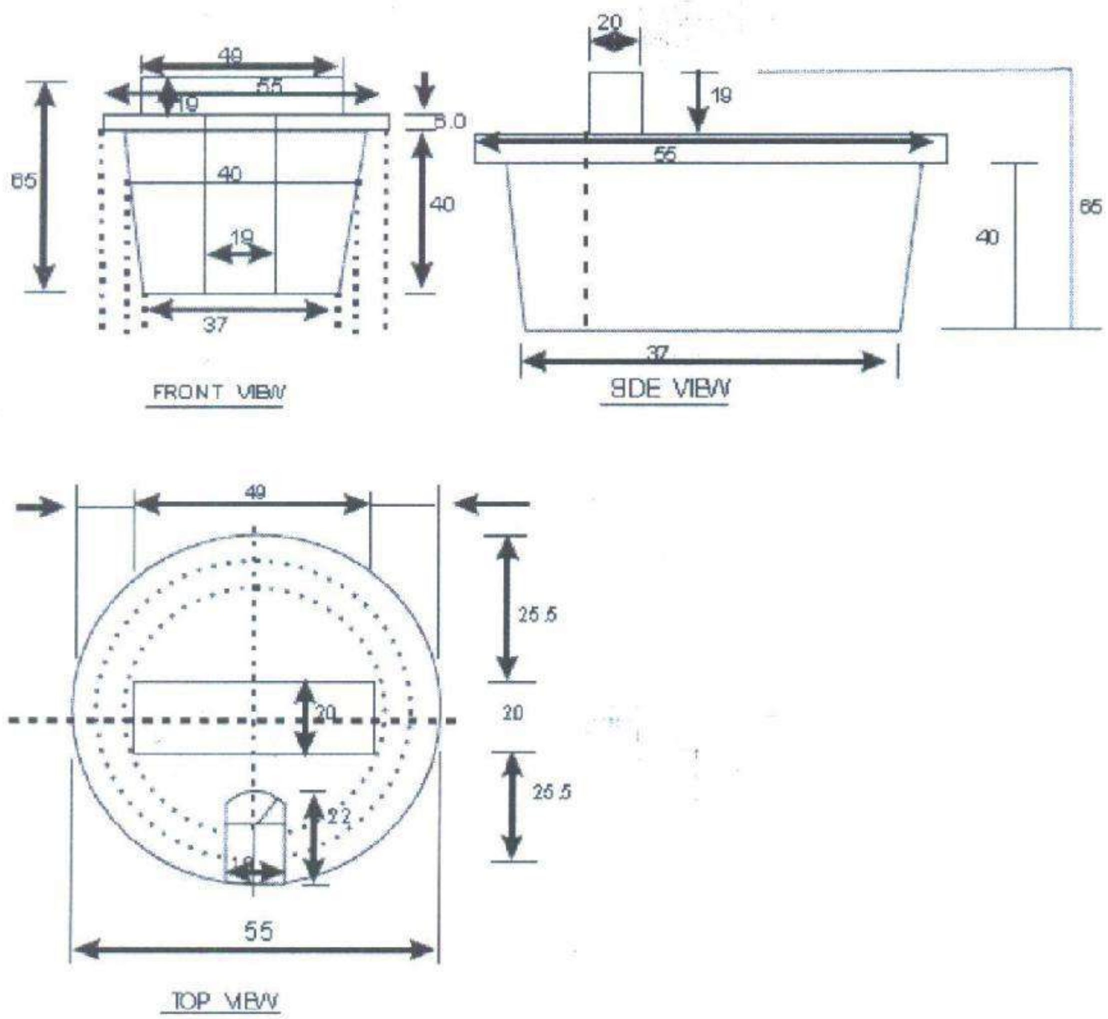
Note: All measurements are in centimetres.

Note:

1. All Dimensions are in mm
2. Dimensions are only for guidance.
3. Tapper should be such that it should be tightly fixed into Type A & Type B HOPE 50 mm OO pipes

Figure-3

Rubber Cork



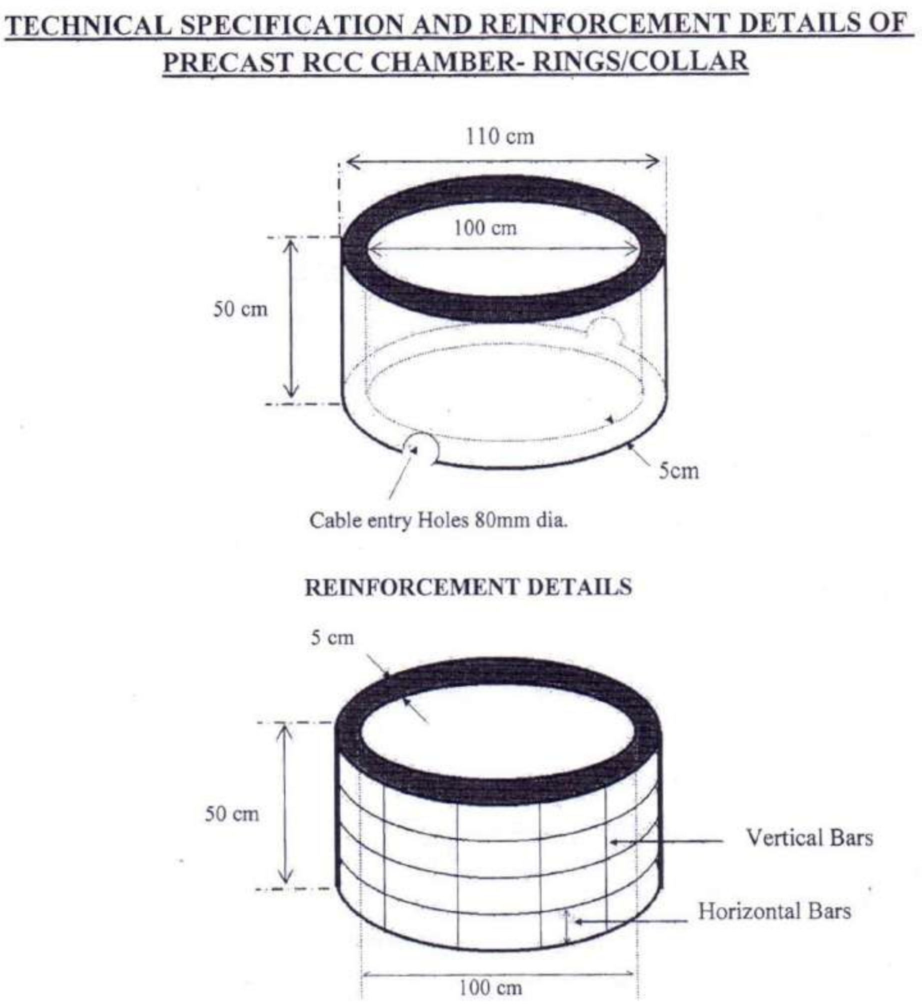
NOTE:

1. ALL DIMENSIONS ARE IN MM.

2. DIMENSIONS ARE ONLY FOR GUIDENCE. TAPPER SHOULD BE SUCH THAT IT SHOULD TIGHTLY FIX. INTO TYPE A & TYPE B HOPE 50 mm OO PIPES.

Figure -4.1

TECHNICAL SPECIFICATION AND REINFORCEMENT DETAILS OF PRECAST RCC CHAMBER – RINGS / COLLAR



Specifications

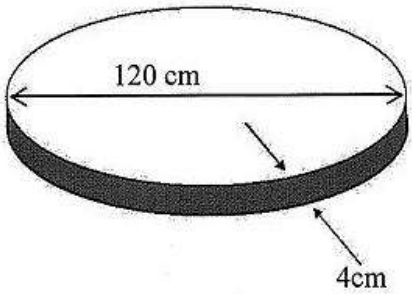
1) Inner Diameter: 100 cm
2) Outer Diameter:110 cm
3) Height: 50 cm (2 rings of 25cm each 0r 1 ring of 50 cm
4) Thickness: 5 cm
5) Two number of 80 mm diameter semi-circular holes for cable entry diametrically opposite to each other at bottom end of the collar.
6) RCC Rings shall be NP-2 Class. The ring may be supplied either as two rings of 25 cm each or as one ring of 50 cm. One single ring shall be preferred.

Reinforcement Details

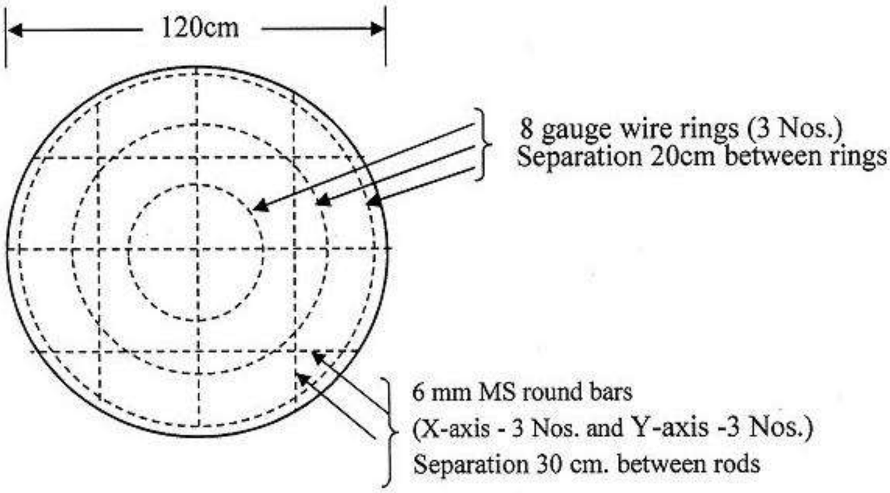
1) 6 mm Horizontal Iron round rings – 4 Nos.
2) 6 mm Vertical bars – 12 mm Nos.
3) 12-gauge GI wire mesh to be wrapped before reinforcing the concrete.
4) Concrete Mix: 1:2:3 (1 cement: 2 sand: 3 graded stone aggregate 20 mm nominal size)
5) Finishing: Smooth

Figure-4.2

TECHNICAL SPECIFICATION AND REINFORCEMENT DETAILS PRECAST RCC CHAMBER-ROUND BASE PLATE



REINFORCEMENT DETAILS

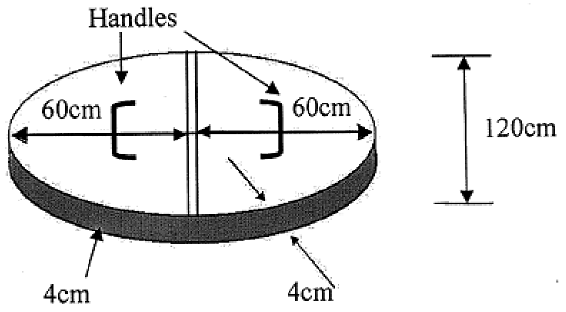


Specifications and Reinforcement Details

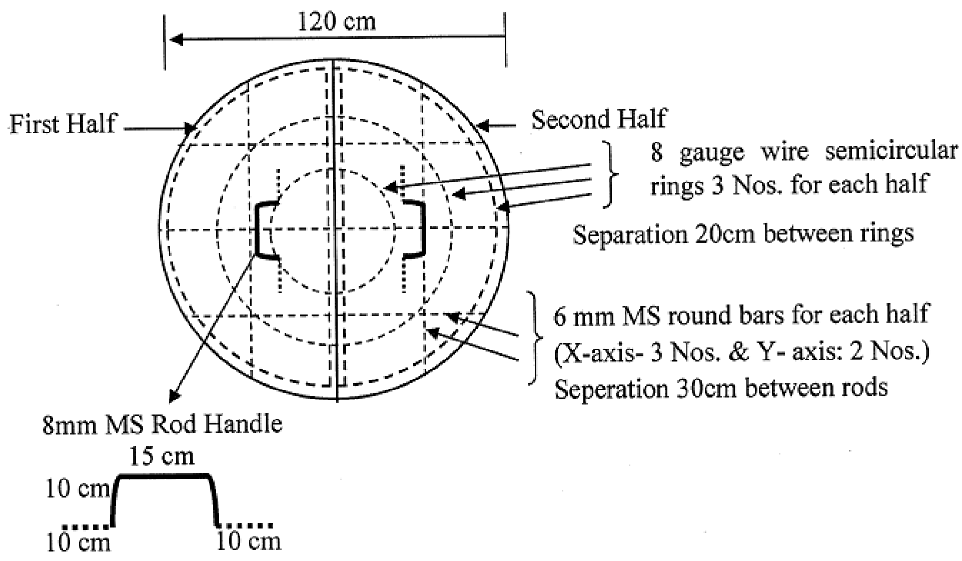
1. 120 cm outer diameter circular plate with 4 cm thickness.
2. 8 gauge wire rings -3 Nos. at equidistance.
3. The separation of 6mm MS round rings shall be about 20cm between the rings.
4. 6 mm MS round bars on X axis - 3Nos and on Y axis - 3Nos.
5. 6 mm MS round bars separation shall be about 30 cm.
6. Concrete Mix – 1:2:3 (1 cement : 2 sand : 3 graded stone aggregate 20 mm nominal size)
7. Finishing: smooth.
8. The base plate can be supplied in two halves also. However in such cases 6 mm MS round bars shall be 3 Nos. on X axis and 2 Nos. on Y axis.

Figure-4.3

TECHNICAL SPECIFICATION AND REINFORCEMENT DETAILS OF PRECAST RCC CHAMBER-ROUND TOP COVER



REINFORCEMENT DETAILS



Specifications and Reinforcement Details

1. Two numbers of semicircular plates of 120 cm diameter (60 cm radius)
2. 8 gauge wire semicircular rings 3 Nos. for each half. Separation between the rings shall 20 cm.
3. 6 mm MS round bars X-axis: 3 Nos. and Y- axis: 2 Nos. for each half. Separation between the bars shall be 30 cm.
4. One handle shall be provided for each half in centre for handling and for smooth opening of joint Chamber. The hooks size shall be of size 15 cm x 10cm of 8 mm MS rod and properly secured with the reinforcement.
5. The word "BBNL OFC" shall be engraved on each half.
6. Concrete Mix : 1:2:3 (1 cement: 2 sand : 3 graded stone aggregate 20mm nominal size)
7. Finishing : smooth

Figure-5

RCC Route Indicator

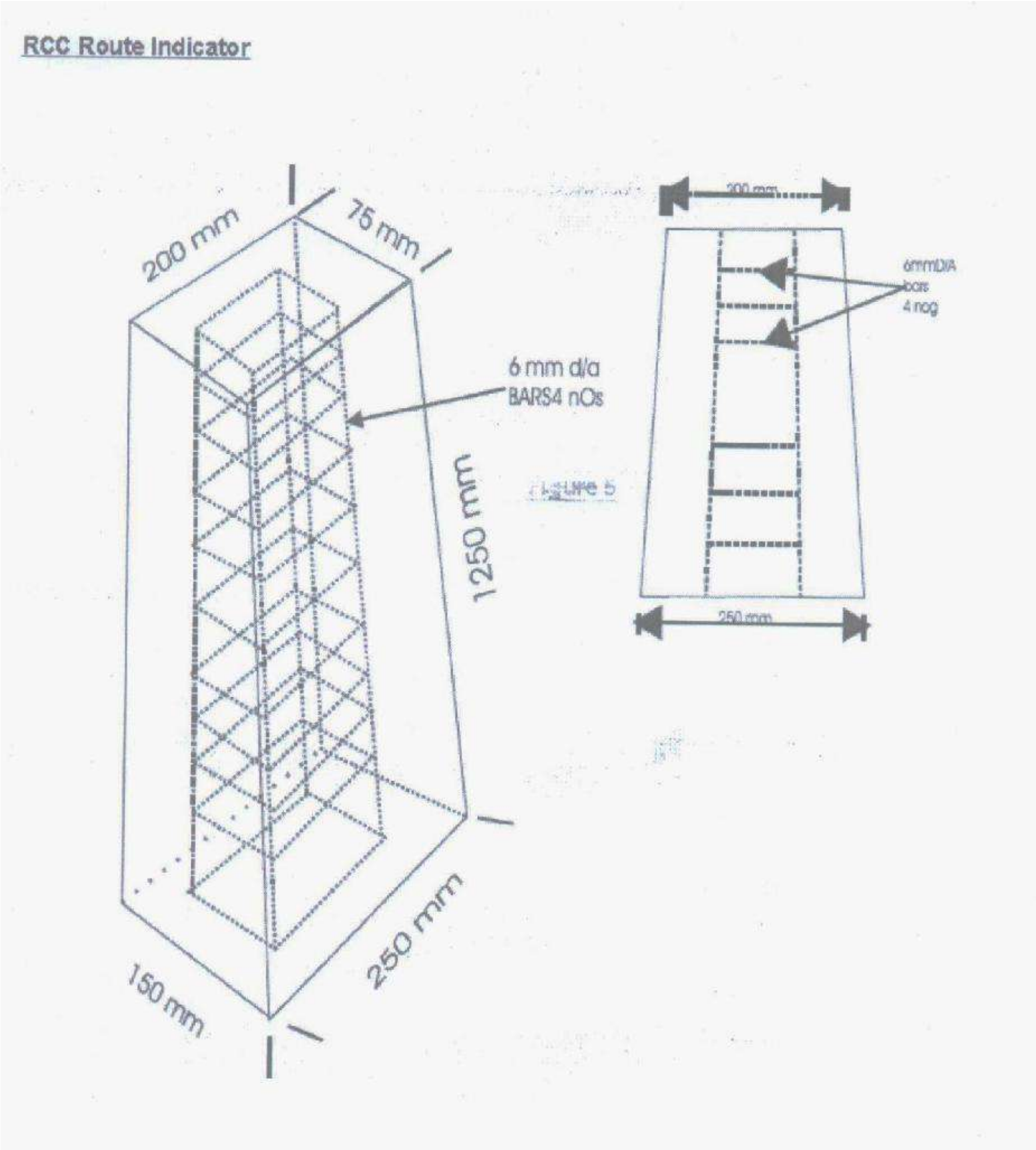
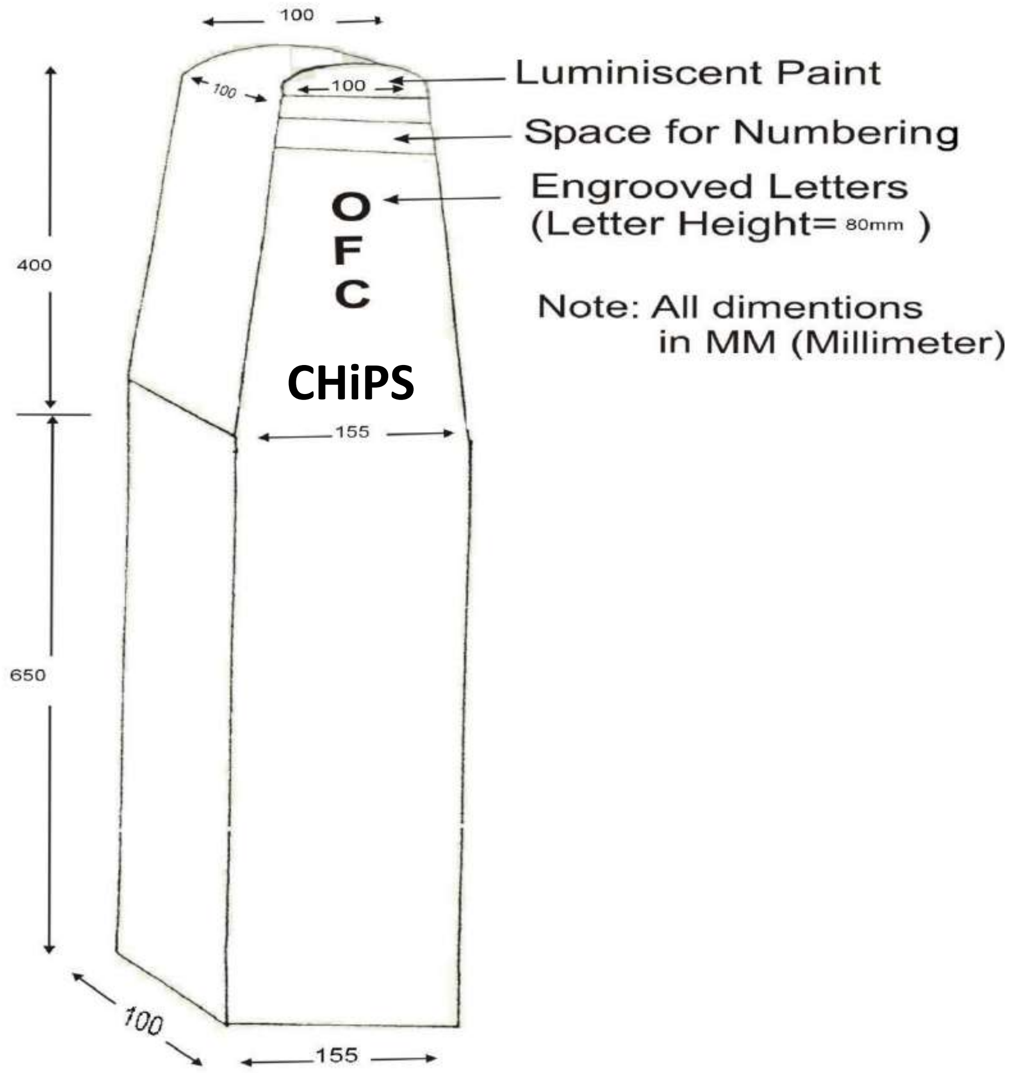


Figure 6

1. Stone OFC Route Indicator



11.19 Annexure-18: Format for Manufacturer Authorization Letter

To,

Chief Executive Officer

Chhattisgarh infotech Promotion Society (CHiPS)

SDC Building, 02nd floor, Near Police Control Room,

Civil Lines, Raipur, Chhattisgarh–492001

Subject: Manufacturer Authorization Letter for Tender Reference No. ………………………………….

Sir,

We, <OEM Name> having our registered office at <OEM address>, hereinafter referred to as OEM are an established manufacturer of the following items quoted by <Bidder Name> having their registered office at <Bidder address>, hereinafter referred to as Bidder.

We <OEM Name> authorize <Bidder's name> to quote our product for above mentioned tender as our Authorized Indian Agent.

We confirm that we have understood the delivery & installation time lines defined in the tender. We confirm that we have worked out all necessary logistics and pricing agreement with <SI name>, and there won't be any delay in delivery, installation and support due to any delay from our side. Our full support as per pre-purchased support contract is extended in all respects for supply and maintenance of our products. We also ensure to provide the required spares and service support as pre-purchased for the supplied equipment for entire contract period. In case of any difficulties in logging complaint at bidder end, user shall have option to log complaint at our call support centre.

We also undertake that in case of default in execution of this Contract by SI, we shall provide necessary extended support to the new partner in accordance with the Contract Terms in identifying another authorized partner with similar certifications/capabilities and extend support to the new partner in accordance with OEM's agreement with the new partner. In case SI is unable to fulfil the obligations given under this Contract, OEM shall be responsible to replace the SI with an alternate Indian Authorized agent to facilitate to get the requisite work done. OEM shall also ensure that the alternate Indian Authorized Agent in this case shall abide by all the terms & conditions laid down under the Contract and during the Award of Work to the SI for the quoted OEM products.

If any product is declared end of sale, we shall proactively ensure that a suitable equivalent or higher roll over product is offered through the existing SI to CHiPS for due approval, contract and order executions thereafter.

We understand that any false information/commitment provided here may result in <OEM's Name> getting blacklisted/debarred from doing business with CHiPS.

Yours sincerely,

Authorized Signature [In full and initials]: _________________

Name and Title of Signatory: ____________________

Name of Firm: _________________

Address: ________________________

Location: _______________________________________ Date: __________________

NOTE:

i. The letter should be submitted on the letter head of the manufacturer / OEM and should be signed by the authorized signatory.

ii. Any deviation would lead to summarily rejection of bid

11.20 Annexure-19: Non-Disclosure Agreement (NDA)

Non- Disclosure AGREEMENT FOR EXCHANGE OF CONFIDENTIAL INFORMATION

This Non-Disclosure Agreement for Exchange of Confidential Information (the "Agreement") is entered into as of _______DD, 2025 (the "Effective Date") by and between Chhattisgarh infotech Promotion Society (CHiPS), hereinafter referred to as **'Nodal Agency'** referred as the Nodal Agency company having its registered office at SDC Building, 02nd floor, Near Police Control Room, Civil Lines, Raipur, Chhattisgarh–492001

And

<***>, a Company incorporated under the Companies Act, 1956, having its registered office at <***> (hereinafter referred to as **'the System Integrator' (SI)** which expression shall, unless the context otherwise requires, include its permitted successors and assigns).

Each of the parties mentioned above are collectively referred to as the 'Parties' and individually as a 'Party'.

Whereas:

1. Nodal Agency is desirous to implement the project of "BharatNet Phase-II in the State of Chhattisgarh"
2. The Nodal Agency and Implementation Agency have entered into a Services
3. Agreement dated <***> (the "MSA") as well as a Service Level Agreement dated <***> (the "SLA") in furtherance of the Project.
4. Whereas in pursuing the Project (the "Business Purpose"), a Party ("Disclosing Party) recognizes that they will disclose certain Confidential Information (as defined hereinafter) to the other Party ("Receiving Party").
5. Whereas such Confidential Information (as defined hereinafter) belongs to Receiving Party as the case may be and is being transferred to the Disclosing Party to be used only for the
6. Business Purpose and hence there is a need to protect such information from unauthorized use and disclosure.

1. Term: This Agreement shall have a term of five (8) years from the Effective Date. Either Party may request for an extension of the Term by giving a renewal notice to the other Party. The Parties may agree to extend the Term of Agreement by an instrument in writing.

2. **SCOPE *OF THE AGREEMENT***

a. *This Agreement shall apply to all confidential and proprietary information disclosed by Disclosing Party to the Receiving Party and other information which the disclosing party identifies in writing or otherwise as confidential before or within (30) thirty days after disclosure to the Receiving Party ("Confidential Information"). Such Confidential Information consists of certain specifications, documents, software, prototypes and/or technical information, and all copies and derivatives containing such Information* *that may be disclosed to the Disclosing Party for and during the Business Purpose, which a party considers proprietary or confidential.*

b. Such Confidential Information may be in any form or medium, tangible or intangible, and may be communicated/disclosed in writing, orally, or through visual observation or by any other means to the Receiving Party.

3. Purpose: The Parties intend to share Confidential Information for a potential business relationship with respect to implementation and O&M of "BharatNet Phase-II project in the State of Chhattisgarh". ("Purpose").

4. Discloser & Recipient: Either Party, including its Affiliates, may disclose Confidential Information under this Agreement for the Purpose and shall be referred to as "Discloser" hereunder. The other Party, including its Affiliates, receiving Confidential Information hereunder shall be referred to as "Recipient". For the purpose of this Agreement, "Affiliates" shall mean any legal entity which, is directly or indirectly controlling, controlled by or under the common control of the Party.

5. Confidential Information: The information disclosed by Discloser to Recipient hereunder relating to Discloser's business, including, without limitation, computer programs, technical drawings, algorithms, know-how, processes, designs, reports, specifications, ideas, trade secrets, inventions, schematics, pricing information, and other technical, business, financial, customer and product development plans, strategies or any other information which is reasonably understood to be confidential or proprietary based on the circumstances of disclosure or the nature of the information itself, such information is hereinafter referred to as "Confidential Information" of the Discloser.

6. Information which is orally or visually disclosed, or is disclosed in writing without being marked as confidential, shall constitute Confidential Information, if Discloser within seven (7) days after such disclosure, delivers to Recipient, a written document(s) describing such Information and referencing the place and date of such oral or visual disclosure and the names of the employees or officers of the Recipient to whom such disclosure was made.

7. Confidential Information shall not include any information that is a) lawfully known by the Recipient at the time of disclosure without any obligation to keep the same confidential; b) or becomes, through no fault of the Recipient, known or available to the public; c) independently developed by the Recipient without use or reference to such Confidential Information; or d) rightfully disclosed to Recipient by a third party without any restrictions on disclosure.

8. Confidentiality Obligation: Discloser shall observe the duty of reasonable care while disclosing any Confidential Information to the Recipient. Recipient agrees that it shall a) not use any such Confidential Information except for the Purpose of this Agreement; b) hold the Confidential Information in confidence and shall take all reasonable precautions to protect such Confidential Information from unauthorized disclosure including all precautions that Recipient employs to protect its own confidential material; c) not divulge any such Confidential Information to any third party without prior approval of Discloser; and d) not copy or reverse engineer any such Confidential Information. Recipient may permit access to Confidential Information to its employees, consultants, vendors and agents, on a need to know basis and to the extent required to meet the Purpose, and shall ensure that they are bound to maintain confidentiality of such Confidential Information to the same extent as provided under this Agreement.

9. Survival, Exception & Return: Confidentiality obligations under this Agreement shall survive for a period of five (5) years following the expiry of this Agreement, provided that the obligations shall be perpetual with regard to any source code or trade secret that may be disclosed hereunder.

10. Recipient may make disclosures to the extent required by law or by order of any court or regulatory body, provided the Recipient promptly notifies the Discloser in writing about such requirement to disclose.

11. Recipient will return to Discloser, upon request, any Confidential Information under its possession or control and/or destroy all documents or media containing any such Confidential Information provided that Recipient may retain a copy of Confidential Information to the extent necessary to meet any statutory requirements.

12. Disclaimer: Parties acknowledges that providing or receiving Confidential Information under this Agreement shall not constitute an offer, acceptance, or promise to enter into or amend any other contract.

13. To the extent permitted by law, Confidential Information is disclosed on "as is" basis, without any express or implied warranties and in particular, without any limitation, as to fitness for the intended Purpose.

14. The ownership of all intellectual property rights (IPRs) in Confidential Information disclosed hereunder shall remain with its original owner and no grant of license or conveyance of any IPRs in such Confidential Information is to be implied from exchange or sharing of any such information under this Agreement.

15. Injunctive Relief: Recipient acknowledges that due to the unique nature of the Discloser's Confidential Information, any breach of its obligations hereunder will result in irreparable harm to the Discloser, and therefore, upon any such breach or threat thereof, the Discloser shall be entitled to appropriate equitable relief including the relief of injunction and/or specific performance, in addition to any other remedies available at law.

16. General: The Parties agree to be bound by any applicable export control regulations while sharing Confidential Information hereunder.

17. This Agreement shall be governed by the laws of India and shall be subject to the exclusive jurisdiction of courts in Delhi.

Neither party may assign or transfer any rights or obligations arising out of this Agreement without the prior written consent of the other party.

No failure or delay in enforcing any right will be deemed a waiver unless made in writing and signed by a duly authorized representative of such Party.

Any notice under this Agreement shall be in writing and shall be sent at the registered addresses of the Parties specified in this Agreement.

This Agreement may be modified only by an amendment executed in writing by a duly authorized representative of both Parties.

This Agreement constitutes the entire agreement between the Parties and supersedes all prior discussions or agreements relating to subject matter hereof.

For Chhattisgarh infotech Promotion Society (CHiPS),

Signature: ____

Name: ____

Designation: ____

Date:

____

For System Integrator' (SI)

Signature: ____

Name: ____

Designation: ____

Date:

____

11.21 Annexure-20: Type of Network & Electrical Equipment

List of Equipment already installed under BharatNet Phase II Project:

S/No	Equipment Name	Location	Model
1	Core Router	SDC	Cisco ASR 9010
2	Block/Aggregation Router (High)	Block	Cisco ASR 9006
3	Block/Aggregation Router (Low)	Block	Cisco ASR 9006
4	Amplifier Router	GP	Cisco ASR 903
5	Edge Router	GP	Cisco ASR 920
6	Hyperflex	SNOC	Cisco HyperFlex HX240c M5

Note: The Bidders may enquire with CHiPS regarding specification of any of the active and passive equipment installed in either GP, Block and SNOC.

**** End of Document ****